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(54) **COMPOSITIONS FOR MODULATING C9ORF72 EXPRESSION**

(71) Applicant: **IONIS PHARMACEUTICALS, INC.**, Carlsbad, CA (US)

(72) Inventors: **Susan M. Freier**, San Diego, CA (US); **Frank Rigo**, Carlsbad, CA (US); **Priyam Singh**, Brisbane, CA (US)

(73) Assignee: **Ionis Pharmaceuticals, Inc.**, Carlsbad, CA (US)

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(57) **ABSTRACT**

Disclosed herein are compositions and methods for reducing expression of C90RF72 mRNA and protein in an animal with C90RF72 specific inhibitors. Also disclosed herein are compositions and methods of selectively inhibiting a C90RF72 pathogenic associated mRNA variant by administering an antisense compound targeting the region beginning at the start site of exon 1A to the start site of exon 1B of a C90RF72 pre-mRNA. Such methods are useful to treat, prevent, or ameliorate neurodegenerative diseases in an individual in need thereof. Such C90RF72 specific inhibitors include antisense compounds.

COMPOSITIONS FOR MODULATING C9ORF72 EXPRESSION

SEQUENCE LISTING

[0001] The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled BIOL0235WOSEQ_ST25.txt created Oct. 11, 2014, which is 444 Kb in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

FIELD

[0002] Provided are compositions and methods for reducing expression of C9ORF72 mRNA and protein in an animal. Such methods are useful to treat, prevent, or ameliorate neurodegenerative diseases, including amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), corticalbasal degeneration syndrome (CBD), atypical Parkinsonian syndrome, and olivopontocerellar degeneration (OPCD).

BACKGROUND

[0003] Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative disease characterized clinically by progressive paralysis leading to death from respiratory failure, typically within two to three years of symptom onset (Rowland and Shneider, N. Engl. J. Med., 2001, 344, 1688-1700). ALS is the third most common neurodegenerative disease in the Western world (Hirtz et al., Neurology, 2007, 68, 326-337), and there are currently no effective therapies. Approximately 10% of cases are familial in nature, whereas the bulk of patients diagnosed with the disease are classified as sporadic as they appear to occur randomly throughout the population (Chio et al., Neurology, 2008, 70, 533-537). There is growing recognition, based on clinical, genetic, and epidemiological data, that ALS and frontotemporal dementia (FTD) represent an overlapping continuum of disease, characterized pathologically by the presence of TDP-43 positive inclusions throughout the central nervous system (Lino and Hodges, J. Clin. Neurosci., 2009, 16, 1131-1135; Neumann et al., Science, 2006, 314, 130-133).

[0004] To date, a number of genes have been discovered as causative for classical familial ALS, for example, SOD1, TARDBP, FUS, OPTN, and VCP (Johnson et al., Neuron, 2010, 68, 857-864; Kwiatkowski et al., Science, 2009, 323, 1205-1208; Maruyama et al., Nature, 2010, 465, 223-226; Rosen et al., Nature, 1993, 362, 59-62; Sreedharan et al., Science, 2008, 319, 1668-1672; Vance et al., Brain, 2009, 129, 868-876). Recently, linkage analysis of kindreds involving multiple cases of ALS, FTD, and ALS-FTD had suggested that there was an important locus for the disease on the short arm of chromosome 9 (Boxer et al., J. Neurol. Neurosurg. Psychiatry, 2011, 82, 196-203; Morita et al., Neurology, 2006, 66, 839-844; Pearson et al. J. Nerol., 2011, 258, 647-655; Vance et al., Brain, 2006, 129, 868-876). The mutation in the C9ORF72 gene is the most common genetic cause of ALS and FTD. The ALS-FTD causing mutation is a large hexanucleotide (GGGGCC) repeat expansion in the first intron of the C9ORF72 gene (Renton et al., Neuron, 2011, 72, 257-268; DeJesus-Hernandez et al., Neuron, 2011, 72, 245-256). A founder haplotype, covering the C9ORF72 gene, is present in the majority of cases linked to this region (Renton et al., Neuron, 2011, 72, 257-268). This locus on chromosome 9p21 accounts for nearly half of familial ALS and nearly one-quarter of all ALS cases in a cohort of 405 Finnish patients (Laaksovirta et al, Lancet Neurol., 2010, 9, 978-985).

[0005] A founder haplotype, covering the C9ORF72 gene, is present in the majority of cases linked to this region.

[0006] There are currently no effective therapies to treat such neurodegenerative diseases. Therefore, it is an object to provide compositions and methods for the treatment of such neurodegenerative diseases.

SUMMARY

[0007] Provided herein are compositions and methods for modulating levels of C9ORF72 mRNA and protein in cells, tissues, and animals. In certain embodiments, C9ORF72 specific inhibitors modulate expression of C9ORF72 mRNA and protein. In certain embodiments, C9ORF72 specific inhibitors are nucleic acids, proteins, or small molecules.

[0008] In certain embodiments, modulation can occur in a cell or tissue. In certain embodiments, the cell or tissue is in an animal. In certain embodiments, the animal is a human. In certain embodiments, C9ORF72 mRNA levels are reduced. In certain embodiments, C9ORF72 protein levels are reduced. In certain embodiments, C9ORF72 associated Repeat Associated Non-ATG Translation (RAN translation) products are reduced. In certain embodiments, the C9ORF72 associated RAN translation products are poly-(glycine-proline), poly-(glycine-alanine), and poly-(glycine-arginine) In certain embodiments, certain C9ORF72 mRNA variants are preferentially reduced. In certain embodiments, the C9ORF72 mRNA variants preferentially reduced are variants processed from a pre-mRNA containing intron 1. In certain embodiments, intron 1 contains a hexanucleotide repeat expansion. In certain embodiments, the C9ORF72 mRNA variant preferentially reduced is a C9ORF72 pathogenic associated mRNA variant. In certain embodiments, the C9ORF72 pathogenic associated mRNA variant is NM_001256054.1 (SEQ ID NO: 1). In certain embodiments, the hexanucleotide repeat expansion is associated with a C9ORF72 associated disease. In certain embodiments, the hexanucleotide repeat expansion is associated with a C9ORF72 hexanucleotide repeat expansion associated disease. In certain embodiments, the hexanucleotide repeat expansion comprises at least 24 GGGGCC repeats. In certain embodiments, the hexanucleotide repeat expansion is associated with nuclear foci. In certain embodiments, C9ORF72 associated RAN translation products are associated with nuclear foci. In certain embodiments, the C9ORF72 associated RAN translation products are poly-(glycine-proline), poly-(glycine-alanine), and poly-(glycine-arginine). In certain embodiments, the compositions and methods described herein are useful for reducing C9ORF72 mRNA levels, C9ORF72 protein levels, C9ORF72 RAN translation products, and nuclear foci. In certain embodiments, the compositions and methods described herein are useful for selectively reducing C9ORF72 pathogenic associated mRNA variants. Such reduction can occur in a time-dependent manner or in a dose-dependent manner.

[0009] Also provided are methods useful for preventing, treating, and ameliorating diseases, disorders, and conditions associated with C9ORF72. In certain embodiments, such diseases, disorders, and conditions associated with C9ORF72 are neurodegenerative diseases. In certain embodiments, the neurodegenerative disease is amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), corticalbasal degen-

eration syndrome (CBD), atypical Parkinsonian syndrome, and olivopontocerellar degeneration (OPCD).

[0010] Such diseases, disorders, and conditions can have one or more risk factors, causes, or outcomes in common. Certain risk factors and causes for development of a neurodegenerative disease, and, in particular, ALS and FTD, include genetic predisposition and older age.

[0011] In certain embodiments, methods of treatment include administering a C9ORF72 specific inhibitor to an individual in need thereof. In certain embodiments, the C9ORF72 specific inhibitor is a nucleic acid. In certain embodiments, the nucleic acid is an antisense compound. In certain embodiments, the antisense compound is a single-stranded antisense oligonucleotide. In certain embodiments, the single-stranded antisense oligonucleotide is complementary to a C9ORF72 nucleic acid.

DETAILED DESCRIPTION

[0012] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed. Herein, the use of the singular includes the plural unless specifically stated otherwise. As used herein, the use of "or" means "and/or" unless stated otherwise. Additionally, as used herein, the use of "and" means "and/or" unless stated otherwise. Furthermore, the use of the term "including" as well as other forms, such as "includes" and "included", is not limiting. Also, terms such as "element" or "component" encompass both elements and components comprising one unit and elements and components that comprise more than one subunit, unless specifically stated otherwise.

[0013] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this disclosure, including, but not limited to, patents, patent applications, published patent applications, articles, books, treatises, and GENBANK Accession Numbers and associated sequence information obtainable through databases such as National Center for Biotechnology Information (NCBI) and other data referred to throughout in the disclosure herein are hereby expressly incorporated by reference for the portions of the document discussed herein, as well as in their entirety.

DEFINITIONS

[0014] Unless specific definitions are provided, the nomenclature utilized in connection with, and the procedures and techniques of, analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well known and commonly used in the art. Standard techniques may be used for chemical synthesis, and chemical analysis.

[0015] Unless otherwise indicated, the following terms have the following meanings:

[0016] "2'-O-methoxyethyl" (also 2'-MOE and 2'-OCH_1CH_2—OCH_3 and MOE) refers to an 0-methoxyethyl modification of the 2' position of a furanose ring. A 2'-O-methoxyethyl modified sugar is a modified sugar.

[0017] "2'-MOE nucleoside" (also 2'-O-methoxyethyl nucleoside) means a nucleoside comprising a MOE modified sugar moiety.

[0018] "2'-substituted nucleoside" means a nucleoside comprising a substituent at the 2'-position of the furanose ring other than H or OH. In certain embodiments, 2'-substituted nucleosides include nucleosides with bicyclic sugar modifications.

[0019] "5-methylcytosine" means a cytosine modified with a methyl group attached to the 5' position. A 5-methylcytosine is a modified nucleobase.

[0020] "About" means within ±7% of a value. For example, if it is stated, "the compounds affected at least about 70% inhibition of C9ORF72", it is implied that the C9ORF72 levels are inhibited within a range of 63% and 77%.

[0021] "Administered concomitantly" refers to the co-administration of two pharmaceutical agents in any manner in which the pharmacological effects of both are manifest in the patient at the same time. Concomitant administration does not require that both pharmaceutical agents be administered in a single pharmaceutical composition, in the same dosage form, or by the same route of administration. The effects of both pharmaceutical agents need not manifest themselves at the same time. The effects need only be overlapping for a period of time and need not be coextensive.

[0022] "Administering" means providing a pharmaceutical agent to an animal, and includes, but is not limited to administering by a medical professional and self-administering.

[0023] "Amelioration" refers to a lessening, slowing, stopping, or reversing of at least one indicator of the severity of a condition or disease. The severity of indicators may be determined by subjective or objective measures, which are known to those skilled in the art.

[0024] "Animal" refers to a human or non-human animal, including, but not limited to, mice, rats, rabbits, dogs, cats, pigs, and non-human primates, including, but not limited to, monkeys and chimpanzees.

[0025] "Antibody" refers to a molecule characterized by reacting specifically with an antigen in some way, where the antibody and the antigen are each defined in terms of the other. Antibody may refer to a complete antibody molecule or any fragment or region thereof, such as the heavy chain, the light chain, Fab region, and Fc region.

[0026] "Antisense activity" means any detectable or measurable activity attributable to the hybridization of an antisense compound to its target nucleic acid. In certain embodiments, antisense activity is a decrease in the amount or expression of a target nucleic acid or protein encoded by such target nucleic acid.

[0027] "Antisense compound" means an oligomeric compound that is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding. Examples of antisense compounds include single-stranded and double-stranded compounds, such as, antisense oligonucleotides, siRNAs, shRNAs, ssRNAs, and occupancy-based compounds.

[0028] "Antisense inhibition" means reduction of target nucleic acid levels in the presence of an antisense compound complementary to a target nucleic acid compared to target nucleic acid levels or in the absence of the antisense compound.

[0029] "Antisense mechanisms" are all those mechanisms involving hybridization of a compound with a target nucleic acid, wherein the outcome or effect of the hybridization is either target degradation or target occupancy with concomitant stalling of the cellular machinery involving, for example, transcription or splicing.

[0030] "Antisense oligonucleotide" means a single-stranded oligonucleotide having a nucleobase sequence that permits hybridization to a corresponding segment of a target nucleic acid.

[0031] "Base complementarity" refers to the capacity for the precise base pairing of nucleobases of an antisense oligonucleotide with corresponding nucleobases in a target nucleic acid (i.e., hybridization), and is mediated by Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen binding between corresponding nucleobases.

[0032] "Bicyclic sugar" means a furanose ring modified by the bridging of two atoms. A bicyclic sugar is a modified sugar.

[0033] "Bicyclic nucleoside" (also BNA) means a nucleoside having a sugar moiety comprising a bridge connecting two carbon atoms of the sugar ring, thereby forming a bicyclic ring system. In certain embodiments, the bridge connects the 4'-carbon and the 2'-carbon of the sugar ring.

[0034] "C9ORF72 associated disease" means any disease associated with any C9ORF72 nucleic acid or expression product thereof. Such diseases may include a neurodegenerative disease. Such neurodegenerative diseases may include ALS and FTD.

[0035] "C9ORF72 associated RAN translation products" means aberrant peptide or di-peptide polymers translated through RAN translation (i.e., repeat-associated, and non-ATG-dependent translation). In certain embodiments, the C9ORF72 associated RAN translation products are any of poly-(glycine-proline), poly-(glycine-alanine), and poly-(glycine-arginine)

[0036] "C9ORF72 hexanucleotide repeat expansion associated disease" means any disease associated with a C9ORF72 nucleic acid containing a hexanucleotide repeat expansion. In certain embodiments, the hexanucleotide repeat expansion may comprise GGGGCC, GGGGGG, GGGGGC, or GGGGCG repeated at least 24 times. Such diseases may include a neurodegenerative disease. Such neurodegenerative diseases may include ALS and FTD.

[0037] "C9ORF72 nucleic acid" means any nucleic acid encoding C9ORF72. For example, in certain embodiments, a C9ORF72 nucleic acid includes a DNA sequence encoding C9ORF72, an RNA sequence transcribed from DNA encoding C9ORF72 including genomic DNA comprising introns and exons (i.e., pre-mRNA), and an mRNA sequence encoding C9ORF72. "C9ORF72 mRNA" means an mRNA encoding a C9ORF72 protein.

[0038] "C9ORF72 pathogenic associated mRNA variant" means the C9ORF72 mRNA variant processed from a C9ORF72 pre-mRNA variant containing the hexanucleotide repeat. A C9ORF72 pre-mRNA contains the hexanucleotide repeat when transcription of the pre-mRNA begins in the region from the start site of exon 1A to the start site of exon 1B, e.g., nucleotides 1107 to 1520 of the genomic sequence (SEQ ID NO: 2, the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, the level of a C9ORF72 pathogenic associated mRNA variant is measured to determine the level of a C9ORF72 pre-mRNA containing the hexanucleotide repeat in a sample.

[0039] "C9ORF72 specific inhibitor" refers to any agent capable of specifically inhibiting the expression of C9ORF72 mRNA and/or C9ORF72 protein at the molecular level. For example, C9ORF72 specific inhibitors include nucleic acids (including antisense compounds), siRNAs, aptamers, antibodies, peptides, small molecules, and other agents capable of inhibiting the expression of C9ORF72 mRNA and/or C9ORF72 protein. Similarly, in certain embodiments, C9ORF72 specific inhibitors may affect other molecular processes in an animal.

[0040] "Cap structure" or "terminal cap moiety" means chemical modifications, which have been incorporated at either terminus of an antisense compound.

[0041] "cEt" or "constrained ethyl" means a bicyclic nucleoside having a sugar moiety comprising a bridge connecting the 4'-carbon and the 2'-carbon, wherein the bridge has the formula: 4'-CH(CH_3)—O-2'.

[0042] "Constrained ethyl nucleoside" (also cEt nucleoside) means a nucleoside comprising a bicyclic sugar moiety comprising a 4'-CH(CH_3)—O-2' bridge.

[0043] "Chemically distinct region" refers to a region of an antisense compound that is in some way chemically different than another region of the same antisense compound. For example, a region having 2'-O-methoxyethyl nucleosides is chemically distinct from a region having nucleosides without 2'-O-methoxyethyl modifications.

[0044] "Chimeric antisense compound" means an antisense compound that has at least two chemically distinct regions, each position having a plurality of subunits.

[0045] "Co-administration" means administration of two or more pharmaceutical agents to an individual. The two or more pharmaceutical agents may be in a single pharmaceutical composition, or may be in separate pharmaceutical compositions. Each of the two or more pharmaceutical agents may be administered through the same or different routes of administration. Co-administration encompasses parallel or sequential administration.

[0046] "Complementarity" means the capacity for pairing between nucleobases of a first nucleic acid and a second nucleic acid.

[0047] "Comprise," "comprises," and "comprising" will be understood to imply the inclusion of a stated step or element or group of steps or elements but not the exclusion of any other step or element or group of steps or elements.

[0048] "Contiguous nucleobases" means nucleobases immediately adjacent to each other. "Designing" or "designed to" refer to the process of designing an oligomeric compound that specifically hybridizes with a selected nucleic acid molecule.

[0049] "Diluent" means an ingredient in a composition that lacks pharmacological activity, but is pharmaceutically necessary or desirable. For example, in drugs that are injected, the diluent may be a liquid, e.g. saline solution.

[0050] "Dose" means a specified quantity of a pharmaceutical agent provided in a single administration, or in a specified time period. In certain embodiments, a dose may be administered in one, two, or more boluses, tablets, or injections. For example, in certain embodiments where subcutaneous administration is desired, the desired dose requires a volume not easily accommodated by a single injection, therefore, two or more injections may be used to achieve the desired dose. In certain embodiments, the pharmaceutical agent is administered by infusion over an extended period of time or continuously. Doses may be stated as the amount of pharmaceutical agent per hour, day, week, or month.

[0051] "Effective amount" in the context of modulating an activity or of treating or preventing a condition means the administration of that amount of pharmaceutical agent to a subject in need of such modulation, treatment, or prophylaxis,

either in a single dose or as part of a series, that is effective for modulation of that effect, or for treatment or prophylaxis or improvement of that condition. The effective amount may vary among individuals depending on the health and physical condition of the individual to be treated, the taxonomic group of the individuals to be treated, the formulation of the composition, assessment of the individual's medical condition, and other relevant factors.

[0052] "Efficacy" means the ability to produce a desired effect.

[0053] "Expression" includes all the functions by which a gene's coded information is converted into structures present and operating in a cell. Such structures include, but are not limited to the products of transcription and translation.

[0054] "Foci" means a nuclear foci comprising a C9ORF72 transcript. In certain embodiments, a foci comprises at least one C9ORF72 transcript. In certain embodiments, C9ORF72 foci comprise transcripts comprising any of the following hexanucleotide repeats: GGGGCC, GGGGGG, GGGGGC, and/or GGGGCG.

[0055] "Fully complementary" or "100% complementary" means each nucleobase of a first nucleic acid has a complementary nucleobase in a second nucleic acid. In certain embodiments, a first nucleic acid is an antisense compound and a target nucleic acid is a second nucleic acid.

[0056] "Gapmer" means a chimeric antisense compound in which an internal region having a plurality of nucleosides that support RNase H cleavage is positioned between external regions having one or more nucleosides, wherein the nucleosides comprising the internal region are chemically distinct from the nucleoside or nucleosides comprising the external regions. The internal region may be referred to as a "gap" and the external regions may be referred to as the "wings."

[0057] "Gap-narrowed" means a chimeric antisense compound having a gap segment of 9 or fewer contiguous 2'-deoxyribonucleosides positioned between and immediately adjacent to 5' and 3' wing segments having from 1 to 6 nucleosides.

[0058] "Gap-widened" means a chimeric antisense compound having a gap segment of 12 or more contiguous 2'-deoxyribonucleosides positioned between and immediately adjacent to 5' and 3' wing segments having from 1 to 6 nucleosides.

[0059] "Hexanucleotide repeat expansion" means a series of six bases (for example, GGGGCC, GGGGGG, GGGGCG, or GGGGGC) repeated at least twice. In certain embodiments, the hexanucleotide repeat expansion may be located in intron 1 of a C9ORF72 nucleic acid. In certain embodiments, a pathogenic hexanucleotide repeat expansion includes at least 24 repeats of GGGGCC, GGGGGG, GGGGCG, or GGGGGC in a C9ORF72 nucleic acid and is associated with disease. In certain embodiments, the repeats are consecutive. In certain embodiments, the repeats are interrupted by 1 or more nucleobases. In certain embodiments, a wild-type hexanucleotide repeat expansion includes 23 or fewer repeats of GGGGCC, GGGGGG, GGGGCG, or GGGGGC in a C9ORF72 nucleic acid. In certain embodiments, the repeats are consecutive. In certain embodiments, the repeats are interrupted by 1 or more nucleobases.

[0060] "Hybridization" means the annealing of complementary nucleic acid molecules. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense compound and a target nucleic acid. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense oligonucleotide and a nucleic acid target.

[0061] "Identifying an animal having a C9ORF72 associated disease" means identifying an animal having been diagnosed with a C9ORF72 associated disease or predisposed to develop a C9ORF72 associated disease. Individuals predisposed to develop a C9ORF72 associated disease include those having one or more risk factors for developing a C9ORF72 associated disease, including, having a personal or family history or genetic predisposition of one or more C9ORF72 associated diseases. Such identification may be accomplished by any method including evaluating an individual's medical history and standard clinical tests or assessments, such as genetic testing.

[0062] "Immediately adjacent" means there are no intervening elements between the immediately adjacent elements.

[0063] "Individual" means a human or non-human animal selected for treatment or therapy.

[0064] "Inhibiting C9ORF72" means reducing the level or expression of a C9ORF72 mRNA and/or protein. In certain embodiments, C9ORF72 mRNA and/or protein levels are inhibited in the presence of an antisense compound targeting C9ORF72, including an antisense oligonucleotide targeting C9ORF72, as compared to expression of C9ORF72 mRNA and/or protein levels in the absence of a C9ORF72 antisense compound, such as an antisense oligonucleotide.

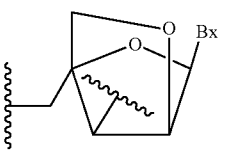
[0065] "Inhibiting the expression or activity" refers to a reduction or blockade of the expression or activity and does not necessarily indicate a total elimination of expression or activity.

[0066] "Internucleoside linkage" refers to the chemical bond between nucleosides.

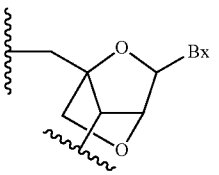
[0067] "Linked nucleosides" means adjacent nucleosides linked together by an internucleoside linkage.

[0068] "Locked nucleic acid" or "LNA" or "LNA nucleosides" means nucleic acid monomers having a bridge connecting two carbon atoms between the 4' and 2'position of the nucleoside sugar unit, thereby forming a bicyclic sugar. Examples of such bicyclic sugar include, but are not limited to A) α-L-Methyleneoxy (4'-CH_2—O-2') LNA, (B) β-D-Methyleneoxy (4'-CH_2—O-2') LNA, (C) Ethyleneoxy (4'-$(CH_2)_2$—O-2') LNA, (D) Aminooxy (4'-CH_2—O—N(R)-2') LNA and (E) Oxyamino (4'-CH_2—N(R)—O-2') LNA, as depicted below.

(A)

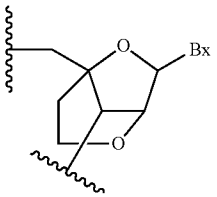


(B)

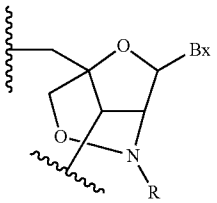


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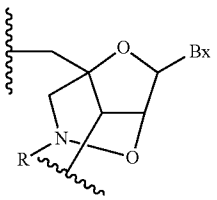
(C)



(D)



(E)



[0069] As used herein, LNA compounds include, but are not limited to, compounds having at least one bridge between the 4' and the 2' position of the sugar wherein each of the bridges independently comprises 1 or from 2 to 4 linked groups independently selected from —[$C(R_1)(R_2)]_n$—, —$C(R_1)$=$C(R_2)$—, —$C(R_1)$=N—, —C(=NR_1)—, —C(=O)—, —C(=S)—, —O—, —$Si(R_1)_2$—, —S(=O)$_x$— and —$N(R_1)$—; wherein: x is 0, 1, or 2; n is 1, 2, 3, or 4; each R_1 and R_2 is, independently, H, a protecting group, hydroxyl, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_5-C_{20} aryl, substituted C_5-C_{20} aryl, a heterocycle radical, a substituted heterocycle radical, heteroaryl, substituted heteroaryl, C_5-C_7 alicyclic radical, substituted C_5-C_7 alicyclic radical, halogen, OJ_1, NJ_1J_2, SJ_1, N_3, $COOJ_1$, acyl (C(=O)—H), substituted acyl, CN, sulfonyl (S(=O)$_2$-J_1), or sulfoxyl (S(=O)-J_1); and each J_1 and J_2 is, independently, H, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_5-C_{20} aryl, substituted C_5-C_{20} aryl, acyl (C(=O)—H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C_1-C_{12} aminoalkyl, substituted C_1-C_{12} aminoalkyl or a protecting group.

[0070] Examples of 4'-2' bridging groups encompassed within the definition of LNA include, but are not limited to one of formulae: —$[C(R_1)(R_2)]_n$—, —$[C(R_1)(R_2)]_n$—O—, —$C(R_1R_2)$—$N(R_1)$—O— or —$C(R_1R_2)$—O—$N(R_1)$—. Furthermore, other bridging groups encompassed with the definition of LNA are 4'-CH_2-2', 4'-$(CH_2)_2$-2', 4'-$(CH_2)_3$-2', 4'-CH_2—O-2', 4'-$(CH_2)_2$—O-2', 4'-CH_2—O—$N(R_1)$-2' and 4'-CH_2—$N(R_1)$—O-2'- bridges, wherein each R_1 and R_2 is, independently, H, a protecting group or C_1-C_{12} alkyl.

[0071] Also included within the definition of LNA according to the invention are LNAs in which the 2'-hydroxyl group of the ribosyl sugar ring is connected to the 4' carbon atom of the sugar ring, thereby forming a methyleneoxy (4'-CH_2—O-2') bridge to form the bicyclic sugar moiety. The bridge can also be a methylene (—CH_2—) group connecting the 2' oxygen atom and the 4' carbon atom, for which the term methyleneoxy (4'-CH_2—O-2') LNA is used. Furthermore; in the case of the bicyclic sugar moiety having an ethylene bridging group in this position, the term ethyleneoxy (4'-CH_2CH_2—O-2') LNA is used. α-L-methyleneoxy (4'-CH_2—O-2'), an isomer of methyleneoxy (4'-CH_2—O-2') LNA is also encompassed within the definition of LNA, as used herein.

[0072] "Mismatch" or "non-complementary nucleobase" refers to the case when a nucleobase of a first nucleic acid is not capable of pairing with the corresponding nucleobase of a second or target nucleic acid.

[0073] "Modified internucleoside linkage" refers to a substitution or any change from a naturally occurring internucleoside bond (i.e., a phosphodiester internucleoside bond).

[0074] "Modified nucleobase" means any nucleobase other than adenine, cytosine, guanine, thymidine, or uracil. An "unmodified nucleobase" means the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C), and uracil (U).

[0075] "Modified nucleoside" means a nucleoside having, independently, a modified sugar moiety and/or modified nucleobase.

[0076] "Modified nucleotide" means a nucleotide having, independently, a modified sugar moiety, modified internucleoside linkage, and/or modified nucleobase.

[0077] "Modified oligonucleotide" means an oligonucleotide comprising at least one modified internucleoside linkage, modified sugar, and/or modified nucleobase.

[0078] "Modified sugar" means substitution and/or any change from a natural sugar moiety.

[0079] "Monomer" means a single unit of an oligomer. Monomers include, but are not limited to, nucleosides and nucleotides, whether naturally occurring or modified.

[0080] "Motif" means the pattern of unmodified and modified nucleoside in an antisense compound.

[0081] "Natural sugar moiety" means a sugar moiety found in DNA (2'-H) or RNA (2'-OH).

[0082] "Naturally occurring internucleoside linkage" means a 3' to 5' phosphodiester linkage.

[0083] "Non-complementary nucleobase" refers to a pair of nucleobases that do not form hydrogen bonds with one another or otherwise support hybridization.

[0084] "Nucleic acid" refers to molecules composed of monomeric nucleotides. A nucleic acid includes, but is not limited to, ribonucleic acids (RNA), deoxyribonucleic acids (DNA), single-stranded nucleic acids, double-stranded nucleic acids, small interfering ribonucleic acids (siRNA), and microRNAs (miRNA).

[0085] "Nucleobase" means a heterocyclic moiety capable of pairing with a base of another nucleic acid.

[0086] "Nucleobase complementarity" refers to a nucleobase that is capable of base pairing with another nucleobase. For example, in DNA, adenine (A) is complementary to thymine (T). For example, in RNA, adenine (A) is complementary to uracil (U). In certain embodiments, complementary nucleobase refers to a nucleobase of an antisense compound that is capable of base pairing with a nucleobase of its target nucleic acid. For example, if a nucleobase at a certain position of an antisense compound is capable of hydrogen bonding with a nucleobase at a certain position of a target nucleic acid, then the position of hydrogen bonding between

the oligonucleotide and the target nucleic acid is considered to be complementary at that nucleobase pair.

[0087] "Nucleobase sequence" means the order of contiguous nucleobases independent of any sugar, linkage, and/or nucleobase modification.

[0088] "Nucleoside" means a nucleobase linked to a sugar.

[0089] "Nucleoside mimetic" includes those structures used to replace the sugar or the sugar and the base and not necessarily the linkage at one or more positions of an oligomeric compound such as for example nucleoside mimetics having morpholino, cyclohexenyl, cyclohexyl, tetrahydropyranyl, bicyclo, or tricyclo sugar mimetics, e.g., non furanose sugar units. Nucleotide mimetic includes those structures used to replace the nucleoside and the linkage at one or more positions of an oligomeric compound such as for example peptide nucleic acids or morpholinos (morpholinos linked by —N(H)—C(=O)—O— or other non-phosphodiester linkage). Sugar surrogate overlaps with the slightly broader term nucleoside mimetic but is intended to indicate replacement of the sugar unit (furanose ring) only. The tetrahydropyranyl rings provided herein are illustrative of an example of a sugar surrogate wherein the furanose sugar group has been replaced with a tetrahydropyranyl ring system. "Mimetic" refers to groups that are substituted for a sugar, a nucleobase, and/or internucleoside linkage. Generally, a mimetic is used in place of the sugar or sugar-internucleoside linkage combination, and the nucleobase is maintained for hybridization to a selected target.

[0090] "Nucleotide" means a nucleoside having a phosphate group covalently linked to the sugar portion of the nucleoside.

[0091] "Off-target effect" refers to an unwanted or deleterious biological effect assocaited with modulation of RNA or protein expression of a gene other than the intended target nucleic acid.

[0092] "Oligomeric compound" or "oligomer" means a polymer of linked monomeric subunits which is capable of hybridizing to at least a region of a nucleic acid molecule.

[0093] "Oligonucleotide" means a polymer of linked nucleosides each of which can be modified or unmodified, independent one from another.

[0094] "Parenteral administration" means administration through injection (e.g., bolus injection) or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intraarterial administration, intraperitoneal administration, or intracranial administration, e.g., intrathecal or intracerebroventricular administration.

[0095] "Peptide" means a molecule formed by linking at least two amino acids by amide bonds. Without limitation, as used herein, peptide refers to polypeptides and proteins.

[0096] "Pharmaceutical agent" means a substance that provides a therapeutic benefit when administered to an individual. For example, in certain embodiments, an antisense oligonucleotide targeted to C9ORF72 is a pharmaceutical agent.

[0097] "Pharmaceutical composition" means a mixture of substances suitable for administering to as subject. For example, a pharmaceutical composition may comprise an antisense oligonucleotide and a sterile aqueous solution.

[0098] "Pharmaceutically acceptable derivative" encompasses pharmaceutically acceptable salts, conjugates, prodrugs or isomers of the compounds described herein.

[0099] "Pharmaceutically acceptable salts" means physiologically and pharmaceutically acceptable salts of antisense compounds, i.e., salts that retain the desired biological activity of the parent oligonucleotide and do not impart undesired toxicological effects thereto.

[0100] "Phosphorothioate linkage" means a linkage between nucleosides where the phosphodiester bond is modified by replacing one of the non-bridging oxygen atoms with a sulfur atom. A phosphorothioate linkage is a modified internucleoside linkage.

[0101] "Portion" means a defined number of contiguous (i.e., linked) nucleobases of a nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of a target nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of an antisense compound.

[0102] "Prevent" or "preventing" refers to delaying or forestalling the onset or development of a disease, disorder, or condition for a period of time from minutes to days, weeks to months, or indefinitely.

[0103] "Prodrug" means a therapeutic agent that is prepared in an inactive form that is converted to an active form within the body or cells thereof by the action of endogenous enzymes or other chemicals or conditions.

[0104] "Prophylactically effective amount" refers to an amount of a pharmaceutical agent that provides a prophylactic or preventative benefit to an animal.

[0105] "Region" is defined as a portion of the target nucleic acid having at least one identifiable structure, function, or characteristic.

[0106] "Ribonucleotide" means a nucleotide having a hydroxy at the 2' position of the sugar portion of the nucleotide. Ribonucleotides may be modified with any of a variety of substituents.

[0107] "Salts" mean a physiologically and pharmaceutically acceptable salts of antisense compounds, i.e., salts that retain the desired biological activity of the parent oligonucleotide and do not impart undesired toxicological effects thereto.

[0108] "Segments" are defined as smaller or sub-portions of regions within a target nucleic acid.

[0109] "Shortened" or "truncated" versions of antisense oligonucleotides taught herein have one, two or more nucleosides deleted.

[0110] "Side effects" means physiological responses attributable to a treatment other than desired effects. In certain embodiments, side effects include, without limitation, injection site reactions, liver function test abnormalities, renal function abnormalities, liver toxicity, renal toxicity, central nervous system abnormalities, and myopathies.

[0111] "Single-stranded oligonucleotide" means an oligonucleotide which is not hybridized to a complementary strand.

[0112] "Sites," as used herein, are defined as unique nucleobase positions within a target nucleic acid.

[0113] "Slows progression" means decrease in the development of the disease.

[0114] "Specifically hybridizable" refers to an antisense compound having a sufficient degree of complementarity between an antisense oligonucleotide and a target nucleic acid to induce a desired effect, while exhibiting minimal or no effects on non-target nucleic acids under conditions in which specific binding is desired, i.e., under physiological conditions in the case of in vivo assays and therapeutic treatments.

[0115] "Stringent hybridization conditions" or "stringent conditions" refer to conditions under which an oligomeric compound will hybridize to its target sequence, but to a minimal number of other sequences.

[0116] "Subject" means a human or non-human animal selected for treatment or therapy.

[0117] "Target" refers to a protein, the modulation of which is desired.

[0118] "Target gene" refers to a gene encoding a target.

[0119] "Targeting" or "targeted" means the process of design and selection of an antisense compound that will specifically hybridize to a target nucleic acid and induce a desired effect.

[0120] "Target nucleic acid," "target RNA," and "target RNA transcript" and "nucleic acid target" all mean a nucleic acid capable of being targeted by antisense compounds.

[0121] "Target region" means a portion of a target nucleic acid to which one or more antisense compounds is targeted.

[0122] "Target segment" means the sequence of nucleotides of a target nucleic acid to which an antisense compound is targeted. "5' target site" refers to the 5'-most nucleotide of a target segment. "3' target site" refers to the 3'-most nucleotide of a target segment.

[0123] "Therapeutically effective amount" means an amount of a pharmaceutical agent that provides a therapeutic benefit to an individual.

[0124] "Treat" or "treating" or "treatment" means administering a composition to effect an alteration or improvement of a disease or condition.

[0125] "Unmodified nucleobases" means the purine bases adenine (A) and guanine (G), and the pyrimidine bases (T), cytosine (C), and uracil (U).

[0126] "Unmodified nucleotide" means a nucleotide composed of naturally occurring nucleobases, sugar moieties, and internucleoside linkages. In certain embodiments, an unmodified nucleotide is an RNA nucleotide (i.e. β-D-ribonucleosides) or a DNA nucleotide (i.e. β-D-deoxyribonucleoside).

[0127] "Wing segment" means a plurality of nucleosides modified to impart to an oligonucleotide properties such as enhanced inhibitory activity, increased binding affinity for a target nucleic acid, or resistance to degradation by in vivo nucleases.

Certain Embodiments

[0128] Certain embodiments provide compositions and methods for decreasing total C9ORF72 mRNA and protein expression.

[0129] Certain embodiments provide compositions and methods for decreasing C9ORF72 pathogenic associated mRNA variants.

[0130] Certain embodiments provide methods for the treatment, prevention, or amelioration of diseases, disorders, and conditions associated with C9ORF72 in an individual in need thereof. Also contemplated are methods for the preparation of a medicament for the treatment, prevention, or amelioration of a disease, disorder, or condition associated with C9ORF72. C9ORF72 associated diseases, disorders, and conditions include neurodegenerative diseases. In certain embodiments, the neurodegenerative disease may be ALS or FTD. In certain embodiments, the neurodegenerative disease may be familial or sporadic.

[0131] Certain embodiments provide compositions and methods for the treatment, prevention, or amelioration of a C9ORF72 hexanucleotide repeat expansion associated disease. In certain embodiments, the hexanucleotide repeat expansion may comprise GGGGCC, GGGGGG, GGGGGC, or GGGGCG.

[0132] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases of any of the nucleobase sequences of SEQ ID NOs: SEQ ID NOs: 20-401 and 441-1545.

[0133] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1107-1520 of SEQ ID NO: 2.

[0134] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1111-1200 of SEQ ID NO: 2.

[0135] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1211-1318 of SEQ ID NO: 2.

[0136] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1326-1540 of SEQ ID NO: 2.

[0137] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1331-1375 of SEQ ID NO: 2.

[0138] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1368-1391 of SEQ ID NO: 2

[0139] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at

least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1398-1424 of SEQ ID NO: 2.

[0140] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1411-1440 of SEQ ID NO: 2.

[0141] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1429-1481 of SEQ ID NO: 2.

[0142] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1502-1539 of SEQ ID NO: 2.

[0143] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1508-1539 of SEQ ID NO: 2.

[0144] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 7860-7906 of SEQ ID NO: 2.

[0145] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 7907-9744 of SEQ ID NO: 2.

[0146] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 7989-8038 of SEQ ID NO: 2.

[0147] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 8020-8135 of SEQ ID NO: 2.

[0148] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 8136-8161 of SEQ ID NO: 2.

[0149] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 8174-8211 of SEQ ID NO: 2.

[0150] Provided herein are compounds, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 8213-8325 of SEQ ID NO: 2.

[0151] In certain embodiments, the nucleobase sequence of the modified oligonucleotide is at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% complementary to SEQ ID NO: 1.

[0152] In certain embodiments, the modified oligonucleotide is a single-stranded modified oligonucleotide.

[0153] In certain embodiments at least one internucleoside linkage of the modified oligonucleotide is a modified internucleoside linkage.

[0154] In certain embodiments, at least one modified internucleoside linkage is a phosphorothioate internucleoside linkage.

[0155] In certain embodiments, each modified internucleoside linkage is a phosphorothioate internucleoside linkage.

[0156] In certain embodiments, at least one internucleoside linkage is a phosphodiester internucleoside linkage.

[0157] In certain embodiments, at least one internucleoside linkage is a phosphorothioate linkage and at least one internucleoside linkage is a phosphodiester linkage.

[0158] In certain embodiments, at least one nucleoside comprises a modified nucleobase.

[0159] In certain embodiments, the modified nucleobase is a 5-methylcytosine.

[0160] In certain embodiments, at least one nucleoside of the modified oligonucleotide comprises a modified sugar.

[0161] In certain embodiments, the at least one modified sugar is a bicyclic sugar.

[0162] In certain embodiments, the bicyclic sugar comprises a chemical link between the 2' and 4' position of the sugar 4'-CH2-N(R)—O-2' bridge wherein R is, independently, H, C1-C12 alkyl, or a protecting group.

[0163] In certain embodiments, the bicyclic sugar comprises a 4'-CH2-N(R)—O-2' bridge wherein R is, independently, H, C1-C12 alkyl, or a protecting group.

[0164] In certain embodiments, at least one modified sugar comprises a 2'-O-methoxyethyl group.

[0165] In certain embodiments, the modified sugar comprises a 2'-$O(CH_2)_2$—OCH_3 group.

[0166] In certain embodiments, the modified oligonucleotide comprises:

[0167] a gap segment consisting of 10 linked deoxynucleosides;

[0168] a 5' wing segment consisting of 5 linked nucleosides; and

[0169] a 3' wing segment consisting of 5 linked nucleosides;

[0170] wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

[0171] In certain embodiments, the modified oligonucleotide comprises:

[0172] a gap segment consisting of 8 linked deoxynucleosides;

[0173] a 5' wing segment consisting of 5 linked nucleosides; and

[0174] a 3' wing segment consisting of 5 linked nucleosides;

[0175] wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

[0176] In certain embodiments, the modified oligonucleotide comprises sugar modifications in any of the following patterns: eeekkddddddkkeee, eekkddddddddkkeee, ekddddddddekekeee, kekeddddddddekeke, and ekekddddddddkekee; wherein,

[0177] e=a 2'-O-methoxyethyl modified nucleoside

[0178] d=a 2'-deoxynucleoside, and

[0179] k=a cEt nucleoside.

[0180] In certain embodiments, the modified oligonucleotide comprises internucleoside linkages in any of the following patterns: soooosssssssssssooss, sooossssssssssooss, sooossssssssooss, and sosssssssssoooss; wherein,

[0181] s=a phosphorothioate linkage, and

[0182] o=a phosphodiester linkage.

[0183] In certain embodiments, the modified oligonucleotide consists of 20 linked nucleosides.

[0184] In certain embodiments, the modified oligonucleotide consists of 19 linked nucleosides.

[0185] In certain embodiments, the modified oligonucleotide consists of 18 linked nucleosides.

[0186] In certain embodiments, the modified oligonucleotide consists of 17 linked nucleosides.

[0187] Provided herein are compositions comprising the compound of any preceding claim or salt thereof and at least one of a pharmaceutically acceptable carrier or diluent.

[0188] Provided herein are methods comprising administering to an animal the compound or composition of any preceding claim.

[0189] In certain embodiments, the animal is a human.

[0190] In certain embodiments, administering the compound prevents, treats, ameliorates, or slows progression of a C9ORF72 associated disease, disorder or condition.

[0191] In certain embodiments, administering the compound prevents, treats, ameliorates, or slows progression of a C9ORF72 hexanucleotide repeat expansion associated disease, disorder or condition.

[0192] In certain embodiments, the disease, disorder or condition is amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), corticalbasal degeneration syndrome (CBD), atypical Parkinsonian syndrome, and olivopontocerellar degeneration (OPCD).

[0193] In certain embodiments, the administering reduces nuclear foci.

[0194] In certain embodiments, the administering reduces expression of C9ORF72 associated RAN translation products.

[0195] In certain embodiments, the C9ORF72 associated RAN translation products are any of poly-(glycine-proline), poly-(glycine-alanine), and poly-(glycine-arginine)

[0196] Provided herein are uses of the compound or composition of any preceding claim for the manufacture of a medicament for treating a neurodegenerative disorder.

[0197] Provided herein are methods of selectively inhibiting a C9ORF72 pathogenic associated mRNA variant by administering an antisense compound targeting the region beginning at the start site of exon 1A to the start site of exon 1B of a C9ORF72 pre-mRNA.

Antisense Compounds

[0198] Oligomeric compounds include, but are not limited to, oligonucleotides, oligonucleosides, oligonucleotide analogs, oligonucleotide mimetics, antisense compounds, antisense oligonucleotides, and siRNAs. An oligomeric compound may be "antisense" to a target nucleic acid, meaning that is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding.

[0199] In certain embodiments, an antisense compound has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted. In certain such embodiments, an antisense oligonucleotide has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.

[0200] In certain embodiments, an antisense compound targeted to a C9ORF72 nucleic acid is 12 to 30 subunits in length. In other words, such antisense compounds are from 12 to 30 linked subunits. In certain embodiments, the antisense compound is 8 to 80, 12 to 50, 15 to 30, 18 to 24, 19 to 22, or 20 linked subunits. In certain embodiments, the antisense compounds are 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, or 80 linked subunits in length, or a range defined by any two of the above values. In some embodiments the antisense compound is an antisense oligonucleotide, and the linked subunits are nucleosides.

[0201] In certain embodiments antisense oligonucleotides targeted to a C9ORF72 nucleic acid may be shortened or truncated. For example, a single subunit may be deleted from the 5' end (5' truncation), or alternatively from the 3' end (3' truncation). A shortened or truncated antisense compound targeted to a C9ORF72 nucleic acid may have two subunits deleted from the 5' end, or alternatively may have two subunits deleted from the 3' end, of the antisense compound.

Alternatively, the deleted nucleosides may be dispersed throughout the antisense compound, for example, in an antisense compound having one nucleoside deleted from the 5' end and one nucleoside deleted from the 3' end.

[0202] When a single additional subunit is present in a lengthened antisense compound, the additional subunit may be located at the 5' or 3' end of the antisense compound. When two or more additional subunits are present, the added subunits may be adjacent to each other, for example, in an antisense compound having two subunits added to the 5' end (5' addition), or alternatively to the 3' end (3' addition), of the antisense compound. Alternatively, the added subunits may be dispersed throughout the antisense compound, for example, in an antisense compound having one subunit added to the 5' end and one subunit added to the 3' end.

[0203] It is possible to increase or decrease the length of an antisense compound, such as an antisense oligonucleotide, and/or introduce mismatch bases without eliminating activity. For example, in Woolf et al. (Proc. Natl. Acad. Sci. USA 89:7305-7309, 1992), a series of antisense oligonucleotides 13-25 nucleobases in length were tested for their ability to induce cleavage of a target RNA in an oocyte injection model. Antisense oligonucleotides 25 nucleobases in length with 8 or 11 mismatch bases near the ends of the antisense oligonucleotides were able to direct specific cleavage of the target mRNA, albeit to a lesser extent than the antisense oligonucleotides that contained no mismatches. Similarly, target specific cleavage was achieved using 13 nucleobase antisense oligonucleotides, including those with 1 or 3 mismatches.

[0204] Gautschi et al (J. Natl. Cancer Inst. 93:463-471, March 2001) demonstrated the ability of an oligonucleotide having 100% complementarity to the bcl-2 mRNA and having 3 mismatches to the bcl-xL mRNA to reduce the expression of both bcl-2 and bcl-xL in vitro and in vivo. Furthermore, this oligonucleotide demonstrated potent anti-tumor activity in vivo.

[0205] Maher and Dolnick (Nuc. Acid. Res. 16:3341-3358, 1988) tested a series of tandem 14 nucleobase antisense oligonucleotides, and a 28 and 42 nucleobase antisense oligonucleotides comprised of the sequence of two or three of the tandem antisense oligonucleotides, respectively, for their ability to arrest translation of human DHFR in a rabbit reticulocyte assay. Each of the three 14 nucleobase antisense oligonucleotides alone was able to inhibit translation, albeit at a more modest level than the 28 or 42 nucleobase antisense oligonucleotides.

Antisense Compound Motifs

[0206] In certain embodiments, antisense compounds targeted to a C9ORF72 nucleic acid have chemically modified subunits arranged in patterns, or motifs, to confer to the antisense compounds properties such as enhanced inhibitory activity, increased binding affinity for a target nucleic acid, or resistance to degradation by in vivo nucleases.

[0207] Chimeric antisense compounds typically contain at least one region modified so as to confer increased resistance to nuclease degradation, increased cellular uptake, increased binding affinity for the target nucleic acid, and/or increased inhibitory activity. A second region of a chimeric antisense compound may optionally serve as a substrate for the cellular endonuclease RNase H, which cleaves the RNA strand of an RNA:DNA duplex.

[0208] Antisense compounds having a gapmer motif are considered chimeric antisense compounds. In a gapmer an internal region having a plurality of nucleotides that supports RNaseH cleavage is positioned between external regions having a plurality of nucleotides that are chemically distinct from the nucleosides of the internal region. In the case of an antisense oligonucleotide having a gapmer motif, the gap segment generally serves as the substrate for endonuclease cleavage, while the wing segments comprise modified nucleosides. In certain embodiments, the regions of a gapmer are differentiated by the types of sugar moieties comprising each distinct region. The types of sugar moieties that are used to differentiate the regions of a gapmer may in some embodiments include β-D-ribonucleosides, β-D-deoxyribonucleosides, 2'-modified nucleosides (such 2'-modified nucleosides may include 2'-MOE, and 2'-O—CH_3, among others), and bicyclic sugar modified nucleosides (such bicyclic sugar modified nucleosides may include those having a 4'-(CH_2)n-O-2' bridge, where n=1 or n=2 and 4'-CH_2—O—CH_2-2'). Preferably, each distinct region comprises uniform sugar moieties. The wing-gap-wing motif is frequently described as "X—Y—Z", where "X" represents the length of the 5' wing region, "Y" represents the length of the gap region, and "Z" represents the length of the 3' wing region. As used herein, a gapmer described as "X—Y—Z" has a configuration such that the gap segment is positioned immediately adjacent to each of the 5' wing segment and the 3' wing segment. Thus, no intervening nucleotides exist between the 5' wing segment and gap segment, or the gap segment and the 3' wing segment. Any of the antisense compounds described herein can have a gapmer motif. In some embodiments, X and Z are the same, in other embodiments they are different. In a preferred embodiment, Y is between 8 and 15 nucleotides. X, Y or Z can be any of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30 or more nucleotides. Thus, gapmers described herein include, but are not limited to, for example 5-10-5, 5-10-4, 4-10-4, 4-10-3, 3-10-3, 2-10-2, 5-9-5, 5-9-4, 4-9-5, 5-8-5, 5-8- 4, 4-8-5, 5-7-5, 4-7-5, 5-7-4, or 4-7-4.

[0209] In certain embodiments, the antisense compound has a "wingmer" motif, having a wing-gap or gap-wing configuration, i.e. an X—Y or Y—Z configuration as described above for the gapmer configuration. Thus, wingmer configurations described herein include, but are not limited to, for example 5-10, 8-4, 4-12, 12-4, 3-14, 16-2, 18-1, 10-3, 2-10, 1-10, 8-2, 2-13, 5-13, 5-8, or 6-8.

[0210] In certain embodiments, antisense compounds targeted to a C9ORF72 nucleic acid possess a 5-10-5 gapmer motif.

[0211] In certain embodiments, antisense compounds targeted to a C9ORF72 nucleic acid possess a 5-8-5 gapmer motif.

[0212] In certain embodiments, antisense compounds targeted to a C9ORF72 nucleic acid possess sugar modifications in any of the following patterns: eeekkddddddkkeee, eekkddddddkkeee, ekddddddddekekeee, kekedddddddekeke, and ekekddddddddkekee; wherein,

[0213] e=a 2'-O-methoxyethyl modified nucleoside

[0214] d=a 2'-deoxynucleoside, and

[0215] k=a cEt nucleoside.

[0216] In certain embodiments, an antisense compound targeted to a C9ORF72 nucleic acid has a gap-narrowed motif. In certain embodiments, a gap-narrowed antisense oligonucleotide targeted to a C9ORF72 nucleic acid has a gap segment of 9, 8, 7, or 6 2'-deoxynucleotides positioned immediately adjacent to and between wing segments of 5, 4, 3, 2, or 1 chemically modified nucleosides. In certain embodiments,

the chemical modification comprises a bicyclic sugar. In certain embodiments, the bicyclic sugar comprises a 4' to 2' bridge selected from among: 4'-$(CH_2)_n$—O-2' bridge, wherein n is 1 or 2; and 4'-CH_2—O—CH_2-2'. In certain embodiments, the bicyclic sugar is comprises a 4'-CH(CH_3)—O-2' bridge. In certain embodiments, the chemical modification comprises a non-bicyclic 2'-modified sugar moiety. In certain embodiments, the non-bicyclic 2'-modified sugar moiety comprises a 2'-O-methylethyl group or a 2'-O-methyl group.

Target Nucleic Acids, Target Regions and Nucleotide Sequences

[0217] Nucleotide sequences that encode C9ORF72 include, without limitation, the following: the complement of GENBANK Accession No. NM_001256054.1 (incorporated herein as SEQ ID NO: 1), GENBANK Accession No. NT_008413.18 truncated from nucleobase 27535000 to 27565000 (incorporated herein as SEQ ID NO: 2), GENBANK Accession No. BQ068108.1 (incorporated herein as SEQ ID NO: 3), GENBANK Accession No. NM_018325.3 (incorporated herein as SEQ ID NO: 4), GENBANK Accession No. DN993522.1 (incorporated herein as SEQ ID NO: 5), GENBANK Accession No. NM_145005.5 (incorporated herein as SEQ ID NO: 6), GENBANK

[0218] Accession No. DB079375.1 (incorporated herein as SEQ ID NO: 7), GENBANK Accession No. BU194591.1 (incorporated herein as SEQ ID NO: 8), Sequence Identifier 4141_014_A (incorporated herein as SEQ ID NO: 9), and Sequence Identifier 4008_73_A (incorporated herein as SEQ ID NO: 10), and GENBANK Accession No. NW_001101662.1 truncated from nucleosides 8522000 to U.S. Pat. No. 8,552,000 (incorporated herein as SEQ ID NO: 19).

[0219] It is understood that the sequence set forth in each SEQ ID NO in the Examples contained herein is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, antisense compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Antisense compounds described by Isis Number (Isis No) indicate a combination of nucleobase sequence and motif.

[0220] In certain embodiments, a target region is a structurally defined region of the target nucleic acid. For example, a target region may encompass a 3' UTR, a 5' UTR, an exon, an intron, an exon/intron junction, a coding region, a translation initiation region, translation termination region, or other defined nucleic acid region. The structurally defined regions for C9ORF72 can be obtained by accession number from sequence databases such as NCBI and such information is incorporated herein by reference. In certain embodiments, a target region may encompass the sequence from a 5' target site of one target segment within the target region to a 3' target site of another target segment within the same target region.

[0221] Targeting includes determination of at least one target segment to which an antisense compound hybridizes, such that a desired effect occurs. In certain embodiments, the desired effect is a reduction in mRNA target nucleic acid levels. In certain embodiments, the desired effect is reduction of levels of protein encoded by the target nucleic acid or a phenotypic change associated with the target nucleic acid.

[0222] A target region may contain one or more target segments. Multiple target segments within a target region may be overlapping. Alternatively, they may be non-overlapping. In certain embodiments, target segments within a target region are separated by no more than about 300 nucleotides. In certain embodiments, target segments within a target region are separated by a number of nucleotides that is, is about, is no more than, is no more than about, 250, 200, 150, 100, 90, 80, 70, 60, 50, 40, 30, 20, or 10 nucleotides on the target nucleic acid, or is a range defined by any two of the preceeding values. In certain embodiments, target segments within a target region are separated by no more than, or no more than about, 5 nucleotides on the target nucleic acid. In certain embodiments, target segments are contiguous. Contemplated are target regions defined by a range having a starting nucleic acid that is any of the 5' target sites or 3' target sites listed herein.

[0223] Suitable target segments may be found within a 5' UTR, a coding region, a 3' UTR, an intron, an exon, or an exon/intron junction. Target segments containing a start codon or a stop codon are also suitable target segments. A suitable target segment may specifically exclude a certain structurally defined region such as the start codon or stop codon.

[0224] The determination of suitable target segments may include a comparison of the sequence of a target nucleic acid to other sequences throughout the genome. For example, the BLAST algorithm may be used to identify regions of similarity amongst different nucleic acids. This comparison can prevent the selection of antisense compound sequences that may hybridize in a non-specific manner to sequences other than a selected target nucleic acid (i.e., non-target or off-target sequences).

[0225] There may be variation in activity (e.g., as defined by percent reduction of target nucleic acid levels) of the antisense compounds within a target region. In certain embodiments, reductions in C9ORF72 mRNA levels are indicative of inhibition of C9ORF72 expression. Reductions in levels of a C9ORF72 protein are also indicative of inhibition of target mRNA expression. Reduction in the presence of expanded C9ORF72 RNA foci are indicative of inhibition of C9ORF72 expression. Further, phenotypic changes are indicative of inhibition of C9ORF72 expression. For example, improved motor function and respiration may be indicative of inhibition of C9ORF72 expression.

Hybridization

[0226] In some embodiments, hybridization occurs between an antisense compound disclosed herein and a C9ORF72 nucleic acid. The most common mechanism of hybridization involves hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) between complementary nucleobases of the nucleic acid molecules.

[0227] Hybridization can occur under varying conditions. Stringent conditions are sequence-dependent and are determined by the nature and composition of the nucleic acid molecules to be hybridized.

[0228] Methods of determining whether a sequence is specifically hybridizable to a target nucleic acid are well known in the art. In certain embodiments, the antisense compounds provided herein are specifically hybridizable with a C9ORF72 nucleic acid.

Complementarity

[0229] An antisense compound and a target nucleic acid are complementary to each other when a sufficient number of nucleobases of the antisense compound can hydrogen bond with the corresponding nucleobases of the target nucleic acid, such that a desired effect will occur (e.g., antisense inhibition of a target nucleic acid, such as a C9ORF72 nucleic acid).

[0230] Non-complementary nucleobases between an antisense compound and a C9ORF72 nucleic acid may be tolerated provided that the antisense compound remains able to specifically hybridize to a target nucleic acid. Moreover, an antisense compound may hybridize over one or more segments of a C9ORF72 nucleic acid such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure, mismatch or hairpin structure).

[0231] In certain embodiments, the antisense compounds provided herein, or a specified portion thereof, are, or are at least, 70%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% complementary to a C9ORF72 nucleic acid, a target region, target segment, or specified portion thereof. Percent complementarity of an antisense compound with a target nucleic acid can be determined using routine methods.

[0232] For example, an antisense compound in which 18 of 20 nucleobases of the antisense compound are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In this example, the remaining noncomplementary nucleobases may be clustered or interspersed with complementary nucleobases and need not be contiguous to each other or to complementary nucleobases. As such, an antisense compound which is 18 nucleobases in length having 4 (four) noncomplementary nucleobases which are flanked by two regions of complete complementarity with the target nucleic acid would have 77.8% overall complementarity with the target nucleic acid and would thus fall within the scope of the present invention. Percent complementarity of an antisense compound with a region of a target nucleic acid can be determined routinely using BLAST programs (basic local alignment search tools) and PowerBLAST programs known in the art (Altschul et al., J. Mol. Biol., 1990, 215, 403 410; Zhang and Madden, Genome Res., 1997, 7, 649 656). Percent homology, sequence identity or complementarity, can be determined by, for example, the Gap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madison Wis.), using default settings, which uses the algorithm of Smith and Waterman (Adv. Appl. Math., 1981, 2, 482 489).

[0233] In certain embodiments, the antisense compounds provided herein, or specified portions thereof, are fully complementary (i.e., 100% complementary) to a target nucleic acid, or specified portion thereof. For example, an antisense compound may be fully complementary to a C9ORF72 nucleic acid, or a target region, or a target segment or target sequence thereof. As used herein, "fully complementary" means each nucleobase of an antisense compound is capable of precise base pairing with the corresponding nucleobases of a target nucleic acid. For example, a 20 nucleobase antisense compound is fully complementary to a target sequence that is 400 nucleobases long, so long as there is a corresponding 20 nucleobase portion of the target nucleic acid that is fully complementary to the antisense compound. Fully complementary can also be used in reference to a specified portion of the first and/or the second nucleic acid. For example, a 20 nucleobase portion of a 30 nucleobase antisense compound can be "fully complementary" to a target sequence that is 400 nucleobases long. The 20 nucleobase portion of the 30 nucleobase oligonucleotide is fully complementary to the target sequence if the target sequence has a corresponding 20 nucleobase portion wherein each nucleobase is complementary to the 20 nucleobase portion of the antisense compound. At the same time, the entire 30 nucleobase antisense compound may or may not be fully complementary to the target sequence, depending on whether the remaining 10 nucleobases of the antisense compound are also complementary to the target sequence.

[0234] The location of a non-complementary nucleobase may be at the 5' end or 3' end of the antisense compound. Alternatively, the non-complementary nucleobase or nucleobases may be at an internal position of the antisense compound. When two or more non-complementary nucleobases are present, they may be contiguous (i.e., linked) or non-contiguous. In one embodiment, a non-complementary nucleobase is located in the wing segment of a gapmer antisense oligonucleotide.

[0235] In certain embodiments, antisense compounds that are, or are up to 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleobases in length comprise no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a C9ORF72 nucleic acid, or specified portion thereof.

[0236] In certain embodiments, antisense compounds that are, or are up to 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleobases in length comprise no more than 6, no more than 5, no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a C9ORF72 nucleic acid, or specified portion thereof.

[0237] The antisense compounds provided herein also include those which are complementary to a portion of a target nucleic acid. As used herein, "portion" refers to a defined number of contiguous (i.e. linked) nucleobases within a region or segment of a target nucleic acid. A "portion" can also refer to a defined number of contiguous nucleobases of an antisense compound. In certain embodiments, the antisense compounds, are complementary to at least an 8 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 9 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 10 nucleobase portion of a target segment. In certain embodiments, the antisense compounds, are complementary to at least an 11 nucleobase portion of a target segment. In certain embodiments, the antisense compounds, are complementary to at least a 12 nucleobase portion of a target segment. In certain embodiments, the antisense compounds, are complementary to at least a 13 nucleobase portion of a target segment. In certain embodiments, the antisense compounds, are complementary to at least a 14 nucleobase portion of a target segment. In certain embodiments, the antisense compounds, are complementary to at least a 15 nucleobase portion of a target segment. Also contemplated are antisense compounds that are complementary to at least a 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more nucleobase portion of a target segment, or a range defined by any two of these values.

Identity

[0238] The antisense compounds provided herein may also have a defined percent identity to a particular nucleotide sequence, SEQ ID NO, or compound represented by a specific Isis number, or portion thereof. As used herein, an antisense compound is identical to the sequence disclosed herein if it has the same nucleobase pairing ability. For example, a RNA which contains uracil in place of thymidine in a disclosed DNA sequence would be considered identical to the DNA sequence since both uracil and thymidine pair with adenine. Shortened and lengthened versions of the antisense compounds described herein as well as compounds having non-identical bases relative to the antisense compounds provided herein also are contemplated. The non-identical bases may be adjacent to each other or dispersed throughout the antisense compound. Percent identity of an antisense compound is calculated according to the number of bases that have identical base pairing relative to the sequence to which it is being compared.

[0239] In certain embodiments, the antisense compounds, or portions thereof, are at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99% or 100% identical to one or more of the antisense compounds or SEQ ID NOs, or a portion thereof, disclosed herein.

[0240] In certain embodiments, a portion of the antisense compound is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

[0241] In certain embodiments, a portion of the antisense oligonucleotide is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

Modifications

[0242] A nucleoside is a base-sugar combination. The nucleobase (also known as base) portion of the nucleoside is normally a heterocyclic base moiety. Nucleotides are nucleosides that further include a phosphate group covalently linked to the sugar portion of the nucleoside. For those nucleosides that include a pentofuranosyl sugar, the phosphate group can be linked to the 2', 3' or 5' hydroxyl moiety of the sugar. Oligonucleotides are formed through the covalent linkage of adjacent nucleosides to one another, to form a linear polymeric oligonucleotide. Within the oligonucleotide structure, the phosphate groups are commonly referred to as forming the internucleoside linkages of the oligonucleotide.

[0243] Modifications to antisense compounds encompass substitutions or changes to internucleoside linkages, sugar moieties, or nucleobases. Modified antisense compounds are often preferred over native forms because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for nucleic acid target, increased stability in the presence of nucleases, or increased inhibitory activity.

[0244] Chemically modified nucleosides may also be employed to increase the binding affinity of a shortened or truncated antisense oligonucleotide for its target nucleic acid. Consequently, comparable results can often be obtained with shorter antisense compounds that have such chemically modified nucleosides.

Modified Internucleoside Linkages

[0245] The naturally occurring internucleoside linkage of RNA and DNA is a 3' to 5' phosphodiester linkage. Antisense compounds having one or more modified, i.e. non-naturally occurring, internucleoside linkages are often selected over antisense compounds having naturally occurring internucleoside linkages because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for target nucleic acids, and increased stability in the presence of nucleases.

[0246] Oligonucleotides having modified internucleoside linkages include internucleoside linkages that retain a phosphorus atom as well as internucleoside linkages that do not have a phosphorus atom. Representative phosphorus containing internucleoside linkages include, but are not limited to, phosphodiesters, phosphotriesters, methylphosphonates, phosphoramidate, and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing linkages are well known.

[0247] In certain embodiments, antisense compounds targeted to a C9ORF72 nucleic acid comprise one or more modified internucleoside linkages. In certain embodiments, the modified internucleoside linkages are interspersed throughout the antisense compound. In certain embodiments, the modified internucleoside linkages are phosphorothioate linkages. In certain embodiments, each internucleoside linkage of an antisense compound is a phosphorothioate internucleoside linkage. In certain embodiments, the antisense compounds targeted to a C9ORF72 nucleic acid comprise at least one phosphodiester linkage and at least one phosphorothioate linkage.

[0248] In certain embodiments, antisense compounds targeted to a C9ORF72 nucleic acid possess internucleoside linkages in any of the following patterns: soooossssssssssooss, soooossssssssooss, soossssssssooss, and sossssssssoooss; wherein,

[0249] s=a phosphorothioate linkage, and

[0250] o=a phosphodiester linkage.

Modified Sugar Moieties

[0251] Antisense compounds can optionally contain one or more nucleosides wherein the sugar group has been modified. Such sugar modified nucleosides may impart enhanced nuclease stability, increased binding affinity, or some other beneficial biological property to the antisense compounds. In certain embodiments, nucleosides comprise chemically modified ribofuranose ring moieties. Examples of chemically modified ribofuranose rings include without limitation, addition of substitutent groups (including 5' and 2' substituent groups, bridging of non-geminal ring atoms to form bicyclic nucleic acids (BNA), replacement of the ribosyl ring oxygen atom with S, N(R), or $C(R_1)(R_2)$ (R, R_1 and R_2 are each independently H, C_1-C_{12} alkyl or a protecting group) and combinations thereof. Examples of chemically modified sugars include 2'-F-5'-methyl substituted nucleoside (see PCT International Application WO 2008/101157 Published on Aug. 21, 2008 for other disclosed 5',2'-bis substituted nucleosides) or replacement of the ribosyl ring oxygen atom with S with further substitution at the 2'-position (see published U.S. Patent Application US2005-0130923, published on Jun. 16, 2005) or alternatively 5'-substitution of a BNA (see PCT International Application WO 2007/134181 Published on

Nov. 22, 2007 wherein LNA is substituted with for example a 5'-methyl or a 5'-vinyl group).

[0252] Examples of nucleosides having modified sugar moieties include without limitation nucleosides comprising 5'-vinyl, 5'-methyl (R or S), 4'-S, 2'-F, 2'-OCH_3, 2'-OCH_2CH_3, 2'-OCH_2CH_2F and 2'-$O(CH_2)_2OCH_3$ substituent groups. The substituent at the 2' position can also be selected from allyl, amino, azido, thio, O-allyl, O—C_1-C_{10} alkyl, OCF_3, OCH_2F, $O(CH_2)_2SCH_3$, $O(CH_2)_2$—O—$N(R_m)(R_n)$, O—CH_2—C(=O)—$N(R_m)(R_n)$, and O—CH_2—C(=O)—$N(R_1)$—$(CH_2)_2$—$N(R_m)(R_n)$, where each R_l, R_m and R_n is, independently, H or substituted or unsubstituted C_1-C_{10} alkyl.

[0253] As used herein, "bicyclic nucleosides" refer to modified nucleosides comprising a bicyclic sugar moiety. Examples of bicyclic nucleosides include without limitation nucleosides comprising a bridge between the 4' and the 2' ribosyl ring atoms. In certain embodiments, antisense compounds provided herein include one or more bicyclic nucleosides comprising a 4' to 2' bridge. Examples of such 4' to 2' bridged bicyclic nucleosides, include but are not limited to one of the formulae: 4'-(CH_2)—O-2' (LNA); 4'-(CH_2)—S-2; 4'-$(CH_2)_2$—O-2' (ENA); 4'-$CH(CH_3)$—O-2' and 4'-$CH(CH_2OCH_3)$—O-2' (and analogs thereof see U.S. Pat. No. 7,399,845, issued on Jul. 15, 2008); 4'-$C(CH_3)(CH_3)$—O-2' (and analogs thereof see published International Application WO/2009/006478, published Jan. 8, 2009); 4'-CH_2—$N(OCH_3)$-2' (and analogs thereof see published International Application WO/2008/150729, published Dec. 11, 2008); 4'-CH_2—O—$N(CH_3)$-2' (see published U.S. Patent Application US2004-0171570, published Sep. 2, 2004); 4'-CH_2—N(R)—O-2', wherein R is H, C_1-C_{12} alkyl, or a protecting group (see U.S. Pat. No. 7,427,672, issued on Sep. 23, 2008); 4'-CH_2—$C(H)(CH_3)$-2' (see Chattopadhyaya et al., *J. Org. Chem.,* 2009, 74, 118-134); and 4'-CH_2—C—(=CH_2)-2' (and analogs thereof see published International Application WO 2008/154401, published on Dec. 8, 2008).

[0254] Further reports related to bicyclic nucleosides can also be found in published literature (see for example: Singh et al., *Chem. Commun.,* 1998, 4, 455-456; Koshkin et al., *Tetrahedron,* 1998, 54, 3607-3630; Wahlestedt et al., *Proc. Natl. Acad. Sci. U.S.A,* 2000, 97, 5633-5638; Kumar et al., *Bioorg. Med. Chem. Lett.,* 1998, 8, 2219-2222; Singh et al., *J. Org. Chem.,* 1998, 63, 10035-10039; Srivastava et al., *J. Am. Chem. Soc.,* 2007, 129(26) 8362-8379; Elayadi et al., *Curr. Opinion Invest. Drugs,* 2001, 2, 558-561; Braasch et al., *Chem. Biol.,* 2001, 8, 1-7; and Orum et al., *Curr. Opinion Mol. Ther.,* 2001, 3, 239-243; U.S. Pat. Nos. 6,268,490; 6,525,191; 6,670,461; 6,770,748; 6,794,499; 7,034,133; 7,053,207; 7,399,845; 7,547,684; and 7,696,345; U.S. Patent Publication No. US2008-0039618; US2009-0012281; U.S. Patent Ser. No. 60/989,574; 61/026,995; 61/026,998; 61/056,564; 61/086,231; 61/097,787; and 61/099,844; Published PCT International applications WO 1994/014226; WO 2004/106356; WO 2005/021570; WO 2007/134181; WO 2008/150729; WO 2008/154401; and WO 2009/006478. Each of the foregoing bicyclic nucleosides can be prepared having one or more stereochemical sugar configurations including for example α-L-ribofuranose and β-D-ribofuranose (see PCT international application PCT/DK98/00393, published on Mar. 25, 1999 as WO 99/14226).

[0255] In certain embodiments, bicyclic sugar moieties of BNA nucleosides include, but are not limited to, compounds having at least one bridge between the 4' and the 2' position of the pentofuranosyl sugar moiety wherein such bridges independently comprises 1 or from 2 to 4 linked groups independently selected from —$[C(R_a)(R_b)]_n$—, —$C(R_a)$=$C(R_b)$—, —$C(R_a)$=N—, —C(=O)—, —C(=NR_a)—, —C(=S)—, —O—, —$Si(R_a)_2$—, —$S(=O)_x$—, and —$N(R_a)$—;

[0256] wherein:

[0257] x is 0, 1, or 2;

[0258] n is 1, 2, 3, or 4;

[0259] each R_a and R_b is, independently, H, a protecting group, hydroxyl, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_5-C_{20} aryl, substituted C_5-C_{20} aryl, heterocycle radical, substituted heterocycle radical, heteroaryl, substituted heteroaryl, C_5-C_7 alicyclic radical, substituted C_5-C_7 alicyclic radical, halogen, OJ_1, NJ_1J_2, SJ_1, N_3, $COOJ_1$, acyl (C(=O)—H), substituted acyl, CN, sulfonyl ($S(=O)_2$-J_1), or sulfoxyl (S(=O)-J_1); and

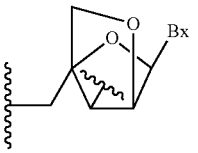
[0260] each J_1 and J_2 is, independently, H, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_5-C_{20} aryl, substituted C_5-C_{20} aryl, acyl (C(=O)—H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C_1-C_{12} aminoalkyl, substituted C_1-C_{12} aminoalkyl or a protecting group.

[0261] In certain embodiments, the bridge of a bicyclic sugar moiety is —$[C(R_a)(R_b)]_n$—, —$[C(R_a)(R_b)]_n$—O—, —$C(R_aR_b)$—N(R)—O— or —$C(R_aR_b)$—O—N(R)—. In certain embodiments, the bridge is 4'-CH_2-2', 4'-$(CH_2)_2$-2', 4'-$(CH_2)_3$-2', 4'-CH_2—O-2', 4'-$(CH_2)_2$—O-2', 4'-CH_2—O—N(R)-2' and 4'-CH_2—N(R)—O-2'- wherein each R is, independently, H, a protecting group or C_1-C_{12} alkyl.

[0262] In certain embodiments, bicyclic nucleosides are further defined by isomeric configuration. For example, a nucleoside comprising a 4'-2' methylene-oxy bridge, may be in the α-L configuration or in the β-D configuration. Previously, α-L-methyleneoxy (4'-CH_2—O-2') BNA's have been incorporated into antisense oligonucleotides that showed antisense activity (Frieden et al., *Nucleic Acids Research,* 2003, 21, 6365-6372).

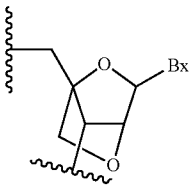
[0263] In certain embodiments, bicyclic nucleosides include, but are not limited to, (A) α-L-methyleneoxy (4'-CH_2—O-2') BNA, (B) β-D-methyleneoxy (4'-CH_2—O-2') BNA, (C) ethyleneoxy (4'-$(CH_2)_2$—O-2) BNA, (D) aminooxy (4'-CH_2—O—N(R)-2') BNA, (E) oxyamino (4'-CH_2—N(R)—O-2') BNA, and (F) methyl(methyleneoxy) (4'-$CH(CH_3)$—O-2') BNA, (G) methylene-thio (4'-CH_2—S-2') BNA, (H) methylene-amino (4'-CH_2—N(R)-2') BNA, (I) methyl carbocyclic (4'-CH_2—$CH(CH_3)$-2') BNA, and (J) propylene carbocyclic (4'-$(CH_2)_3$-2') BNA as depicted below.

(A)

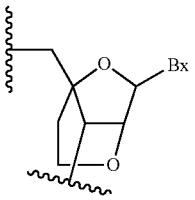


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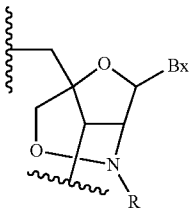
(B)



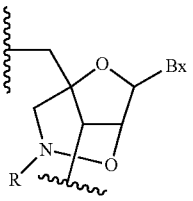
(C)



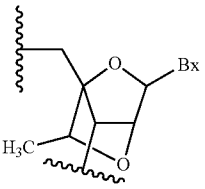
(D)



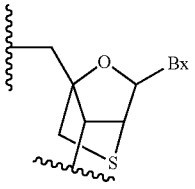
(E)



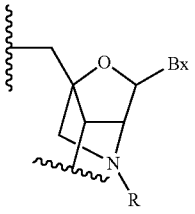
(F)



(G)

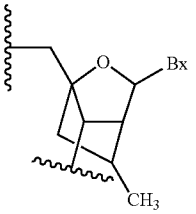


(H)

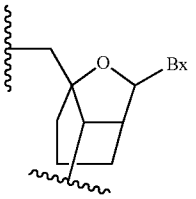


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(I)



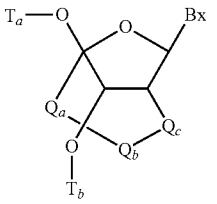
(J)



wherein Bx is the base moiety and R is independently H, a protecting group or C_1-C_{12} alkyl.

[0264] In certain embodiments, bicyclic nucleosides are provided having Formula I:

I



wherein:

[0265] Bx is a heterocyclic base moiety;

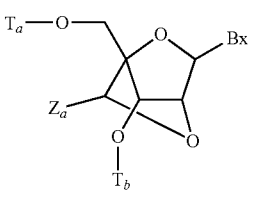
[0266] $-Q_a-Q_b-Q_c-$ is $—CH_2—N(R_c)—CH_2—$, $—C(=O)—N(R_c)—CH_2—$, $—CH_2—O—N(R_c)—$, $—CH_2—N(R_c)—O—$ or $—N(R_c)—O—CH_2$;

[0267] R_c is C_1-C_{12} alkyl or an amino protecting group; and

[0268] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium.

[0269] In certain embodiments, bicyclic nucleosides are provided having Formula II:

II



wherein:

[0270] Bx is a heterocyclic base moiety;

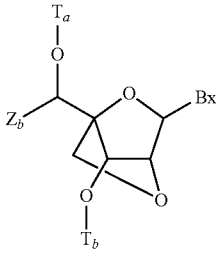
[0271] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0272] Z_a is C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_1-C_6 alkyl, substituted C_2-C_6 alkenyl, substituted C_2-C_6 alkynyl, acyl, substituted acyl, substituted amide, thiol or substituted thio.

[0273] In one embodiment, each of the substituted groups is, independently, mono or poly substituted with substituent groups independently selected from halogen, oxo, hydroxyl, OJ_c, NJ_cJ_d, SJ_c, N_3, $OC(=X)J_c$, and $NJ_eC(=X)NJ_cJ_d$, wherein each J_c, J_d and J_e is, independently, H, C_1-C_6 alkyl, or substituted C_1-C_6 alkyl and X is O or NJ_c.

[0274] In certain embodiments, bicyclic nucleosides are provided having Formula III:

III



wherein:

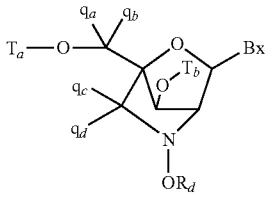
[0275] Bx is a heterocyclic base moiety;

[0276] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0277] Z_b is C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_1-C_6 alkyl, substituted C_2-C_6 alkenyl, substituted C_2-C_6 alkynyl or substituted acyl (C(=O)—).

[0278] In certain embodiments, bicyclic nucleosides are provided having Formula IV:

IV



wherein:

[0279] Bx is a heterocyclic base moiety;

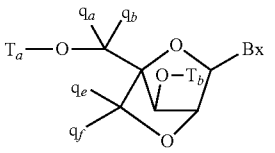
[0280] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0281] R_d is C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0282] each q_a, q_b, q_c and q_d is, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl, C_1-C_6 alkoxyl, substituted C_1-C_6 alkoxyl, acyl, substituted acyl, C_1-C_6 aminoalkyl or substituted C_1-C_6 aminoalkyl;

[0283] In certain embodiments, bicyclic nucleosides are provided having Formula V:

V



wherein:

[0284] Bx is a heterocyclic base moiety;

[0285] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0286] q_a, q_b, q_e and q_f are each, independently, hydrogen, halogen, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_1-C_{12} alkoxy, substituted C_1-C_{12} alkoxy, OJ_j, SJ_j, SOJ_j, SO_2J_j, NJ_jJ_k, N_3, CN, $C(=O)OJ_j$, $C(=O)NJ_jJ_k$, $C(=O)J_j$, $O—C(=O)NJ_jJ_k$, $N(H)C(=NH)NJ_jJ_k$, $N(H)C(=O)NJ_jJ_k$ or $N(H)C(=S)NJ_jJ_k$;

[0287] or q_e and q_f together are $=C(q_g)(q_h)$;

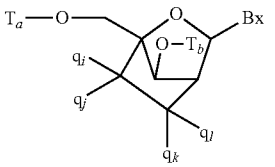
[0288] q_g and q_h are each, independently, H, halogen, C_1-C_{12} alkyl or substituted C_1-C_{12} alkyl.

[0289] The synthesis and preparation of the methyleneoxy (4'-CH_2—O-2') BNA monomers adenine, cytosine, guanine, 5-methyl-cytosine, thymine and uracil, along with their oligomerization, and nucleic acid recognition properties have been described (Koshkin et al., *Tetrahedron,* 1998, 54, 3607-3630). BNAs and preparation thereof are also described in WO 98/39352 and WO 99/14226.

[0290] Analogs of methyleneoxy (4'-CH_2—O-2') BNA and 2'-thio-BNAs, have also been prepared (Kumar et al., *Bioorg. Med. Chem. Lett.,* 1998, 8, 2219-2222). Preparation of locked nucleoside analogs comprising oligodeoxyribonucleotide duplexes as substrates for nucleic acid polymerases has also been described (Wengel et al., WO 99/14226). Furthermore, synthesis of 2'-amino-BNA, a novel comformationally restricted high-affinity oligonucleotide analog has been described in the art (Singh et al., *J. Org. Chem.,* 1998, 63, 10035-10039). In addition, 2'-amino- and 2'-methylamino-BNA's have been prepared and the thermal stability of their duplexes with complementary RNA and DNA strands has been previously reported.

[0291] In certain embodiments, bicyclic nucleosides are provided having Formula VI:

VI



wherein:

[0292] Bx is a heterocyclic base moiety;

[0293] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0294] each q_i, q_j, q_k and q_l is, independently, H, halogen, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_1-C_{12} alkoxyl, substituted C_1-C_{12} alkoxyl, OJ_j, SJ_j, SOJ_j, SO_2J_j, NJ_jJ_k, N_3, CN, C(═O)OJ_j, C(═O)NJ_jJ_k, C(═O)J_j, O—C(═O)NJ_jJ_k, N(H)C(═NH)NJ_jJ_k, N(H)C(═O)NJ_jJ_k or N(H)C(═S)NJ_jJ_k; and

[0295] q_i and q_j or q_l and q_k together are ═C(q_g)(q_h), wherein q_g and q_h are each, independently, H, halogen, C_1-C_{12} alkyl or substituted C_1-C_{12} alkyl.

[0296] One carbocyclic bicyclic nucleoside having a 4'-$(CH_2)_3$-2' bridge and the alkenyl analog bridge 4'-CH═CH—CH_2-2' have been described (Freier et al., *Nucleic Acids Research,* 1997, 25(22), 4429-4443 and Albaek et al., *J. Org. Chem.,* 2006, 71, 7731-7740). The synthesis and preparation of carbocyclic bicyclic nucleosides along with their oligomerization and biochemical studies have also been described (Srivastava et al., *J. Am. Chem. Soc.,* 2007, 129(26), 8362-8379).

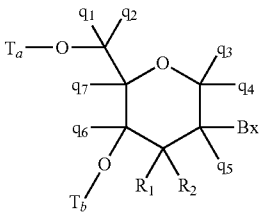
[0297] As used herein, "4'-2' bicyclic nucleoside" or "4' to 2' bicyclic nucleoside" refers to a bicyclic nucleoside comprising a furanose ring comprising a bridge connecting two carbon atoms of the furanose ring connects the 2' carbon atom and the 4' carbon atom of the sugar ring.

[0298] As used herein, "monocylic nucleosides" refer to nucleosides comprising modified sugar moieties that are not bicyclic sugar moieties. In certain embodiments, the sugar moiety, or sugar moiety analogue, of a nucleoside may be modified or substituted at any position.

[0299] As used herein, "2'-modified sugar" means a furanosyl sugar modified at the 2' position. In certain embodiments, such modifications include substituents selected from: a halide, including, but not limited to substituted and unsubstituted alkoxy, substituted and unsubstituted thioalkyl, substituted and unsubstituted amino alkyl, substituted and unsubstituted alkyl, substituted and unsubstituted allyl, and substituted and unsubstituted alkynyl. In certain embodiments, 2' modifications are selected from substituents including, but not limited to: $O[(CH_2)_nO]_mCH_3$, $O(CH_2)_nNH_2$, $O(CH_2)_nCH_3$, $O(CH_2)_nF$, $O(CH_2)_nONH_2$, $OCH_2C(═O)N(H)CH_3$, and $O(CH_2)_nON[(CH_2)_nCH_3]_2$, where n and m are from 1 to about 10. Other 2'- substituent groups can also be selected from: C_1-C_{12} alkyl, substituted alkyl, alkenyl, alkynyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH_3, OCN, Cl, Br, CN, F, CF_3, OCF_3, $SOCH_3$, SO_2CH_3, ONO_2, NO_2, N_3, NH_2, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving pharmacokinetic properties, or a group for improving the pharmacodynamic properties of an antisense compound, and other substituents having similar properties. In certain embodiments, modified nucleosides comprise a 2'-MOE side chain (Baker et al., *J. Biol. Chem.,* 1997, 272, 11944-12000). Such 2'-MOE substitution have been described as having improved binding affinity compared to unmodified nucleosides and to other modified nucleosides, such as 2'-O-methyl, O-propyl, and O-aminopropyl. Oligonucleotides having the 2'-MOE substituent also have been shown to be antisense inhibitors of gene expression with promising features for in vivo use (Martin, *Helv. Chim. Acta,* 1995, 78, 486-504; Altmann et al., *Chimia,* 1996, 50, 168-176; Altmann et al., *Biochem. Soc. Trans.,* 1996, 24, 630-637; and Altmann et al., *Nucleosides Nucleotides,* 1997, 16, 917-926).

[0300] As used herein, a "modified tetrahydropyran nucleoside" or "modified THP nucleoside" means a nucleoside having a six-membered tetrahydropyran "sugar" substituted in for the pentofuranosyl residue in normal nucleosides (a sugar surrogate). Modified THP nucleosides include, but are not limited to, what is referred to in the art as hexitol nucleic acid (HNA), anitol nucleic acid (ANA), manitol nucleic acid (MNA) (see Leumann, *Bioorg. Med. Chem.,* 2002, 10, 841-854), fluoro HNA (F-HNA) or those compounds having Formula VII:

VII



wherein independently for each of said at least one tetrahydropyran nucleoside analog of Formula VII:

[0301] Bx is a heterocyclic base moiety;

[0302] T_a and T_b are each, independently, an internucleoside linking group linking the tetrahydropyran nucleoside analog to the antisense compound or one of T_a and T_b is an internucleoside linking group linking the tetrahydropyran nucleoside analog to the antisense compound and the other of T_a and T_b is H, a hydroxyl protecting group, a linked conjugate group or a 5' or 3'-terminal group;

[0303] q_1, q_2, q_3, q_4, q_5, q_6 and q_7 are each independently, H, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl; and each of R_1 and R_2 is selected from hydrogen, hydroxyl, halogen, substituted or unsubstituted alkoxy, NJ_1J_2, SJ_1, N_3, OC(═X)J_1, OC(═X)NJ_1J_2, NJ_3C(═X)NJ_1J_2 and CN, wherein X is O, S or NJ_1 and each J_1, J_2 and J_3 is, independently, H or C_1-C_6 alkyl.

[0304] In certain embodiments, the modified THP nucleosides of Formula VII are provided wherein q_1, q_2, q_3, q_4, q_5, q_6 and q_7 are each H. In certain embodiments, at least one of q_1, q_2, q_3, q_4, q_5, q_6 and q_7 is other than H. In certain embodiments, at least one of q_1, q_2, q_3, q_4, q_5, q_6 and q_7 is methyl. In certain embodiments, THP nucleosides of Formula VII are provided wherein one of R_1 and R_2 is fluoro. In certain embodiments, R_1 is fluoro and R_2 is H; R_1 is methoxy and R_2 is H, and R_1 is H and R_2 is methoxyethoxy.

[0305] As used herein, "2'-modified" or "2'-substituted" refers to a nucleoside comprising a sugar comprising a substituent at the 2' position other than H or OH. 2'-modified nucleosides, include, but are not limited to, bicyclic nucleosides wherein the bridge connecting two carbon atoms of the sugar ring connects the 2' carbon and another carbon of the sugar ring; and nucleosides with non-bridging 2'substituents, such as allyl, amino, azido, thio, O-allyl, O—C_1-C_{10} alkyl, —OCF_3, O—$(CH_2)_2$—O—CH_3, 2'-O$(CH_2)_2SCH_3$, O—$(CH_2)_2$—O—N(R_m)(R_n), or O—CH_2—C(═O)—N(R_m)(R_n), where each R_m and R_n is, independently, H or substituted or unsubstituted C_1-C_{10} alkyl. 2'-modified nucleosides may further comprise other modifications, for example at other positions of the sugar and/or at the nucleobase.

[0306] As used herein, "2'-F" refers to a nucleoside comprising a sugar comprising a fluoro group at the 2' position.

[0307] As used herein, "2'-OMe" or "2'-OCH_3" or "2'-O-methyl" each refers to a nucleoside comprising a sugar comprising an —OCH_3 group at the 2' position of the sugar ring.

[0308] As used herein, "MOE" or "2'-MOE" or "2'-$OCH_2CH_2OCH_3$" or "2'-O-methoxyethyl" each refers to a nucleoside comprising a sugar comprising a —$OCH_2CH_2OCH_3$ group at the 2' position of the sugar ring.

[0309] As used herein, "oligonucleotide" refers to a compound comprising a plurality of linked nucleosides. In certain embodiments, one or more of the plurality of nucleosides is modified. In certain embodiments, an oligonucleotide comprises one or more ribonucleosides (RNA) and/or deoxyribonucleosides (DNA).

[0310] Many other bicyclo and tricyclo sugar surrogate ring systems are also known in the art that can be used to modify nucleosides for incorporation into antisense compounds (see for example review article: Leumann, *Bioorg. Med. Chem.,* 2002, 10, 841-854).

Such ring systems can undergo various additional substitutions to enhance activity.

[0311] Methods for the preparations of modified sugars are well known to those skilled in the art.

[0312] In nucleotides having modified sugar moieties, the nucleobase moieties (natural, modified or a combination thereof) are maintained for hybridization with an appropriate nucleic acid target.

[0313] In certain embodiments, antisense compounds comprise one or more nucleosides having modified sugar moieties. In certain embodiments, the modified sugar moiety is 2'-MOE. In certain embodiments, the 2'-MOE modified nucleosides are arranged in a gapmer motif In certain embodiments, the modified sugar moiety is a bicyclic nucleoside having a (4'-$CH(CH_3)$—O-2') bridging group. In certain embodiments, the (4'-$CH(CH_3)$—O-2') modified nucleosides are arranged throughout the wings of a gapmer motif.

Compositions and Methods for Formulating Pharmaceutical Compositions

[0314] Antisense oligonucleotides may be admixed with pharmaceutically acceptable active or inert substances for the preparation of pharmaceutical compositions or formulations. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

[0315] An antisense compound targeted to a C9ORF72 nucleic acid can be utilized in pharmaceutical compositions by combining the antisense compound with a suitable pharmaceutically acceptable diluent or carrier. A pharmaceutically acceptable diluent includes phosphate-buffered saline (PBS). PBS is a diluent suitable for use in compositions to be delivered parenterally. Accordingly, in one embodiment, employed in the methods described herein is a pharmaceutical composition comprising an antisense compound targeted to a C9ORF72 nucleic acid and a pharmaceutically acceptable diluent. In certain embodiments, the pharmaceutically acceptable diluent is PBS. In certain embodiments, the antisense compound is an antisense oligonucleotide.

[0316] Pharmaceutical compositions comprising antisense compounds encompass any pharmaceutically acceptable salts, esters, or salts of such esters, or any other oligonucleotide which, upon administration to an animal, including a human, is capable of providing (directly or indirectly) the biologically active metabolite or residue thereof. Accordingly, for example, the disclosure is also drawn to pharmaceutically acceptable salts of antisense compounds, prodrugs, pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts.

[0317] A prodrug can include the incorporation of additional nucleosides at one or both ends of an antisense compound which are cleaved by endogenous nucleases within the body, to form the active antisense compound.

Conjugated Antisense Compounds

[0318] Antisense compounds may be covalently linked to one or more moieties or conjugates which enhance the activity, cellular distribution or cellular uptake of the resulting antisense oligonucleotides. Typical conjugate groups include cholesterol moieties and lipid moieties. Additional conjugate groups include carbohydrates, phospholipids, biotin, phenazine, folate, phenanthridine, anthraquinone, acridine, fluoresceins, rhodamines, coumarins, and dyes.

[0319] Antisense compounds can also be modified to have one or more stabilizing groups that are generally attached to one or both termini of antisense compounds to enhance properties such as, for example, nuclease stability. Included in stabilizing groups are cap structures. These terminal modifications protect the antisense compound having terminal nucleic acid from exonuclease degradation, and can help in delivery and/or localization within a cell. The cap can be present at the 5'-terminus (5'-cap), or at the 3'-terminus (3'-cap), or can be present on both termini. Cap structures are well known in the art and include, for example, inverted deoxy abasic caps. Further 3' and 5'-stabilizing groups that can be used to cap one or both ends of an antisense compound to impart nuclease stability include those disclosed in WO 03/004602 published on Jan. 16, 2003.

Cell Culture and Antisense Compounds Treatment

[0320] The effects of antisense compounds on the level, activity or expression of C90RF72 nucleic acids can be tested in vitro in a variety of cell types. Cell types used for such analyses are available from commercial vendors (e.g. American Type Culture Collection, Manassas, Va.; Zen-Bio, Inc., Research Triangle Park, N.C.; Clonetics Corporation, Walkersville, Md.) and are cultured according to the vendor's instructions using commercially available reagents (e.g. Invitrogen Life Technologies, Carlsbad, Calif.). Illustrative cell types include, but are not limited to, HepG2 cells, Hep3B cells, and primary hepatocytes.

In Vitro Testing of Antisense Oligonucleotides

[0321] Described herein are methods for treatment of cells with antisense oligonucleotides, which can be modified appropriately for treatment with other antisense compounds.

[0322] In general, cells are treated with antisense oligonucleotides when the cells reach approximately 60-80% confluency in culture.

[0323] One reagent commonly used to introduce antisense oligonucleotides into cultured cells includes the cationic lipid transfection reagent LIPOFECTIN (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotides are mixed with LIPOFECTIN in OPTI-MEM 1 (Invitrogen, Carlsbad, Calif.) to achieve the desired final concentration of antisense oligo-

nucleotide and a LIPOFECTIN concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0324] Another reagent used to introduce antisense oligonucleotides into cultured cells includes LIPOFECTAMINE (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotide is mixed with LIPOFECTAMINE in OPTI-MEM 1 reduced serum medium (Invitrogen, Carlsbad, Calif.) to achieve the desired concentration of antisense oligonucleotide and a LIPOFECTAMINE concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0325] Another technique used to introduce antisense oligonucleotides into cultured cells includes electroporation.

[0326] Cells are treated with antisense oligonucleotides by routine methods. Cells are typically harvested 16-24 hours after antisense oligonucleotide treatment, at which time RNA or protein levels of target nucleic acids are measured by methods known in the art and described herein. In general, when treatments are performed in multiple replicates, the data are presented as the average of the replicate treatments.

[0327] The concentration of antisense oligonucleotide used varies from cell line to cell line. Methods to determine the optimal antisense oligonucleotide concentration for a particular cell line are well known in the art. Antisense oligonucleotides are typically used at concentrations ranging from 1 nM to 300 nM when transfected with LIPOFECTAMINE. Antisense oligonucleotides are used at higher concentrations ranging from 625 to 20,000 nM when transfected using electroporation.

RNA Isolation

[0328] RNA analysis can be performed on total cellular RNA or poly(A)+mRNA. Methods of RNA isolation are well known in the art. RNA is prepared using methods well known in the art, for example, using the TRIZOL Reagent (Invitrogen, Carlsbad, Calif.) according to the manufacturer's recommended protocols.

Analysis of Inhibition of Target Levels or Expression

[0329] Inhibition of levels or expression of a C9ORF72 nucleic acid can be assayed in a variety of ways known in the art. For example, target nucleic acid levels can be quantitated by, e.g., Northern blot analysis, competitive polymerase chain reaction (PCR), or quantitative real-time PCR. RNA analysis can be performed on total cellular RNA or poly(A)+ mRNA. Methods of RNA isolation are well known in the art. Northern blot analysis is also routine in the art. Quantitative real-time PCR can be conveniently accomplished using the commercially available ABI PRISM 7600, 7700, or 7900 Sequence Detection System, available from PE-Applied Biosystems, Foster City, Calif. and used according to manufacturer's instructions.

Quantitative Real-Time PCR Analysis of Target RNA Levels

[0330] Quantitation of target RNA levels may be accomplished by quantitative real-time PCR using the ABI PRISM 7600, 7700, or 7900 Sequence Detection System (PE-Applied Biosystems, Foster City, Calif.) according to manufacturer's instructions. Methods of quantitative real-time PCR are well known in the art.

[0331] Prior to real-time PCR, the isolated RNA is subjected to a reverse transcriptase (RT) reaction, which produces complementary DNA (cDNA) that is then used as the substrate for the real-time PCR amplification. The RT and real-time PCR reactions are performed sequentially in the same sample well. RT and real-time PCR reagents are obtained from Invitrogen (Carlsbad, Calif.). RT real-time-PCR reactions are carried out by methods well known to those skilled in the art.

[0332] Gene (or RNA) target quantities obtained by real time PCR are normalized using either the expression level of a gene whose expression is constant, such as cyclophilin A, or by quantifying total RNA using RIBOGREEN (Invitrogen, Inc. Carlsbad, Calif.). Cyclophilin A expression is quantified by real time PCR, by being run simultaneously with the target, multiplexing, or separately. Total RNA is quantified using RIBOGREEN RNA quantification reagent (Invetrogen, Inc. Eugene, Oreg.). Methods of RNA quantification by RIBOGREEN are taught in Jones, L. J., et al, (Analytical Biochemistry, 1998, 265, 368-374). A CYTOFLUOR 4000 instrument (PE Applied Biosystems) is used to measure RIBOGREEN fluorescence.

[0333] Probes and primers are designed to hybridize to a C9ORF72 nucleic acid. Methods for designing real-time PCR probes and primers are well known in the art, and may include the use of software such as PRIMER EXPRESS Software (Applied Biosystems, Foster City, Calif.).

Analysis of Protein Levels

[0334] Antisense inhibition of C9ORF72 nucleic acids can be assessed by measuring C9ORF72 protein levels. Protein levels of C9ORF72 can be evaluated or quantitated in a variety of ways well known in the art, such as immunoprecipitation, Western blot analysis (immunoblotting), enzyme-linked immunosorbent assay (ELISA), quantitative protein assays, protein activity assays (for example, caspase activity assays), immunohistochemistry, immunocytochemistry or fluorescence-activated cell sorting (FACS). Antibodies directed to a target can be identified and obtained from a variety of sources, such as the MSRS catalog of antibodies (Aerie Corporation, Birmingham, Mich.), or can be prepared via conventional monoclonal or polyclonal antibody generation methods well known in the art. Antibodies useful for the detection of mouse, rat, monkey, and human C9ORF72 are commercially available.

In Vivo Testing of Antisense Compounds

[0335] Antisense compounds, for example, antisense oligonucleotides, are tested in animals to assess their ability to inhibit expression of C9ORF72 and produce phenotypic changes, such as, improved motor function and respiration. In certain embodiments, motor function is measured by rotarod, grip strength, pole climb, open field performance, balance beam, hindpaw footprint testing in the animal. In certain embodiments, respiration is measured by whole body plethysmograph, invasive resistance, and compliance measurements in the animal. Testing may be performed in normal animals, or in experimental disease models. For administration to animals, antisense oligonucleotides are formulated in a pharmaceutically acceptable diluent, such as phosphate-buffered saline (PBS) or artificial cerebrospinal fluid (aCSF). Administration includes parenteral routes of administration, such as intraperitoneal, intravenous, and subcutaneous, as well as central routes of administration such as intracerebroventricular or intrathecal. Calculation of antisense oligonucleotide dosage and dosing frequency is within the abilities of those skilled in the art, and depends upon factors such as

route of administration and animal body weight. Following a period of treatment with antisense oligonucleotides, RNA is isolated from CNS tissue or CSF and changes in C9ORF72 nucleic acid expression are measured.

Targeting C9ORF72

[0336] Antisense oligonucleotides described herein may hybridize to a C9ORF72 nucleic acid in any stage of RNA processing. For example, described herein are antisense oligonucleotides that are complementary to a pre-mRNA or a mature mRNA. Additionally, antisense oligonucleotides described herein may hybridize to any element of a C9ORF72 nucleic acid. For example, described herein are antisense oligonucleotides that are complementary to an exon, an intron, the 5' UTR, the 3' UTR, a repeat region, a hexanucleotide repeat expansion, a splice junction, an exon:exon splice junction, an exonic splicing silencer (ESS), an exonic splicing enhancer (ESE), exon 1a, exon 1b, exon 1c, exon 1d, exon 1e, exon 2, exon 3, exon 4, exon 5, exon 6, exon 7, exon 8, exon 9, exon 10, exonl 1, intron 1, intron 2, intron 3, intron 4, intron 5, intron 6, intron 7, intron 8, intron 9, or intron 10 of a C9ORF72 nucleic acid.

[0337] In certain embodiments, antisense oligonucleotides described herein hybridize to all variants of C9ORF72. In certain embodiments, the antisense oligonucleotides described herein selectively hybridize to certain variants of C9ORF72. In certain embodiments, the antisense oligonucleotides described herein selectively hybridize to variants of C9ORF72 containing a hexanucleotide repeat expansion. In certain embodiments, the antisense oligonucleotides described herein selectively hybridize to pre-mRNA variants containing the hexanucleotide repeat. In certain embodiments, pre-mRNA variants of C9ORF72 containing a hexanucleotide repeat expansion include SEQ ID NO: 1-3 and 6-10. In certain embodiments, such hexanucleotide repeat expansion comprises at least 24 repeats of any of GGGGCC, GGGGGG, GGGGGC, or GGGGCG.

[0338] In certain embodiments, the antisense oligonucleotides described herein inhibit expression of all variants of C9ORF72. In certain embodiments, the antisense oligonucleotides described herein inhibit expression of all variants of C9ORF72 equally. In certain embodiments, the antisense oligonucleotides described herein preferentially inhibit expression of one or more variants of C9ORF72. In certain embodiments, the antisense oligonucleotides described herein preferentially inhibit expression of variants of C9ORF72 containing a hexanucleotide repeat expansion. In certain embodiments, the antisense oligonucleotides described herein selectively inhibit expression of pre-mRNA variants containing the hexanucleotide repeat. In certain embodiments, the antisense oligonucleotides described herein selectively inhibit expression of C9ORF72 pathogenic associated mRNA variants. In certain embodiments, pre-mRNA variants of C9ORF72 containing a hexanucleotide repeat expansion include SEQ ID NO: 1-3 and 6-10. In certain embodiments, such hexanucleotide repeat expansion comprises at least 24 repeats of any of GGGGCC, GGGGGG, GGGGGC, or GGGGCG. In certain embodiments, the hexanucleotide repeat expansion forms nuclear foci. In certain embodiments, antisense oligonucleotides described herein are useful for reducing nuclear foci. Nuclear foci may be reduced in terms of percent of cells with foci as well as number of foci per cell.

Selective Inhibition of Certain Pathogenic Associated Variants

[0339] In certain examples herein, primer probe set RTS3905 detects an mRNA variant (e.g. NM_001256054.1) processed from a pre-mRNA variant containing the hexanucleotide repeat. The mRNA variant processed from a pre-mRNA variant containing the hexanucleotide repeat (i.e., the "C9ORF72 pathogenic associated mRNA variant"). A pre-mRNA contains the hexanucleotide repeat when transcription of the pre-mRNA begins in the region from the start site of exon 1A to the start site of exon 1B, e.g., nucleotides 1107 to 1520 of the genomic sequence (SEQ ID NO: 2, the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). Oligonucleotides were designed in this region to selectively target the pre-mRNA variant containing the hexanucleotide repeat. RTS3905 measures an mRNA product (i.e. the C9ORF72 pathogenic associated mRNA variant) of the pre-mRNA variant containing the hexanucleotide repeat and, therefore, measures the reduction of the pre-mRNA variant containing the hexanucleotide repeat.

C90RF72 Features

[0340] Antisense oligonucleotides described herein may hybridize to any C9ORF72 variant at any state of processing within any element of the C9ORF72 gene. For example, antisense oligonucleotides described herein may hybridize to an exon, an intron, the 5' UTR, the 3' UTR, a repeat region, a hexanucleotide repeat expansion, a splice junction, an exon: exon splice junction, an exonic splicing silencer (ESS), an exonic splicing enhancer (ESE), exon 1a, exon 1b, exon 1c, exon 1d, exon 1e, exon 2, exon 3, exon 4, exon 5, exon 6, exon 7, exon 8, exon 9, exon 10, exon 11, intron 1, intron 2, intron 3, intron 4, intron 5, intron 6, intron 7, intron 8, intron 9, or intron 10. For example, antisense oligonucleotides may target any of the exons characterized below in Tables 1-5 for the various C9ORF72 variants described below. Antisense oligonucleotides described herein may also target variants not characterized below and such variants are characterized in GENBANK. Moreover, antisense oligonucleotides described herein may also target elements other than exons and such elements are characterized in GENBANK.

TABLE 1

Functional Segments for NM_001256054.1 (SEQ ID NO: 1)

Exon Number	mRNA start site	mRNA stop site	Start site in reference to SEQ ID NO: 2	Stop site in reference to SEQ ID NO: 2
exon 1C	1	158	1137	1294
exon 2	159	646	7839	8326
exon 3	647	706	9413	9472
exon 4	707	802	12527	12622
exon 5	803	867	13354	13418
exon 6	868	940	14704	14776
exon 7	941	1057	16396	16512
exon 8	1058	1293	18207	18442
exon 9	1294	1351	24296	24353
exon 10	1352	1461	26337	26446
exon 11	1462	3339	26581	28458

TABLE 2

Functional Segments for NM_018325.3 (SEQ ID NO: 4)

Exon Number	mRNA start site	mRNA stop site	Start site in reference to SEQ ID NO: 2	Stop site in reference to SEQ ID NO: 2
exon 1B	1	63	1510	1572
exon 2	64	551	7839	8326
exon 3	552	611	9413	9472
exon 4	612	707	12527	12622
exon 5	708	772	13354	13418
exon 6	773	845	14704	14776
exon 7	846	962	16396	16512
exon 8	963	1198	18207	18442
exon 9	1199	1256	24296	24353
exon 10	1257	1366	26337	26446
exon 11	1367	3244	26581	28458

TABLE 3

Functional Segments for NM_145005.5 (SEQ ID NO: 6)

Exon Number	mRNA start site	mRNA stop site	Start site in reference to SEQ ID NO: 2	Stop site in reference to SEQ ID NO: 2
exon 1A	1	80	1137	1216
exon 2	81	568	7839	8326
exon 3	569	628	9413	9472
exon 4	629	724	12527	12622
exon 5B (exon 5 into intron 5)	725	1871	13354	14500

TABLE 4

Functional Segments for DB079375.1 (SEQ ID NO: 7)

Exon Number	mRNA start site	mRNA stop site	Start site in reference to SEQ ID NO: 2	Stop site in reference to SEQ ID NO: 2
exon 1E	1	35	1135	1169
exon 2	36	524	7839	8326
exon 3 (EST ends before end of full exon)	525	562	9413	9450

TABLE 5

Functional Segments for BU194591.1 (SEQ ID NO: 8)

Exon Number	mRNA start site	mRNA stop site	Start site in reference to SEQ ID NO: 2	Stop site in reference to SEQ ID NO: 2
exon 1D	1	36	1241	1279
exon 2	37	524	7839	8326
exon 3	525	584	9413	9472
exon 4	585	680	12527	12622
exon 5B (exon 5 into intron 5)	681	798	13354	13465

Certain Indications

[0341] In certain embodiments, provided herein are methods of treating an individual comprising administering one or more pharmaceutical compositions described herein. In certain embodiments, the individual has a neurodegenerative disease. In certain embodiments, the individual is at risk for developing a neurodegenerative disease, including, but not limited to, ALS or FTD. In certain embodiments, the individual has been identified as having a C9ORF72 associated disease. In certain embodiments, the individual has been identified as having a C9ORF72 hexanucleotide repeat expansion associated disease. In certain embodiments, provided herein are methods for prophylactically reducing C9ORF72 expression in an individual. Certain embodiments include treating an individual in need thereof by administering to an individual a therapeutically effective amount of an antisense compound targeted to a C9ORF72 nucleic acid.

[0342] In one embodiment, administration of a therapeutically effective amount of an antisense compound targeted to a C9ORF72 nucleic acid is accompanied by monitoring of C9ORF72 levels in an individual, to determine an individual's response to administration of the antisense compound. An individual's response to administration of the antisense compound may be used by a physician to determine the amount and duration of therapeutic intervention.

[0343] In certain embodiments, administration of an antisense compound targeted to a C9ORF72 nucleic acid results in reduction of C9ORF72 expression by at least 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or a range defined by any two of these values. In certain embodiments, administration of an antisense compound targeted to a C9ORF72 nucleic acid results in improved motor function and respiration in an animal. In certain embodiments, administration of a C9ORF72 antisense compound improves motor function and respiration by at least 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or a range defined by any two of these values.

[0344] In certain embodiments, pharmaceutical compositions comprising an antisense compound targeted to C9ORF72 are used for the preparation of a medicament for treating a patient suffering or susceptible to a neurodegenerative disease including ALS and FTD.

Certain Combination Therapies

[0345] In certain embodiments, one or more pharmaceutical compositions described herein are co-administered with one or more other pharmaceutical agents. In certain embodiments, such one or more other pharmaceutical agents are designed to treat the same disease, disorder, or condition as the one or more pharmaceutical compositions described herein. In certain embodiments, such one or more other pharmaceutical agents are designed to treat a different disease, disorder, or condition as the one or more pharmaceutical compositions described herein. In certain embodiments, such one or more other pharmaceutical agents are designed to treat an undesired side effect of one or more pharmaceutical compositions described herein. In certain embodiments, one or more pharmaceutical compositions described herein are co-administered with another pharmaceutical agent to treat an undesired effect of that other pharmaceutical agent. In certain embodiments, one or more pharmaceutical compositions described herein are co-administered with another pharmaceutical agent to produce a combinational effect. In certain embodiments, one or more pharmaceutical compositions described herein are co-administered with another pharmaceutical agent to produce a synergistic effect.

[0346] In certain embodiments, one or more pharmaceutical compositions described herein and one or more other pharmaceutical agents are administered at the same time. In certain embodiments, one or more pharmaceutical compositions described herein and one or more other pharmaceutical agents are administered at different times. In certain embodiments, one or more pharmaceutical compositions described herein and one or more other pharmaceutical agents are prepared together in a single formulation. In certain embodiments, one or more pharmaceutical compositions described herein and one or more other pharmaceutical agents are prepared separately.

[0347] In certain embodiments, pharmaceutical agents that may be co-administered with a pharmaceutical composition described herein include Riluzole (Rilutek), Lioresal (Lioresal), and Dexpramipexole.

[0348] In certain embodiments, pharmaceutical agents that may be co-administered with a C9ORF72 specific inhibitor described herein include, but are not limited to, an additional C9ORF72 inhibitor. In certain embodiments, the co-administered pharmaceutical agent is administered prior to administration of a pharmaceutical composition described herein. In certain embodiments, the co-administered pharmaceutical agent is administered following administration of a pharmaceutical composition described herein. In certain embodiments the co-administered pharmaceutical agent is administered at the same time as a pharmaceutical composition described herein. In certain embodiments the dose of a co-administered pharmaceutical agent is the same as the dose that would be administered if the co-administered pharmaceutical agent was administered alone. In certain embodiments the dose of a co-administered pharmaceutical agent is lower than the dose that would be administered if the co-administered pharmaceutical agent was administered alone. In certain embodiments the dose of a co-administered pharmaceutical agent is greater than the dose that would be administered if the co-administered pharmaceutical agent was administered alone.

[0349] In certain embodiments, the co-administration of a second compound enhances the effect of a first compound, such that co-administration of the compounds results in an effect that is greater than the effect of administering the first compound alone. In other embodiments, the co-administration results in effects that are additive of the effects of the compounds when administered alone. In certain embodiments, the co-administration results in effects that are supra-additive of the effects of the compounds when administered alone. In certain embodiments, the first compound is an antisense compound. In certain embodiments, the second compound is an antisense compound.

Certain Amplicon Regions

[0350] Certain antisense oligonucleotides described herein may target the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of these compounds.

Certain Human Therapeutics

[0351] The human C9ORF72 antisense oligonucleotides described herein are being evaluated as possible human therapeutics. Various parameters of potency, efficacy, and/or tolerability are being examined. Such parameters include in vitro inhibition of total C9ORF72 RNA expression, in vitro inhibition of C9ORF72 pathogenic associated RNA variant expression, in vitro dose response (IC50), in vivo inhibition of total or pathogenic RNA and/or protein in a transgenic animal containing a human C9ORF72 transgene in relevant tissues (e.g., brain and/or spinal cord), tolerability in mouse, tolerability in rat, and/or tolerability in a primate. Tolerability markers that may be measured in elude blood and serum chemistry parameters, CSF chemistry parameters, body and organ weights, general observations and/or behavioral tests, and/or biochemical markers such as GFAP and/or AIF1. Acute or long term tolerability may be measured.

Certain Hotspot Regions

1. Nucleobases 1107-1520 of SEQ ID NO: 2

[0352] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1107-1520 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1107-1520 are a hotspot region. In certain embodiments, nucleobases 1107-1520 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 17, 18, or 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers or 5-8-5 MOE gapmers. In certain embodiments, the antisense oligonucleotides are 17-mer Deoxy, MOE and cEt oligonucleotides. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleotide linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0353] In certain embodiments, nucleobases 1107-1520 are targeted by the following ISIS numbers: 619042-619333, 672581-672714, 672735-672865, 672885-673015, 673035-673165, 673185-673315, and 673335-673465.

[0354] In certain embodiments, nucleobases 1107-1520 are targeted by the following SEQ ID NOs: 21-31, 33-50, 52, 54-134, 138-248, 251-319, 325, 744-877, and 898-1028.

[0355] In certain embodiments, nucleobases 1107-1520 are targeted by the following ISIS numbers: 619042-619333.

[0356] In certain embodiments, nucleobases 1107-1520 are targeted by the following SEQ ID NOs: 21-31, 33-50, 52, 54-134, 138-248, 251-319, and 325.

[0357] In certain embodiments, antisense oligonucleotides targeting nucleobases 1107-1520 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at

least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0358] In certain embodiments, antisense oligonucleotides targeting nucleobases 1107-1520 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C90RF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

2. Nucleobases 1111-1200 of SEQ ID NO: 2

[0359] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1111-1200 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1111-1200 are a hotspot region. In certain embodiments, nucleobases 1111-1200 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0360] In certain embodiments, nucleobases 1111-1200 are targeted by the following ISIS numbers: 619042-619095.

[0361] In certain embodiments, nucleobases 1111-1200 are targeted by the following SEQ ID NOs: 21, 26-31, 33-50, 52, 54-60, 75, 81, and 87-96.

[0362] In certain embodiments, antisense oligonucleotides targeting nucleobases 1111-1200 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0363] In certain embodiments, antisense oligonucleotides targeting nucleobases 1111-1200 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

3. Nucleobases 1211-1318 of SEQ ID NO: 2

[0364] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1211-1318 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1211-1318 are a hotspot region. In certain embodiments, nucleobases 1211-1318 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucle-

otides are linked by phosphodiester internucleotide linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0365] In certain embodiments, nucleobases 1211-1318 are targeted by the following ISIS numbers: 619096-619172.

[0366] In certain embodiments, nucleobases 1211-1318 are targeted by the following SEQ ID NOs: 22-25, 70-74, 76-80, 82-86, 99-134, and 138-159.

[0367] In certain embodiments, antisense oligonucleotides targeting nucleobases 1211-1318 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0368] In certain embodiments, antisense oligonucleotides targeting nucleobases 1211-1318 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C90RF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

4. Nucleobases 1326-1540 of SEQ ID NO: 2

[0369] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1326-1540 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1326-1540 are a hotspot region. In certain embodiments, nucleobases 1326-1540 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 17, 18, or 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers or 5-8-5 MOE gapmers. In certain embodiments, the antisense oligonucleotides are 17-mer Deoxy, MOE and cEt oligonucleotides. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleotide linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0370] In certain embodiments, nucleobases 1326-1540 are targeted by the following ISIS numbers: 619173-619354, and 672581-673484.

[0371] In certain embodiments, nucleobases 1326-1540 are targeted by the following SEQ ID NOs: 97, 98, 160-248, 251-322, 325-343, and 744-1047.

[0372] In certain embodiments, nucleobases 1326-1540 are targeted by the following ISIS numbers: 619173-619354.

[0373] In certain embodiments, nucleobases 1326-1540 are targeted by the following SEQ ID NOs: 97, 98, 160-248, 251-322, and 325-343.

[0374] In certain embodiments, antisense oligonucleotides targeting nucleobases 1326-1540 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0375] In certain embodiments, antisense oligonucleotides targeting nucleobases 1326-1540 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at

least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

5. Nucleobases 1331-1375 of SEQ ID NO: 2

[0376] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1331-1375 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1331-1375 are a hotspot region. In certain embodiments, nucleobases 1331-1375 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooosssssssssssooss.

[0377] In certain embodiments, nucleobases 1331-1375 are targeted by the following ISIS numbers: 619178-619203.

[0378] In certain embodiments, nucleobases 1331-1375 are targeted by the following SEQ ID NOs: 165-190.

[0379] In certain embodiments, antisense oligonucleotides targeting nucleobases 1331-1375 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

6. Nucleobases 1368-1391 of SEQ ID NO: 2

[0380] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1368-1391 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1368-1391 are a hotspot region. In certain embodiments, nucleobases 1368-1391 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleotide linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooosssssssssssooss.

[0381] In certain embodiments, nucleobases 1368-1391 are targeted by the following ISIS numbers: 619215-619219.

[0382] In certain embodiments, nucleobases 1368-1391 are targeted by the following SEQ ID NOs: 202-206.

[0383] In certain embodiments, antisense oligonucleotides targeting nucleobases 1368-1391 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

7. Nucleobases 1398-1424 of SEQ ID NO: 2

[0384] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1398-1424 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1398-1424 are a hotspot region. In certain embodiments, nucleobases 1398-1424 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleotide linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooosssssssssssooss.

[0385] In certain embodiments, nucleobases 1398-1424 are targeted by the following ISIS numbers: 619245-619252.

[0386] In certain embodiments, nucleobases 1398-1424 are targeted by the following SEQ ID NOs: 232-239.

[0387] In certain embodiments, antisense oligonucleotides targeting nucleobases 1398-1424 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

8. Nucleobases 1411-1440 of SEQ ID NO: 2

[0388] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1411-1440 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1411-1440 are a hotspot region. In certain embodiments, nucleobases 1411-1440 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleotide linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooosssssssssssooss.

[0389] In certain embodiments, nucleobases 1411-1440 are targeted by the following ISIS numbers: 619258-619268.

[0390] In certain embodiments, nucleobases 1411-1440 are targeted by the following SEQ ID NOs: 244-248, 251-255, and 325.

[0391] In certain embodiments, antisense oligonucleotides targeting nucleobases 1411-1440 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

9. Nucleobases 1429-1481 of SEQ ID NO: 2

[0392] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1429-1481 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1429-1481 are a hotspot region. In certain embodiments, nucleobases 1429-1481 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain

embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooossssssssssooss.

[0393] In certain embodiments, nucleobases 1429-1481 are targeted by the following ISIS numbers: 619276-619303.

[0394] In certain embodiments, nucleobases 1429-1481 are targeted by the following SEQ ID NOs: 98 and 263-289.

[0395] In certain embodiments, antisense oligonucleotides targeting nucleobases 1429-1481 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

10. Nucleobases 1502-1539 of SEQ ID NO: 2

[0396] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1502-1539 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1502-1539 are a hotspot region. In certain embodiments, nucleobases 1502-1539 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooossssssssssooss.

[0397] In certain embodiments, nucleobases 1502-1539 are targeted by the following ISIS numbers: 619335-619353.

[0398] In certain embodiments, nucleobases 1502-1539 are targeted by the following SEQ ID NOs: 321, 322, and 326-342.

[0399] In certain embodiments, antisense oligonucleotides targeting nucleobases 1502-1539 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

11. Nucleobases 1508-1539 of SEQ ID NO: 2

[0400] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 1508-1539 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 1508-1539 are a hotspot region. In certain embodiments, nucleobases 1508-1539 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleotide linkages (e.g., the antisense oligonucleotides have "mixed backbones"). In certain embodiments, the antisense oligonucleotides comprise the following sugar modification pattern: soooossssssssssooss.

[0401] In certain embodiments, nucleobases 1508-1539 are targeted by the following ISIS numbers: 619341-619353.

[0402] In certain embodiments, nucleobases 1508-1539 are targeted by the following SEQ ID NOs: 330-342.

[0403] In certain embodiments, antisense oligonucleotides targeting nucleobases 1508-1539 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least

62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo

12. Nucleobases 7860-7906 of SEQ ID NO: 2

[0404] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 7860-7906 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 7860-7906 are a hotspot region. In certain embodiments, nucleobases 7860-7906 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0405] In certain embodiments, nucleobases 7860-7906 are targeted by the following ISIS numbers: 655135-655144.

[0406] In certain embodiments, nucleobases 7860-7906 are targeted by the following SEQ ID NOs: 445-454.

[0407] In certain embodiments, antisense oligonucleotides targeting nucleobases 7860-7906 achieve at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, or at least 89% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

13. Nucleobases 7907-7944 of SEQ ID NO: 2

[0408] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 7907-7944 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 7907-7944 are a hotspot region. In certain embodiments, nucleobases 7907-7944 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0409] In certain embodiments, nucleobases 7907-7944 are targeted by the following ISIS numbers: 655150-655156.

[0410] In certain embodiments, nucleobases 7907-7944 are targeted by the following SEQ ID NOs: 460-467.

[0411] In certain embodiments, antisense oligonucleotides targeting nucleobases 7907-7944 achieve at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, or at least 91% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

14. Nucleobases 7989-8038 of SEQ ID NO: 2

[0412] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 7989-8038 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 7989-8038 are a hotspot region. In certain embodiments, nucleobases 7989-8038 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester Internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0413] In certain embodiments, nucleobases 7989-8038 are targeted by the following ISIS numbers: 619411, 619412, 619420, 625183, 627833, and 655173-655180.

[0414] In certain embodiments, nucleobases 7989-8038 are targeted by the following SEQ ID NOs: 20, 51, 53, and 484-493.

[0415] In certain embodiments, antisense oligonucleotides targeting nucleobases 7989-8038 achieve at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, or at least 76% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0416] In certain embodiments, antisense oligonucleotides targeting nucleobases 7989-8038 achieve at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

15. Nucleobases 8020-8135 of SEQ ID NO: 2

[0417] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 8020-8135 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 8020-8135 are a hotspot region. In certain embodiments, nucleobases 8020-8135 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0418] In certain embodiments, nucleobases 8020-8135 are targeted by the following ISIS numbers: 619413, 619414, 625255, 627834, 655181-655208.

[0419] In certain embodiments, nucleobases 8020-8135 are targeted by the following SEQ ID NOs: 135, 136, 494-511, and 517-528.

[0420] In certain embodiments, antisense oligonucleotides targeting nucleobases 8020-8135 achieve at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, at least 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, or at least 54% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0421] In certain embodiments, antisense oligonucleotides targeting nucleobases 8020-8135 achieve at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

16. Nucleobases 8136-8161 of SEQ ID NO: 2

[0422] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 8136-8161 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 8136-8161 are a hotspot region. In certain embodiments, nucleobases 8136-8161 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0423] In certain embodiments, nucleobases 8136-8161 are targeted by the following ISIS numbers: 655215-655217.

[0424] In certain embodiments, nucleobases 8136-8161 are targeted by the following SEQ ID NOs: 535-537.

[0425] In certain embodiments, antisense oligonucleotides targeting nucleobases 8136-8161 achieve at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, or at least 41% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0426] In certain embodiments, antisense oligonucleotides targeting nucleobases 8136-8161 achieve at least 91%, at least 92%, at least 93%, at least 94%, or at least 95% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

17. Nucleobases 8174-8211 of SEQ ID NO: 2

[0427] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 8174-8211 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 8174-8211 are a hotspot region. In certain embodiments, nucleobases 8174-8211 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0428] In certain embodiments, nucleobases 8174-8211 are targeted by the following ISIS numbers: 655228-655234.

[0429] In certain embodiments, nucleobases 8174-8211 are targeted by the following SEQ ID NOs: 548-554.

[0430] In certain embodiments, antisense oligonucleotides targeting nucleobases 8174-8211 achieve at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at

least 62%, or at least 63% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0431] In certain embodiments, antisense oligonucleotides targeting nucleobases 8174-8211 achieve at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 100% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

18. Nucleobases 8213-8325 of SEQ ID NO: 2

[0432] In certain embodiments, antisense oligonucleotides are designed to target nucleobases 8213-8325 of SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000). In certain embodiments, nucleobases 8213-8325 are a hotspot region. In certain embodiments, nucleobases 8213-8325 are targeted by antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are 20 nucleobases in length. In certain embodiments, the antisense oligonucleotides are gapmers. In certain embodiments, the gapmers are 5-10-5 MOE gapmers. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphodiester internucleoside linkages. In certain embodiments, the nucleosides of the antisense oligonucleotides are linked by phosphorothioate and phosphodiester internucleoside linkages (e.g., the antisense oligonucleotides have "mixed backbones").

[0433] In certain embodiments, nucleobases 8213-8325 are targeted by the following ISIS numbers: 655235-655270.

[0434] In certain embodiments, nucleobases 8213-8325 are targeted by the following SEQ ID NOs: 555-590.

[0435] In certain embodiments, antisense oligonucleotides targeting nucleobases 8213-8325 achieve at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 6%, at least 7%, at least 8%, at least 9%, at least 10%, at least 11%, at least 12%, at least 13%, at least 14%, at least 15%, at least 16%, at least 17%, at least 18%, at least 19%, at least 20%, at least 21%, at least 22%, at least 23%, at least 24%, at least 25%, at least 26%, at least 27%, at least 28%, at least 29%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, at least 35%, at least 36%, at least 37%, at least 38%, at least 39%, at least 40%, at least 41%, at least 42%, at least 43%, at least 44%, at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, or at least 51% reduction of total C9ORF72 mRNA and/or protein levels in vitro and/or in vivo.

[0436] In certain embodiments, antisense oligonucleotides targeting nucleobases 8213-8325 achieve at least 45%, at least 46%, at least 47%, at least 48%, at least 49%, at least 50%, at least 51%, at least 52%, at least 53%, at least 54%, at least 55%, at least 56%, at least 57%, at least 58%, at least 59%, at least 60%, at least 61%, at least 62%, at least 63%, at least 64%, at least 65%, at least 66%, at least 67%, at least 68%, at least 69%, at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, or at least 98% reduction of C9ORF72 pathogenic associated mRNA variant levels in vitro and/or in vivo.

EXAMPLES

Non-Limiting Disclosure and Incorporation by Reference

[0437] While certain compounds, compositions, and methods described herein have been described with specificity in accordance with certain embodiments, the following examples serve only to illustrate the compounds described herein and are not intended to limit the same. Each of the references recited in the present application is incorporated herein by reference in its entirety.

Example 1

Antisense Inhibition of a Human C9ORF72 mRNA Variant in HepG2 Cells by MOE Gapmers

[0438] Antisense oligonucleotides targeting a C9ORF72 nucleic acid were designed and tested for their effects on C9ORF72 mRNA in vitro. ISIS 576816, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. ISIS 576816 is a 5-10-5 MOE gapmer with phosphorothioate linkages throughout. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 4,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 (forward primer sequence GGGTCTAGCAAGAGCAGGTG, designated herein as SEQ ID NO: 13; reverse primer sequence GTCTTGGCAACAGCTGGAGAT, designated herein as SEQ ID NO: 14; probe sequence TGATGTC-GACTCTTTGCCCACCGC, designated herein as SEQ ID NO: 15—a TAQ-man primer probe set) was used. RTS3905 detects an mRNA variant (e.g. NM_001256054.1) processed from a pre-mRNA variant containing the hexanucleotide repeat. The mRNA variant processed from a pre-mRNA variant containing the hexanucleotide repeat is herein the "C9ORF72 pathogenic associated mRNA variant." A pre-mRNA contains the hexanucleotide repeat when transcription of the pre-mRNA begins in the region from the start site of exon 1A to the start site of exon 1B (generally nucleotides 1107 to 1520 of the genomic sequence: SEQ ID NO: 2, the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000. Therefore, oligonucleotides were designed in this region selectively target the pre-mRNA variant containing the hexanucleotide repeat. RTS3905 measures an mRNA product (i.e. the C9ORF72 pathogenic associated mRNA variant) of the pre-mRNA variant containing the hexanucleotide repeat and, therefore, measures the reduction of the pre-mRNA variant containing the hexanucleotide repeat. The levels of the C9ORF72 pathogenic associated mRNA variant were normalized to the total RNA content of the cell, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72, relative to untreated control cells. The oligonucleotide marked with an asterisk (*) targets the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of these oligonucleotides.

[0439] The chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment comprises ten 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and on the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment comprises a

2'-MOE group. All cytosine residues throughout each oligonucleotide are 5-methylcytosines. The internucleoside linkages for the gapmers are mixed phosphorothioate and phosphodiester linkages. The internucleoside linkages for each gapmer are presented in the Linkage column, where 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage.

[0440] "Start site" indicates the 5'-most nucleoside to which the gapmer is targeted in the human gene sequence. "Stop site" indicates the 3'-most nucleoside to which the gapmer is targeted human gene sequence. Each antisense oligonucleotide listed in the Table below is targeted to either human C9ORF72 mRNA sequence, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_001256054.1) or the human C9ORF72 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000), or both. 'n/a' indicates that the antisense oligonucleotide does not target that particular gene sequence

TABLE 6

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
576816	310	7990	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	40	20
619060	10	1146	CTTTCCTAGCGGGACACCGT	soooossssssssssooss	34	21
619111	100	1236	AAAAGAGAAGCAACCGGGCA	soooossssssssssooss	34	22
619112	101	1237	CAAAAGAGAAGCAACCGGGC	soooossssssssssooss	38	23
619113	102	1238	CCAAAAGAGAAGCAACCGGG	soooossssssssssooss	44	24
619114	103	1239	CCCAAAAGAGAAGCAACCGG	soooossssssssssooss	100	25
619061	11	1147	TCTTTCCTAGCGGGACACCG	soooossssssssssooss	100	26
619062	12	1148	CTCTTTCCTAGCGGGACACC	soooossssssssssooss	20	27
619063	13	1149	TCTCTTTCCTAGCGGGACAC	soooossssssssssooss	14	28
619064	14	1150	CTCTCTTTCCTAGCGGGACA	soooossssssssssooss	9	29
619065	15	1151	CCTCTCTTTCCTAGCGGGAC	soooossssssssssooss	12	30
619066	16	1152	ACCTCTCTTTCCTAGCGGGA	soooossssssssssooss	0	31
619410*	160	7840	ATCCAAATGCTCCGGAGATA	soooossssssssssooss	96	32
619067	17	1153	CACCTCTCTTTCCTAGCGGG	soooossssssssssooss	0	33
619068	18	1154	GCACCTCTCTTTCCTAGCGG	soooossssssssssooss	24	34
619069	19	1155	CGCACCTCTCTTTCCTAGCG	soooossssssssssooss	39	35
619052	2	1138	GCGGGACACCGTAGGTTACG	soooossssssssssooss	30	36
619070	20	1156	ACGCACCTCTCTTTCCTAGC	soooossssssssssooss	14	37
619071	21	1157	GACGCACCTCTCTTTCCTAG	soooossssssssssooss	7	38
619072	22	1158	TGACGCACCTCTCTTTCCTA	soooossssssssssooss	50	39
619073	23	1159	TTGACGCACCTCTCTTTCCT	soooossssssssssooss	0	40
619074	24	1160	TTTGACGCACCTCTCTTTCC	soooossssssssssooss	55	41
619075	25	1161	GTTTGACGCACCTCTCTTTC	soooossssssssssooss	12	42
619076	26	1162	TGTTTGACGCACCTCTCTTT	soooossssssssssooss	65	43
619077	27	1163	CTGTTTGACGCACCTCTCTT	soooossssssssssooss	28	44
619078	28	1164	GCTGTTTGACGCACCTCTCT	soooossssssssssooss	18	45
619079	29	1165	CGCTGTTTGACGCACCTCTC	soooossssssssssooss	14	46

TABLE 6-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619053	3	1139	AGCGGGACACCGTAGGTTAC	soooossssssssssooss	23	47
619080	30	1166	TCGCTGTTTGACGCACCTCT	soooossssssssssooss	100	48
619081	31	1167	GTCGCTGTTTGACGCACCTC	soooossssssssssooss	23	49
619082	32	1168	TGTCGCTGTTTGACGCACCT	soooossssssssssooss	0	50
619411	324	8004	TGGAGCCCAAATGTGCCTTA	soooossssssssssooss	99	51
619083	33	1169	TTGTCGCTGTTTGACGCACC	soooossssssssssooss	36	52
619412	332	8012	TCTGTCTTTGGAGCCCAAAT	soooossssssssssooss	100	53
619084	34	1170	CTTGTCGCTGTTTGACGCAC	soooossssssssssooss	28	54
619085	35	1171	ACTTGTCGCTGTTTGACGCA	soooossssssssssooss	55	55
619086	36	1172	AACTTGTCGCTGTTTGACGC	soooossssssssssooss	29	56
619087	37	1173	GAACTTGTCGCTGTTTGACG	soooossssssssssooss	21	57
619088	38	1174	GGAACTTGTCGCTGTTTGAC	soooossssssssssooss	100	58
619089	39	1175	CGGAACTTGTCGCTGTTTGA	soooossssssssssooss	67	59
619054	4	1140	TAGCGGGACACCGTAGGTTA	soooossssssssssooss	59	60
619090	40	1176	GCGGAACTTGTCGCTGTTTG	soooossssssssssooss	8	61
619091	41	1177	GGCGGAACTTGTCGCTGTTT	soooossssssssssooss	38	62
619092	42	1178	GGGCGGAACTTGTCGCTGTT	soooossssssssssooss	16	63
619093	43	1179	TGGGCGGAACTTGTCGCTGT	soooossssssssssooss	22	64
619094	44	1180	GTGGGCGGAACTTGTCGCTG	soooossssssssssooss	24	65
619095	45	1181	CGTGGGCGGAACTTGTCGCT	soooossssssssssooss	100	66
619055	5	1141	CTAGCGGGACACCGTAGGTT	soooossssssssssooss	100	67
619056	6	1142	CCTAGCGGGACACCGTAGGT	soooossssssssssooss	11	68
619057	7	1143	TCCTAGCGGGACACCGTAGG	soooossssssssssooss	22	69
619096	75	1211	GACGGCTGACACACCAAGCG	soooossssssssssooss	100	70
619097	76	1212	GGACGGCTGACACACCAAGC	soooossssssssssooss	88	71
619098	77	1213	GGGACGGCTGACACACCAAG	soooossssssssssooss	29	72
619099	78	1214	AGGGACGGCTGACACACCAA	soooossssssssssooss	83	73
619100	79	1215	CAGGGACGGCTGACACACCA	soooossssssssssooss	23	74
619058	8	1144	TTCCTAGCGGGACACCGTAG	soooossssssssssooss	32	75
619101	80	1216	GCAGGGACGGCTGACACACC	soooossssssssssooss	18	76
619102	81	1217	AGCAGGGACGGCTGACACAC	soooossssssssssooss	39	77
619103	82	1218	CAGCAGGGACGGCTGACACA	soooossssssssssooss	39	78
619104	83	1219	GCAGCAGGGACGGCTGACAC	soooossssssssssooss	22	79
619105	84	1220	GGCAGCAGGGACGGCTGACA	soooossssssssssooss	43	80

TABLE 6-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619059	9	1145	TTTCCTAGCGGGACACCGTA	soooossssssssssooss	4	81
619106	95	1231	AGAAGCAACCGGGCAGCAGG	soooossssssssssooss	54	82
619107	96	1232	GAGAAGCAACCGGGCAGCAG	soooossssssssssooss	100	83
619108	97	1233	AGAGAAGCAACCGGGCAGCA	soooossssssssssooss	32	84
619109	98	1234	AAGAGAAGCAACCGGGCAGC	soooossssssssssooss	30	85
619110	99	1235	AAAGAGAAGCAACCGGGCAG	soooossssssssssooss	44	86
619042	n/a	1111	GTTTTCTATGTGCGATGACG	soooossssssssssooss	6	87
619043	n/a	1112	TGTTTTCTATGTGCGATGAC	soooossssssssssooss	45	88
619044	n/a	1113	CTGTTTTCTATGTGCGATGA	soooossssssssssooss	23	89
619045	n/a	1114	TCTGTTTTCTATGTGCGATG	soooossssssssssooss	10	90
619046	n/a	1115	GTCTGTTTTCTATGTGCGAT	soooossssssssssooss	11	91
619047	n/a	1116	TGTCTGTTTTCTATGTGCGA	soooossssssssssooss	28	92
619048	n/a	1117	CTGTCTGTTTTCTATGTGCG	soooossssssssssooss	34	93
619049	n/a	1118	TCTGTCTGTTTTCTATGTGC	soooossssssssssooss	72	94
619050	n/a	1119	GTCTGTCTGTTTTCTATGTG	soooossssssssssooss	37	95
619051	n/a	1120	CGTCTGTCTGTTTTCTATGT	soooossssssssssooss	1	96
619253	n/a	1406	TACAGGCTGCGGTTGTTTCC	soooossssssssssooss	100	97
619293	n/a	1446	CCCGGCCCCTAGCGCGCGAC	soooossssssssssooss	98	98

Example 2

Antisense Inhibition of a Human C9ORF72 mRNA Variant in HepG2 Cells by MOE Gapmers

[0441] Additional antisense oligonucleotides targeting a C9ORF72 nucleic acid were designed and tested for their effects on C9ORF72 mRNA in vitro. ISIS 576816, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 1,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant, which is the product of a pre-mRNA containing a hexanucleotide repeat. The levels of the C9ORF72 pathogenic associated mRNA variant were normalized to the total RNA content of the cell, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72, relative to untreated control cells. The oligonucleotides marked with as asterisk (*) targets the region of the primer probe set. Additional assays may be used to measure the potency and efficacy of these oligonucleotides. 'n.d.' indicates that there was no signal reading in the assay for that particular oligonucleotide.

[0442] The chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment comprises ten 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and on the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment comprises a 2'-MOE group. All cytosine residues throughout each oligonucleotide are 5-methylcytosines. The internucleoside linkages for the gapmers are mixed phosphorothioate and phosphodiester linkages. The internucleoside linkages for each gapmer are presented in the Linkage column, where 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage.

[0443] "Start site" indicates the 5'-most nucleoside to which the gapmer is targeted in the human gene sequence. "Stop site" indicates the 3'-most nucleoside to which the

gapmer is targeted human gene sequence. The gapmers of the Tables below also target either human C9ORF72 mRNA sequence, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_001256054.1) or the human C9ORF72 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000), or both. Some of the gapmers of Table 10 are targeted to GENBANK Accession No. NM_145005.5 (incorporated herein as SEQ ID NO: 3), GENBANK Accession No. DB079375.1 (incorporated herein as SEQ ID NO: 4), or GENBANK Accession No. BU194591.1 (incorporated herein as SEQ ID NO: 5). 'n/a' indicates that the antisense oligonucleotide does not target that particular gene sequence.

TABLE 7

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
576816	310	7990	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	70	20
619115	104	1240	CCCCAAAAGAGAAGCAACCG	soooossssssssssooss	10	99
619116	105	1241	CCCCCAAAAGAGAAGCAACC	soooossssssssssooss	19	100
619117	106	1242	GCCCCCAAAAGAGAAGCAAC	soooossssssssssooss	0	101
619118*	107	1243	CGCCCCCAAAAGAGAAGCAA	soooossssssssssooss	17	102
619119*	108	1244	CCGCCCCCAAAAGAGAAGCA	soooossssssssssooss	14	103
619120*	109	1245	CCCGCCCCCAAAAGAGAAGC	soooossssssssssooss	4	104
619121*	110	1246	CCCCGCCCCCAAAAGAGAAG	soooossssssssssooss	19	105
619122*	111	1247	ACCCCGCCCCCAAAAGAGAA	soooossssssssssooss	47	106
619123*	112	1248	GACCCCGCCCCCAAAAGAGA	soooossssssssssooss	11	107
619124*	113	1249	AGACCCCGCCCCCAAAAGAG	soooossssssssssooss	16	108
619125*	114	1250	TAGACCCCGCCCCCAAAAGA	soooossssssssssooss	15	109
619126*	115	1251	CTAGACCCCGCCCCCAAAAG	soooossssssssssooss	43	110
619127*	116	1252	GCTAGACCCCGCCCCCAAAA	soooossssssssssooss	67	111
619128*	117	1253	TGCTAGACCCCGCCCCCAAA	soooossssssssssooss	85	112
619129*	118	1254	TTGCTAGACCCCGCCCCCAA	soooossssssssssooss	41	113
619130*	119	1255	CTTGCTAGACCCCGCCCCCA	soooossssssssssooss	62	114
619131*	120	1256	TCTTGCTAGACCCCGCCCCC	soooossssssssssooss	95	115
619132*	121	1257	CTCTTGCTAGACCCCGCCCC	soooossssssssssooss	81	116
619133*	122	1258	GCTCTTGCTAGACCCCGCCC	soooossssssssssooss	90	117
619134*	123	1259	TGCTCTTGCTAGACCCCGCC	soooossssssssssooss	85	118
619135*	124	1260	CTGCTCTTGCTAGACCCCGC	soooossssssssssooss	81	119
619136*	125	1261	CCTGCTCTTGCTAGACCCCG	soooossssssssssooss	78	120
619137*	126	1262	ACCTGCTCTTGCTAGACCCC	soooossssssssssooss	85	121
619138*	127	1263	CACCTGCTCTTGCTAGACCC	soooossssssssssooss	81	122
619139*	128	1264	ACACCTGCTCTTGCTAGACC	soooossssssssssooss	77	123
619140*	129	1265	CACACCTGCTCTTGCTAGAC	soooossssssssssooss	86	124
619141*	130	1266	CCACACCTGCTCTTGCTAGA	soooossssssssssooss	90	125
619142*	131	1267	CCCACACCTGCTCTTGCTAG	soooossssssssssooss	98	126

TABLE 7-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619143*	132	1268	ACCCACACCTGCTCTTGCTA	soooossssssssssooss	94	127
619144*	133	1269	AACCCACACCTGCTCTTGCT	soooossssssssssooss	93	128
619145*	134	1270	AAACCCACACCTGCTCTTGC	soooossssssssssooss	95	129
619146*	135	1271	TAAACCCACACCTGCTCTTG	soooossssssssssooss	78	130
619147*	136	1272	CTAAACCCACACCTGCTCTT	soooossssssssssooss	64	131
619148*	137	1273	CCTAAACCCACACCTGCTCT	soooossssssssssooss	84	132
619149*	138	1274	TCCTAAACCCACACCTGCTC	soooossssssssssooss	87	133
619150*	139	1275	CTCCTAAACCCACACCTGCT	soooossssssssssooss	89	134
619413	340	8020	GTACCTGTTCTGTCTTTGGA	soooossssssssssooss	70	135
619414	353	8033	CCATCACTGAGAAGTACCTG	soooossssssssssooss	78	136
619415	940	16395	GGCATAATGTTCTGACTATC	soooossssssssssooss	67	137
619151*	n/a	1276	CCTCCTAAACCCACACCTGC	soooossssssssssooss	64	138
619152*	n/a	1277	ACCTCCTAAACCCACACCTG	soooossssssssssooss	45	139
619153*	n/a	1278	CACCTCCTAAACCCACACCT	soooossssssssssooss	36	140
619154*	n/a	1279	ACACCTCCTAAACCCACACC	soooossssssssssooss	26	141
619155*	n/a	1280	CACACCTCCTAAACCCACAC	soooossssssssssooss	50	142
619156*	n/a	1281	ACACACCTCCTAAACCCACA	soooossssssssssooss	53	143
619157*	n/a	1282	CACACACCTCCTAAACCCAC	soooossssssssssooss	44	144
619158*	n/a	1283	ACACACACCTCCTAAACCCA	soooossssssssssooss	65	145
619159*	n/a	1284	AACACACACCTCCTAAACCC	soooossssssssssooss	9	146
619160*	n/a	1285	AAACACACACCTCCTAAACC	soooossssssssssooss	0	147
619161*	n/a	1286	AAAACACACACCTCCTAAAC	soooossssssssssooss	15	148
619162*	n/a	1287	AAAAACACACACCTCCTAAA	soooossssssssssooss	10	149
619163*	n/a	1288	CAAAAACACACACCTCCTAA	soooossssssssssooss	7	150
619164*	n/a	1289	ACAAAAACACACACCTCCTA	soooossssssssssooss	55	151
619165*	n/a	1290	AACAAAAACACACACCTCCT	soooossssssssssooss	24	152
619166*	n/a	1291	AAACAAAAACACACACCTCC	soooossssssssssooss	19	153
619167*	n/a	1292	AAAACAAAAACACACACCTC	soooossssssssssooss	8	154
619168*	n/a	1294	GAAAAACAAAAACACACACC	soooossssssssssooss	17	155
619169*	n/a	1295	GGAAAAACAAAAACACACAC	soooossssssssssooss	26	156
619170	n/a	1297	TGGGAAAAACAAAAACACAC	soooossssssssssooss	30	157
619171	n/a	1298	GTGGGAAAAACAAAAACACA	soooossssssssssooss	23	158
619172	n/a	1299	GGTGGGAAAAACAAAAACAC	soooossssssssssooss	22	159
619173	n/a	1326	CTGTGAGAGCAAGTAGTGGG	soooossssssssssooss	43	160

TABLE 7-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619174	n/a	1327	ACTGTGAGAGCAAGTAGTGG	soooossssssssssooss	36	161
619175	n/a	1328	TACTGTGAGAGCAAGTAGTG	soooossssssssssooss	24	162
619176	n/a	1329	GTACTGTGAGAGCAAGTAGT	soooossssssssssooss	58	163
619177	n/a	1330	AGTACTGTGAGAGCAAGTAG	soooossssssssssooss	22	164
619178	n/a	1331	GAGTACTGTGAGAGCAAGTA	soooossssssssssooss	100	165
619179	n/a	1332	CGAGTACTGTGAGAGCAAGT	soooossssssssssooss	62	166
619180	n/a	1333	GCGAGTACTGTGAGAGCAAG	soooossssssssssooss	63	167
619181	n/a	1334	AGCGAGTACTGTGAGAGCAA	soooossssssssssooss	36	168
619182	n/a	1335	CAGCGAGTACTGTGAGAGCA	soooossssssssssooss	41	169
619183	n/a	1336	TCAGCGAGTACTGTGAGAGC	soooossssssssssooss	66	170
619184	n/a	1337	CTCAGCGAGTACTGTGAGAG	soooossssssssssooss	28	171
619185	n/a	1338	CCTCAGCGAGTACTGTGAGA	soooossssssssssooss	37	172
619186	n/a	1339	CCCTCAGCGAGTACTGTGAG	soooossssssssssooss	43	173
619187	n/a	1340	ACCCTCAGCGAGTACTGTGA	soooossssssssssooss	84	174
619253	n/a	1406	TACAGGCTGCGGTTGTTTCC	soooossssssssssooss	31	97
619293	n/a	1446	CCCGGCCCCTAGCGCGCGAC	soooossssssssssooss	63	98

TABLE 8

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
576816	310	7990	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	82	20
619188	n/a	1341	CACCCTCAGCGAGTACTGTG	soooossssssssssooss	56	175
619189	n/a	1342	TCACCCTCAGCGAGTACTGT	soooossssssssssooss	66	176
619190	n/a	1343	TTCACCCTCAGCGAGTACTG	soooossssssssssooss	57	177
619191	n/a	1344	GTTCACCCTCAGCGAGTACT	soooossssssssssooss	83	178
619192	n/a	1345	TGTTCACCCTCAGCGAGTAC	soooossssssssssooss	66	179
619193	n/a	1346	TTGTTCACCCTCAGCGAGTA	soooossssssssssooss	57	180
619194	n/a	1347	CTTGTTCACCCTCAGCGAGT	soooossssssssssooss	48	181
619195	n/a	1348	TCTTGTTCACCCTCAGCGAG	soooossssssssssooss	46	182
619196	n/a	1349	TTCTTGTTCACCCTCAGCGA	soooossssssssssooss	66	183
619197	n/a	1350	TTTCTTGTTCACCCTCAGCG	soooossssssssssooss	31	184

TABLE 8-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619198	n/a	1351	TTTTCTTGTTCACCCTCAGC	soooossssssssssooss	47	185
619199	n/a	1352	CTTTTCTTGTTCACCCTCAG	soooossssssssssooss	53	186
619200	n/a	1353	TCTTTTCTTGTTCACCCTCA	soooossssssssssooss	46	187
619201	n/a	1354	GTCTTTTCTTGTTCACCCTC	soooossssssssssooss	71	188
619202	n/a	1355	GGTCTTTTCTTGTTCACCCT	soooossssssssssooss	73	189
619203	n/a	1356	AGGTCTTTTCTTGTTCACCC	soooossssssssssooss	79	190
619204	n/a	1357	CAGGTCTTTTCTTGTTCACC	soooossssssssssooss	0	191
619205	n/a	1358	TCAGGTCTTTTCTTGTTCAC	soooossssssssssooss	68	192
619206	n/a	1359	ATCAGGTCTTTTCTTGTTCA	soooossssssssssooss	52	193
619207	n/a	1360	TATCAGGTCTTTTCTTGTTC	soooossssssssssooss	47	194
619208	n/a	1361	TTATCAGGTCTTTTCTTGTT	soooossssssssssooss	37	195
619209	n/a	1362	TTTATCAGGTCTTTTCTTGT	soooossssssssssooss	31	196
619210	n/a	1363	CTTTATCAGGTCTTTTCTTG	soooossssssssssooss	24	197
619211	n/a	1364	TCTTTATCAGGTCTTTTCTT	soooossssssssssooss	37	198
619212	n/a	1365	ATCTTTATCAGGTCTTTTCT	soooossssssssssooss	34	199
619213	n/a	1366	AATCTTTATCAGGTCTTTTC	soooossssssssssooss	38	200
619214	n/a	1367	TAATCTTTATCAGGTCTTTT	soooossssssssssooss	32	201
619215	n/a	1368	TTAATCTTTATCAGGTCTTT	soooossssssssssooss	55	202
619216	n/a	1369	GTTAATCTTTATCAGGTCTT	soooossssssssssooss	72	203
619217	n/a	1370	GGTTAATCTTTATCAGGTCT	soooossssssssssooss	85	204
619218	n/a	1371	TGGTTAATCTTTATCAGGTC	soooossssssssssooss	82	205
619219	n/a	1372	CTGGTTAATCTTTATCAGGT	soooossssssssssooss	62	206
619220	n/a	1373	TCTGGTTAATCTTTATCAGG	soooossssssssssooss	19	207
619221	n/a	1374	TTCTGGTTAATCTTTATCAG	soooossssssssssooss	31	208
619222	n/a	1375	CTTCTGGTTAATCTTTATCA	soooossssssssssooss	40	209
619223	n/a	1376	TCTTCTGGTTAATCTTTATC	soooossssssssssooss	41	210
619224	n/a	1377	TTCTTCTGGTTAATCTTTAT	soooossssssssssooss	11	211
619225	n/a	1378	TTTCTTCTGGTTAATCTTTA	soooossssssssssooss	46	212
619226	n/a	1379	TTTTCTTCTGGTTAATCTTT	soooossssssssssooss	14	213
619227	n/a	1380	GTTTTCTTCTGGTTAATCTT	soooossssssssssooss	50	214
619228	n/a	1381	TGTTTTCTTCTGGTTAATCT	soooossssssssssooss	49	215
619229	n/a	1382	TTGTTTTCTTCTGGTTAATC	soooossssssssssooss	31	216
619230	n/a	1383	CTTGTTTTCTTCTGGTTAAT	soooossssssssssooss	16	217

TABLE 8-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619231	n/a	1384	CCTTGTTTTCTTCTGGTTAA	soooossssssssssooss	23	218
619232	n/a	1385	TCCTTGTTTTCTTCTGGTTA	soooossssssssssooss	52	219
619233	n/a	1386	CTCCTTGTTTTCTTCTGGTT	soooossssssssssooss	32	220
619234	n/a	1387	CCTCCTTGTTTTCTTCTGGT	soooossssssssssooss	59	221
619235	n/a	1388	CCCTCCTTGTTTTCTTCTGG	soooossssssssssooss	48	222
619236	n/a	1389	TCCCTCCTTGTTTTCTTCTG	soooossssssssssooss	31	223
619237	n/a	1390	TTCCCTCCTTGTTTTCTTCT	soooossssssssssooss	24	224
619238	n/a	1391	TTTCCCTCCTTGTTTTCTTC	soooossssssssssooss	42	225
619239	n/a	1392	GTTTCCCTCCTTGTTTTCTT	soooossssssssssooss	46	226
619240	n/a	1393	TGTTTCCCTCCTTGTTTTCT	soooossssssssssooss	81	227
619241	n/a	1394	TTGTTTCCCTCCTTGTTTTC	soooossssssssssooss	53	228
619242	n/a	1395	GTTGTTTCCCTCCTTGTTTT	soooossssssssssooss	28	229
619243	n/a	1396	GGTTGTTTCCCTCCTTGTTT	soooossssssssssooss	40	230
619244	n/a	1397	CGGTTGTTTCCCTCCTTGTT	soooossssssssssooss	31	231
619245	n/a	1398	GCGGTTGTTTCCCTCCTTGT	soooossssssssssooss	69	232
619246	n/a	1399	TGCGGTTGTTTCCCTCCTTG	soooossssssssssooss	73	233
619247	n/a	1400	CTGCGGTTGTTTCCCTCCTT	soooossssssssssooss	90	234
619248	n/a	1401	GCTGCGGTTGTTTCCCTCCT	soooossssssssssooss	40	235
619249	n/a	1402	GGCTGCGGTTGTTTCCCTCC	soooossssssssssooss	50	236
619250	n/a	1403	AGGCTGCGGTTGTTTCCCTC	soooossssssssssooss	61	237
619251	n/a	1404	CAGGCTGCGGTTGTTTCCCT	soooossssssssssooss	72	238
619252	n/a	1405	ACAGGCTGCGGTTGTTTCCC	soooossssssssssooss	69	239
619253	n/a	1406	TACAGGCTGCGGTTGTTTCC	soooossssssssssooss	67	97
619254	n/a	1407	CTACAGGCTGCGGTTGTTTC	soooossssssssssooss	43	240
619255	n/a	1408	GCTACAGGCTGCGGTTGTTT	soooossssssssssooss	54	241
619256	n/a	1409	TGCTACAGGCTGCGGTTGTT	soooossssssssssooss	36	242
619257	n/a	1410	TTGCTACAGGCTGCGGTTGT	soooossssssssssooss	29	243
619258	n/a	1411	CTTGCTACAGGCTGCGGTTG	soooossssssssssooss	50	244
619259	n/a	1412	GCTTGCTACAGGCTGCGGTT	soooossssssssssooss	76	245
619260	n/a	1413	AGCTTGCTACAGGCTGCGGT	soooossssssssssooss	80	246
619261	n/a	1414	GAGCTTGCTACAGGCTGCGG	soooossssssssssooss	54	247
619262	n/a	1415	AGAGCTTGCTACAGGCTGCG	soooossssssssssooss	62	248
619293	n/a	1446	CCCGGCCCCTAGCGCGCGAC	soooossssssssssooss	64	98

TABLE 8-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619416	1937	24657 27056	AAAAAACAGTAGTTGTGGTC	soooossssssssssooss	78	249
619417	1988	27107	GCCAACTCAGATTTCACCTT	soooossssssssssooss	86	250

TABLE 9

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
576816	310	7990	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	71	20
619264	n/a	1417	CCAGAGCTTGCTACAGGCTG	soooossssssssssooss	54	251
619265	n/a	1418	TCCAGAGCTTGCTACAGGCT	soooossssssssssooss	69	252
619266	n/a	1419	TTCCAGAGCTTGCTACAGGC	soooossssssssssooss	70	253
619267	n/a	1420	GTTCCAGAGCTTGCTACAGG	soooossssssssssooss	49	254
619268	n/a	1421	AGTTCCAGAGCTTGCTACAG	soooossssssssssooss	95	255
619269	n/a	1422	GAGTTCCAGAGCTTGCTACA	soooossssssssssooss	36	256
619270	n/a	1423	TGAGTTCCAGAGCTTGCTAC	soooossssssssssooss	15	257
619271	n/a	1424	CTGAGTTCCAGAGCTTGCTA	soooossssssssssooss	31	258
619272	n/a	1425	CCTGAGTTCCAGAGCTTGCT	soooossssssssssooss	41	259
619273	n/a	1426	TCCTGAGTTCCAGAGCTTGC	soooossssssssssooss	36	260
619274	n/a	1427	CTCCTGAGTTCCAGAGCTTG	soooossssssssssooss	25	261
619275	n/a	1428	ACTCCTGAGTTCCAGAGCTT	soooossssssssssooss	50	262
619276	n/a	1429	GACTCCTGAGTTCCAGAGCT	soooossssssssssooss	74	263
619277	n/a	1430	CGACTCCTGAGTTCCAGAGC	soooossssssssssooss	69	264
619278	n/a	1431	GCGACTCCTGAGTTCCAGAG	soooossssssssssooss	98	265
619279	n/a	1432	CGCGACTCCTGAGTTCCAGA	soooossssssssssooss	69	266
619280	n/a	1433	GCGCGACTCCTGAGTTCCAG	soooossssssssssooss	75	267
619281	n/a	1434	CGCGCGACTCCTGAGTTCCA	soooossssssssssooss	67	268
619282	n/a	1435	GCGCGCGACTCCTGAGTTCC	soooossssssssssooss	55	269
619283	n/a	1436	AGCGCGCGACTCCTGAGTTC	soooossssssssssooss	62	270
619284	n/a	1437	TAGCGCGCGACTCCTGAGTT	soooossssssssssooss	100	271
619285	n/a	1438	CTAGCGCGCGACTCCTGAGT	soooossssssssssooss	68	272

TABLE 9-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619286	n/a	1439	CCTAGCGCGCGACTCCTGAG	soooossssssssssooss	41	273
619287	n/a	1440	CCCTAGCGCGCGACTCCTGA	soooossssssssssooss	70	274
619288	n/a	1441	CCCCTAGCGCGCGACTCCTG	soooossssssssssooss	68	275
619289	n/a	1442	GCCCCTAGCGCGCGACTCCT	soooossssssssssooss	52	276
619290	n/a	1443	GGCCCCTAGCGCGCGACTCC	soooossssssssssooss	49	277
619291	n/a	1444	CGGCCCCTAGCGCGCGACTC	soooossssssssssooss	69	278
619292	n/a	1445	CCGGCCCCTAGCGCGCGACT	soooossssssssssooss	76	279
619293	n/a	1446	CCCGGCCCCTAGCGCGCGAC	soooossssssssssooss	52	98
619294	n/a	1447	CCCCGGCCCCTAGCGCGCGA	soooossssssssssooss	62	280
619295	n/a	1448	GCCCCGGCCCCTAGCGCGCG	soooossssssssssooss	68	281
619296	n/a	1449	GGCCCCGGCCCCTAGCGCGC	soooossssssssssooss	56	282
619297	n/a	1450	CGGCCCCGGCCCCTAGCGCG	soooossssssssssooss	71	283
619298	n/a	1451	CCGGCCCCGGCCCCTAGCGC	soooossssssssssooss	73	284
619299	n/a	1452	CCCGGCCCCGGCCCCTAGCG	soooossssssssssooss	51	285
619300	n/a	1453	CCCCGGCCCCGGCCCCTAGC	soooossssssssssooss	49	286
619301	n/a	1454	GCCCCGGCCCCGGCCCCTAG	soooossssssssssooss	60	287
619302	n/a	1455	GGCCCCGGCCCCGGCCCCTA	soooossssssssssooss	48	288
619303	n/a	1462	ACGCCCCGGCCCCGGCCCCG	soooossssssssssooss	49	289
619304	n/a	1463	CACGCCCCGGCCCCGGCCCC	soooossssssssssooss	25	290
619305	n/a	1464	CCACGCCCCGGCCCCGGCCC	soooossssssssssooss	66	291
619306	n/a	1465	ACCACGCCCCGGCCCCGGCC	soooossssssssssooss	70	292
619307	n/a	1466	GACCACGCCCCGGCCCCGGC	soooossssssssssooss	73	293
619308	n/a	1467	CGACCACGCCCCGGCCCCGG	soooossssssssssooss	55	294
619309	n/a	1468	CCGACCACGCCCCGGCCCCG	soooossssssssssooss	70	295
619310	n/a	1469	CCCGACCACGCCCCGGCCCC	soooossssssssssooss	50	296
619311	n/a	1470	CCCCGACCACGCCCCGGCCC	soooossssssssssooss	45	297
619312	n/a	1471	GCCCCGACCACGCCCCGGCC	soooossssssssssooss	36	298
619313	n/a	1472	CGCCCCGACCACGCCCCGGC	soooossssssssssooss	58	299
619314	n/a	1473	CCGCCCCGACCACGCCCCGG	soooossssssssssooss	100	300
619315	n/a	1474	CCCGCCCCGACCACGCCCCG	soooossssssssssooss	44	301
619316	n/a	1475	GCCCGCCCCGACCACGCCCC	soooossssssssssooss	97	302
619317	n/a	1476	GGCCCGCCCCGACCACGCCC	soooossssssssssooss	76	303
619318	n/a	1477	GGGCCCGCCCCGACCACGCC	soooossssssssssooss	44	304
619319	n/a	1478	CGGGCCCGCCCCGACCACGC	soooossssssssssooss	40	305

TABLE 9-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1 and 2

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619320	n/a	1479	CCGGGCCCGCCCCGACCACG	soooossssssssssooss	50	306
619321	n/a	1480	CCCGGGCCCGCCCCGACCAC	soooossssssssssooss	22	307
619322	n/a	1481	CCCCGGGCCCGCCCCGACCA	soooossssssssssooss	56	308
619323	n/a	1482	CCCCCGGGCCCGCCCCGACC	soooossssssssssooss	40	309
619324	n/a	1483	GCCCCCGGGCCCGCCCCGAC	soooossssssssssooss	65	310
619325	n/a	1484	CGCCCCCGGGCCCGCCCCGA	soooossssssssssooss	25	311
619326	n/a	1486	CCCGCCCCCGGGCCCGCCCC	soooossssssssssooss	41	312
619327	n/a	1487	GCCCGCCCCCGGGCCCGCCC	soooossssssssssooss	36	313
619328	n/a	1488	GGCCCGCCCCCGGGCCCGCC	soooossssssssssooss	14	314
619329	n/a	1495	CGCCCCGGGCCCGCCCCCGG	soooossssssssssooss	33	315
619330	n/a	1497	CCCGCCCCGGGCCCGCCCCC	soooossssssssssooss	35	316
619331	n/a	1498	CCCCGCCCCGGGCCCGCCCC	soooossssssssssooss	42	317
619332	n/a	1499	GCCCCGCCCCGGGCCCGCCC	soooossssssssssooss	47	318
619333	n/a	1500	AGCCCCGCCCCGGGCCCGCC	soooossssssssssooss	53	319
619334	n/a	1501	CAGCCCCGCCCCGGGCCCGC	soooossssssssssooss	32	320
619335	n/a	1502	GCAGCCCCGCCCCGGGCCCG	soooossssssssssooss	58	321
619336	n/a	1503	CGCAGCCCCGCCCCGGGCCC	soooossssssssssooss	75	322
619418	n/a	27155	CTACACACCAAAGAATGCCA	soooossssssssssooss	71	323
619419	n/a	15587	GGAATAAGGTCACTAGTTCG	soooossssssssssooss	72	324
619420	n/a	7990	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	100	20
619263	n/a	1416	CAGAGCTTGCTACAGGCTGC	soooossssssssssooss	63	325
619253	n/a	1406	TACAGGCTGCGGTTGTTTCC	soooossssssssssooss	50	97

TABLE 10

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1-5

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 3 Start Site	SEQ ID NO: 4 Start Site	SEQ ID NO: 5 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
576816	310	7990	232	188	188	GCCTTACTCTAGGACCAAGAsssssssssssssssssss		69	20
619253	n/a	1406	n/a	n/a	n/a	TACAGGCTGCGGTTGTTTCCsoooossssssssssooss		19	97
619293	n/a	1446	n/a	n/a	n/a	CCCGGCCCCTAGCGCGCGACsoooossssssssssooss		56	98
619337	n/a	1504	n/a	n/a	n/a	CCGCAGCCCCGCCCCGGGCCsoooossssssssssooss		55	326

TABLE 10-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1-5

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 3 Start Site	SEQ ID NO: 4 Start Site	SEQ ID NO: 5 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619338	n/a	1505	n/a	n/a	n/a	ACCGCAGCCCCGCCCCGGGC	soooossssssssssooss	62	327
619339	n/a	1506	n/a	n/a	n/a	AACCGCAGCCCCGCCCCGGG	soooossssssssssooss	68	328
619340	n/a	1507	n/a	n/a	n/a	CAACCGCAGCCCCGCCCCGG	soooossssssssssooss	14	329
619341	n/a	1508	n/a	n/a	n/a	GCAACCGCAGCCCCGCCCCG	soooossssssssssooss	95	330
619342	n/a	1509	n/a	n/a	n/a	CGCAACCGCAGCCCCGCCCC	soooossssssssssooss	56	331
619343	n/a	1510	n/a	n/a	n/a	CCGCAACCGCAGCCCCGCCC	soooossssssssssooss	58	332
619344	n/a	1511	n/a	n/a	n/a	ACCGCAACCGCAGCCCCGCC	soooossssssssssooss	51	333
619345	n/a	1512	n/a	n/a	n/a	CACCGCAACCGCAGCCCCGC	soooossssssssssooss	97	334
619346	n/a	1513	n/a	n/a	n/a	GCACCGCAACCGCAGCCCCG	soooossssssssssooss	49	335
619347	n/a	1514	n/a	n/a	n/a	GGCACCGCAACCGCAGCCCC	soooossssssssssooss	100	336
619348	n/a	1515	n/a	n/a	n/a	AGGCACCGCAACCGCAGCCC	soooossssssssssooss	46	337
619349	n/a	1516	n/a	n/a	n/a	CAGGCACCGCAACCGCAGCC	soooossssssssssooss	100	338
619350	n/a	1517	n/a	n/a	n/a	GCAGGCACCGCAACCGCAGC	soooossssssssssooss	55	339
619351	n/a	1518	n/a	n/a	n/a	CGCAGGCACCGCAACCGCAG	soooossssssssssooss	63	340
619352	n/a	1519	n/a	n/a	n/a	GCGCAGGCACCGCAACCGCA	soooossssssssssooss	42	341
619353	n/a	1520	n/a	n/a	n/a	GGCGCAGGCACCGCAACCGC	soooossssssssssooss	95	342
619354	n/a	1521	n/a	n/a	n/a	GGGCGCAGGCACCGCAACCG	soooossssssssssooss	23	343
619355*	140	n/a	n/a	n/a	n/a	TCTCCTAAACCCACACCTGC	soooossssssssssooss	94	384
619356*	141	n/a	n/a	n/a	n/a	ATCTCCTAAACCCACACCTG	soooossssssssssooss	69	385
619357	142	n/a	n/a	n/a	n/a	TATCTCCTAAACCCACACCT	soooossssssssssooss	n.d.	386
619358*	143	n/a	n/a	n/a	n/a	ATATCTCCTAAACCCACACC	soooossssssssssooss	53	387
619359	144	n/a	n/a	n/a	n/a	GATATCTCCTAAACCCACAC	soooossssssssssooss	n.d.	388
619360*	145	n/a	n/a	n/a	n/a	AGATATCTCCTAAACCCACA	soooossssssssssooss	30	389
619361*	146	n/a	n/a	n/a	n/a	GAGATATCTCCTAAACCCAC	soooossssssssssooss	48	390
619362*	147	n/a	n/a	n/a	n/a	GGAGATATCTCCTAAACCCA	soooossssssssssooss	60	391
619363*	148	n/a	n/a	n/a	n/a	CGGAGATATCTCCTAAACCC	soooossssssssssooss	26	392
619364*	149	n/a	n/a	n/a	n/a	CCGGAGATATCTCCTAAACC	soooossssssssssooss	97	393
619365*	150	n/a	n/a	n/a	n/a	TCCGGAGATATCTCCTAAAC	soooossssssssssooss	60	394
619366*	151	n/a	n/a	n/a	n/a	CTCCGGAGATATCTCCTAAA	soooossssssssssooss	34	395
619367	152	n/a	n/a	n/a	n/a	GCTCCGGAGATATCTCCTAA	soooossssssssssooss	n.d.	396
619368*	153	n/a	n/a	n/a	n/a	TGCTCCGGAGATATCTCCTA	soooossssssssssooss	95	397
619369	154	n/a	n/a	n/a	n/a	ATGCTCCGGAGATATCTCCT	soooossssssssssooss	n.d.	398
619370*	155	n/a	n/a	n/a	n/a	AATGCTCCGGAGATATCTCC	soooossssssssssooss	59	399

TABLE 10-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1-5

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 3 Start Site	SEQ ID NO: 4 Start Site	SEQ ID NO: 5 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619371*	n/a	n/a	n/a	17	n/a	TCTCTCTTTCCTAGCGGGAC	soooossssssssssooss	0	344
619372*	n/a	n/a	n/a	18	n/a	ATCTCTCTTTCCTAGCGGGA	soooossssssssssooss	8	345
619373*	n/a	n/a	n/a	19	n/a	TATCTCTCTTTCCTAGCGGG	soooossssssssssooss	6	346
619374*	n/a	n/a	n/a	20	n/a	ATATCTCTCTTTCCTAGCGG	soooossssssssssooss	0	347
619375*	n/a	n/a	n/a	21	n/a	GATATCTCTCTTTCCTAGCG	soooossssssssssooss	8	348
619376*	n/a	n/a	n/a	22	n/a	AGATATCTCTCTTTCCTAGC	soooossssssssssooss	0	349
619377*	n/a	n/a	n/a	23	n/a	GAGATATCTCTCTTTCCTAG	soooossssssssssooss	89	350
619378*	n/a	n/a	n/a	24	n/a	GGAGATATCTCTCTTTCCTA	soooossssssssssooss	1	351
619379*	n/a	n/a	n/a	25	n/a	CGGAGATATCTCTCTTTCCT	soooossssssssssooss	0	352
619380	n/a	n/a	n/a	26	n/a	CCGGAGATATCTCTCTTTCC	soooossssssssssooss	n.d.	353
619381*	n/a	n/a	n/a	27	n/a	TCCGGAGATATCTCTCTTTC	soooossssssssssooss	28	354
619382*	n/a	n/a	n/a	28	n/a	CTCCGGAGATATCTCTCTTT	soooossssssssssooss	28	355
619383*	n/a	n/a	n/a	29	n/a	GCTCCGGAGATATCTCTCTT	soooossssssssssooss	91	356
619384*	n/a	n/a	n/a	30	n/a	TGCTCCGGAGATATCTCTCT	soooossssssssssooss	61	357
619385*	n/a	n/a	n/a	31	n/a	ATGCTCCGGAGATATCTCTC	soooossssssssssooss	69	358
619386*	n/a	n/a	n/a	32	n/a	AATGCTCCGGAGATATCTCT	soooossssssssssooss	75	359
619387	156	n/a	n/a	33	n/a	AAATGCTCCGGAGATATCTC	soooossssssssssooss	n.d.	400
619388*	n/a	n/a	n/a	n/a	18	TCTGCTCTTGCTAGACCCCG	soooossssssssssooss	50	360
619389	n/a	n/a	n/a	n/a	19	ATCTGCTCTTGCTAGACCCC	soooossssssssssooss	n.d.	361
619390*	n/a	n/a	n/a	n/a	20	TATCTGCTCTTGCTAGACCC	soooossssssssssooss	5	362
619391*	n/a	n/a	n/a	n/a	21	ATATCTGCTCTTGCTAGACC	soooossssssssssooss	0	363
619392*	n/a	n/a	n/a	n/a	22	GATATCTGCTCTTGCTAGAC	soooossssssssssooss	0	364
619393*	n/a	n/a	n/a	n/a	23	AGATATCTGCTCTTGCTAGA	soooossssssssssooss	6	365
619394*	n/a	n/a	n/a	n/a	24	GAGATATCTGCTCTTGCTAG	soooossssssssssooss	57	366
619395*	n/a	n/a	n/a	n/a	25	GGAGATATCTGCTCTTGCTA	soooossssssssssooss	6	367
619396*	n/a	n/a	n/a	n/a	26	CGGAGATATCTGCTCTTGCT	soooossssssssssooss	0	368
619397*	n/a	n/a	n/a	n/a	27	CCGGAGATATCTGCTCTTGC	soooossssssssssooss	0	369
619398*	n/a	n/a	n/a	n/a	28	TCCGGAGATATCTGCTCTTG	soooossssssssssooss	22	370
619399*	n/a	n/a	n/a	n/a	29	CTCCGGAGATATCTGCTCTT	soooossssssssssooss	14	371
619400*	n/a	n/a	n/a	n/a	30	GCTCCGGAGATATCTGCTCT	soooossssssssssooss	46	372
619401*	n/a	n/a	n/a	n/a	31	TGCTCCGGAGATATCTGCTC	soooossssssssssooss	40	373
619402*	n/a	n/a	n/a	n/a	32	ATGCTCCGGAGATATCTGCT	soooossssssssssooss	79	374
619403*	n/a	n/a	n/a	n/a	33	AATGCTCCGGAGATATCTGC	soooossssssssssooss	65	375
619404*	n/a	n/a	n/a	n/a	34	AAATGCTCCGGAGATATCTG	soooossssssssssooss	22	376

TABLE 10-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1-5

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 3 Start Site	SEQ ID NO: 4 Start Site	SEQ ID NO: 5 Start Site	Sequence	Linkage	% inhibition	SEQ ID NO
619405*	n/a	n/a	75	n/a	n/a	TGCTCCGGAGATATCAAGCG	sooooossssssssssooss	18	377
619406*	n/a	n/a	76	n/a	n/a	ATGCTCCGGAGATATCAAGC	sooooossssssssssooss	21	378
619407*	n/a	n/a	77	n/a	n/a	AATGCTCCGGAGATATCAAG	sooooossssssssssooss	98	379
619408	n/a	n/a	78	n/a	n/a	AAATGCTCCGGAGATATCAA	sooooossssssssssooss	n.d.	380
619409*	n/a	n/a	79	n/a	n/a	CAAATGCTCCGGAGATATCA	sooooossssssssssooss	98.	381
619421	3132	28251	n/a	n/a	n/a	GGGACACTACAAGGTAGTAT	sooooossssssssssooss	80	401
619422*	n/a	3452	n/a	n/a	n/a	GGTAACTTCAAACTCTTGGG	sooooossssssssssooss	85.	382
619423	n/a	13642	1013	n/a	n/a	GCCATGATTTCTTGTCTGGG	sooooossssssssssooss	67	383

Example 3

Dose-Dependent Antisense Inhibition of a Human C9ORF72 mRNA Variant in HepG2 Cells

[0444] Antisense oligonucleotides from the study described above exhibiting significant in vitro inhibition of C9ORF72 mRNA were selected and tested at various doses in HepG2 cells. ISIS 576816 and ISIS 577061, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, were used as benchmark oligonucleotides. ISIS 576816 and ISIS 577061 are 5-10-5 MOE gapmers with phosphorothioate linkages throughout. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 78.1 nM, 312.5 nM, 1,250.0 nM, or 5,000.0 nM concentrations of antisense oligonucleotide. After a treatment period of approximately 16 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human C9ORF72 primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant. The levels of the C9ORF72 pathogenic associated mRNA variant were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of variant C9ORF72 levels, relative to untreated control cells. 'n.d.' means no data.

[0445] The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented in the Tables below. As illustrated, the C9ORF72 pathogenic associated mRNA variant levels were reduced in a dose-dependent manner in some of the antisense oligonucleotide treated cells.

TABLE 11

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant

ISIS No	78.1 nM	312.5 nM	1250.0 nM	5000.0 nM	IC_{50} (μM)
576816	13	47	92	95	0.4
577061	0	1	23	57	4.2
619049	11	46	34	18	>5.0
619055	14	13	0	0	>5.0
619061	0	2	0	0	>5.0
619080	7	4	0	0	>5.0
619088	0	7	16	16	>5.0
619089	2	0	0	17	>5.0
619095	10	0	0	27	>5.0
619096	16	0	10	27	>5.0
619097	23	55	41	38	>5.0
619099	10	36	46	23	>5.0
619107	15	0	4	33	>5.0
619114	0	0	8	31	>5.0
619253	26	33	66	86	0.5
619293	26	71	n.d.	95	0.2
619410	42	63	86	n.d.	0.1
619411	28	20	96	n.d.	0.3
619412	39	66	93	97	0.1

TABLE 12

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant

ISIS No	78.1 nM	312.5 nM	1250.0 nM	5000.0 nM	IC_{50} (μM)
576816	39	84	96	90	0.1
577061	29	0	25	71	3.7
619173	18	46	84	98	0.4
619174	21	36	77	n.d.	0.4
619176	16	29	57	95	0.7
619178	5	31	86	97	0.5
619179	0	16	65	96	0.8
619180	4	24	66	96	0.7
619181	10	34	72	95	0.6

TABLE 12-continued

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant

ISIS No	78.1 nM	312.5 nM	1250.0 nM	5000.0 nM	IC_{50} (μM)
619182	23	32	74	92	0.5
619183	0	26	44	96	1.0
619185	14	34	70	93	0.5
619186	8	32	71	98	0.6
619187	18	36	81	95	0.4
619253	17	22	61	97	0.7
619293	0	49	86	99	0.6
619413	8	48	84	95	0.4
619415	26	67	90	n.d.	0.2

TABLE 13

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant

ISIS No	78.1 nM	312.5 nM	1250.0 nM	5000.0 nM	IC_{50} (μM)
576816	0	71	97	n.d.	0.3
577061	0	12	46	82	1.4
619191	66	84	94	98	<0.07
619201	16	64	95	97	0.3
619202	35	78	95	95	0.1
619203	29	55	92	95	0.2
619216	61	55	92	96	<0.07
619217	44	86	83	n.d.	0.1
619218	35	87	93	n.d.	0.1
619240	0	43	52	78	1.0
619245	0	39	85	n.d.	0.5
619246	34	74	96	99	0.1
619247	0	46	92	93	0.6
619251	40	91	93	96	<0.07
619252	24	67	87	n.d.	0.2
619259	7	76	85	98	0.3
619260	16	80	92	99	0.2
619416	13	63	91	92	0.3
619417	45	88	91	97	<0.07

TABLE 14

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant

ISIS No	78.1 nM	312.5 nM	1250.0 nM	5000.0 nM	IC_{50} (μM)
576816	38	61	95	98	0.1
577061	0	8	13	77	2.5
619266	0	23	73	98	0.8
619268	5	29	77	97	0.6
619276	31	82	90	91	0.1
619278	35	83	97	99	0.1
619280	66	80	97	96	<0.07
619284	22	55	88	98	0.3
619292	37	79	85	94	0.1
619297	0	68	82	91	0.6
619298	47	89	93	91	<0.07
619307	37	71	96	98	0.1
619314	21	0	61	97	0.8
619316	13	37	71	91	0.5
619317	7	17	68	87	0.8
619336	17	51	64	90	0.5
619418	43	78	89	95	0.1
619419	43	68	94	95	0.1
619420	66	88	100	n.d.	<0.07

TABLE 15

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant

ISIS No	78.1 nM	312.5 nM	1250.0 nM	5000.0 nM	IC_{50} (μM)
576816	8	48	91	91	0.4
577061	0	13	33	64	2.6
619293	32	77	85	95	0.1
619337	28	42	79	95	0.3
619338	11	55	83	91	0.4
619339	6	42	83	90	0.5
619341	17	14	66	83	0.8
619342	37	57	85	97	0.2
619343	25	51	87	96	0.3
619344	17	27	75	91	0.5
619345	0	18	50	88	1.2
619346	0	22	76	90	0.8
619347	1	8	41	76	1.7
619349	10	12	51	78	1.3
619350	16	46	84	91	0.4
619351	20	21	69	87	0.6
619353	0	13	52	76	1.4
619421	27	78	93	92	0.1
619423	0	53	87	93	0.5

TABLE 16

Gapmers chosen for further analysis

Isis No	IC_{50} (μM)	Sequence	SEQ ID NO: 32 Start Site	Location	SEQ ID NO
619173	0.4	CTGTGAGAGCAAGTAGTGGG	1326	intron	160
619174	0.4	ACTGTGAGAGCAAGTAGTGG	1327	intron	161
619178	0.5	GAGTACTGTGAGAGCAAGTA	1331	intron	165
619179	0.8	CGAGTACTGTGAGAGCAAGT	1332	intron	166
619181	0.5	AGCGAGTACTGTGAGAGCAA	1334	intron	168

TABLE 16-continued

Gapmers chosen for further analysis

Isis No	IC_{50} (μM)	Sequence	SEQ ID NO: 32 Start Site	Location	SEQ ID NO
619182	0.5	CAGCGAGTACTGTGAGAGCA	1335	intron	169
619185	0.5	CCTCAGCGAGTACTGTGAGA	1338	intron	172
619186	0.6	CCCTCAGCGAGTACTGTGAG	1339	intron	173
619187	0.4	ACCCTCAGCGAGTACTGTGA	1340	intron	174
619191	<0.07	GTTCACCCTCAGCGAGTACT	1344	intron	178
619201	0.3	GTCTTTTCTTGTTCACCCTC	1354	intron	188
619202	0.1	GGTCTTTTCTTGTTCACCCT	1355	intron	189
619203	0.2	AGGTCTTTTCTTGTTCACCC	1356	intron	190
619216	<0.07	GTTAATCTTTATCAGGTCTT	1369	intron	203
619217	0.1	GGTTAATCTTTATCAGGTCT	1370	intron	204
619218	0.1	TGGTTAATCTTTATCAGGTC	1371	intron	205
619245	0.5	GCGGTTGTTTCCCTCCTTGT	1398	intron	232
619246	0.1	TGCGGTTGTTTCCCTCCTTG	1399	intron	233
619251	<0.07	CAGGCTGCGGTTGTTTCCCT	1404	intron	238
619252	0.2	ACAGGCTGCGGTTGTTTCCC	1405	intron	239
619253	0.5	TACAGGCTGCGGTTGTTTCC	1406	intron	97
619259	0.3	GCTTGCTACAGGCTGCGGTT	1412	intron	245
619260	0.2	AGCTTGCTACAGGCTGCGGT	1413	intron	246
619276	0.1	GACTCCTGAGTTCCAGAGCT	1429	intron	263
619278	0.1	GCGACTCCTGAGTTCCAGAG	1431	intron	265
619280	<0.07	GCGCGACTCCTGAGTTCCAG	1433	intron	267
619284	0.3	TAGCGCGCGACTCCTGAGTT	1437	Intron	271
619292	0.1	CCGGCCCCTAGCGCGCGACT	1445	intron	279
619293	0.2	CCCGGCCCCTAGCGCGCGAC	1446	intron	98
619298	<0.07	CCGGCCCCGGCCCCTAGCGC	1451	intron	284
619307	0.1	GACCACGCCCCGGCCCCGGC	1466	intron	293
619337	0.3	CCGCAGCCCCGCCCCGGGCC	1504	intron: exon1B junction	326
619338	0.4	ACCGCAGCCCCGCCCCGGGC	1505	intron: exon1B junction	327
619339	0.5	AACCGCAGCCCCGCCCCGGG	1506	intron: exon1B junction	328
619342	0.2	CGCAACCGCAGCCCCGCCCC	1509	intron: exon1B junction	331
619343	0.3	CCGCAACCGCAGCCCCGCCC	1510	intron: exon1B junction	332
619344	0.5	ACCGCAACCGCAGCCCCGCC	1511	intron: exon1B junction	333

TABLE 16-continued

Gapmers chosen for further analysis

Isis No	IC_{50} (μM)	Sequence	SEQ ID NO: 32 Start Site	Location	SEQ ID NO
619350	0.4	GCAGGCACCGCAACCGCAGC	1517	intron: exon1B junction	339
619351	0.6	CGCAGGCACCGCAACCGCAG	1518	intron: exon1B junction	340

Example 4

Antisense Inhibition of C9ORF72 by Human-Rhesus Cross-Reactive Antisense Oligonucleotides in LLC-MK2 Cells

[0446] Antisense oligonucleotides targeting a human C9ORF72 nucleic acid and fully cross-reactive with a rhesus C9ORF72 nucleic acid were designed and were tested for their effects on rhesus C9ORF72 mRNA in vitro. ISIS 576816, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in the tables below. Cultured rhesus LLC-MK2 cells at a density of 20,000 cells per well were transfected using electroporation with 4,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3750 (forward sequence TGTGACAGTTGGAATGCAGTGA, designated herein as SEQ ID NO: 16; reverse sequence GCCACTTAAAGCAATCTCTGTCTTG, designated herein as SEQ ID NO: 17; probe sequence TCGACTCTTTGCCCACCGCCA, designated herein as SEQ ID NO: 18—a TAQ-man primer probe set) was used to measure total C9ORF72 mRNA levels. RTS3750 targets exon 2 of the mRNA transcripts and, therefore, measures total mRNA transcripts. C9ORF72 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72, relative to untreated control cells. The oligonucleotides marked with as asterisk (*) target the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of these oligonucleotides. 'n.d.' indicates that there was no signal reading in the assay for that particular oligonucleotide. The antisense oligonucleotides were also tested in HepG2 cells in a series of experiments that had similar culture conditions. The results for each experiment are also presented in tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 4,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant, which is the product of a pre-mRNA containing a hexanucleotide repeat. The levels of the C9ORF72 pathogenic associated mRNA variant were normalized to the total RNA content of the cell, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72, relative to untreated control cells. 'n.d.' means no data.

[0447] The chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment comprises ten 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and on the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment comprises a 2'-MOE group. All cytosine residues throughout each oligonucleotide are 5-methylcytosines. The internucleoside linkages for the gapmers are mixed phosphorothioate and phosphodiester linkages. The internucleoside linkages for each gapmer are presented in the Linkage column, where 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage.

[0448] "Start site" indicates the 5'-most nucleoside to which the gapmer is targeted in the human gene sequence. "Stop site" indicates the 3'-most nucleoside to which the gapmer is targeted human gene sequence. Each antisense oligonucleotide listed in the Tables below is targeted to either human C9ORF72 mRNA sequence, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_001256054.1), the human C9ORF72 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000), GENBANK Accession No. NM_018325.3 (incorporated herein as SEQ ID NO: 6), or all three. 'n/a' indicates that the antisense oligonucleotide does not target that particular gene sequence.

TABLE 17

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
576816	310	7990	215	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	51	91	20
619343	n/a	1510	1	CCGCAACCGCAGCCCCGCCC	soooossssssssssooss	0	96	332

TABLE 17-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
619420	310	7990	215	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	59	100	20
625183	309	7989	214	CCTTACTCTAGGACCAAGAA	soooossssssssssooss	27	92	484
625249	239	7919	144	AAAGCAATCTCTGTCTTGGC	soooossssssssssooss	78	n.d.	465
625255	364	8044	269	AAGTTATTTCTCCATCACTG	soooossssssssssooss	40	97	502
627833	321	8001	226	AGCCCAAATGTGCCTTACTC	soooossssssssssooss	52	n.d.	487
627834	382	8062	287	GAGTGTGGTTGGCAAGAAAA	soooossssssssssooss	0	61	506
655126	n/a	1548	39	CGCCACCGCCTGCGCCTCCG	soooossssssssssooss	0	96	512
655127	n/a	1551	42	ACTCGCCACCGCCTGCGCCT	soooossssssssssooss	11	93	513
655128	n/a	n/a	45	TCCACTCGCCACCGCCTGCG	soooossssssssssooss	0	60	514
655129	n/a	n/a	48	ATATCCACTCGCCACCGCCT	soooossssssssssooss	1	79	515
655130	n/a	n/a	51	GAGATATCCACTCGCCACCG	soooossssssssssooss	8	45	516
655131*	167	7847	72	TCACATTATCCAAATGCTCC	soooossssssssssooss	36	88	441
655132*	170	7850	75	CTGTCACATTATCCAAATGC	soooossssssssssooss	20	79	442
655133*	173	7853	78	CAACTGTCACATTATCCAAA	soooossssssssssooss	31	82	443
655134*	176	7856	81	TTCCAACTGTCACATTATCC	soooossssssssssooss	43	90	444
655135*	180	7860	85	TGCATTCCAACTGTCACATT	soooossssssssssooss	69	94	445
655136*	183	7863	88	CACTGCATTCCAACTGTCAC	soooossssssssssooss	75	98	446
655137*	186	7866	91	CATCACTGCATTCCAACTGT	soooossssssssssooss	71	95	447
655138	189	7869	94	CGACATCACTGCATTCCAAC	soooossssssssssooss	85	n.d.	448
655139	192	7872	97	AGTCGACATCACTGCATTCC	soooossssssssssooss	89	n.d.	449
655140*	195	7875	100	AAGAGTCGACATCACTGCAT	soooossssssssssooss	81	100	450
655141*	198	7878	103	GCAAAGAGTCGACATCACTG	soooossssssssssooss	68	98	451
655142*	201	7881	106	TGGGCAAAGAGTCGACATCA	soooossssssssssooss	60	98	452
655143*	204	7884	109	CGGTGGGCAAAGAGTCGACA	soooossssssssssooss	69	99	453
655144*	207	7887	112	TGGCGGTGGGCAAAGAGTCG	soooossssssssssooss	60	97	454
655145*	211	7891	116	GAGATGGCGGTGGGCAAAGA	soooossssssssssooss	27	93	455
655146*	214	7894	119	CTGGAGATGGCGGTGGGCAA	soooossssssssssooss	56	99	456
655147*	218	7898	123	ACAGCTGGAGATGGCGGTGG	soooossssssssssooss	17	84	457
655148	221	7901	126	GCAACAGCTGGAGATGGCGG	soooossssssssssooss	34	n.d.	458
655149	224	7904	129	TTGGCAACAGCTGGAGATGG	soooossssssssssooss	27	n.d.	459
655150*	227	7907	132	GTCTTGGCAACAGCTGGAGA	soooossssssssssooss	52	99	460
655151	230	7910	135	TCTGTCTTGGCAACAGCTGG	soooossssssssssooss	60	n.d.	461
655152*	232	7912	137	TCTCTGTCTTGGCAACAGCT	soooossssssssssooss	65	98	462

TABLE 17-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
655153*	236	7916	141	GCAATCTCTGTCTTGGCAAC	soooossssssssssooss	91	99	463
655154*	237	7917	142	AGCAATCTCTGTCTTGGCAA	soooossssssssssooss	87	99	464
655155*	242	7922	147	CTTAAAGCAATCTCTGTCTT	soooossssssssssooss	80	76	466
655156*	245	7925	150	CCACTTAAAGCAATCTCTGT	soooossssssssssooss	74	86	467
655157	267	7947	172	AGCTGCTAATAAAGGTGATT	soooossssssssssooss	27	90	468
655158	270	7950	175	AGTAGCTGCTAATAAAGGTG	soooossssssssssooss	17	87	469
655159	273	7953	178	AAAAGTAGCTGCTAATAAAG	soooossssssssssooss	6	48	470
655160	276	7956	181	AGCAAAAGTAGCTGCTAATA	soooossssssssssooss	3	87	471
655161	279	7959	184	GTAAGCAAAAGTAGCTGCTA	soooossssssssssooss	30	99	472
655162	282	7962	187	CCAGTAAGCAAAAGTAGCTG	soooossssssssssooss	24	85	473
655163	285	7965	190	GTCCCAGTAAGCAAAAGTAG	soooossssssssssooss	15	70	474
655164	286	7966	191	TGTCCCAGTAAGCAAAAGTA	soooossssssssssooss	5	60	475
655165	288	7968	193	ATTGTCCCAGTAAGCAAAAG	soooossssssssssooss	0	46	476
655166	290	7970	195	ATATTGTCCCAGTAAGCAAA	soooossssssssssooss	0	54	477
655167	291	7971	196	AATATTGTCCCAGTAAGCAA	soooossssssssssooss	16	60	478
655168	294	7974	199	AAGAATATTGTCCCAGTAAG	soooossssssssssooss	0	52	479
655169	297	7977	202	ACCAAGAATATTGTCCCAGT	soooossssssssssooss	20	82	480
655170	300	7980	205	AGGACCAAGAATATTGTCCC	soooossssssssssooss	17	54	481
655171	303	7983	208	TCTAGGACCAAGAATATTGT	soooossssssssssooss	0	36	482
655172	306	7986	211	TACTCTAGGACCAAGAATAT	soooossssssssssooss	10	43	483
655173	312	7992	217	GTGCCTTACTCTAGGACCAA	soooossssssssssooss	75	99	485
655174	315	7995	220	AATGTGCCTTACTCTAGGAC	soooossssssssssooss	47	98	486
655175	327	8007	232	CTTTGGAGCCCAAATGTGCC	soooossssssssssooss	21	86	488
655176	330	8010	235	TGTCTTTGGAGCCCAAATGT	soooossssssssssooss	21	88	489
655177	333	8013	238	TTCTGTCTTTGGAGCCCAAA	soooossssssssssooss	42	97	490
655178	334	8014	239	GTTCTGTCTTTGGAGCCCAA	soooossssssssssooss	66	99	491
655179	336	8016	241	CTGTTCTGTCTTTGGAGCCC	soooossssssssssooss	68	95	492
655180	339	8019	244	TACCTGTTCTGTCTTTGGAG	soooossssssssssooss	29	92	493
655181	342	8022	247	AAGTACCTGTTCTGTCTTTG	soooossssssssssooss	26	76	494
655182	345	8025	250	GAGAAGTACCTGTTCTGTCT	soooossssssssssooss	38	94	495
655183	348	8028	253	ACTGAGAAGTACCTGTTCTG	soooossssssssssooss	23	89	496
655184	350	8030	255	TCACTGAGAAGTACCTGTTC	soooossssssssssooss	19	83	497
655185	351	8031	256	ATCACTGAGAAGTACCTGTT	soooossssssssssooss	40	87	498

TABLE 17-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
655186	354	8034	259	TCCATCACTGAGAAGTACCT	soooossssssssssooss	46	97	499
655187	358	8038	263	TTTCTCCATCACTGAGAAGT	soooossssssssssooss	36	94	500
655188	361	8041	266	TTATTTCTCCATCACTGAGA	soooossssssssssooss	9	82	501
655189	367	8047	272	GAAAAGTTATTTCTCCATCA	soooossssssssssooss	37	96	503
655190	376	8056	281	GGTTGGCAAGAAAAGTTATT	soooossssssssssooss	15	74	504
655191	379	8059	284	TGTGGTTGGCAAGAAAAGTT	soooossssssssssooss	15	62	505
655192	385	8065	290	TTAGAGTGTGGTTGGCAAGA	soooossssssssssooss	12	63	507
655193	388	8068	293	CATTTAGAGTGTGGTTGGCA	soooossssssssssooss	7	89	508
655194	389	8069	294	CCATTTAGAGTGTGGTTGGC	soooossssssssssooss	31	93	509
655195	394	8074	299	TTTCTCCATTTAGAGTGTGG	soooossssssssssooss	19	91	510
655196	397	8077	302	GGATTTCTCCATTTAGAGTG	soooossssssssssooss	8	82	511

TABLE 18

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
576816	310	7990	215	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	55	97	20
619420	310	7990	215	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	74	97	20
655197	403	8083	308	TTCGAAGGATTTCTCCATTT	soooossssssssssooss	14	87	517
655198	406	8086	311	CATTTCGAAGGATTTCTCCA	soooossssssssssooss	23	83	518
655199	409	8089	314	CTGCATTTCGAAGGATTTCT	soooossssssssssooss	42	97	519
655200	412	8092	317	TCTCTGCATTTCGAAGGATT	soooossssssssssooss	21	87	520
655201	415	8095	320	CACTCTCTGCATTTCGAAGG	soooossssssssssooss	25	89	521
655202	418	8098	323	CACCACTCTCTGCATTTCGA	soooossssssssssooss	54	96	522
655203	420	8100	325	AGCACCACTCTCTGCATTTC	soooossssssssssooss	51	90	523
655204	421	8101	326	TAGCACCACTCTCTGCATTT	soooossssssssssooss	28	94	524
655205	424	8104	329	CTATAGCACCACTCTCTGCA	soooossssssssssooss	17	90	525
655206	427	8107	332	CATCTATAGCACCACTCTCT	soooossssssssssooss	15	83	526
655207	433	8113	338	ACTTTACATCTATAGCACCA	soooossssssssssooss	20	85	527
655208	436	8116	341	AAAACTTTACATCTATAGCA	soooossssssssssooss	25	72	528

TABLE 18-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
655209	443	8123	348	AAGACAAAAAACTTTACATC	soooossssssssssooss	10	45	529
655210	444	8124	349	CAAGACAAAAAACTTTACAT	soooossssssssssooss	0	56	530
655211	446	8126	351	GACAAGACAAAAAACTTTAC	soooossssssssssooss	32	84	531
655212	449	8129	354	TCAGACAAGACAAAAAACTT	soooossssssssssooss	10	74	532
655213	451	8131	356	TTTCAGACAAGACAAAAAAC	soooossssssssssooss	7	54	533
655214	453	8133	358	CTTTTCAGACAAGACAAAAA	soooossssssssssooss	7	57	534
655215	456	8136	361	TCCCTTTTCAGACAAGACAA	soooossssssssssooss	32	91	535
655216	459	8139	364	CACTCCCTTTTCAGACAAGA	soooossssssssssooss	39	91	536
655217	462	8142	367	AATCACTCCCTTTTCAGACA	soooossssssssssooss	41	95	537
655218	465	8145	370	AATAATCACTCCCTTTTCAG	soooossssssssssooss	13	63	538
655219	467	8147	372	ACAATAATCACTCCCTTTTC	soooossssssssssooss	2	60	539
655220	468	8148	373	AACAATAATCACTCCCTTTT	soooossssssssssooss	11	73	540
655221	471	8151	376	TGAAACAATAATCACTCCCT	soooossssssssssooss	17	83	541
655222	474	8154	379	TAATGAAACAATAATCACTC	soooossssssssssooss	1	79	542
655223	477	8157	382	GATTAATGAAACAATAATCA	soooossssssssssooss	12	59	543
655224	482	8162	387	TCAAAGATTAATGAAACAAT	soooossssssssssooss	0	9	544
655225	485	8165	390	CCATCAAAGATTAATGAAAC	soooossssssssssooss	17	81	545
655226	488	8168	393	TTTCCATCAAAGATTAATGA	soooossssssssssooss	14	87	546
655227	491	8171	396	CAGTTTCCATCAAAGATTAA	soooossssssssssooss	7	69	547
655228	494	8174	399	TTCCAGTTTCCATCAAAGAT	soooossssssssssooss	25	88	548
655229	497	8177	402	CCATTCCAGTTTCCATCAAA	soooossssssssssooss	44	90	549
655230	500	8180	405	TCCCCATTCCAGTTTCCATC	soooossssssssssooss	48	94	550
655231	503	8183	408	CGATCCCCATTCCAGTTTCC	soooossssssssssooss	54	96	551
655232	506	8186	411	CTGCGATCCCCATTCCAGTT	soooossssssssssooss	61	96	552
655233	509	8189	414	GTGCTGCGATCCCCATTCCA	soooossssssssssooss	63	100	553
655234	512	8192	417	TATGTGCTGCGATCCCCATT	soooossssssssssooss	34	94	554
655235	533	8213	438	GGAAGTATAATTGATAGTCC	soooossssssssssooss	25	91	555
655236	536	8216	441	TGTGGAAGTATAATTGATAG	soooossssssssssooss	13	66	556
655237	539	8219	444	GTCTGTGGAAGTATAATTGA	soooossssssssssooss	24	83	557
655238	542	8222	447	TCTGTCTGTGGAAGTATAAT	soooossssssssssooss	51	92	558
655239	545	8225	450	AGTTCTGTCTGTGGAAGTAT	soooossssssssssooss	42	96	559
655240	548	8228	453	CTAAGTTCTGTCTGTGGAAG	soooossssssssssooss	16	88	560
655241	551	8231	456	AAACTAAGTTCTGTCTGTGG	soooossssssssssooss	25	93	561

TABLE 18-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
655242	554	8234	459	TAGAAACTAAGTTCTGTCTG	soooossssssssssooss	29	94	562
655243	557	8237	462	AGGTAGAAACTAAGTTCTGT	soooossssssssssooss	43	85	563
655244	560	8240	465	GGGAGGTAGAAACTAAGTTC	soooossssssssssooss	27	77	564
655245	563	8243	468	AGTGGGAGGTAGAAACTAAG	soooossssssssssooss	22	77	565
655246	566	8246	471	TGAAGTGGGAGGTAGAAACT	soooossssssssssooss	16	45	566
655247	569	8249	474	CTATGAAGTGGGAGGTAGAA	soooossssssssssooss	14	60	567
655248	572	8252	477	ACTCTATGAAGTGGGAGGTA	soooossssssssssooss	30	81	568
655249	574	8254	479	ACACTCTATGAAGTGGGAGG	soooossssssssssooss	34	94	569
655250	575	8255	480	CACACTCTATGAAGTGGGAG	soooossssssssssooss	46	98	570
655251	576	8256	481	ACACACTCTATGAAGTGGGA	soooossssssssssooss	20	94	571
655252	578	8258	483	ACACACACTCTATGAAGTGG	soooossssssssssooss	28	97	572
655253	581	8261	486	TCAACACACACTCTATGAAG	soooossssssssssooss	12	62	573
655254	584	8264	489	CTATCAACACACACTCTATG	soooossssssssssooss	6	68	574
655255	587	8267	492	AATCTATCAACACACACTCT	soooossssssssssooss	16	87	575
655256	590	8270	495	GTTAATCTATCAACACACAC	soooossssssssssooss	28	95	576
655257	592	8272	497	GTGTTAATCTATCAACACAC	soooossssssssssooss	15	76	577
655258	593	8273	498	TGTGTTAATCTATCAACACA	soooossssssssssooss	14	53	578
655259	595	8275	500	TATGTGTTAATCTATCAACA	soooossssssssssooss	11	78	579
655260	596	8276	501	ATATGTGTTAATCTATCAAC	soooossssssssssooss	25	79	580
655261	599	8279	504	ATTATATGTGTTAATCTATC	soooossssssssssooss	11	71	581
655262	602	8282	507	CGGATTATATGTGTTAATCT	soooossssssssssooss	21	91	582
655263	605	8285	510	TTCCGGATTATATGTGTTAA	soooossssssssssooss	15	88	583
655264	608	8288	513	CCTTTCCGGATTATATGTGT	soooossssssssssooss	12	83	584
655265	611	8291	516	CTTCCTTTCCGGATTATATG	soooossssssssssooss	1	69	585
655266	614	8294	519	ATTCTTCCTTTCCGGATTAT	soooossssssssssooss	15	76	586
655267	617	8297	522	CATATTCTTCCTTTCCGGAT	soooossssssssssooss	17	69	587
655268	620	8300	525	ATCCATATTCTTCCTTTCCG	soooossssssssssooss	21	67	588
655269	624	8304	529	ATGCATCCATATTCTTCCTT	soooossssssssssooss	5	73	589
655270	626	8306	531	TTATGCATCCATATTCTTCC	soooossssssssssooss	23	71	590
655271	629	n/a	534	TCCTTATGCATCCATATTCT	soooossssssssssooss	15	64	591
655272	632	n/a	537	CTTTCCTTATGCATCCATAT	soooossssssssssooss	24	66	592
655273	635	n/a	540	TGTCTTTCCTTATGCATCCA	soooossssssssssooss	37	76	593

TABLE 19

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
576816	310	7990	215	GCCTTACTCTAGGACCAAGA	sssssssssssssssss	53	95	20
619253	n/a	1406	n/a	TACAGGCTGCGGTTGTTTCC	soooossssssssssooss	0	72	97
619293	n/a	1446	n/a	CCCGGCCCCTAGCGCGCGAC	soooossssssssssooss	0	95	98
619422	n/a	3452	n/a	GGTAACTTCAAACTCTTGGG	soooossssssssssooss	0	93	382
655329	n/a	2330	n/a	AGGACCTCCCTCCTGTTTCT	soooossssssssssooss	0	84	594
655330	n/a	2490	n/a	AGAAGTAATGCCAGACAGAT	soooossssssssssooss	0	75	595
655331	n/a	2901	n/a	CTTTGTTTCTCTGAAAGCAA	soooossssssssssooss	16	75	596
655332	n/a	3576	n/a	GTGGTTGGTCCACTGCTATT	soooossssssssssooss	28	94	597
655333	n/a	3801	n/a	TTGAGGGAAGCCAAGATTCA	soooossssssssssooss	10	77	598
655334	n/a	3975	n/a	AGAGCTGTACAATTATTTTA	soooossssssssssooss	13	90	599
655335	n/a	4725	n/a	GGTAATGACACTACTGCTGT	soooossssssssssooss	27	96	600
655336	n/a	5970	n/a	GATCCTAATCCTGTCTATGC	soooossssssssssooss	0	70	601
655337	n/a	7382	n/a	ACTTGTGGGTTGAATTGTGT	soooossssssssssooss	4	66	602
655338	n/a	8310	n/a	TACCTTATGCATCCATATTC	soooossssssssssooss	11	62	603
655339	n/a	8409	n/a	GATGTTCACTGCATATAATT	soooossssssssssooss	31	88	604
655340	n/a	8464	n/a	CCAGATGTATTTGTATCTAA	soooossssssssssooss	41	96	605
655341	n/a	8523	n/a	TAATGTGGAGCTACCATTTC	soooossssssssssooss	6	72	606
655342	n/a	8587	n/a	GCTCCCAAGAAGAATCCAGG	soooossssssssssooss	44	74	607
655343	n/a	8658	n/a	ACTTACACATAGTAGTAAGC	soooossssssssssooss	16	88	608
655344	n/a	8716	n/a	AAAGAGACCAAAGGCTACAT	soooossssssssssooss	4	82	609
655345	n/a	8785	n/a	GGAATTCTCTTGGGAACCAT	soooossssssssssooss	9	80	610
619420	310	7990	215	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	51	98	20
655274	638	n/a	543	TCTTGTCTTTCCTTATGCAT	soooossssssssssooss	0	73	611
655275	641	n/a	546	TTTTCTTGTCTTTCCTTATG	soooossssssssssooss	10	61	612
625344	647	9413	552	TGGACATTTTCTTGTCTTTC	soooossssssssssooss	32	94	613
655276	650	9416	555	TTCTGGACATTTTCTTGTCT	soooossssssssssooss	8	79	614
655277	653	9419	558	ATCTTCTGGACATTTTCTTG	soooossssssssssooss	0	60	615
655278	655	9421	560	TAATCTTCTGGACATTTTCT	soooossssssssssooss	0	66	616
655279	659	9425	564	AAGATAATCTTCTGGACATT	soooossssssssssooss	5	72	617
655280	662	9428	567	TCTAAGATAATCTTCTGGAC	soooossssssssssooss	2	81	618
655281	665	9431	570	CCTTCTAAGATAATCTTCTG	soooossssssssssooss	0	19	619
655282	668	9434	573	GTGCCTTCTAAGATAATCTT	soooossssssssssooss	4	20	620
655283	671	9437	576	TCTGTGCCTTCTAAGATAAT	soooossssssssssooss	9	27	621

TABLE 19-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
655284	674	9440	579	CTCTCTGTGCCTTCTAAGAT	soooossssssssssooss	0	35	622
655285	677	9443	582	ATTCTCTCTGTGCCTTCTAA	soooossssssssssooss	7	40	623
655286	680	9446	585	TCCATTCTCTCTGTGCCTTC	soooossssssssssooss	18	65	624
655287	683	9449	588	TCTTCCATTCTCTCTGTGCC	soooossssssssssooss	14	67	625
655288	686	9452	591	TGATCTTCCATTCTCTCTGT	soooossssssssssooss	7	65	626
655289	691	n/a	596	GACCCTGATCTTCCATTCTC	soooossssssssssooss	12	89	627
655290	694	n/a	599	TCTGACCCTGATCTTCCATT	soooossssssssssooss	13	81	628
655291	697	n/a	602	TACTCTGACCCTGATCTTCC	soooossssssssssooss	0	82	629
655292	700	n/a	605	TAATACTCTGACCCTGATCT	soooossssssssssooss	0	72	630
655293	703	n/a	608	GAATAATACTCTGACCCTGA	soooossssssssssooss	0	77	631
655294	709	12529	614	GCATTGGAATAATACTCTGA	soooossssssssssooss	0	79	632
655295	712	12532	617	TAAGCATTGGAATAATACTC	soooossssssssssooss	8	79	633
655296	715	12535	620	CAGTAAGCATTGGAATAATA	soooossssssssssooss	0	66	634
655297	718	12538	623	CTCCAGTAAGCATTGGAATA	soooossssssssssooss	12	78	635
655298	721	12541	626	CTTCTCCAGTAAGCATTGGA	soooossssssssssooss	0	81	636
655299	724	12544	629	TCACTTCTCCAGTAAGCATT	soooossssssssssooss	0	70	637
655300	727	12547	632	GAATCACTTCTCCAGTAAGC	soooossssssssssooss	0	75	638
655301	730	12550	635	CAGGAATCACTTCTCCAGTA	soooossssssssssooss	20	85	639
655302	733	12553	638	TTACAGGAATCACTTCTCCA	soooossssssssssooss	0	50	640
655303	736	12556	641	CCATTACAGGAATCACTTCT	soooossssssssssooss	0	57	641
655304	744	12564	649	AAGCAGTTCCATTACAGGAA	soooossssssssssooss	0	83	642
655305	747	12567	652	TGAAAGCAGTTCCATTACAG	soooossssssssssooss	0	55	643
655306	750	12570	655	AGATGAAAGCAGTTCCATTA	soooossssssssssooss	0	82	644
655307	753	12573	658	CATAGATGAAAGCAGTTCCA	soooossssssssssooss	2	83	645
655308	756	12576	661	TTTCATAGATGAAAGCAGTT	soooossssssssssooss	0	59	646
655309	762	12582	667	GTGTGATTTCATAGATGAAA	soooossssssssssooss	0	39	647
655310	766	12586	671	CACTGTGTGATTTCATAGAT	soooossssssssssooss	10	31	648
655311	769	12589	674	GAACACTGTGTGATTTCATA	soooossssssssssooss	0	80	649
655312	772	12592	677	CAGGAACACTGTGTGATTTC	soooossssssssssooss	0	53	650
655313	778	12598	683	TTTCTTCAGGAACACTGTGT	soooossssssssssooss	0	45	651
655314	781	12601	686	CTATTTCTTCAGGAACACTG	soooossssssssssooss	0	56	652
655315	784	n/a	689	TATCTATTTCTTCAGGAACA	soooossssssssssooss	0	75	653
655316	787	n/a	692	CTATATCTATTTCTTCAGGA	soooossssssssssooss	0	75	654

TABLE 19-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (HepG2) Primer Probe RTS3905	SEQ ID NO
655317	790	n/a	695	CAGCTATATCTATTTCTTCA	soooossssssssssooss	23	89	655
655318	793	n/a	698	TATCAGCTATATCTATTTCT	soooossssssssssooss	7	80	656
655319	796	n/a	701	CTGTATCAGCTATATCTATT	soooossssssssssooss	34	86	657
655320	799	n/a	704	GTACTGTATCAGCTATATCT	soooossssssssssooss	42	88	658
655321	802	n/a	707	TGAGTACTGTATCAGCTATA	soooossssssssssooss	31	87	659
655322	805	13356	710	CATTGAGTACTGTATCAGCT	soooossssssssssooss	3	84	660
655323	808	13359	713	CATCATTGAGTACTGTATCA	soooossssssssssooss	5	88	661
655324	811	13362	716	CATCATCATTGAGTACTGTA	soooossssssssssooss	17	90	662
655325	814	13365	719	TATCATCATCATTGAGTACT	soooossssssssssooss	2	84	663
655326	817	13368	722	CAATATCATCATCATTGAGT	soooossssssssssooss	0	74	664
655327	822	13373	727	GTCACCAATATCATCATCAT	soooossssssssssooss	6	78	665
655328	848	13399	753	TTGAGAAGAAAGCCTTCATG	soooossssssssssooss	0	42	666
619421	3132	28251	3037	GGGACACTACAAGGTAGTAT	soooossssssssssooss	0	94	401

TABLE 20

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (hepG2) Primer Probe RTS3905	SEQ ID NO
576816	310	7990	215	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	49	97	20
655346	n/a	8861	n/a	ACTATACTGAAATGTAAATA	soooossssssssssooss	4	7	667
655347	n/a	8915	n/a	TATCAAACTGGAACACAGGA	soooossssssssssooss	16	53	668
655348	n/a	8965	n/a	TGGGCAAAAGCCTTTTAAAA	soooossssssssssooss	13	63	669
655349	n/a	9017	n/a	GCAAACATAGTAAAAAATTA	soooossssssssssooss	6	7	670
655350	n/a	9067	n/a	TTCTCCTGATTTTAAGAGTT	soooossssssssssooss	39	90	671
655351	n/a	9117	n/a	AAGAATGACTTGCACTTTTC	soooossssssssssooss	25	100	672
655352	n/a	9173	n/a	AAAGATAACTTCACAGAAAA	soooossssssssssooss	13	10	673
655353	n/a	9286	n/a	CTTTCTACTTTAGGGAAAAA	soooossssssssssooss	16	64	674
655354	n/a	9336	n/a	TTTTTCAATAGACATGTTCT	soooossssssssssooss	19	81	675
655355	n/a	9403	n/a	CTTGTCTTTCCTGAGCAAGA	soooossssssssssooss	8	54	676
655356	n/a	9455	n/a	ACCTGATCTTCCATTCTCTC	soooossssssssssooss	22	81	677

TABLE 20-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (hepG2) Primer Probe RTS3905	SEQ ID NO
655357	n/a	9576	n/a	CTCCATAAAAGCTCCATTAA	soooossssssssssooss	34	52	678
655358	n/a	9640	n/a	TGTTTACTGATTTAACTCTT	soooossssssssssooss	34	83	679
655359	n/a	9696	n/a	AACAGAAAAAAAAAGGGAGC	soooossssssssssooss	18	22	680
655360	n/a	9772	n/a	GTACCTTAAAGAACATATCA	soooossssssssssooss	34	100	681
655361	n/a	9920	n/a	AAATGTAAATTGCATGAGTC	soooossssssssssooss	7	100	682
655362	n/a	9970	n/a	GGGTAAGAAATATCACTGAC	soooossssssssssooss	43	63	683
655363	n/a	10055	n/a	AACCATGCTTCTCAAACTCT	soooossssssssssooss	29	75	684
655364	n/a	10122	n/a	AAGAACTTCTCTGCTTTACA	soooossssssssssooss	13	68	685
655365	n/a	10172	n/a	AATGGAAGTAAAAGTGAAGA	soooossssssssssooss	7	13	686
655366	n/a	10233	n/a	AACAGCCATGTTTAAAATAT	soooossssssssssooss	15	37	687
655367	n/a	10283	n/a	TTAAAGTATCATCTGTCTCA	soooossssssssssooss	34	74	688
655368	n/a	10364	n/a	CAATTTGGTAAAGGAGATCA	soooossssssssssooss	6	48	689
655369	n/a	10418	n/a	ACACAGAATAACTGTCTCTG	soooossssssssssooss	16	64	690
655370	n/a	10491	n/a	GCTTATTGACCAGCAAATAA	soooossssssssssooss	22	69	691
655371	n/a	10615	n/a	CCCAGTAAAAGCAGAATTTT	soooossssssssssooss	24	57	692
655372	n/a	10665	n/a	ATTAATAGTAGTCAACTTAA	soooossssssssssooss	20	10	693
655373	n/a	10730	n/a	ACTTGAACTTCTCAGCAGTA	soooossssssssssooss	25	62	694
655374	n/a	10786	n/a	AGAAGAGGCTCTAAAAGAAA	soooossssssssssooss	7	59	695
655375	n/a	10970	n/a	AAAGGCAACTCCTCCTTTTC	soooossssssssssooss	13	78	696
655376	n/a	11020	n/a	ACAGTATTGTTCAAAATAAA	soooossssssssssooss	18	100	697
655377	n/a	11070	n/a	CAGATGACAGCTACAACTGA	soooossssssssssooss	16	49	698
655378	n/a	11121	n/a	GAATAATGACTAGATCCGTG	soooossssssssssooss	26	88	699
655379	n/a	11171	n/a	ATAATTATCATGCCTGTTTA	soooossssssssssooss	22	80	700
655380	n/a	11221	n/a	CTTTAGTAACCTCCACAACT	soooossssssssssooss	19	26	701
655381	n/a	11313	n/a	GAATTTAAATGTGATGCTAC	soooossssssssssooss	15	49	702
655382	n/a	11385	n/a	TGTCAGACCCAGGGCCATTT	soooossssssssssooss	40	83	703
655383	n/a	11445	n/a	ACTTATTTTATGAAATGATT	soooossssssssssooss	7	3	704
655384	n/a	11513	n/a	TTTTAGCTAAACATATTTTT	soooossssssssssooss	2	1	705
655385	n/a	11591	n/a	CAGTCTCATCAGTTTTGTGA	soooossssssssssooss	33	83	706
655386	n/a	11641	n/a	TGTAAAGTGTCTCAAATATG	soooossssssssssooss	21	39	707
655387	n/a	11711	n/a	CTTGAAATTGTAATTTTGAA	soooossssssssssooss	14	42	708
655388	n/a	11798	n/a	AATCAAAATCAGCACATATA	soooossssssssssooss	8	46	709
655389	n/a	11871	n/a	ACCAAATAGGTAAGGAAAAC	soooossssssssssooss	9	32	710

TABLE 20-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (hepG2) Primer Probe RTS3905	SEQ ID NO
655390	n/a	11978	n/a	AAAGATCTCCTTTAAAATTT	soooossssssssssooss	11	29	711
655391	n/a	12053	n/a	CTTTAGAGAGTATGGAATCA	soooossssssssssooss	22	71	712
655392	n/a	12334	n/a	CAAAGCTCACTTTTATTCTT	soooossssssssssooss	25	70	713
655393	n/a	12384	n/a	ACACAGTATCAAACAAGTCT	soooossssssssssooss	29	45	714
655394	n/a	12459	n/a	AAGCTGGGCAATAAAAAATA	soooossssssssssooss	0	24	715
655395	n/a	12513	n/a	CTGACCCTGCACAATAAAGT	soooossssssssssooss	17	0	716
655396	n/a	12604	n/a	CATCTATTTCTTCAGGAACA	soooossssssssssooss	21	74	717
655397	n/a	12681	n/a	GAATATTAATAATATACATA	soooossssssssssooss	0	16	718
655398	n/a	12765	n/a	AGGATTTTGTGTGTGCTTAT	soooossssssssssooss	30	65	719
655399	n/a	12855	n/a	TTTTAGGAATTATAAAAGTA	soooossssssssssooss	7	61	720
655400	n/a	12924	n/a	ACACAGTTTTGTTTCAAAAG	soooossssssssssooss	13	61	721
655401	n/a	12978	n/a	GGAAACTAAATTTGTGACTA	soooossssssssssooss	21	56	722
655402	n/a	13028	n/a	CTCTTAACACTCATAGTGTG	soooossssssssssooss	13	52	723
655403	n/a	13084	n/a	GAGACTAACCTAAATGACAA	soooossssssssssooss	20	35	724
655404	n/a	13159	n/a	CAAATGTGAAAGCTGGTCAA	soooossssssssssooss	8	100	725
655405	n/a	13237	n/a	TAACACACTGCCTTCATTTC	soooossssssssssooss	17	32	726
655406	n/a	13337	n/a	TATCTAAAATGCATCAAAAA	soooossssssssssooss	2	6	727
655407	n/a	13400	n/a	CTTGAGAAGAAAGCCTTCAT	soooossssssssssooss	23	42	728
655408	n/a	13471	n/a	CCAAATCTTGTCATAGGTGA	soooossssssssssooss	42	95	729
655409	n/a	13550	n/a	TAACACAAATTTAAGCAACA	soooossssssssssooss	21	55	730
655410	n/a	13603	n/a	AAATAGCAAATGGAATAACA	soooossssssssssooss	14	55	731
655411	n/a	13662	n/a	AAACCAGAATCAAGCAAGGG	soooossssssssssooss	27	73	732
655412	n/a	13722	n/a	CATCTACAGTACAACTTAAT	soooossssssssssooss	13	100	733
655413	n/a	13773	n/a	AGATCAGTATAAATATGAAT	soooossssssssssooss	1	43	734
655414	n/a	13823	n/a	GTTTAAGGGCACAAACTCTT	soooossssssssssooss	27	78	735
655415	n/a	13884	n/a	AGGTGTATAGAGAATTCAGG	soooossssssssssooss	43	89	736
655416	n/a	13955	n/a	TACTCAATGCTTATAACAAC	soooossssssssssooss	26	84	737
655417	n/a	14089	n/a	GGAACTAACATGTAGGCACT	soooossssssssssooss	66	100	738
655418	n/a	14213	n/a	CATAAAAGTGAATACTTTAT	soooossssssssssooss	13	0	739
655419	n/a	14281	n/a	AGGCTCTTAGGTTAAACACA	soooossssssssssooss	16	79	740
655420	n/a	14331	n/a	GCTGACACTGAACAGATACA	soooossssssssssooss	52	84	741
655421	n/a	14392	n/a	CATGTAGAGAGATTAAGTGA	soooossssssssssooss	27	42	742

TABLE 20-continued

Percent inhibition of human C9ORF72 compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2, and 6

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 6 Start Site	Sequence	Linkage	% inhibition (LLC-MK2) Primer Probe RTS3750	% inhibition (hepG2) Primer Probe RTS3905	SEQ ID NO
655422	n/a	14452	n/a	ATCATTTAATTAATGTATTT	soooossssssssssooss	13	0	743
619420	310	7990	215	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	76	98	20

Example 5

Dose-Dependent Antisense Inhibition of Human C9ORF72 mRNA in LLC-MK2

[0449] Antisense oligonucleotides from the study described above exhibiting significant in vitro inhibition of C9ORF72 mRNA were selected and tested at various doses in LLC-MK2 cells. ISIS 576816, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.33 μM, 1.00 μM, 3.00 μM, or 9.00 μM concentrations of antisense oligonucleotide. After a treatment period of approximately 16 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human C9ORF72 primer probe set RTS3750 was used to measure total C9ORF72 mRNA levels. C9ORF72 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72 levels, relative to untreated control cells.

[0450] As shown in Table 21, total C9ORF72 mRNA levels were reduced in a dose-dependent manner in the antisense oligonucleotide treated cells.

TABLE 21

Dose-dependent inhibition of total C9ORF72 mRNA transcript levels in LLC-MK2 cells

ISIS No	0.33 μM	1.00 μM	3.00 μM	9.00 μM
576816	0	33	55	66
619411	15	43	67	87
619412	9	30	55	84
619413	17	27	58	79
619414	13	49	75	83
619415	15	41	62	57
619416	29	47	70	81
619420	17	49	70	85
619423	25	52	71	82
627833	4	31	63	82
655173	37	72	86	90
655178	9	35	71	86
655179	30	50	52	84
655202	1	12	41	72
655231	0	28	43	71
655232	17	45	64	76
655233	19	30	62	80
655420	24	28	49	78

[0451] The antisense oligonucleotides were also selected and tested at various doses in HepG2 cells. ISIS 576816, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.11 μM, 0.33 μM, or 3.00 μM concentrations of antisense oligonucleotide. After a treatment period of approximately 16 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human C9ORF72 primer probe set RTS3750 was used to measure total C9ORF72 mRNA levels. C9ORF72 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72 levels, relative to untreated control cells.

[0452] As shown in Table 22, total C9ORF72 mRNA levels were reduced in a dose-dependent manner in the antisense oligonucleotide treated cells.

TABLE 22

Dose-dependent inhibition of total C9ORF72 mRNA transcript levels in HepG2 cells

ISIS No	0.11 μM	0.33 μM	1.00 μM	3.00 μM
576816	16	32	71	90
619411	11	41	71	91
619412	24	44	79	92
619413	0	18	59	84
619414	16	51	76	92
619415	16	38	72	85
619416	16	36	47	80
619420	15	47	75	91
619423	24	50	81	89
627833	0	29	69	90
655173	24	56	88	96
655178	25	48	81	92
655179	16	43	79	87
655202	17	19	61	84
655231	14	45	68	84
655232	13	38	69	79
655233	17	30	65	86
655420	19	35	60	86

Example 6

Design of Mixed Backbone 5-8-5 MOE Gapmers and Deoxy, MOE, and cEt Oligonucleotides Targeting Human C9ORF72

[0453] Additional antisense oligonucleotides were designed targeting a C9ORF72 nucleic acid. The newly

designed chimeric antisense oligonucleotides in the Tables below were designed as 5-8-5 MOE gapmers, 5-10-5 MOE gapmers, or deoxy, MOE, and cEt gapmers.

[0454] The 5-8-5 MOE gapmers are 18 nucleosides in length, wherein the central gap segment comprises of eight 2'-deoxynucleosides and is flanked by wing segments on the 5' direction and the 3' direction comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment has a 2'-MOE modification. The linkages between the nucleosides are described in the Linkage column; 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage. All cytosine residues throughout each gapmer are 5-methylcytosines.

[0455] The 5-10-5 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises of ten 2'-deoxynucleosides and is flanked by wing segments on the 5' direction and the 3' direction comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment has a 2'-MOE modification. The linkages between the nucleosides are described in the Linkage column; 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage. All cytosine residues throughout each gapmer are 5-methylcytosines.

[0456] The deoxy, MOE, and cEt oligonucleotides are 17 nucleosides in length wherein the nucleoside has either a MOE sugar modification, a cEt sugar modification, or a deoxyribose sugar. The 'Chemistry' column describes the sugar modifications of each oligonucleotide; 'k' indicates a cEt nucleoside; d' indicates deoxyribonucleosides, the number indicates the number of deoxyribonucleosides; and 'e' indicates a 2'-MOE nucleoside. The internucleoside linkages throughout each gapmer are either phosphorothioate linkages or phosphodiester linkages. The linkages between the nucleosides are described in the Linkage column; 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage. All cytosine residues throughout each gapmer are 5-methylcytosines.

[0457] "Start site" indicates the 5'-most nucleoside to which the antisense oligonucleotide is targeted in the human gene sequence. "Stop site" indicates the 3'-most nucleoside to which the antisense oligonucleotide is targeted in the human gene sequence. Each antisense oligonucleotide listed in the Tables 23-29 below is targeted to the human C9ORF72 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18 truncated from nucleosides 27535000 to 27565000. Table 30 presents a 5-10-5 gapmer that is targeted to C9ORF72 mRNA sequence, SEQ ID NO: 1 (GENBANK Accession No. NM_001256054.1).

TABLE 23

5-8-5 MOE gapmers targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO:2 Start Site	SEQ ID NO:2 Stop Site	SEQ ID NO
672581	GTGAGAGCAAGTAGTGGG	sooossssssssooss	1326	1343	744
672582	TGTGAGAGCAAGTAGTGG	sooossssssssooss	1327	1344	745
672583	CTGTGAGAGCAAGTAGTG	sooossssssssooss	1328	1345	746
672584	ACTGTGAGAGCAAGTAGT	sooossssssssooss	1329	1346	747
672585	TACTGTGAGAGCAAGTAG	sooossssssssooss	1330	1347	748
672586	GTACTGTGAGAGCAAGTA	sooossssssssooss	1331	1348	749
672587	AGTACTGTGAGAGCAAGT	sooossssssssooss	1332	1349	750
672588	GAGTACTGTGAGAGCAAG	sooossssssssooss	1333	1350	751
672589	CGAGTACTGTGAGAGCAA	sooossssssssooss	1334	1351	752
672590	GCGAGTACTGTGAGAGCA	sooossssssssooss	1335	1352	753
672591	AGCGAGTACTGTGAGAGC	sooossssssssooss	1336	1353	754
672592	CAGCGAGTACTGTGAGAG	sooossssssssooss	1337	1354	755
672593	TCAGCGAGTACTGTGAGA	sooossssssssooss	1338	1355	756
672594	CTCAGCGAGTACTGTGAG	sooossssssssooss	1339	1356	757
672595	CCTCAGCGAGTACTGTGA	sooossssssssooss	1340	1357	758
672596	CCCTCAGCGAGTACTGTG	sooossssssssooss	1341	1358	759
672597	ACCCTCAGCGAGTACTGT	sooossssssssooss	1342	1359	760
672598	CACCCTCAGCGAGTACTG	sooossssssssooss	1343	1360	761
672599	TCACCCTCAGCGAGTACT	sooossssssssooss	1344	1361	762
672600	TTCACCCTCAGCGAGTAC	sooossssssssooss	1345	1362	763

TABLE 23-continued

5-8-5 MOE gapmers targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO:2 Start Site	SEQ ID NO:2 Stop Site	SEQ ID NO
672601	GTTCACCCTCAGCGAGTA	sooooooooooooooss	1346	1363	764
672602	TGTTCACCCTCAGCGAGT	sooooooooooooooss	1347	1364	765
672603	TTGTTCACCCTCAGCGAG	sooooooooooooooss	1348	1365	766
672604	CTTGTTCACCCTCAGCGA	sooooooooooooooss	1349	1366	767
672605	TCTTGTTCACCCTCAGCG	sooooooooooooooss	1350	1367	768
672606	TTCTTGTTCACCCTCAGC	sooooooooooooooss	1351	1368	769
672607	TTTCTTGTTCACCCTCAG	sooooooooooooooss	1352	1369	770
672608	TTTTCTTGTTCACCCTCA	sooooooooooooooss	1353	1370	771
672609	CTTTTCTTGTTCACCCTC	sooooooooooooooss	1354	1371	772
672610	TCTTTTCTTGTTCACCCT	sooooooooooooooss	1355	1372	773
672611	GTCTTTTCTTGTTCACCC	sooooooooooooooss	1356	1373	774
672612	GGTCTTTTCTTGTTCACC	sooooooooooooooss	1357	1374	775
672613	AGGTCTTTTCTTGTTCAC	sooooooooooooooss	1358	1375	776
672614	CAGGTCTTTTCTTGTTCA	sooooooooooooooss	1359	1376	777
672615	TCAGGTCTTTTCTTGTTC	sooooooooooooooss	1360	1377	778
672616	ATCAGGTCTTTTCTTGTT	sooooooooooooooss	1361	1378	779
672617	TATCAGGTCTTTTCTTGT	sooooooooooooooss	1362	1379	780
672618	TTATCAGGTCTTTTCTTG	sooooooooooooooss	1363	1380	781
672619	TTTATCAGGTCTTTTCTT	sooooooooooooooss	1364	1381	782
672620	AATCTTTATCAGGTCTTT	sooooooooooooooss	1368	1385	783
672621	TAATCTTTATCAGGTCTT	sooooooooooooooss	1369	1386	784
672622	TTAATCTTTATCAGGTCT	sooooooooooooooss	1370	1387	785
672623	GTTAATCTTTATCAGGTC	sooooooooooooooss	1371	1388	786
672624	GGTTAATCTTTATCAGGT	sooooooooooooooss	1372	1389	787
672625	TGGTTAATCTTTATCAGG	sooooooooooooooss	1373	1390	788
672626	CTGGTTAATCTTTATCAG	sooooooooooooooss	1374	1391	789
672627	TCTGGTTAATCTTTATCA	sooooooooooooooss	1375	1392	790
672628	TTCTGGTTAATCTTTATC	sooooooooooooooss	1376	1393	791
672629	TCCCTCCTTGTTTTCTTC	sooooooooooooooss	1391	1408	792
672630	TTTCCCTCCTTGTTTTCT	sooooooooooooooss	1393	1410	793
672631	GTTTCCCTCCTTGTTTTC	sooooooooooooooss	1394	1411	794
672632	TGTTTCCCTCCTTGTTTT	sooooooooooooooss	1395	1412	795
672633	TTGTTTCCCTCCTTGTTT	sooooooooooooooss	1396	1413	796
672634	GTTGTTTCCCTCCTTGTT	sooooooooooooooss	1397	1414	797
672635	GGTTGTTTCCCTCCTTGT	sooooooooooooooss	1398	1415	798

TABLE 23-continued

5-8-5 MOE gapmers targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO:2 Start Site	SEQ ID NO:2 Stop Site	SEQ ID NO
672636	CGGTTGTTTCCCTCCTTG	sooooooooooooooss	1399	1416	799
672637	GCGGTTGTTTCCCTCCTT	sooooooooooooooss	1400	1417	800
672638	TGCGGTTGTTTCCCTCCT	sooooooooooooooss	1401	1418	801
672639	CTGCGGTTGTTTCCCTCC	sooooooooooooooss	1402	1419	802
672640	GCTGCGGTTGTTTCCCTC	sooooooooooooooss	1403	1420	803
672641	GGCTGCGGTTGTTTCCCT	sooooooooooooooss	1404	1421	804
672642	AGGCTGCGGTTGTTTCCC	sooooooooooooooss	1405	1422	805
672643	CAGGCTGCGGTTGTTTCC	sooooooooooooooss	1406	1423	806
672644	ACAGGCTGCGGTTGTTTC	sooooooooooooooss	1407	1424	807
672645	TACAGGCTGCGGTTGTTT	sooooooooooooooss	1408	1425	808
672646	CTACAGGCTGCGGTTGTT	sooooooooooooooss	1409	1426	809
672647	GCTACAGGCTGCGGTTGT	sooooooooooooooss	1410	1427	810
672648	TGCTACAGGCTGCGGTTG	sooooooooooooooss	1411	1428	811
672649	TTGCTACAGGCTGCGGTT	sooooooooooooooss	1412	1429	812
672650	CTTGCTACAGGCTGCGGT	sooooooooooooooss	1413	1430	813
672651	GCTTGCTACAGGCTGCGG	sooooooooooooooss	1414	1431	814
672652	AGCTTGCTACAGGCTGCG	sooooooooooooooss	1415	1432	815
672653	GAGCTTGCTACAGGCTGC	sooooooooooooooss	1416	1433	816
672654	AGAGCTTGCTACAGGCTG	sooooooooooooooss	1417	1434	817
672655	CAGAGCTTGCTACAGGCT	sooooooooooooooss	1418	1435	818
672656	CCAGAGCTTGCTACAGGC	sooooooooooooooss	1419	1436	819
672657	TCCAGAGCTTGCTACAGG	sooooooooooooooss	1420	1437	820
672658	TTCCAGAGCTTGCTACAG	sooooooooooooooss	1421	1438	821
672659	GTTCCAGAGCTTGCTACA	sooooooooooooooss	1422	1439	822
672660	AGTTCCAGAGCTTGCTAC	sooooooooooooooss	1423	1440	823
672661	GAGTTCCAGAGCTTGCTA	sooooooooooooooss	1424	1441	824
672662	TGAGTTCCAGAGCTTGCT	sooooooooooooooss	1425	1442	825
672663	CTGAGTTCCAGAGCTTGC	sooooooooooooooss	1426	1443	826
672664	CCTGAGTTCCAGAGCTTG	sooooooooooooooss	1427	1444	827
672665	TCCTGAGTTCCAGAGCTT	sooooooooooooooss	1428	1445	828
672666	CTCCTGAGTTCCAGAGCT	sooooooooooooooss	1429	1446	829
672667	ACTCCTGAGTTCCAGAGC	sooooooooooooooss	1430	1447	830
672668	GACTCCTGAGTTCCAGAG	sooooooooooooooss	1431	1448	831
672669	CGACTCCTGAGTTCCAGA	sooooooooooooooss	1432	1449	832
672670	GCGACTCCTGAGTTCCAG	sooooooooooooooss	1433	1450	833
672671	CGCGACTCCTGAGTTCCA	sooooooooooooooss	1434	1451	834

TABLE 23-continued

5-8-5 MOE gapmers targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO:2 Start Site	SEQ ID NO:2 Stop Site	SEQ ID NO
672672	GCGCGACTCCTGAGTTCC	sooooooooooooooss	1435	1452	835
672673	CGCGCGACTCCTGAGTTC	sooooooooooooooss	1436	1453	836
672674	GCGCGCGACTCCTGAGTT	sooooooooooooooss	1437	1454	837
672675	AGCGCGCGACTCCTGAGT	sooooooooooooooss	1438	1455	838
672676	TAGCGCGCGACTCCTGAG	sooooooooooooooss	1439	1456	839
672677	CTAGCGCGCGACTCCTGA	sooooooooooooooss	1440	1457	840
672678	CCTAGCGCGCGACTCCTG	sooooooooooooooss	1441	1458	841
672679	CCCTAGCGCGCGACTCCT	sooooooooooooooss	1442	1459	842
672680	CCCCTAGCGCGCGACTCC	sooooooooooooooss	1443	1460	843
672681	GCCCCTAGCGCGCGACTC	sooooooooooooooss	1444	1461	844
672682	GGCCCCTAGCGCGCGACT	sooooooooooooooss	1445	1462	845
672683	CGGCCCCTAGCGCGCGAC	sooooooooooooooss	1446	1463	846
672684	CCGGCCCCTAGCGCGCGA	sooooooooooooooss	1447	1464	847
672685	CCCGGCCCCTAGCGCGCG	sooooooooooooooss	1448	1465	848
672686	CCCCGGCCCCTAGCGCGC	sooooooooooooooss	1449	1466	849
672687	GCCCCGGCCCCTAGCGCG	sooooooooooooooss	1450	1467	850
672688	GGCCCCGGCCCCTAGCGC	sooooooooooooooss	1451	1468	851
672689	CGGCCCCGGCCCCTAGCG	sooooooooooooooss	1452	1469	852
672690	CCGGCCCCGGCCCCTAGC	sooooooooooooooss	1453	1470	853
672691	CCCGGCCCCGGCCCCTAG	sooooooooooooooss	1454	1471	854
672692	CCCCGGCCCCGGCCCCTA	sooooooooooooooss	1455	1472	855
672693	ACGCCCCGGCCCCGGCCC	sooooooooooooooss	1464	1481	856
672694	CACGCCCCGGCCCCGGCC	sooooooooooooooss	1465	1482	857
672695	CCACGCCCCGGCCCCGGC	sooooooooooooooss	1466	1483	858
672696	ACCACGCCCCGGCCCCGG	sooooooooooooooss	1467	1484	859
672697	GACCACGCCCCGGCCCCG	sooooooooooooooss	1468	1485	860
672698	CGACCACGCCCCGGCCCC	sooooooooooooooss	1469	1486	861
672699	CCGACCACGCCCCGGCCC	sooooooooooooooss	1470	1487	862
672700	CCCGACCACGCCCCGGCC	sooooooooooooooss	1471	1488	863
672701	CCCCGACCACGCCCCGGC	sooooooooooooooss	1472	1489	864
672702	GCCCCGACCACGCCCCGG	sooooooooooooooss	1473	1490	865
672703	CGCCCCGACCACGCCCCG	sooooooooooooooss	1474	1491	866
672704	CCGCCCCGACCACGCCCC	sooooooooooooooss	1475	1492	867
672705	CCCGCCCCGACCACGCCC	sooooooooooooooss	1476	1493	868
672706	GCCCGCCCCGACCACGCC	sooooooooooooooss	1477	1494	869

TABLE 23-continued

5-8-5 MOE gapmers targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO:2 Start Site	SEQ ID NO:2 Stop Site	SEQ ID NO
672707	GGCCCGCCCCGACCACGC	sooossssssssooss	1478	1495	870
672708	GGGCCCGCCCCGACCACG	sooossssssssooss	1479	1496	871
672709	CGGGCCCGCCCCGACCAC	sooossssssssooss	1480	1497	872
672710	CCGGGCCCGCCCCGACCA	sooossssssssooss	1481	1498	873
672711	CCCGGGCCCGCCCCGACC	sooossssssssooss	1482	1499	874
672712	CCCCGGGCCCGCCCCGAC	sooossssssssooss	1483	1500	875
672713	AGCCCCGCCCCGGGCCCG	sooossssssssooss	1502	1519	876
672714	CAGCCCCGCCCCGGGCCC	sooossssssssooss	1503	1520	877
672715	GCAGCCCCGCCCCGGGCC	sooossssssssooss	1504	1521	878
672716	CGCAGCCCCGCCCCGGGC	sooossssssssooss	1505	1522	879
672717	CCGCAGCCCCGCCCCGGG	sooossssssssooss	1506	1523	880
672718	ACCGCAGCCCCGCCCCGG	sooossssssssooss	1507	1524	881
672719	AACCGCAGCCCCGCCCCG	sooossssssssooss	1508	1525	882
672720	CAACCGCAGCCCCGCCCC	sooossssssssooss	1509	1526	883
672721	GCAACCGCAGCCCCGCCC	sooossssssssooss	1510	1527	884
672722	CGCAACCGCAGCCCCGCC	sooossssssssooss	1511	1528	885
672723	CCGCAACCGCAGCCCCGC	sooossssssssooss	1512	1529	886
672724	ACCGCAACCGCAGCCCCG	sooossssssssooss	1513	1530	887
672725	CACCGCAACCGCAGCCCC	sooossssssssooss	1514	1531	888
672726	GCACCGCAACCGCAGCCC	sooossssssssooss	1515	1532	889
672727	GGCACCGCAACCGCAGCC	sooossssssssooss	1516	1533	890
672728	AGGCACCGCAACCGCAGC	sooossssssssooss	1517	1534	891
672729	CAGGCACCGCAACCGCAG	sooossssssssooss	1518	1535	892
672730	GCAGGCACCGCAACCGCA	sooossssssssooss	1519	1536	893
672731	CGCAGGCACCGCAACCGC	sooossssssssooss	1520	1537	894
672732	GCGCAGGCACCGCAACCG	sooossssssssooss	1521	1538	895
672733	GGCGCAGGCACCGCAACC	sooossssssssooss	1522	1539	896
672734	GGGCGCAGGCACCGCAAC	sooossssssssooss	1523	1540	897

TABLE 24

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672735	TGAGAGCAAGTAGTGGG	eeekk-d7-kkeee	soossssssssooss	1326	1342	898
672736	GTGAGAGCAAGTAGTGG	eeekk-d7-kkeee	soossssssssooss	1327	1343	899

TABLE 24-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672737	TGTGAGAGCAAGTAGTG	eeekk-d7-kkeee	soossssssssooss	1328	1344	900
672738	CTGTGAGAGCAAGTAGT	eeekk-d7-kkeee	soossssssssooss	1329	1345	901
672739	ACTGTGAGAGCAAGTAG	eeekk-d7-kkeee	soossssssssooss	1330	1346	902
672740	TACTGTGAGAGCAAGTA	eeekk-d7-kkeee	soossssssssooss	1331	1347	903
672741	GTACTGTGAGAGCAAGT	eeekk-d7-kkeee	soossssssssooss	1332	1348	904
672742	AGTACTGTGAGAGCAAG	eeekk-d7-kkeee	soossssssssooss	1333	1349	905
672743	GAGTACTGTGAGAGCAA	eeekk-d7-kkeee	soossssssssooss	1334	1350	906
672744	CGAGTACTGTGAGAGCA	eeekk-d7-kkeee	soossssssssooss	1335	1351	907
672745	GCGAGTACTGTGAGAGC	eeekk-d7-kkeee	soossssssssooss	1336	1352	908
672746	AGCGAGTACTGTGAGAG	eeekk-d7-kkeee	soossssssssooss	1337	1353	909
672747	CAGCGAGTACTGTGAGA	eeekk-d7-kkeee	soossssssssooss	1338	1354	910
672748	TCAGCGAGTACTGTGAG	eeekk-d7-kkeee	soossssssssooss	1339	1355	911
672749	CTCAGCGAGTACTGTGA	eeekk-d7-kkeee	soossssssssooss	1340	1356	912
672750	CCTCAGCGAGTACTGTG	eeekk-d7-kkeee	soossssssssooss	1341	1357	913
672751	CCCTCAGCGAGTACTGT	eeekk-d7-kkeee	soossssssssooss	1342	1358	914
672752	ACCCTCAGCGAGTACTG	eeekk-d7-kkeee	soossssssssooss	1343	1359	915
672753	CACCCTCAGCGAGTACT	eeekk-d7-kkeee	soossssssssooss	1344	1360	916
672754	TCACCCTCAGCGAGTAC	eeekk-d7-kkeee	soossssssssooss	1345	1361	917
672755	TTCACCCTCAGCGAGTA	eeekk-d7-kkeee	soossssssssooss	1346	1362	918
672756	GTTCACCCTCAGCGAGT	eeekk-d7-kkeee	soossssssssooss	1347	1363	919
672757	TGTTCACCCTCAGCGAG	eeekk-d7-kkeee	soossssssssooss	1348	1364	920
672758	TTGTTCACCCTCAGCGA	eeekk-d7-kkeee	soossssssssooss	1349	1365	921
672759	CTTGTTCACCCTCAGCG	eeekk-d7-kkeee	soossssssssooss	1350	1366	922
672760	TCTTGTTCACCCTCAGC	eeekk-d7-kkeee	soossssssssooss	1351	1367	923
672761	TTCTTGTTCACCCTCAG	eeekk-d7-kkeee	soossssssssooss	1352	1368	924
672762	TTTCTTGTTCACCCTCA	eeekk-d7-kkeee	soossssssssooss	1353	1369	925
672763	TTTTCTTGTTCACCCTC	eeekk-d7-kkeee	soossssssssooss	1354	1370	926
672764	CTTTTCTTGTTCACCCT	eeekk-d7-kkeee	soossssssssooss	1355	1371	927
672765	TCTTTTCTTGTTCACCC	eeekk-d7-kkeee	soossssssssooss	1356	1372	928
672766	GTCTTTTCTTGTTCACC	eeekk-d7-kkeee	soossssssssooss	1357	1373	929
672767	GGTCTTTTCTTGTTCAC	eeekk-d7-kkeee	soossssssssooss	1358	1374	930
672768	AGGTCTTTTCTTGTTCA	eeekk-d7-kkeee	soossssssssooss	1359	1375	931
672769	CAGGTCTTTTCTTGTTC	eeekk-d7-kkeee	soossssssssooss	1360	1376	932
672770	TCAGGTCTTTTCTTGTT	eeekk-d7-kkeee	soossssssssooss	1361	1377	933
672771	ATCAGGTCTTTTCTTGT	eeekk-d7-kkeee	soossssssssooss	1362	1378	934

TABLE 24-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672772	TATCAGGTCTTTTCTTG	eeekk-d7-kkeee	soossssssssooss	1363	1379	935
672773	TTATCAGGTCTTTTCTT	eeekk-d7-kkeee	soossssssssooss	1364	1380	936
672774	ATCTTTATCAGGTCTTT	eeekk-d7-kkeee	soossssssssooss	1368	1384	937
672775	AATCTTTATCAGGTCTT	eeekk-d7-kkeee	soossssssssooss	1369	1385	938
672776	TAATCTTTATCAGGTCT	eeekk-d7-kkeee	soossssssssooss	1370	1386	939
672777	TTAATCTTTATCAGGTC	eeekk-d7-kkeee	soossssssssooss	1371	1387	940
672778	GTTAATCTTTATCAGGT	eeekk-d7-kkeee	soossssssssooss	1372	1388	941
672779	GGTTAATCTTTATCAGG	eeekk-d7-kkeee	soossssssssooss	1373	1389	942
672780	TGGTTAATCTTTATCAG	eeekk-d7-kkeee	soossssssssooss	1374	1390	943
672781	CTGGTTAATCTTTATCA	eeekk-d7-kkeee	soossssssssooss	1375	1391	944
672782	TCTGGTTAATCTTTATC	eeekk-d7-kkeee	soossssssssooss	1376	1392	945
672783	CCCTCCTTGTTTTCTTC	eeekk-d7-kkeee	soossssssssooss	1391	1407	946
672784	TCCCTCCTTGTTTTCTT	eeekk-d7-kkeee	soossssssssooss	1392	1408	947
672785	TTCCCTCCTTGTTTTCT	eeekk-d7-kkeee	soossssssssooss	1393	1409	948
672786	TTTCCCTCCTTGTTTTC	eeekk-d7-kkeee	soossssssssooss	1394	1410	949
672787	GTTTCCCTCCTTGTTTT	eeekk-d7-kkeee	soossssssssooss	1395	1411	950
672788	TGTTTCCCTCCTTGTTT	eeekk-d7-kkeee	soossssssssooss	1396	1412	951
672789	TTGTTTCCCTCCTTGTT	eeekk-d7-kkeee	soossssssssooss	1397	1413	952
672790	GGTTGTTTCCCTCCTTG	eeekk-d7-kkeee	soossssssssooss	1399	1415	953
672791	CGGTTGTTTCCCTCCTT	eeekk-d7-kkeee	soossssssssooss	1400	1416	954
672792	GCGGTTGTTTCCCTCCT	eeekk-d7-kkeee	soossssssssooss	1401	1417	955
672793	TGCGGTTGTTTCCCTCC	eeekk-d7-kkeee	soossssssssooss	1402	1418	956
672794	CTGCGGTTGTTTCCCTC	eeekk-d7-kkeee	soossssssssooss	1403	1419	957
672795	GCTGCGGTTGTTTCCCT	eeekk-d7-kkeee	soossssssssooss	1404	1420	958
672796	GGCTGCGGTTGTTTCCC	eeekk-d7-kkeee	soossssssssooss	1405	1421	959
672797	AGGCTGCGGTTGTTTCC	eeekk-d7-kkeee	soossssssssooss	1406	1422	960
672798	CAGGCTGCGGTTGTTTC	eeekk-d7-kkeee	soossssssssooss	1407	1423	961
672799	ACAGGCTGCGGTTGTTT	eeekk-d7-kkeee	soossssssssooss	1408	1424	962
672800	TACAGGCTGCGGTTGTT	eeekk-d7-kkeee	soossssssssooss	1409	1425	963
672801	CTACAGGCTGCGGTTGT	eeekk-d7-kkeee	soossssssssooss	1410	1426	964
672802	GCTACAGGCTGCGGTTG	eeekk-d7-kkeee	soossssssssooss	1411	1427	965
672803	TGCTACAGGCTGCGGTT	eeekk-d7-kkeee	soossssssssooss	1412	1428	966
672804	TTGCTACAGGCTGCGGT	eeekk-d7-kkeee	soossssssssooss	1413	1429	967
672805	CTTGCTACAGGCTGCGG	eeekk-d7-kkeee	soossssssssooss	1414	1430	968
672806	GCTTGCTACAGGCTGCG	eeekk-d7-kkeee	soossssssssooss	1415	1431	969

TABLE 24-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672807	AGCTTGCTACAGGCTGC	eeekk-d7-kkeee	soossssssssooss	1416	1432	970
672808	GAGCTTGCTACAGGCTG	eeekk-d7-kkeee	soossssssssooss	1417	1433	971
672809	AGAGCTTGCTACAGGCT	eeekk-d7-kkeee	soossssssssooss	1418	1434	972
672810	CAGAGCTTGCTACAGGC	eeekk-d7-kkeee	soossssssssooss	1419	1435	973
672811	CCAGAGCTTGCTACAGG	eeekk-d7-kkeee	soossssssssooss	1420	1436	974
672812	TCCAGAGCTTGCTACAG	eeekk-d7-kkeee	soossssssssooss	1421	1437	975
672813	TTCCAGAGCTTGCTACA	eeekk-d7-kkeee	soossssssssooss	1422	1438	976
672814	GTTCCAGAGCTTGCTAC	eeekk-d7-kkeee	soossssssssooss	1423	1439	977
672815	AGTTCCAGAGCTTGCTA	eeekk-d7-kkeee	soossssssssooss	1424	1440	978
672816	GAGTTCCAGAGCTTGCT	eeekk-d7-kkeee	soossssssssooss	1425	1441	979
672817	TGAGTTCCAGAGCTTGC	eeekk-d7-kkeee	soossssssssooss	1426	1442	980
672818	CTGAGTTCCAGAGCTTG	eeekk-d7-kkeee	soossssssssooss	1427	1443	981
672819	CCTGAGTTCCAGAGCTT	eeekk-d7-kkeee	soossssssssooss	1428	1444	982
672820	TCCTGAGTTCCAGAGCT	eeekk-d7-kkeee	soossssssssooss	1429	1445	983
672821	CTCCTGAGTTCCAGAGC	eeekk-d7-kkeee	soossssssssooss	1430	1446	984
672822	ACTCCTGAGTTCCAGAG	eeekk-d7-kkeee	soossssssssooss	1431	1447	985
672823	GACTCCTGAGTTCCAGA	eeekk-d7-kkeee	soossssssssooss	1432	1448	986
672824	CGACTCCTGAGTTCCAG	eeekk-d7-kkeee	soossssssssooss	1433	1449	987
672825	GCGACTCCTGAGTTCCA	eeekk-d7-kkeee	soossssssssooss	1434	1450	988
672826	CGCGACTCCTGAGTTCC	eeekk-d7-kkeee	soossssssssooss	1435	1451	989
672827	GCGCGACTCCTGAGTTC	eeekk-d7-kkeee	soossssssssooss	1436	1452	990
672828	CGCGCGACTCCTGAGTT	eeekk-d7-kkeee	soossssssssooss	1437	1453	991
672829	GCGCGCGACTCCTGAGT	eeekk-d7-kkeee	soossssssssooss	1438	1454	992
672830	AGCGCGCGACTCCTGAG	eeekk-d7-kkeee	soossssssssooss	1439	1455	993
672831	TAGCGCGCGACTCCTGA	eeekk-d7-kkeee	soossssssssooss	1440	1456	994
672832	CTAGCGCGCGACTCCTG	eeekk-d7-kkeee	soossssssssooss	1441	1457	995
672833	CCTAGCGCGCGACTCCT	eeekk-d7-kkeee	soossssssssooss	1442	1458	996
672834	CCCTAGCGCGCGACTCC	eeekk-d7-kkeee	soossssssssooss	1443	1459	997
672835	CCCCTAGCGCGCGACTC	eeekk-d7-kkeee	soossssssssooss	1444	1460	998
672836	GCCCCTAGCGCGCGACT	eeekk-d7-kkeee	soossssssssooss	1445	1461	999
672837	GGCCCCTAGCGCGCGAC	eeekk-d7-kkeee	soossssssssooss	1446	1462	1000
672838	CGGCCCCTAGCGCGCGA	eeekk-d7-kkeee	soossssssssooss	1447	1463	1001
672839	CCGGCCCCTAGCGCGCG	eeekk-d7-kkeee	soossssssssooss	1448	1464	1002
672840	CCCGGCCCCTAGCGCGC	eeekk-d7-kkeee	soossssssssooss	1449	1465	1003
672841	CCCCGGCCCCTAGCGCG	eeekk-d7-kkeee	soossssssssooss	1450	1466	1004

TABLE 24-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672842	GCCCCGGCCCCTAGCGC	eeekk-d7-kkeee	soossssssssooss	1451	1467	1005
672843	GGCCCCGGCCCCTAGCG	eeekk-d7-kkeee	soossssssssooss	1452	1468	1006
672844	CGGCCCCGGCCCCTAGC	eeekk-d7-kkeee	soossssssssooss	1453	1469	1007
672845	CCGGCCCCGGCCCCTAG	eeekk-d7-kkeee	soossssssssooss	1454	1470	1008
672846	CCCGGCCCCGGCCCCTA	eeekk-d7-kkeee	soossssssssooss	1455	1471	1009
672847	ACGCCCCGGCCCCGGCC	eeekk-d7-kkeee	soossssssssooss	1465	1481	1010
672848	CACGCCCCGGCCCCGGC	eeekk-d7-kkeee	soossssssssooss	1466	1482	1011
672849	CCACGCCCCGGCCCCGG	eeekk-d7-kkeee	soossssssssooss	1467	1483	1012
672850	ACCACGCCCCGGCCCCG	eeekk-d7-kkeee	soossssssssooss	1468	1484	1013
672851	GACCACGCCCCGGCCCC	eeekk-d7-kkeee	soossssssssooss	1469	1485	1014
672852	CGACCACGCCCCGGCCC	eeekk-d7-kkeee	soossssssssooss	1470	1486	1015
672853	CCGACCACGCCCCGGCC	eeekk-d7-kkeee	soossssssssooss	1471	1487	1016
672854	CCCGACCACGCCCCGGC	eeekk-d7-kkeee	soossssssssooss	1472	1488	1017
672855	CCCCGACCACGCCCCGG	eeekk-d7-kkeee	soossssssssooss	1473	1489	1018
672856	GCCCCGACCACGCCCCG	eeekk-d7-kkeee	soossssssssooss	1474	1490	1019
672857	CGCCCCGACCACGCCCC	eeekk-d7-kkeee	soossssssssooss	1475	1491	1020
672858	CCGCCCCGACCACGCCC	eeekk-d7-kkeee	soossssssssooss	1476	1492	1021
672859	CCCGCCCCGACCACGCC	eeekk-d7-kkeee	soossssssssooss	1477	1493	1022
672860	GCCCGCCCCGACCACGC	eeekk-d7-kkeee	soossssssssooss	1478	1494	1023
672861	GGCCCGCCCCGACCACG	eeekk-d7-kkeee	soossssssssooss	1479	1495	1024
672862	GGGCCCGCCCCGACCAC	eeekk-d7-kkeee	soossssssssooss	1480	1496	1025
672863	CGGGCCCGCCCCGACCA	eeekk-d7-kkeee	soossssssssooss	1481	1497	1026
672864	CCGGGCCCGCCCCGACC	eeekk-d7-kkeee	soossssssssooss	1482	1498	1027
672865	CCCGGGCCCGCCCCGAC	eeekk-d7-kkeee	soossssssssooss	1483	1499	1028
672866	GCAGCCCCGCCCCGGGC	eeekk-d7-kkeee	soossssssssooss	1505	1521	1029
672867	CGCAGCCCCGCCCCGGG	eeekk-d7-kkeee	soossssssssooss	1506	1522	1030
672868	CCGCAGCCCCGCCCCGG	eeekk-d7-kkeee	soossssssssooss	1507	1523	1031
672869	ACCGCAGCCCCGCCCCG	eeekk-d7-kkeee	soossssssssooss	1508	1524	1032
672870	AACCGCAGCCCCGCCCC	eeekk-d7-kkeee	soossssssssooss	1509	1525	1033
672871	CAACCGCAGCCCCGCCC	eeekk-d7-kkeee	soossssssssooss	1510	1526	1034
672872	GCAACCGCAGCCCCGCC	eeekk-d7-kkeee	soossssssssooss	1511	1527	1035
672873	CGCAACCGCAGCCCCGC	eeekk-d7-kkeee	soossssssssooss	1512	1528	1036
672874	CCGCAACCGCAGCCCCG	eeekk-d7-kkeee	soossssssssooss	1513	1529	1037
672875	ACCGCAACCGCAGCCCC	eeekk-d7-kkeee	soossssssssooss	1514	1530	1038
672876	CACCGCAACCGCAGCCC	eeekk-d7-kkeee	soossssssssooss	1515	1531	1039

TABLE 24-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672877	GCACCGCAACCGCAGCC	eeekk-d7-kkeee	soossssssssooss	1516	1532	1040
672878	GGCACCGCAACCGCAGC	eeekk-d7-kkeee	soossssssssooss	1517	1533	1041
672879	AGGCACCGCAACCGCAG	eeekk-d7-kkeee	soossssssssooss	1518	1534	1042
672880	CAGGCACCGCAACCGCA	eeekk-d7-kkeee	soossssssssooss	1519	1535	1043
672881	GCAGGCACCGCAACCGC	eeekk-d7-kkeee	soossssssssooss	1520	1536	1044
672882	CGCAGGCACCGCAACCG	eeekk-d7-kkeee	soossssssssooss	1521	1537	1045
672883	GCGCAGGCACCGCAACC	eeekk-d7-kkeee	soossssssssooss	1522	1538	1046
672884	GGCGCAGGCACCGCAAC	eeekk-d7-kkeee	soossssssssooss	1523	1539	1047

TABLE 25

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672885	TGAGAGCAAGTAGTGGG	eekk-d8-kkeee	soossssssssooss	1326	1342	898
672886	GTGAGAGCAAGTAGTGG	eekk-d8-kkeee	soossssssssooss	1327	1343	899
672887	TGTGAGAGCAAGTAGTG	eekk-d8-kkeee	soossssssssooss	1328	1344	900
672888	CTGTGAGAGCAAGTAGT	eekk-d8-kkeee	soossssssssooss	1329	1345	901
672889	ACTGTGAGAGCAAGTAG	eekk-d8-kkeee	soossssssssooss	1330	1346	902
672890	TACTGTGAGAGCAAGTA	eekk-d8-kkeee	soossssssssooss	1331	1347	903
672891	GTACTGTGAGAGCAAGT	eekk-d8-kkeee	soossssssssooss	1332	1348	904
672892	AGTACTGTGAGAGCAAG	eekk-d8-kkeee	soossssssssooss	1333	1349	905
672893	GAGTACTGTGAGAGCAA	eekk-d8-kkeee	soossssssssooss	1334	1350	906
672894	CGAGTACTGTGAGAGCA	eekk-d8-kkeee	soossssssssooss	1335	1351	907
672895	GCGAGTACTGTGAGAGC	eekk-d8-kkeee	soossssssssooss	1336	1352	908
672896	AGCGAGTACTGTGAGAG	eekk-d8-kkeee	soossssssssooss	1337	1353	909
672897	CAGCGAGTACTGTGAGA	eekk-d8-kkeee	soossssssssooss	1338	1354	910
672898	TCAGCGAGTACTGTGAG	eekk-d8-kkeee	soossssssssooss	1339	1355	911
672899	CTCAGCGAGTACTGTGA	eekk-d8-kkeee	soossssssssooss	1340	1356	912
672900	CCTCAGCGAGTACTGTG	eekk-d8-kkeee	soossssssssooss	1341	1357	913
672901	CCCTCAGCGAGTACTGT	eekk-d8-kkeee	soossssssssooss	1342	1358	914
672902	ACCCTCAGCGAGTACTG	eekk-d8-kkeee	soossssssssooss	1343	1359	915
672903	CACCCTCAGCGAGTACT	eekk-d8-kkeee	soossssssssooss	1344	1360	916
672904	TCACCCTCAGCGAGTAC	eekk-d8-kkeee	soossssssssooss	1345	1361	917

TABLE 25-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672905	TTCACCCTCAGCGAGTA	eekk-d8-kkeee	soossssssssooss	1346	1362	918
672906	GTTCACCCTCAGCGAGT	eekk-d8-kkeee	soossssssssooss	1347	1363	919
672907	TGTTCACCCTCAGCGAG	eekk-d8-kkeee	soossssssssooss	1348	1364	920
672908	TTGTTCACCCTCAGCGA	eekk-d8-kkeee	soossssssssooss	1349	1365	921
672909	CTTGTTCACCCTCAGCG	eekk-d8-kkeee	soossssssssooss	1350	1366	922
672910	TCTTGTTCACCCTCAGC	eekk-d8-kkeee	soossssssssooss	1351	1367	923
672911	TTCTTGTTCACCCTCAG	eekk-d8-kkeee	soossssssssooss	1352	1368	924
672912	TTTCTTGTTCACCCTCA	eekk-d8-kkeee	soossssssssooss	1353	1369	925
672913	TTTTCTTGTTCACCCTC	eekk-d8-kkeee	soossssssssooss	1354	1370	926
672914	CTTTTCTTGTTCACCCT	eekk-d8-kkeee	soossssssssooss	1355	1371	927
672915	TCTTTTCTTGTTCACCC	eekk-d8-kkeee	soossssssssooss	1356	1372	928
672916	GTCTTTTCTTGTTCACC	eekk-d8-kkeee	soossssssssooss	1357	1373	929
672917	GGTCTTTTCTTGTTCAC	eekk-d8-kkeee	soossssssssooss	1358	1374	930
672918	AGGTCTTTTCTTGTTCA	eekk-d8-kkeee	soossssssssooss	1359	1375	931
672919	CAGGTCTTTTCTTGTTC	eekk-d8-kkeee	soossssssssooss	1360	1376	932
672920	TCAGGTCTTTTCTTGTT	eekk-d8-kkeee	soossssssssooss	1361	1377	933
672921	ATCAGGTCTTTTCTTGT	eekk-d8-kkeee	soossssssssooss	1362	1378	934
672922	TATCAGGTCTTTTCTTG	eekk-d8-kkeee	soossssssssooss	1363	1379	935
672923	TTATCAGGTCTTTTCTT	eekk-d8-kkeee	soossssssssooss	1364	1380	936
672924	ATCTTTATCAGGTCTTT	eekk-d8-kkeee	soossssssssooss	1368	1384	937
672925	AATCTTTATCAGGTCTT	eekk-d8-kkeee	soossssssssooss	1369	1385	938
672926	TAATCTTTATCAGGTCT	eekk-d8-kkeee	soossssssssooss	1370	1386	939
672927	TTAATCTTTATCAGGTC	eekk-d8-kkeee	soossssssssooss	1371	1387	940
672928	GTTAATCTTTATCAGGT	eekk-d8-kkeee	soossssssssooss	1372	1388	941
672929	GGTTAATCTTTATCAGG	eekk-d8-kkeee	soossssssssooss	1373	1389	942
672930	TGGTTAATCTTTATCAG	eekk-d8-kkeee	soossssssssooss	1374	1390	943
672931	CTGGTTAATCTTTATCA	eekk-d8-kkeee	soossssssssooss	1375	1391	944
672932	TCTGGTTAATCTTTATC	eekk-d8-kkeee	soossssssssooss	1376	1392	945
672933	CCCTCCTTGTTTTCTTC	eekk-d8-kkeee	soossssssssooss	1391	1407	946
672934	TCCCTCCTTGTTTTCTT	eekk-d8-kkeee	soossssssssooss	1392	1408	947
672935	TTCCCTCCTTGTTTTCT	eekk-d8-kkeee	soossssssssooss	1393	1409	948
672936	TTTCCCTCCTTGTTTTC	eekk-d8-kkeee	soossssssssooss	1394	1410	949
672937	GTTTCCCTCCTTGTTTT	eekk-d8-kkeee	soossssssssooss	1395	1411	950
672938	TGTTTCCCTCCTTGTTT	eekk-d8-kkeee	soossssssssooss	1396	1412	951
672939	TTGTTTCCCTCCTTGTT	eekk-d8-kkeee	soossssssssooss	1397	1413	952

TABLE 25-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672940	GGTTGTTTCCCTCCTTG	eekk-d8-kkeee	soossssssssooss	1399	1415	953
672941	CGGTTGTTTCCCTCCTT	eekk-d8-kkeee	soossssssssooss	1400	1416	954
672942	GCGGTTGTTTCCCTCCT	eekk-d8-kkeee	soossssssssooss	1401	1417	955
672943	TGCGGTTGTTTCCCTCC	eekk-d8-kkeee	soossssssssooss	1402	1418	956
672944	CTGCGGTTGTTTCCCTC	eekk-d8-kkeee	soossssssssooss	1403	1419	957
672945	GCTGCGGTTGTTTCCCT	eekk-d8-kkeee	soossssssssooss	1404	1420	958
672946	GGCTGCGGTTGTTTCCC	eekk-d8-kkeee	soossssssssooss	1405	1421	959
672947	AGGCTGCGGTTGTTTCC	eekk-d8-kkeee	soossssssssooss	1406	1422	960
672948	CAGGCTGCGGTTGTTTC	eekk-d8-kkeee	soossssssssooss	1407	1423	961
672949	ACAGGCTGCGGTTGTTT	eekk-d8-kkeee	soossssssssooss	1408	1424	962
672950	TACAGGCTGCGGTTGTT	eekk-d8-kkeee	soossssssssooss	1409	1425	963
672951	CTACAGGCTGCGGTTGT	eekk-d8-kkeee	soossssssssooss	1410	1426	964
672952	GCTACAGGCTGCGGTTG	eekk-d8-kkeee	soossssssssooss	1411	1427	965
672953	TGCTACAGGCTGCGGTT	eekk-d8-kkeee	soossssssssooss	1412	1428	966
672954	TTGCTACAGGCTGCGGT	eekk-d8-kkeee	soossssssssooss	1413	1429	967
672955	CTTGCTACAGGCTGCGG	eekk-d8-kkeee	soossssssssooss	1414	1430	968
672956	GCTTGCTACAGGCTGCG	eekk-d8-kkeee	soossssssssooss	1415	1431	969
672957	AGCTTGCTACAGGCTGC	eekk-d8-kkeee	soossssssssooss	1416	1432	970
672958	GAGCTTGCTACAGGCTG	eekk-d8-kkeee	soossssssssooss	1417	1433	971
672959	AGAGCTTGCTACAGGCT	eekk-d8-kkeee	soossssssssooss	1418	1434	972
672960	CAGAGCTTGCTACAGGC	eekk-d8-kkeee	soossssssssooss	1419	1435	973
672961	CCAGAGCTTGCTACAGG	eekk-d8-kkeee	soossssssssooss	1420	1436	974
672962	TCCAGAGCTTGCTACAG	eekk-d8-kkeee	soossssssssooss	1421	1437	975
672963	TTCCAGAGCTTGCTACA	eekk-d8-kkeee	soossssssssooss	1422	1438	976
672964	GTTCCAGAGCTTGCTAC	eekk-d8-kkeee	soossssssssooss	1423	1439	977
672965	AGTTCCAGAGCTTGCTA	eekk-d8-kkeee	soossssssssooss	1424	1440	978
672966	GAGTTCCAGAGCTTGCT	eekk-d8-kkeee	soossssssssooss	1425	1441	979
672967	TGAGTTCCAGAGCTTGC	eekk-d8-kkeee	soossssssssooss	1426	1442	980
672968	CTGAGTTCCAGAGCTTG	eekk-d8-kkeee	soossssssssooss	1427	1443	981
672969	CCTGAGTTCCAGAGCTT	eekk-d8-kkeee	soossssssssooss	1428	1444	982
672970	TCCTGAGTTCCAGAGCT	eekk-d8-kkeee	soossssssssooss	1429	1445	983
672971	CTCCTGAGTTCCAGAGC	eekk-d8-kkeee	soossssssssooss	1430	1446	984
672972	ACTCCTGAGTTCCAGAG	eekk-d8-kkeee	soossssssssooss	1431	1447	985
672973	GACTCCTGAGTTCCAGA	eekk-d8-kkeee	soossssssssooss	1432	1448	986

TABLE 25-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
672974	CGACTCCTGAGTTCCAG	eekk-d8-kkeee	soossssssssooss	1433	1449	987
672975	GCGACTCCTGAGTTCCA	eekk-d8-kkeee	soossssssssooss	1434	1450	988
672976	CGCGACTCCTGAGTTCC	eekk-d8-kkeee	soossssssssooss	1435	1451	989
672977	GCGCGACTCCTGAGTTC	eekk-d8-kkeee	soossssssssooss	1436	1452	990
672978	CGCGCGACTCCTGAGTT	eekk-d8-kkeee	soossssssssooss	1437	1453	991
672979	GCGCGCGACTCCTGAGT	eekk-d8-kkeee	soossssssssooss	1438	1454	992
672980	AGCGCGCGACTCCTGAG	eekk-d8-kkeee	soossssssssooss	1439	1455	993
672981	TAGCGCGCGACTCCTGA	eekk-d8-kkeee	soossssssssooss	1440	1456	994
672982	CTAGCGCGCGACTCCTG	eekk-d8-kkeee	soossssssssooss	1441	1457	995
672983	CCTAGCGCGCGACTCCT	eekk-d8-kkeee	soossssssssooss	1442	1458	996
672984	CCCTAGCGCGCGACTCC	eekk-d8-kkeee	soossssssssooss	1443	1459	997
672985	CCCCTAGCGCGCGACTC	eekk-d8-kkeee	soossssssssooss	1444	1460	998
672986	GCCCCTAGCGCGCGACT	eekk-d8-kkeee	soossssssssooss	1445	1461	999
672987	GGCCCCTAGCGCGCGAC	eekk-d8-kkeee	soossssssssooss	1446	1462	1000
672988	CGGCCCCTAGCGCGCGA	eekk-d8-kkeee	soossssssssooss	1447	1463	1001
672989	CCGGCCCCTAGCGCGCG	eekk-d8-kkeee	soossssssssooss	1448	1464	1002
672990	CCCGGCCCCTAGCGCGC	eekk-d8-kkeee	soossssssssooss	1449	1465	1003
672991	CCCCGGCCCCTAGCGCG	eekk-d8-kkeee	soossssssssooss	1450	1466	1004
672992	GCCCCGGCCCCTAGCGC	eekk-d8-kkeee	soossssssssooss	1451	1467	1005
672993	GGCCCCGGCCCCTAGCG	eekk-d8-kkeee	soossssssssooss	1452	1468	1006
672994	CGGCCCCGGCCCCTAGC	eekk-d8-kkeee	soossssssssooss	1453	1469	1007
672995	CCGGCCCCGGCCCCTAG	eekk-d8-kkeee	soossssssssooss	1454	1470	1008
672996	CCCGGCCCCGGCCCCTA	eekk-d8-kkeee	soossssssssooss	1455	1471	1009
672997	ACGCCCCGGCCCCGGCC	eekk-d8-kkeee	soossssssssooss	1465	1481	1010
672998	CACGCCCCGGCCCCGGC	eekk-d8-kkeee	soossssssssooss	1466	1482	1011
672999	CCACGCCCCGGCCCCGG	eekk-d8-kkeee	soossssssssooss	1467	1483	1012
673000	ACCACGCCCCGGCCCCG	eekk-d8-kkeee	soossssssssooss	1468	1484	1013
673001	GACCACGCCCCGGCCCC	eekk-d8-kkeee	soossssssssooss	1469	1485	1014
673002	CGACCACGCCCCGGCCC	eekk-d8-kkeee	soossssssssooss	1470	1486	1015
673003	CCGACCACGCCCCGGCC	eekk-d8-kkeee	soossssssssooss	1471	1487	1016
673004	CCCGACCACGCCCCGGC	eekk-d8-kkeee	soossssssssooss	1472	1488	1017
673005	CCCCGACCACGCCCCGG	eekk-d8-kkeee	soossssssssooss	1473	1489	1018
673006	GCCCCGACCACGCCCCG	eekk-d8-kkeee	soossssssssooss	1474	1490	1019
673007	CGCCCCGACCACGCCCC	eekk-d8-kkeee	soossssssssooss	1475	1491	1020
673008	CCGCCCCGACCACGCCC	eekk-d8-kkeee	soossssssssooss	1476	1492	1021

TABLE 25-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673009	CCCGCCCCGACCACGCC	eekk-d8-kkeee	soosssssssssooss	1477	1493	1022
673010	GCCCGCCCCGACCACGC	eekk-d8-kkeee	soosssssssssooss	1478	1494	1023
673011	GGCCCGCCCCGACCACG	eekk-d8-kkeee	soosssssssssooss	1479	1495	1024
673012	GGGCCCGCCCCGACCAC	eekk-d8-kkeee	soosssssssssooss	1480	1496	1025
673013	CGGGCCCGCCCCGACCA	eekk-d8-kkeee	soosssssssssooss	1481	1497	1026
673014	CCGGGCCCGCCCCGACC	eekk-d8-kkeee	soosssssssssooss	1482	1498	1027
673015	CCCGGGCCCGCCCCGAC	eekk-d8-kkeee	soosssssssssooss	1483	1499	1028
673016	GCAGCCCCGCCCCGGGC	eekk-d8-kkeee	soosssssssssooss	1505	1521	1029
673017	CGCAGCCCCGCCCCGGG	eekk-d8-kkeee	soosssssssssooss	1506	1522	1030
673018	CCGCAGCCCCGCCCCGG	eekk-d8-kkeee	soosssssssssooss	1507	1523	1031
673019	ACCGCAGCCCCGCCCCG	eekk-d8-kkeee	soosssssssssooss	1508	1524	1032
673020	AACCGCAGCCCCGCCCC	eekk-d8-kkeee	soosssssssssooss	1509	1525	1033
673021	CAACCGCAGCCCCGCCC	eekk-d8-kkeee	soosssssssssooss	1510	1526	1034
673022	GCAACCGCAGCCCCGCC	eekk-d8-kkeee	soosssssssssooss	1511	1527	1035
673023	CGCAACCGCAGCCCCGC	eekk-d8-kkeee	soosssssssssooss	1512	1528	1036
673024	CCGCAACCGCAGCCCCG	eekk-d8-kkeee	soosssssssssooss	1513	1529	1037
673025	ACCGCAACCGCAGCCCC	eekk-d8-kkeee	soosssssssssooss	1514	1530	1038
673026	CACCGCAACCGCAGCCC	eekk-d8-kkeee	soosssssssssooss	1515	1531	1039
673027	GCACCGCAACCGCAGCC	eekk-d8-kkeee	soosssssssssooss	1516	1532	1040
673028	GGCACCGCAACCGCAGC	eekk-d8-kkeee	soosssssssssooss	1517	1533	1041
673029	AGGCACCGCAACCGCAG	eekk-d8-kkeee	soosssssssssooss	1518	1534	1042
673030	CAGGCACCGCAACCGCA	eekk-d8-kkeee	soosssssssssooss	1519	1535	1043
673031	GCAGGCACCGCAACCGC	eekk-d8-kkeee	soosssssssssooss	1520	1536	1044
673032	CGCAGGCACCGCAACCG	eekk-d8-kkeee	soosssssssssooss	1521	1537	1045
673033	GCGCAGGCACCGCAACC	eekk-d8-kkeee	soosssssssssooss	1522	1538	1046
673034	GGCGCAGGCACCGCAAC	eekk-d8-kkeee	soosssssssssooss	1523	1539	1047

TABLE 26

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673035	TGAGAGCAAGTAGTGGG	ek-d8-ekekeee	sosssssssssoooss	1326	1342	898
673036	GTGAGAGCAAGTAGTGG	ek-d8-ekekeee	sosssssssssoooss	1327	1343	899

TABLE 26-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673037	TGTGAGAGCAAGTAGTG	ek-d8-ekekeee	sosssssssssooooss	1328	1344	900
673038	CTGTGAGAGCAAGTAGT	ek-d8-ekekeee	sosssssssssooooss	1329	1345	901
673039	ACTGTGAGAGCAAGTAG	ek-d8-ekekeee	sosssssssssooooss	1330	1346	902
673040	TACTGTGAGAGCAAGTA	ek-d8-ekekeee	sosssssssssooooss	1331	1347	903
673041	GTACTGTGAGAGCAAGT	ek-d8-ekekeee	sosssssssssooooss	1332	1348	904
673042	AGTACTGTGAGAGCAAG	ek-d8-ekekeee	sosssssssssooooss	1333	1349	905
673043	GAGTACTGTGAGAGCAA	ek-d8-ekekeee	sosssssssssooooss	1334	1350	906
673044	CGAGTACTGTGAGAGCA	ek-d8-ekekeee	sosssssssssooooss	1335	1351	907
673045	GCGAGTACTGTGAGAGC	ek-d8-ekekeee	sosssssssssooooss	1336	1352	908
673046	AGCGAGTACTGTGAGAG	ek-d8-ekekeee	sosssssssssooooss	1337	1353	909
673047	CAGCGAGTACTGTGAGA	ek-d8-ekekeee	sosssssssssooooss	1338	1354	910
673048	TCAGCGAGTACTGTGAG	ek-d8-ekekeee	sosssssssssooooss	1339	1355	911
673049	CTCAGCGAGTACTGTGA	ek-d8-ekekeee	sosssssssssooooss	1340	1356	912
673050	CCTCAGCGAGTACTGTG	ek-d8-ekekeee	sosssssssssooooss	1341	1357	913
673051	CCCTCAGCGAGTACTGT	ek-d8-ekekeee	sosssssssssooooss	1342	1358	914
673052	ACCCTCAGCGAGTACTG	ek-d8-ekekeee	sosssssssssooooss	1343	1359	915
673053	CACCCTCAGCGAGTACT	ek-d8-ekekeee	sosssssssssooooss	1344	1360	916
673054	TCACCCTCAGCGAGTAC	ek-d8-ekekeee	sosssssssssooooss	1345	1361	917
673055	TTCACCCTCAGCGAGTA	ek-d8-ekekeee	sosssssssssooooss	1346	1362	918
673056	GTTCACCCTCAGCGAGT	ek-d8-ekekeee	sosssssssssooooss	1347	1363	919
673057	TGTTCACCCTCAGCGAG	ek-d8-ekekeee	sosssssssssooooss	1348	1364	920
673058	TTGTTCACCCTCAGCGA	ek-d8-ekekeee	sosssssssssooooss	1349	1365	921
673059	CTTGTTCACCCTCAGCG	ek-d8-ekekeee	sosssssssssooooss	1350	1366	922
673060	TCTTGTTCACCCTCAGC	ek-d8-ekekeee	sosssssssssooooss	1351	1367	923
673061	TTCTTGTTCACCCTCAG	ek-d8-ekekeee	sosssssssssooooss	1352	1368	924
673062	TTTCTTGTTCACCCTCA	ek-d8-ekekeee	sosssssssssooooss	1353	1369	925
673063	TTTTCTTGTTCACCCTC	ek-d8-ekekeee	sosssssssssooooss	1354	1370	926
673064	CTTTTCTTGTTCACCCT	ek-d8-ekekeee	sosssssssssooooss	1355	1371	927
673065	TCTTTTCTTGTTCACCC	ek-d8-ekekeee	sosssssssssooooss	1356	1372	928
673066	GTCTTTTCTTGTTCACC	ek-d8-ekekeee	sosssssssssooooss	1357	1373	929
673067	GGTCTTTTCTTGTTCAC	ek-d8-ekekeee	sosssssssssooooss	1358	1374	930
673068	AGGTCTTTTCTTGTTCA	ek-d8-ekekeee	sosssssssssooooss	1359	1375	931
673069	CAGGTCTTTTCTTGTTC	ek-d8-ekekeee	sosssssssssooooss	1360	1376	932
673070	TCAGGTCTTTTCTTGTT	ek-d8-ekekeee	sosssssssssooooss	1361	1377	933

TABLE 26-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673071	ATCAGGTCTTTTCTTGT	ek-d8-ekekeee	sosssssssssooooss	1362	1378	934
673072	TATCAGGTCTTTTCTTG	ek-d8-ekekeee	sosssssssssooooss	1363	1379	935
673073	TTATCAGGTCTTTTCTT	ek-d8-ekekeee	sosssssssssooooss	1364	1380	936
673074	ATCTTTATCAGGTCTTT	ek-d8-ekekeee	sosssssssssooooss	1368	1384	937
673075	AATCTTTATCAGGTCTT	ek-d8-ekekeee	sosssssssssooooss	1369	1385	938
673076	TAATCTTTATCAGGTCT	ek-d8-ekekeee	sosssssssssooooss	1370	1386	939
673077	TTAATCTTTATCAGGTC	ek-d8-ekekeee	sosssssssssooooss	1371	1387	940
673078	GTTAATCTTTATCAGGT	ek-d8-ekekeee	sosssssssssooooss	1372	1388	941
673079	GGTTAATCTTTATCAGG	ek-d8-ekekeee	sosssssssssooooss	1373	1389	942
673080	TGGTTAATCTTTATCAG	ek-d8-ekekeee	sosssssssssooooss	1374	1390	943
673081	CTGGTTAATCTTTATCA	ek-d8-ekekeee	sosssssssssooooss	1375	1391	944
673082	TCTGGTTAATCTTTATC	ek-d8-ekekeee	sosssssssssooooss	1376	1392	945
673083	CCCTCCTTGTTTTCTTC	ek-d8-ekekeee	sosssssssssooooss	1391	1407	946
673084	TCCCTCCTTGTTTTCTT	ek-d8-ekekeee	sosssssssssooooss	1392	1408	947
673085	TTCCCTCCTTGTTTTCT	ek-d8-ekekeee	sosssssssssooooss	1393	1409	948
673086	TTTCCCTCCTTGTTTTC	ek-d8-ekekeee	sosssssssssooooss	1394	1410	949
673087	GTTTCCCTCCTTGTTTT	ek-d8-ekekeee	sosssssssssooooss	1395	1411	950
673088	TGTTTCCCTCCTTGTTT	ek-d8-ekekeee	sosssssssssooooss	1396	1412	951
673089	TTGTTTCCCTCCTTGTT	ek-d8-ekekeee	sosssssssssooooss	1397	1413	952
673090	GGTTGTTTCCCTCCTTG	ek-d8-ekekeee	sosssssssssooooss	1399	1415	953
673091	CGGTTGTTTCCCTCCTT	ek-d8-ekekeee	sosssssssssooooss	1400	1416	954
673092	GCGGTTGTTTCCCTCCT	ek-d8-ekekeee	sosssssssssooooss	1401	1417	955
673093	TGCGGTTGTTTCCCTCC	ek-d8-ekekeee	sosssssssssooooss	1402	1418	956
673094	CTGCGGTTGTTTCCCTC	ek-d8-ekekeee	sosssssssssooooss	1403	1419	957
673095	GCTGCGGTTGTTTCCCT	ek-d8-ekekeee	sosssssssssooooss	1404	1420	958
673096	GGCTGCGGTTGTTTCCC	ek-d8-ekekeee	sosssssssssooooss	1405	1421	959
673097	AGGCTGCGGTTGTTTCC	ek-d8-ekekeee	sosssssssssooooss	1406	1422	960
673098	CAGGCTGCGGTTGTTTC	ek-d8-ekekeee	sosssssssssooooss	1407	1423	961
673099	ACAGGCTGCGGTTGTTT	ek-d8-ekekeee	sosssssssssooooss	1408	1424	962
673100	TACAGGCTGCGGTTGTT	ek-d8-ekekeee	sosssssssssooooss	1409	1425	963
673101	CTACAGGCTGCGGTTGT	ek-d8-ekekeee	sosssssssssooooss	1410	1426	964
673102	GCTACAGGCTGCGGTTG	ek-d8-ekekeee	sosssssssssooooss	1411	1427	965
673103	TGCTACAGGCTGCGGTT	ek-d8-ekekeee	sosssssssssooooss	1412	1428	966
673104	TTGCTACAGGCTGCGGT	ek-d8-ekekeee	sosssssssssooooss	1413	1429	967
673105	CTTGCTACAGGCTGCGG	ek-d8-ekekeee	sosssssssssooooss	1414	1430	968

TABLE 26-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673106	GCTTGCTACAGGCTGCG	ek-d8-ekekeee	sosssssssssooooss	1415	1431	969
673107	AGCTTGCTACAGGCTGC	ek-d8-ekekeee	sosssssssssooooss	1416	1432	970
673108	GAGCTTGCTACAGGCTG	ek-d8-ekekeee	sosssssssssooooss	1417	1433	971
673109	AGAGCTTGCTACAGGCT	ek-d8-ekekeee	sosssssssssooooss	1418	1434	972
673110	CAGAGCTTGCTACAGGC	ek-d8-ekekeee	sosssssssssooooss	1419	1435	973
673111	CCAGAGCTTGCTACAGG	ek-d8-ekekeee	sosssssssssooooss	1420	1436	974
673112	TCCAGAGCTTGCTACAG	ek-d8-ekekeee	sosssssssssooooss	1421	1437	975
673113	TTCCAGAGCTTGCTACA	ek-d8-ekekeee	sosssssssssooooss	1422	1438	976
673114	GTTCCAGAGCTTGCTAC	ek-d8-ekekeee	sosssssssssooooss	1423	1439	977
673115	AGTTCCAGAGCTTGCTA	ek-d8-ekekeee	sosssssssssooooss	1424	1440	978
673116	GAGTTCCAGAGCTTGCT	ek-d8-ekekeee	sosssssssssooooss	1425	1441	979
673117	TGAGTTCCAGAGCTTGC	ek-d8-ekekeee	sosssssssssooooss	1426	1442	980
673118	CTGAGTTCCAGAGCTTG	ek-d8-ekekeee	sosssssssssooooss	1427	1443	981
673119	CCTGAGTTCCAGAGCTT	ek-d8-ekekeee	sosssssssssooooss	1428	1444	982
673120	TCCTGAGTTCCAGAGCT	ek-d8-ekekeee	sosssssssssooooss	1429	1445	983
673121	CTCCTGAGTTCCAGAGC	ek-d8-ekekeee	sosssssssssooooss	1430	1446	984
673122	ACTCCTGAGTTCCAGAG	ek-d8-ekekeee	sosssssssssooooss	1431	1447	985
673123	GACTCCTGAGTTCCAGA	ek-d8-ekekeee	sosssssssssooooss	1432	1448	986
673124	CGACTCCTGAGTTCCAG	ek-d8-ekekeee	sosssssssssooooss	1433	1449	987
673125	GCGACTCCTGAGTTCCA	ek-d8-ekekeee	sosssssssssooooss	1434	1450	988
673126	CGCGACTCCTGAGTTCC	ek-d8-ekekeee	sosssssssssooooss	1435	1451	989
673127	GCGCGACTCCTGAGTTC	ek-d8-ekekeee	sosssssssssooooss	1436	1452	990
673128	CGCGCGACTCCTGAGTT	ek-d8-ekekeee	sosssssssssooooss	1437	1453	991
673129	GCGCGCGACTCCTGAGT	ek-d8-ekekeee	sosssssssssooooss	1438	1454	992
673130	AGCGCGCGACTCCTGAG	ek-d8-ekekeee	sosssssssssooooss	1439	1455	993
673131	TAGCGCGCGACTCCTGA	ek-d8-ekekeee	sosssssssssooooss	1440	1456	994
673132	CTAGCGCGCGACTCCTG	ek-d8-ekekeee	sosssssssssooooss	1441	1457	995
673133	CCTAGCGCGCGACTCCT	ek-d8-ekekeee	sosssssssssooooss	1442	1458	996
673134	CCCTAGCGCGCGACTCC	ek-d8-ekekeee	sosssssssssooooss	1443	1459	997
673135	CCCCTAGCGCGCGACTC	ek-d8-ekekeee	sosssssssssooooss	1444	1460	998
673136	GCCCCTAGCGCGCGACT	ek-d8-ekekeee	sosssssssssooooss	1445	1461	999
673137	GGCCCCTAGCGCGCGAC	ek-d8-ekekeee	sosssssssssooooss	1446	1462	1000
673138	CGGCCCCTAGCGCGCGA	ek-d8-ekekeee	sosssssssssooooss	1447	1463	1001
673139	CCGGCCCCTAGCGCGCG	ek-d8-ekekeee	sosssssssssooooss	1448	1464	1002

TABLE 26-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673140	CCCGGCCCCTAGCGCGC	ek-d8-ekekeee	sosssssssssooooss	1449	1465	1003
673141	CCCCGGCCCCTAGCGCG	ek-d8-ekekeee	sosssssssssooooss	1450	1466	1004
673142	GCCCCGGCCCCTAGCGC	ek-d8-ekekeee	sosssssssssooooss	1451	1467	1005
673143	GGCCCCGGCCCCTAGCG	ek-d8-ekekeee	sosssssssssooooss	1452	1468	1006
673144	CGGCCCCGGCCCCTAGC	ek-d8-ekekeee	sosssssssssooooss	1453	1469	1007
673145	CCGGCCCCGGCCCCTAG	ek-d8-ekekeee	sosssssssssooooss	1454	1470	1008
673146	CCCGGCCCCGGCCCCTA	ek-d8-ekekeee	sosssssssssooooss	1455	1471	1009
673147	ACGCCCCGGCCCCGGCC	ek-d8-ekekeee	sosssssssssooooss	1465	1481	1010
673148	CACGCCCCGGCCCCGGC	ek-d8-ekekeee	sosssssssssooooss	1466	1482	1011
673149	CCACGCCCCGGCCCCGG	ek-d8-ekekeee	sosssssssssooooss	1467	1483	1012
673150	ACCACGCCCCGGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1468	1484	1013
673151	GACCACGCCCCGGCCCC	ek-d8-ekekeee	sosssssssssooooss	1469	1485	1014
673152	CGACCACGCCCCGGCCC	ek-d8-ekekeee	sosssssssssooooss	1470	1486	1015
673153	CCGACCACGCCCCGGCC	ek-d8-ekekeee	sosssssssssooooss	1471	1487	1016
673154	CCCGACCACGCCCCGGC	ek-d8-ekekeee	sosssssssssooooss	1472	1488	1017
673155	CCCCGACCACGCCCCGG	ek-d8-ekekeee	sosssssssssooooss	1473	1489	1018
673156	GCCCCGACCACGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1474	1490	1019
673157	CGCCCCGACCACGCCCC	ek-d8-ekekeee	sosssssssssooooss	1475	1491	1020
673158	CCGCCCCGACCACGCCC	ek-d8-ekekeee	sosssssssssooooss	1476	1492	1021
673159	CCCGCCCCGACCACGCC	ek-d8-ekekeee	sosssssssssooooss	1477	1493	1022
673160	GCCCGCCCCGACCACGC	ek-d8-ekekeee	sosssssssssooooss	1478	1494	1023
673161	GGCCCGCCCCGACCACG	ek-d8-ekekeee	sosssssssssooooss	1479	1495	1024
673162	GGGCCCGCCCCGACCAC	ek-d8-ekekeee	sosssssssssooooss	1480	1496	1025
673163	CGGGCCCGCCCCGACCA	ek-d8-ekekeee	sosssssssssooooss	1481	1497	1026
673164	CCGGGCCCGCCCCGACC	ek-d8-ekekeee	sosssssssssooooss	1482	1498	1027
673165	CCCGGGCCCGCCCCGAC	ek-d8-ekekeee	sosssssssssooooss	1483	1499	1028
673166	GCAGCCCCGCCCCGGGC	ek-d8-ekekeee	sosssssssssooooss	1505	1521	1029
673167	CGCAGCCCCGCCCCGGG	ek-d8-ekekeee	sosssssssssooooss	1506	1522	1030
673168	CCGCAGCCCCGCCCCGG	ek-d8-ekekeee	sosssssssssooooss	1507	1523	1031
673169	ACCGCAGCCCCGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1508	1524	1032
673170	AACCGCAGCCCCGCCCC	ek-d8-ekekeee	sosssssssssooooss	1509	1525	1033
673171	CAACCGCAGCCCCGCCC	ek-d8-ekekeee	sosssssssssooooss	1510	1526	1034
673172	GCAACCGCAGCCCCGCC	ek-d8-ekekeee	sosssssssssooooss	1511	1527	1035
673173	CGCAACCGCAGCCCCGC	ek-d8-ekekeee	sosssssssssooooss	1512	1528	1036
673174	CCGCAACCGCAGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1513	1529	1037

TABLE 26-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673175	ACCGCAACCGCAGCCCC	ek-d8-ekekeee	sosssssssssooooss	1514	1530	1038
673176	CACCGCAACCGCAGCCC	ek-d8-ekekeee	sosssssssssooooss	1515	1531	1039
673177	GCACCGCAACCGCAGCC	ek-d8-ekekeee	sosssssssssooooss	1516	1532	1040
673178	GGCACCGCAACCGCAGC	ek-d8-ekekeee	sosssssssssooooss	1517	1533	1041
673179	AGGCACCGCAACCGCAG	ek-d8-ekekeee	sosssssssssooooss	1518	1534	1042
673180	CAGGCACCGCAACCGCA	ek-d8-ekekeee	sosssssssssooooss	1519	1535	1043
673181	GCAGGCACCGCAACCGC	ek-d8-ekekeee	sosssssssssooooss	1520	1536	1044
673182	CGCAGGCACCGCAACCG	ek-d8-ekekeee	sosssssssssooooss	1521	1537	1045
673183	GCGCAGGCACCGCAACC	ek-d8-ekekeee	sosssssssssooooss	1522	1538	1046
673184	GGCGCAGGCACCGCAAC	ek-d8-ekekeee	sosssssssssooooss	1523	1539	1047

TABLE 27

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673185	TGAGAGCAAGTAGTGGG	keke-d8-ekeke	soosssssssssooss	1326	1342	898
673186	GTGAGAGCAAGTAGTGG	keke-d8-ekeke	soosssssssssooss	1327	1343	899
673187	TGTGAGAGCAAGTAGTG	keke-d8-ekeke	soosssssssssooss	1328	1344	900
673188	CTGTGAGAGCAAGTAGT	keke-d8-ekeke	soosssssssssooss	1329	1345	901
673189	ACTGTGAGAGCAAGTAG	keke-d8-ekeke	soosssssssssooss	1330	1346	902
673190	TACTGTGAGAGCAAGTA	keke-d8-ekeke	soosssssssssooss	1331	1347	903
673191	GTACTGTGAGAGCAAGT	keke-d8-ekeke	soosssssssssooss	1332	1348	904
673192	AGTACTGTGAGAGCAAG	keke-d8-ekeke	soosssssssssooss	1333	1349	905
673193	GAGTACTGTGAGAGCAA	keke-d8-ekeke	soosssssssssooss	1334	1350	906
673194	CGAGTACTGTGAGAGCA	keke-d8-ekeke	soosssssssssooss	1335	1351	907
673195	GCGAGTACTGTGAGAGC	keke-d8-ekeke	soosssssssssooss	1336	1352	908
673196	AGCGAGTACTGTGAGAG	keke-d8-ekeke	soosssssssssooss	1337	1353	909
673197	CAGCGAGTACTGTGAGA	keke-d8-ekeke	soosssssssssooss	1338	1354	910
673198	TCAGCGAGTACTGTGAG	keke-d8-ekeke	soosssssssssooss	1339	1355	911
673199	CTCAGCGAGTACTGTGA	keke-d8-ekeke	soosssssssssooss	1340	1356	912
673200	CCTCAGCGAGTACTGTG	keke-d8-ekeke	soosssssssssooss	1341	1357	913
673201	CCCTCAGCGAGTACTGT	keke-d8-ekeke	soosssssssssooss	1342	1358	914

TABLE 27-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673202	ACCCTCAGCGAGTACTG	keke-d8-ekeke	soossssssssooss	1343	1359	915
673203	CACCCTCAGCGAGTACT	keke-d8-ekeke	soossssssssooss	1344	1360	916
673204	TCACCCTCAGCGAGTAC	keke-d8-ekeke	soossssssssooss	1345	1361	917
673205	TTCACCCTCAGCGAGTA	keke-d8-ekeke	soossssssssooss	1346	1362	918
673206	GTTCACCCTCAGCGAGT	keke-d8-ekeke	soossssssssooss	1347	1363	919
673207	TGTTCACCCTCAGCGAG	keke-d8-ekeke	soossssssssooss	1348	1364	920
673208	TTGTTCACCCTCAGCGA	keke-d8-ekeke	soossssssssooss	1349	1365	921
673209	CTTGTTCACCCTCAGCG	keke-d8-ekeke	soossssssssooss	1350	1366	922
673210	TCTTGTTCACCCTCAGC	keke-d8-ekeke	soossssssssooss	1351	1367	923
673211	TTCTTGTTCACCCTCAG	keke-d8-ekeke	soossssssssooss	1352	1368	924
673212	TTTCTTGTTCACCCTCA	keke-d8-ekeke	soossssssssooss	1353	1369	925
673213	TTTTCTTGTTCACCCTC	keke-d8-ekeke	soossssssssooss	1354	1370	926
673214	CTTTTCTTGTTCACCCT	keke-d8-ekeke	soossssssssooss	1355	1371	927
673215	TCTTTTCTTGTTCACCC	keke-d8-ekeke	soossssssssooss	1356	1372	928
673216	GTCTTTTCTTGTTCACC	keke-d8-ekeke	soossssssssooss	1357	1373	929
673217	GGTCTTTTCTTGTTCAC	keke-d8-ekeke	soossssssssooss	1358	1374	930
673218	AGGTCTTTTCTTGTTCA	keke-d8-ekeke	soossssssssooss	1359	1375	931
673219	CAGGTCTTTTCTTGTTC	keke-d8-ekeke	soossssssssooss	1360	1376	932
673220	TCAGGTCTTTTCTTGTT	keke-d8-ekeke	soossssssssooss	1361	1377	933
673221	ATCAGGTCTTTTCTTGT	keke-d8-ekeke	soossssssssooss	1362	1378	934
673222	TATCAGGTCTTTTCTTG	keke-d8-ekeke	soossssssssooss	1363	1379	935
673223	TTATCAGGTCTTTTCTT	keke-d8-ekeke	soossssssssooss	1364	1380	936
673224	ATCTTTATCAGGTCTTT	keke-d8-ekeke	soossssssssooss	1368	1384	937
673225	AATCTTTATCAGGTCTT	keke-d8-ekeke	soossssssssooss	1369	1385	938
673226	TAATCTTTATCAGGTCT	keke-d8-ekeke	soossssssssooss	1370	1386	939
673227	TTAATCTTTATCAGGTC	keke-d8-ekeke	soossssssssooss	1371	1387	940
673228	GTTAATCTTTATCAGGT	keke-d8-ekeke	soossssssssooss	1372	1388	941
673229	GGTTAATCTTTATCAGG	keke-d8-ekeke	soossssssssooss	1373	1389	942
673230	TGGTTAATCTTTATCAG	keke-d8-ekeke	soossssssssooss	1374	1390	943
673231	CTGGTTAATCTTTATCA	keke-d8-ekeke	soossssssssooss	1375	1391	944
673232	TCTGGTTAATCTTTATC	keke-d8-ekeke	soossssssssooss	1376	1392	945
673233	CCCTCCTTGTTTTCTTC	keke-d8-ekeke	soossssssssooss	1391	1407	946
673234	TCCCTCCTTGTTTTCTT	keke-d8-ekeke	soossssssssooss	1392	1408	947
673235	TTCCCTCCTTGTTTTCT	keke-d8-ekeke	soossssssssooss	1393	1409	948

TABLE 27-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673236	TTTCCCTCCTTGTTTTC	keke-d8-ekeke	soossssssssooss	1394	1410	949
673237	GTTTCCCTCCTTGTTTT	keke-d8-ekeke	soossssssssooss	1395	1411	950
673238	TGTTTCCCTCCTTGTTT	keke-d8-ekeke	soossssssssooss	1396	1412	951
673239	TTGTTTCCCTCCTTGTT	keke-d8-ekeke	soossssssssooss	1397	1413	952
673240	GGTTGTTTCCCTCCTTG	keke-d8-ekeke	soossssssssooss	1399	1415	953
673241	CGGTTGTTTCCCTCCTT	keke-d8-ekeke	soossssssssooss	1400	1416	954
673242	GCGGTTGTTTCCCTCCT	keke-d8-ekeke	soossssssssooss	1401	1417	955
673243	TGCGGTTGTTTCCCTCC	keke-d8-ekeke	soossssssssooss	1402	1418	956
673244	CTGCGGTTGTTTCCCTC	keke-d8-ekeke	soossssssssooss	1403	1419	957
673245	GCTGCGGTTGTTTCCCT	keke-d8-ekeke	soossssssssooss	1404	1420	958
673246	GGCTGCGGTTGTTTCCC	keke-d8-ekeke	soossssssssooss	1405	1421	959
673247	AGGCTGCGGTTGTTTCC	keke-d8-ekeke	soossssssssooss	1406	1422	960
673248	CAGGCTGCGGTTGTTTC	keke-d8-ekeke	soossssssssooss	1407	1423	961
673249	ACAGGCTGCGGTTGTTT	keke-d8-ekeke	soossssssssooss	1408	1424	962
673250	TACAGGCTGCGGTTGTT	keke-d8-ekeke	soossssssssooss	1409	1425	963
673251	CTACAGGCTGCGGTTGT	keke-d8-ekeke	soossssssssooss	1410	1426	964
673252	GCTACAGGCTGCGGTTG	keke-d8-ekeke	soossssssssooss	1411	1427	965
673253	TGCTACAGGCTGCGGTT	keke-d8-ekeke	soossssssssooss	1412	1428	966
673254	TTGCTACAGGCTGCGGT	keke-d8-ekeke	soossssssssooss	1413	1429	967
673255	CTTGCTACAGGCTGCGG	keke-d8-ekeke	soossssssssooss	1414	1430	968
673256	GCTTGCTACAGGCTGCG	keke-d8-ekeke	soossssssssooss	1415	1431	969
673257	AGCTTGCTACAGGCTGC	keke-d8-ekeke	soossssssssooss	1416	1432	970
673258	GAGCTTGCTACAGGCTG	keke-d8-ekeke	soossssssssooss	1417	1433	971
673259	AGAGCTTGCTACAGGCT	keke-d8-ekeke	soossssssssooss	1418	1434	972
673260	CAGAGCTTGCTACAGGC	keke-d8-ekeke	soossssssssooss	1419	1435	973
673261	CCAGAGCTTGCTACAGG	keke-d8-ekeke	soossssssssooss	1420	1436	974
673262	TCCAGAGCTTGCTACAG	keke-d8-ekeke	soossssssssooss	1421	1437	975
673263	TTCCAGAGCTTGCTACA	keke-d8-ekeke	soossssssssooss	1422	1438	976
673264	GTTCCAGAGCTTGCTAC	keke-d8-ekeke	soossssssssooss	1423	1439	977
673265	AGTTCCAGAGCTTGCTA	keke-d8-ekeke	soossssssssooss	1424	1440	978
673266	GAGTTCCAGAGCTTGCT	keke-d8-ekeke	soossssssssooss	1425	1441	979
673267	TGAGTTCCAGAGCTTGC	keke-d8-ekeke	soossssssssooss	1426	1442	980
673268	CTGAGTTCCAGAGCTTG	keke-d8-ekeke	soossssssssooss	1427	1443	981
673269	CCTGAGTTCCAGAGCTT	keke-d8-ekeke	soossssssssooss	1428	1444	982

TABLE 27-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673270	TCCTGAGTTCCAGAGCT	keke-d8-ekeke	soossssssssooss	1429	1445	983
673271	CTCCTGAGTTCCAGAGC	keke-d8-ekeke	soossssssssooss	1430	1446	984
673272	ACTCCTGAGTTCCAGAG	keke-d8-ekeke	soossssssssooss	1431	1447	985
673273	GACTCCTGAGTTCCAGA	keke-d8-ekeke	soossssssssooss	1432	1448	986
673274	CGACTCCTGAGTTCCAG	keke-d8-ekeke	soossssssssooss	1433	1449	987
673275	GCGACTCCTGAGTTCCA	keke-d8-ekeke	soossssssssooss	1434	1450	988
673276	CGCGACTCCTGAGTTCC	keke-d8-ekeke	soossssssssooss	1435	1451	989
673277	GCGCGACTCCTGAGTTC	keke-d8-ekeke	soossssssssooss	1436	1452	990
673278	CGCGCGACTCCTGAGTT	keke-d8-ekeke	soossssssssooss	1437	1453	991
673279	GCGCGCGACTCCTGAGT	keke-d8-ekeke	soossssssssooss	1438	1454	992
673280	AGCGCGCGACTCCTGAG	keke-d8-ekeke	soossssssssooss	1439	1455	993
673281	TAGCGCGCGACTCCTGA	keke-d8-ekeke	soossssssssooss	1440	1456	994
673282	CTAGCGCGCGACTCCTG	keke-d8-ekeke	soossssssssooss	1441	1457	995
673283	CCTAGCGCGCGACTCCT	keke-d8-ekeke	soossssssssooss	1442	1458	996
673284	CCCTAGCGCGCGACTCC	keke-d8-ekeke	soossssssssooss	1443	1459	997
673285	CCCCTAGCGCGCGACTC	keke-d8-ekeke	soossssssssooss	1444	1460	998
673286	GCCCCTAGCGCGCGACT	keke-d8-ekeke	soossssssssooss	1445	1461	999
673287	GGCCCCTAGCGCGCGAC	keke-d8-ekeke	soossssssssooss	1446	1462	1000
673288	CGGCCCCTAGCGCGCGA	keke-d8-ekeke	soossssssssooss	1447	1463	1001
673289	CCGGCCCCTAGCGCGCG	keke-d8-ekeke	soossssssssooss	1448	1464	1002
673290	CCCGGCCCCTAGCGCGC	keke-d8-ekeke	soossssssssooss	1449	1465	1003
673291	CCCCGGCCCCTAGCGCG	keke-d8-ekeke	soossssssssooss	1450	1466	1004
673292	GCCCCGGCCCCTAGCGC	keke-d8-ekeke	soossssssssooss	1451	1467	1005
673293	GGCCCCGGCCCCTAGCG	keke-d8-ekeke	soossssssssooss	1452	1468	1006
673294	CGGCCCCGGCCCCTAGC	keke-d8-ekeke	soossssssssooss	1453	1469	1007
673295	CCGGCCCCGGCCCCTAG	keke-d8-ekeke	soossssssssooss	1454	1470	1008
673296	CCCGGCCCCGGCCCCTA	keke-d8-ekeke	soossssssssooss	1455	1471	1009
673297	ACGCCCCGGCCCCGGCC	keke-d8-ekeke	soossssssssooss	1465	1481	1010
673298	CACGCCCCGGCCCCGGC	keke-d8-ekeke	soossssssssooss	1466	1482	1011
673299	CCACGCCCCGGCCCCGG	keke-d8-ekeke	soossssssssooss	1467	1483	1012
673300	ACCACGCCCCGGCCCCG	keke-d8-ekeke	soossssssssooss	1468	1484	1013
673301	GACCACGCCCCGGCCCC	keke-d8-ekeke	soossssssssooss	1469	1485	1014
673302	CGACCACGCCCCGGCCC	keke-d8-ekeke	soossssssssooss	1470	1486	1015
673303	CCGACCACGCCCCGGCC	keke-d8-ekeke	soossssssssooss	1471	1487	1016

TABLE 27-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673304	CCCGACCACGCCCCGGC	keke-d8-ekeke	soossssssssooss	1472	1488	1017
673305	CCCCGACCACGCCCCGG	keke-d8-ekeke	soossssssssooss	1473	1489	1018
673306	GCCCCGACCACGCCCCG	keke-d8-ekeke	soossssssssooss	1474	1490	1019
673307	CGCCCCGACCACGCCCC	keke-d8-ekeke	soossssssssooss	1475	1491	1020
673308	CCGCCCCGACCACGCCC	keke-d8-ekeke	soossssssssooss	1476	1492	1021
673309	CCCGCCCCGACCACGCC	keke-d8-ekeke	soossssssssooss	1477	1493	1022
673310	GCCCGCCCCGACCACGC	keke-d8-ekeke	soossssssssooss	1478	1494	1023
673311	GGCCCGCCCCGACCACG	keke-d8-ekeke	soossssssssooss	1479	1495	1024
673312	GGGCCCGCCCCGACCAC	keke-d8-ekeke	soossssssssooss	1480	1496	1025
673313	CGGGCCCGCCCCGACCA	keke-d8-ekeke	soossssssssooss	1481	1497	1026
673314	CCGGGCCCGCCCCGACC	keke-d8-ekeke	soossssssssooss	1482	1498	1027
673315	CCCGGGCCCGCCCCGAC	keke-d8-ekeke	soossssssssooss	1483	1499	1028
673316	GCAGCCCCGCCCCGGGC	keke-d8-ekeke	soossssssssooss	1505	1521	1029
673317	CGCAGCCCCGCCCCGGG	keke-d8-ekeke	soossssssssooss	1506	1522	1030
673318	CCGCAGCCCCGCCCCGG	keke-d8-ekeke	soossssssssooss	1507	1523	1031
673319	ACCGCAGCCCCGCCCCG	keke-d8-ekeke	soossssssssooss	1508	1524	1032
673320	AACCGCAGCCCCGCCCC	keke-d8-ekeke	soossssssssooss	1509	1525	1033
673321	CAACCGCAGCCCCGCCC	keke-d8-ekeke	soossssssssooss	1510	1526	1034
673322	GCAACCGCAGCCCCGCC	keke-d8-ekeke	soossssssssooss	1511	1527	1035
673323	CGCAACCGCAGCCCCGC	keke-d8-ekeke	soossssssssooss	1512	1528	1036
673324	CCGCAACCGCAGCCCCG	keke-d8-ekeke	soossssssssooss	1513	1529	1037
673325	ACCGCAACCGCAGCCCC	keke-d8-ekeke	soossssssssooss	1514	1530	1038
673326	CACCGCAACCGCAGCCC	keke-d8-ekeke	soossssssssooss	1515	1531	1039
673327	GCACCGCAACCGCAGCC	keke-d8-ekeke	soossssssssooss	1516	1532	1040
673328	GGCACCGCAACCGCAGC	keke-d8-ekeke	soossssssssooss	1517	1533	1041
673329	AGGCACCGCAACCGCAG	keke-d8-ekeke	soossssssssooss	1518	1534	1042
673330	CAGGCACCGCAACCGCA	keke-d8-ekeke	soossssssssooss	1519	1535	1043
673331	GCAGGCACCGCAACCGC	keke-d8-ekeke	soossssssssooss	1520	1536	1044
673332	CGCAGGCACCGCAACCG	keke-d8-ekeke	soossssssssooss	1521	1537	1045
673333	GCGCAGGCACCGCAACC	keke-d8-ekeke	soossssssssooss	1522	1538	1046
673334	GGCGCAGGCACCGCAAC	keke-d8-ekeke	soossssssssooss	1523	1539	1047

TABLE 28

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673335	TGAGAGCAAGTAGTGGG	ekek-d8-kekee	soossssssssooss	1326	1342	898
673336	GTGAGAGCAAGTAGTGG	ekek-d8-kekee	soossssssssooss	1327	1343	899
673337	TGTGAGAGCAAGTAGTG	ekek-d8-kekee	soossssssssooss	1328	1344	900
673338	CTGTGAGAGCAAGTAGT	ckek-d8-kekee	soossssssssooss	1329	1345	901
673339	ACTGTGAGAGCAAGTAG	ekek-d8-kekee	soossssssssooss	1330	1346	902
673340	TACTGTGAGAGCAAGTA	ekek-d8-kekee	soossssssssooss	1331	1347	903
673341	GTACTGTGAGAGCAAGT	ekek-d8-kekee	soossssssssooss	1332	1348	904
673342	AGTACTGTGAGAGCAAG	ekek-d8-kekee	soossssssssooss	1333	1349	905
673343	GAGTACTGTGAGAGCAA	ckek-d8-kekee	soossssssssooss	1334	1350	906
673344	CGAGTACTGTGAGAGCA	ekek-d8-kekee	soossssssssooss	1335	1351	907
673345	GCGAGTACTGTGAGAGC	ekek-d8-kekee	soossssssssooss	1336	1352	908
673346	AGCGAGTACTGTGAGAG	ekek-d8-kekee	soossssssssooss	1337	1353	909
673347	CAGCGAGTACTGTGAGA	ekek-d8-kekee	soossssssssooss	1338	1354	910
673348	TCAGCGAGTACTGTGAG	ekek-d8-kekee	soossssssssooss	1339	1355	911
673349	CTCAGCGAGTACTGTGA	ekek-d8-kekee	soossssssssooss	1340	1356	912
673350	CCTCAGCGAGTACTGTG	ekek-d8-kekee	soossssssssooss	1341	1357	913
673351	CCCTCAGCGAGTACTGT	ekek-d8-kekee	soossssssssooss	1342	1358	914
673352	ACCCTCAGCGAGTACTG	ekek-d8-kekee	soossssssssooss	1343	1359	915
673353	CACCCTCAGCGAGTACT	ekek-d8-kekee	soossssssssooss	1344	1360	916
673354	TCACCCTCAGCGAGTAC	ekek-d8-kekee	soossssssssooss	1345	1361	917
673355	TTCACCCTCAGCGAGTA	ekek-d8-kekee	soossssssssooss	1346	1362	918
673356	GTTCACCCTCAGCGAGT	ekek-d8-kekee	soossssssssooss	1347	1363	919
673357	TGTTCACCCTCAGCGAG	ekek-d8-kekee	soossssssssooss	1348	1364	920
673358	TTGTTCACCCTCAGCGA	ekek-d8-kekee	soossssssssooss	1349	1365	921
673359	CTTGTTCACCCTCAGCG	ekek-d8-kekee	soossssssssooss	1350	1366	922
673360	TCTTGTTCACCCTCAGC	ekek-d8-kekee	soossssssssooss	1351	1367	923
673361	TTCTTGTTCACCCTCAG	ekek-d8-kekee	soossssssssooss	1352	1368	924
673362	TTTCTTGTTCACCCTCA	ekek-d8-kekee	soossssssssooss	1353	1369	925
673363	TTTTCTTGTTCACCCTC	ekek-d8-kekee	soossssssssooss	1354	1370	926
673364	CTTTTCTTGTTCACCCT	ekek-d8-kekee	soossssssssooss	1355	1371	927
673365	TCTTTTCTTGTTCACCC	ekek-d8-kekee	soossssssssooss	1356	1372	928
673366	GTCTTTTCTTGTTCACC	ekek-d8-kekee	soossssssssooss	1357	1373	929
673367	GGTCTTTTCTTGTTCAC	ekek-d8-kekee	soossssssssooss	1358	1374	930
673368	AGGTCTTTTCTTGTTCA	ekek-d8-kekee	soossssssssooss	1359	1375	931
673369	CAGGTCTTTTCTTGTTC	ekek-d8-kekee	soossssssssooss	1360	1376	932

TABLE 28-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673370	TCAGGTCTTTTCTTGTT	ekek-d8-kekee	soosssssssssooss	1361	1377	933
673371	ATCAGGTCTTTTCTTGT	ekek-d8-kekee	soosssssssssooss	1362	1378	934
673372	TATCAGGTCTTTTCTTG	ekek-d8-kekee	soosssssssssooss	1363	1379	935
673373	TTATCAGGTCTTTTCTT	ekek-d8-kekee	soosssssssssooss	1364	1380	936
673374	ATCTTTATCAGGTCTTT	ekek-d8-kekee	soosssssssssooss	1368	1384	937
673375	AATCTTTATCAGGTCTT	ekek-d8-kekee	soosssssssssooss	1369	1385	938
673376	TAATCTTTATCAGGTCT	ekek-d8-kekee	soosssssssssooss	1370	1386	939
673377	TTAATCTTTATCAGGTC	ekek-d8-kekee	soosssssssssooss	1371	1387	940
673378	GTTAATCTTTATCAGGT	ekek-d8-kekee	soosssssssssooss	1372	1388	941
673379	GGTTAATCTTTATCAGG	ekek-d8-kekee	soosssssssssooss	1373	1389	942
673380	TGGTTAATCTTTATCAG	ekek-d8-kekee	soosssssssssooss	1374	1390	943
673381	CTGGTTAATCTTTATCA	ekek-d8-kekee	soosssssssssooss	1375	1391	944
673382	TCTGGTTAATCTTTATC	ekek-d8-kekee	soosssssssssooss	1376	1392	945
673383	CCCTCCTTGTTTTCTTC	ekek-d8-kekee	soosssssssssooss	1391	1407	946
673384	TCCCTCCTTGTTTTCTT	ekek-d8-kekee	soosssssssssooss	1392	1408	947
673385	TTCCCTCCTTGTTTTCT	ekek-d8-kekee	soosssssssssooss	1393	1409	948
673386	TTTCCCTCCTTGTTTTC	ekek-d8-kekee	soosssssssssooss	1394	1410	949
673387	GTTTCCCTCCTTGTTTT	ekek-d8-kekee	soosssssssssooss	1395	1411	950
673388	TGTTTCCCTCCTTGTTT	ekek-d8-kekee	soosssssssssooss	1396	1412	951
673389	TTGTTTCCCTCCTTGTT	ekek-d8-kekee	soosssssssssooss	1397	1413	952
673390	GGTTGTTTCCCTCCTTG	ekek-d8-kekee	soosssssssssooss	1399	1415	953
673391	CGGTTGTTTCCCTCCTT	ekek-d8-kekee	soosssssssssooss	1400	1416	954
673392	GCGGTTGTTTCCCTCCT	ekek-d8-kekee	soosssssssssooss	1401	1417	955
673393	TGCGGTTGTTTCCCTCC	ekek-d8-kekee	soosssssssssooss	1402	1418	956
673394	CTGCGGTTGTTTCCCTC	ekek-d8-kekee	soosssssssssooss	1403	1419	957
673395	GCTGCGGTTGTTTCCCT	ekek-d8-kekee	soosssssssssooss	1404	1420	958
673396	GGCTGCGGTTGTTTCCC	ekek-d8-kekee	soosssssssssooss	1405	1421	959
673397	AGGCTGCGGTTGTTTCC	ekek-d8-kekee	soosssssssssooss	1406	1422	960
673398	CAGGCTGCGGTTGTTTC	ekek-d8-kekee	soosssssssssooss	1407	1423	961
673399	ACAGGCTGCGGTTGTTT	ekek-d8-kekee	soosssssssssooss	1408	1424	962
673400	TACAGGCTGCGGTTGTT	ekek-d8-kekee	soosssssssssooss	1409	1425	963
673401	CTACAGGCTGCGGTTGT	ekek-d8-kekee	soosssssssssooss	1410	1426	964
673402	GCTACAGGCTGCGGTTG	ekek-d8-kekee	soosssssssssooss	1411	1427	965
673403	TGCTACAGGCTGCGGTT	ekek-d8-kekee	soosssssssssooss	1412	1428	966

TABLE 28-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673404	TTGCTACAGGCTGCGGT	ekek-d8-kekee	soossssssssooss	1413	1429	967
673405	CTTGCTACAGGCTGCGG	ekek-d8-kekee	soossssssssooss	1414	1430	968
673406	GCTTGCTACAGGCTGCG	ekek-d8-kekee	soossssssssooss	1415	1431	969
673407	AGCTTGCTACAGGCTGC	ekek-d8-kekee	soossssssssooss	1416	1432	970
673408	GAGCTTGCTACAGGCTG	ekek-d8-kekee	soossssssssooss	1417	1433	971
673409	AGAGCTTGCTACAGGCT	ekek-d8-kekee	soossssssssooss	1418	1434	972
673410	CAGAGCTTGCTACAGGC	ekek-d8-kekee	soossssssssooss	1419	1435	973
673411	CCAGAGCTTGCTACAGG	ekek-d8-kekee	soossssssssooss	1420	1436	974
673412	TCCAGAGCTTGCTACAG	ekek-d8-kekee	soossssssssooss	1421	1437	975
673413	TTCCAGAGCTTGCTACA	ekek-d8-kekee	soossssssssooss	1422	1438	976
673414	GTTCCAGAGCTTGCTAC	ekek-d8-kekee	soossssssssooss	1423	1439	977
673415	AGTTCCAGAGCTTGCTA	ekek-d8-kekee	soossssssssooss	1424	1440	978
673416	GAGTTCCAGAGCTTGCT	ekek-d8-kekee	soossssssssooss	1425	1441	979
673417	TGAGTTCCAGAGCTTGC	ekek-d8-kekee	soossssssssooss	1426	1442	980
673418	CTGAGTTCCAGAGCTTG	ekek-d8-kekee	soossssssssooss	1427	1443	981
673419	CCTGAGTTCCAGAGCTT	ekek-d8-kekee	soossssssssooss	1428	1444	982
673420	TCCTGAGTTCCAGAGCT	ekek-d8-kekee	soossssssssooss	1429	1445	983
673421	CTCCTGAGTTCCAGAGC	ekek-d8-kekee	soossssssssooss	1430	1446	984
673422	ACTCCTGAGTTCCAGAG	ekek-d8-kekee	soossssssssooss	1431	1447	985
673423	GACTCCTGAGTTCCAGA	ekek-d8-kekee	soossssssssooss	1432	1448	986
673424	CGACTCCTGAGTTCCAG	ekek-d8-kekee	soossssssssooss	1433	1449	987
673425	GCGACTCCTGAGTTCCA	ekek-d8-kekee	soossssssssooss	1434	1450	988
673426	CGCGACTCCTGAGTTCC	ekek-d8-kekee	soossssssssooss	1435	1451	989
673427	GCGCGACTCCTGAGTTC	ekek-d8-kekee	soossssssssooss	1436	1452	990
673428	CGCGCGACTCCTGAGTT	ekek-d8-kekee	soossssssssooss	1437	1453	991
673429	GCGCGCGACTCCTGAGT	ekek-d8-kekee	soossssssssooss	1438	1454	992
673430	AGCGCGCGACTCCTGAG	ekek-d8-kekee	soossssssssooss	1439	1455	993
673431	TAGCGCGCGACTCCTGA	ekek-d8-kekee	soossssssssooss	1440	1456	994
673432	CTAGCGCGCGACTCCTG	ekek-d8-kekee	soossssssssooss	1441	1457	995
673433	CCTAGCGCGCGACTCCT	ekek-d8-kekee	soossssssssooss	1442	1458	996
673434	CCCTAGCGCGCGACTCC	ekek-d8-kekee	soossssssssooss	1443	1459	997
673435	CCCCTAGCGCGCGACTC	ekek-d8-kekee	soossssssssooss	1444	1460	998
673436	GCCCCTAGCGCGCGACT	ekek-d8-kekee	soossssssssooss	1445	1461	999
673437	GGCCCCTAGCGCGCGAC	ekek-d8-kekee	soossssssssooss	1446	1462	1000
673438	CGGCCCCTAGCGCGCGA	ekek-d8-kekee	soossssssssooss	1447	1463	1001

TABLE 28-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673439	CCGGCCCCTAGCGCGCG	ekek-d8-kekee	soosssssssssooss	1448	1464	1002
673440	CCCGGCCCCTAGCGCGC	ekek-d8-kekee	soosssssssssooss	1449	1465	1003
673441	CCCCGGCCCCTAGCGCG	ekek-d8-kekee	soosssssssssooss	1450	1466	1004
673442	GCCCCGGCCCCTAGCGC	ekek-d8-kekee	soosssssssssooss	1451	1467	1005
673443	GGCCCCGGCCCCTAGCG	ekek-d8-kekee	soosssssssssooss	1452	1468	1006
673444	CGGCCCCGGCCCCTAGC	ekek-d8-kekee	soosssssssssooss	1453	1469	1007
673445	CCGGCCCCGGCCCCTAG	ekek-d8-kekee	soosssssssssooss	1454	1470	1008
673446	CCCGGCCCCGGCCCCTA	ekek-d8-kekee	soosssssssssooss	1455	1471	1009
673447	ACGCCCCGGCCCCGGCC	ekek-d8-kekee	soosssssssssooss	1465	1481	1010
673448	CACGCCCCGGCCCCGGC	ekek-d8-kekee	soosssssssssooss	1466	1482	1011
673449	CCACGCCCCGGCCCCGG	ekek-d8-kekee	soosssssssssooss	1467	1483	1012
673450	ACCACGCCCCGGCCCCG	ekek-d8-kekee	soosssssssssooss	1468	1484	1013
673451	GACCACGCCCCGGCCCC	ekek-d8-kekee	soosssssssssooss	1469	1485	1014
673452	CGACCACGCCCCGGCCC	ekek-d8-kekee	soosssssssssooss	1470	1486	1015
673453	CCGACCACGCCCCGGCC	ekek-d8-kekee	soosssssssssooss	1471	1487	1016
673454	CCCGACCACGCCCCGGC	ekek-d8-kekee	soosssssssssooss	1472	1488	1017
673455	CCCCGACCACGCCCCGG	ekek-d8-kekee	soosssssssssooss	1473	1489	1018
673456	GCCCCGACCACGCCCCG	ekek-d8-kekee	soosssssssssooss	1474	1490	1019
673457	CGCCCCGACCACGCCCC	ekek-d8-kekee	soosssssssssooss	1475	1491	1020
673458	CCGCCCCGACCACGCCC	ekek-d8-kekee	soosssssssssooss	1476	1492	1021
673459	CCCGCCCCGACCACGCC	ekek-d8-kekee	soosssssssssooss	1477	1493	1022
673460	GCCCGCCCCGACCACGC	ekek-d8-kekee	soosssssssssooss	1478	1494	1023
673461	GGCCCGCCCCGACCACG	ekek-d8-kekee	soosssssssssooss	1479	1495	1024
673462	GGGCCCGCCCCGACCAC	ekek-d8-kekee	soosssssssssooss	1480	1496	1025
673463	CGGGCCCGCCCCGACCA	ekek-d8-kekee	soosssssssssooss	1481	1497	1026
673464	CCGGGCCCGCCCCGACC	ekek-d8-kekee	soosssssssssooss	1482	1498	1027
673465	CCCGGGCCCGCCCCGAC	ekek-d8-kekee	soosssssssssooss	1483	1499	1028
673466	GCAGCCCCGCCCCGGGC	ekek-d8-kekee	soosssssssssooss	1505	1521	1029
673467	CGCAGCCCCGCCCCGGG	ekek-d8-kekee	soosssssssssooss	1506	1522	1030
673468	CCGCAGCCCCGCCCCGG	ekek-d8-kekee	soosssssssssooss	1507	1523	1031
673469	ACCGCAGCCCCGCCCCG	ekek-d8-kekee	soosssssssssooss	1508	1524	1032
673470	AACCGCAGCCCCGCCCC	ekek-d8-kekee	soosssssssssooss	1509	1525	1033
673471	CAACCGCAGCCCCGCCC	ekek-d8-kekee	soosssssssssooss	1510	1526	1034
673472	GCAACCGCAGCCCCGCC	ekek-d8-kekee	soosssssssssooss	1511	1527	1035

TABLE 28-continued

Deoxy, MOE and cEt oligonucleotides targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	SEQ ID NO
673473	CGCAACCGCAGCCCCGC	ekek-d8-kekee	soosssssssssooss	1512	1528	1036
673474	CCGCAACCGCAGCCCCG	ekek-d8-kekee	soosssssssssooss	1513	1529	1037
673475	ACCGCAACCGCAGCCCC	ekek-d8-kekee	soosssssssssooss	1514	1530	1038
673476	CACCGCAACCGCAGCCC	ekek-d8-kekee	soosssssssssooss	1515	1531	1039
673477	GCACCGCAACCGCAGCC	ekek-d8-kekee	soosssssssssooss	1516	1532	1040
673478	GGCACCGCAACCGCAGC	ekek-d8-kekee	soosssssssssooss	1517	1533	1041
673479	AGGCACCGCAACCGCAG	ekek-d8-kekee	soosssssssssooss	1518	1534	1042
673480	CAGGCACCGCAACCGCA	ekek-d8-kekee	soosssssssssooss	1519	1535	1043
673481	GCAGGCACCGCAACCGC	ekek-d8-kekee	soosssssssssooss	1520	1536	1044
673482	CGCAGGCACCGCAACCG	ekek-d8-kekee	soosssssssssooss	1521	1537	1045
673483	GCGCAGGCACCGCAACC	ekek-d8-kekee	soosssssssssooss	1522	1538	1046
673484	GGCGCAGGCACCGCAAC	ekek-d8-kekee	soosssssssssooss	1523	1539	1047

TABLE 29

5-10-5 MOE gapmers targeting SEQ ID NO: 2

ISIS NO	Start Sequence	Stop Linkage	SEQ ID NO: 2 Site	SEQ ID NO: 2 Site	SEQ ID NO
653222	CCACTCGCCACCGCCTGCGC	soooossssssssssooss	1553	1572	1048
653223	TGCATTCCTAAGCAATGTGT	soooossssssssssooss	5325	5344	1049
655016	CCCGGCCCCGGCCCCGGCCC	soooossssssssssooss	1458	1477	1050
655017	CCCCGGCCCCGGCCCCGGCC	soooossssssssssooss	1459	1478	1051
671081	TTACATCTATAGCACCACTC	soooossssssssssooss	8110	8129	1052
671082	TCACTCCCTTTTCAGACAAG	soooossssssssssooss	8140	8159	1053
671083	AACTAAGTTCTGTCTGTGGA	soooossssssssssooss	8230	8249	1054
671084	ATACAGGACTAAAGTGCTTC	soooossssssssssooss	14316	14335	1055

TABLE 30

5-10-5 MOE gapmers targeting SEQ ID NO: 1

ISIS NO	Sequence	Linkage	SEQ ID NO: 1 Start Site	SEQ ID NO: 1 Stop Site	SEQ ID NO
672561	CTCTGACCCTGATCTTCCAT	soooossssssssssooss	695	714	1056

Example 7

In Vivo Rodent Inhibition and Tolerability with Treatment of C9ORF72 Antisense Oligonucleotides

[0458] In order to assess the tolerability of inhibition of C9ORF72 expression in vivo, antisense oligonucleotides targeting a murine C9ORF72 nucleic acid were designed and assessed in mouse and rat models.

[0459] ISIS 571883 was designed as a 5-10-5 MOE gapmer, 20 nucleosides in length, wherein the central gap segment comprises ten 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and on the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment has a MOE modification. The internucleoside linkages are phosphorothioate linkages. All cytosine residues throughout the gapmer are 5-methylcytosines. ISIS 571883 has a target start site of nucleoside 33704 on the murine C9ORF72 genomic sequence, designated herein as SEQ ID NO: 11 (the complement of GENBANK Accession No. NT_166289.1 truncated from nucleosides 3587000 to 3625000).

[0460] ISIS 603538 was designed as a 5-10-5 MOE gapmer, 20 nucleosides in length, wherein the central gap segment comprises ten 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and on the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment has a MOE modification. The internucleoside linkages are either phosphorothioate linkages or phosphate ester linkages (Gs Ao Co Co Gs Cs Ts Ts Gs As Gs Ts Ts Ts Gs Co Co Ao Cs A; wherein 's' denotes a phosphorothioate internucleoside linkage, 'o' denotes a phosphate ester linkage; and A, G, C, T denote the relevant nucleosides). All cytosine residues throughout the gapmer are 5-methylcytosines. ISIS 603538 has a target start site of nucleoside 2872 on the rat C9ORF72 mRNA sequence, designated herein as SEQ ID NO: 12 (GENBANK Accession No. NM_001007702.1).

Mouse Experiment 1

[0461] Groups of 4 C57BL/6 mice each were injected with 50 μg, 100 μg, 300 μg, 500 μg, or 700 μg of ISIS 571883 administered via an intracerebroventricular bolus injection. A control group of four C57/BL6 mice were similarly treated with PBS. Animals were anesthetized with 3% isofluorane and placed in a stereotactic frame. After sterilizing the surgical site, each mouse was injected −0.2 mm anterio-posterior from the bregma na d 3 mm dorsoventral to the bregma with the above-mentioned doses of ISIS 571883 using a Hamilton syringe. The incision was closed with sutures. The mice were allowed to recover for 14 days, after which animals were euthanized according to a humane protocol approved by the Institutional Animal Care and Use Committee. Brain and spinal cord tissue were harvested and snap frozen in liquid nitrogen. Prior to freezing, brain tissue was cut transversely five sections using a mouse brain matrix.

RNA Analysis

[0462] RNA was extracted from a 2-3 mm brain section posterior to the injection site, from brain frontal cortex and from the lumbar section of the spinal cord tissue for analysis of C9ORF72 mRNA expression. C9ORF72 mRNA expression was measured by RT-PCR. The data is presented in Table 31. The results indicate that treatment with increasing doses of ISIS 571883 resulted in dose-dependent inhibition of C9ORF72 mRNA expression.

[0463] The induction of the microglial marker AIF-1 as a measure of CNS toxicity was also assessed. The data is presented in Table 32. The results indicate that treatment with increasing doses of ISIS 571883 did not result in significant increases in AIF-1 mRNA expression. Hence, the injection of ISIS 571883 was deemed tolerable in this model.

TABLE 31

Percentage inhibition of C9ORF72 mRNA expression compared to the PBS control

Dose (μg)	Posterior brain	Cortex	Spinal cord
50	22	8	46
100	22	12	47
300	55	47	67
500	61	56	78
700	65	65	79

TABLE 32

Percentage expression of AIF-1 mRNA expression compared to the PBS control

Dose (μg)	Posterior brain	Spinal cord
50	102	89
100	105	111
300	107	98
500	131	124
700	122	116

Mouse Experiment 2

[0464] Groups of 4 C57BL/6 mice each were injected with 500 μg of ISIS 571883 administered via an intracerebroventricular bolus injection in a procedure similar to that described above. A control group of four C57/BL6 mice were similarly treated with PBS. The mice were tested at regular time points after ICV administration.

Behavior Analysis

[0465] Two standard assays to assess motor behavior were employed; the rotarod assay and grip strength assay. In case of the rotarod assays, the time of latency to fall was measured. The data for the assays is presented in Tables 33 and 34. The results indicate that there were no significant changes in the motor behavior of the mice as a result of antisense inhibition of ISIS 571883 or due to the ICV injection. Hence, antisense inhibition of C9ORF72 was deemed tolerable in this model.

TABLE 33

Latency to fall (sec) in the rotarod assay

Weeks after injection	PBS	ISIS 571883
0	66	66
4	91	70
8	94	84

TABLE 34

Mean hindlimb grip strength (g) in the grip strength assay

Weeks after injection	PBS	ISIS 571883
0	57	63
1	65	51
2	51	52
3	51	51
4	59	72
5	60	64
6	61	72
7	67	68
8	66	70
9	63	61
10	48	46

Rat Experiment

[0466] Groups of 4 Sprague-Dawley rats each were injected with 700 μg, 1,000 μg, or 3,000 μg of ISIS 603538 administered via an intrathecal bolus injection. A control group of four Sprague-Dawley rats were similarly treated with PBS Animals were anesthetized with 3% isofluorane and placed in a stereotactic frame. After sterilizing the surgical site, each rat was injected with 30 μL of ASO solution administered via 8 cm intrathecal catheter 2 cm into the spinal canal with a 50 μL flush. The rats were allowed to recover for 4 weeks, after which animals were euthanized according to a humane protocol approved by the Institutional Animal Care and Use Committee.

RNA Analysis

[0467] RNA was extracted from a 2-3 mm brain section posterior to the injection site, from brain frontal cortex, and from the cervical and lumbar sections of the spinal cord tissue for analysis of C9ORF72 mRNA expression. C9ORF72 mRNA expression was measured by RT-PCR. The data is presented in Table 35. The results indicate that treatment with increasing doses of ISIS 603538 resulted in dose-dependent inhibition of C9ORF72 mRNA expression.

[0468] The induction of the microglial marker AIF-1 as a measure of CNS toxicity was also assessed. The data is presented in Table 36. The results indicate that treatment with increasing doses of ISIS 603538 did not result in significant increases in AIF-1 mRNA expression. Hence, the injection of ISIS 603538 was deemed tolerable in this model.

TABLE 35

Percentage inhibition of C9ORF72 mRNA expression compared to the PBS control

Dose (μg)	Brain (1 mm section)	Cortex	Spinal cord (lumbar)	Spinal cord (cervical)
700	21	4	86	74
1000	53	49	88	82
3000	64	62	88	80

TABLE 36

Percentage expression of AIF-1 mRNA expression compared to the PBS control

Dose (μg)	Brain (1 mm section)	Cortex	Spinal cord (lumbar)	Spinal cord (cervical)
700	97	119	98	89
1000	105	113	122	96
3000	109	141	156	115

Body Weight Analysis

[0469] Body weights of the rats were measured at regular time point intervals. The data is presented in Table 37. The results indicate that treatment with increasing doses of ISIS 603538 did not have any significant changes in the body weights of the rats.

TABLE 37

Body weights of the rats (% initial body weight)

	Dose (μg)	Week 1	Week 2	Week 3	Week 4	Week 5
PBS		100	94	103	105	109
ISIS	700	100	94	98	103	107
603538	1000	100	95	97	101	103
	3000	100	92	98	102	105

Example 8

Antisense Inhibition of C9ORF72 by 5-8-5 MOE Gapmers with Mixed Backbones and Deoxy, MOE and cEt Antisense Oligonucleotides with Mixed Backbones

[0470] Antisense oligonucleotides described in Example 6 hereinabove (see Table 23 and Table 24 hereinabove) were tested in HepG2 cells in a series of experiments that had similar culture conditions. ISIS 576816, previously tested in PCT/US2013/065073 (claiming priority to U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide for study with deoxy, MOE, and cEt antisense oligonucleotides. The results for each experiment are presented in tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant, which is the product of a pre-mRNA containing a hexanucleotide repeat. The levels of the C9ORF72 pathogenic associated mRNA variant were normalized to the total RNA content of the cell, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72, relative to untreated control cells.

TABLE 38

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by 5-8-5 MOE gapmers with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672581	GTGAGAGCAAGTAGTGGG	sooossssssssooss	1326	1343	11	744
672582	TGTGAGAGCAAGTAGTGG	sooossssssssooss	1327	1344	32	745
672583	CTGTGAGAGCAAGTAGTG	sooossssssssooss	1328	1345	14	746
672584	ACTGTGAGAGCAAGTAGT	sooossssssssooss	1329	1346	1	747
672585	TACTGTGAGAGCAAGTAG	sooossssssssooss	1330	1347	14	748
672586	GTACTGTGAGAGCAAGTA	sooossssssssooss	1331	1348	22	749
672587	AGTACTGTGAGAGCAAGT	sooossssssssooss	1332	1349	0	750
672588	GAGTACTGTGAGAGCAAG	sooossssssssooss	1333	1350	8	751
672589	CGAGTACTGTGAGAGCAA	sooossssssssooss	1334	1351	15	752
672590	GCGAGTACTGTGAGAGCA	sooossssssssooss	1335	1352	13	753
672591	AGCGAGTACTGTGAGAGC	sooossssssssooss	1336	1353	32	754
672592	CAGCGAGTACTGTGAGAG	sooossssssssooss	1337	1354	39	755
672593	TCAGCGAGTACTGTGAGA	sooossssssssooss	1338	1355	15	756
672594	CTCAGCGAGTACTGTGAG	sooossssssssooss	1339	1356	14	757
672595	CCTCAGCGAGTACTGTGA	sooossssssssooss	1340	1357	40	758
672596	CCCTCAGCGAGTACTGTG	sooossssssssooss	1341	1358	28	759
672597	ACCCTCAGCGAGTACTGT	sooossssssssooss	1342	1359	30	760
672598	CACCCTCAGCGAGTACTG	sooossssssssooss	1343	1360	46	761
672599	TCACCCTCAGCGAGTACT	sooossssssssooss	1344	1361	40	762
672600	TTCACCCTCAGCGAGTAC	sooossssssssooss	1345	1362	25	763
672601	GTTCACCCTCAGCGAGTA	sooossssssssooss	1346	1363	15	764
672602	TGTTCACCCTCAGCGAGT	sooossssssssooss	1347	1364	35	765
672603	TTGTTCACCCTCAGCGAG	sooossssssssooss	1348	1365	22	766
672604	CTTGTTCACCCTCAGCGA	sooossssssssooss	1349	1366	6	767
672605	TCTTGTTCACCCTCAGCG	sooossssssssooss	1350	1367	26	768
672606	TTCTTGTTCACCCTCAGC	sooossssssssooss	1351	1368	11	769
672607	TTTCTTGTTCACCCTCAG	sooossssssssooss	1352	1369	9	770
672608	TTTTCTTGTTCACCCTCA	sooossssssssooss	1353	1370	36	771
672609	CTTTTCTTGTTCACCCTC	sooossssssssooss	1354	1371	25	772
672610	TCTTTTCTTGTTCACCCT	sooossssssssooss	1355	1372	16	773
672611	GTCTTTTCTTGTTCACCC	sooossssssssooss	1356	1373	45	774
672612	GGTCTTTTCTTGTTCACC	sooossssssssooss	1357	1374	15	775
672613	AGGTCTTTTCTTGTTCAC	sooossssssssooss	1358	1375	10	776
672614	CAGGTCTTTTCTTGTTCA	sooossssssssooss	1359	1376	25	777
672615	TCAGGTCTTTTCTTGTTC	sooossssssssooss	1360	1377	21	778

TABLE 38-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by 5-8-5 MOE gapmers with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672616	ATCAGGTCTTTTCTTGTT	sooossssssssooss	1361	1378	15	779
672617	TATCAGGTCTTTTCTTGT	sooossssssssooss	1362	1379	27	780
672618	TTATCAGGTCTTTTCTTG	sooossssssssooss	1363	1380	14	781
672619	TTTATCAGGTCTTTTCTT	sooossssssssooss	1364	1381	46	782
672620	AATCTTTATCAGGTCTTT	sooossssssssooss	1368	1385	31	783
672621	TAATCTTTATCAGGTCTT	sooossssssssooss	1369	1386	12	784
672622	TTAATCTTTATCAGGTCT	sooossssssssooss	1370	1387	27	785
672623	GTTAATCTTTATCAGGTC	sooossssssssooss	1371	1388	9	786
672624	GGTTAATCTTTATCAGGT	sooossssssssooss	1372	1389	53	787
672625	TGGTTAATCTTTATCAGG	sooossssssssooss	1373	1390	17	788
672626	CTGGTTAATCTTTATCAG	sooossssssssooss	1374	1391	11	789
672627	TCTGGTTAATCTTTATCA	sooossssssssooss	1375	1392	16	790
672628	TTCTGGTTAATCTTTATC	sooossssssssooss	1376	1393	22	791
672629	TCCCTCCTTGTTTTCTTC	sooossssssssooss	1391	1408	0	792
672630	TTTCCCTCCTTGTTTTCT	sooossssssssooss	1393	1410	8	793
672631	GTTTCCCTCCTTGTTTTC	sooossssssssooss	1394	1411	0	794
672632	TGTTTCCCTCCTTGTTTT	sooossssssssooss	1395	1412	25	795
672633	TTGTTTCCCTCCTTGTTT	sooossssssssooss	1396	1413	0	796
672634	GTTGTTTCCCTCCTTGTT	sooossssssssooss	1397	1414	10	797
672635	GGTTGTTTCCCTCCTTGT	sooossssssssooss	1398	1415	15	798
672636	CGGTTGTTTCCCTCCTTG	sooossssssssooss	1399	1416	49	799
672637	GCGGTTGTTTCCCTCCTT	sooossssssssooss	1400	1417	49	800
672638	TGCGGTTGTTTCCCTCCT	sooossssssssooss	1401	1418	23	801
672639	CTGCGGTTGTTTCCCTCC	sooossssssssooss	1402	1419	21	802
672640	GCTGCGGTTGTTTCCCTC	sooossssssssooss	1403	1420	52	803
672641	GGCTGCGGTTGTTTCCCT	sooossssssssooss	1404	1421	23	804
672642	AGGCTGCGGTTGTTTCCC	sooossssssssooss	1405	1422	35	805
672643	CAGGCTGCGGTTGTTTCC	sooossssssssooss	1406	1423	22	806
672644	ACAGGCTGCGGTTGTTTC	sooossssssssooss	1407	1424	27	807
672645	TACAGGCTGCGGTTGTTT	sooossssssssooss	1408	1425	21	808
672646	CTACAGGCTGCGGTTGTT	sooossssssssooss	1409	1426	18	809
672647	GCTACAGGCTGCGGTTGT	sooossssssssooss	1410	1427	16	810
672648	TGCTACAGGCTGCGGTTG	sooossssssssooss	1411	1428	10	811
672649	TTGCTACAGGCTGCGGTT	sooossssssssooss	1412	1429	0	812

TABLE 38-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by 5-8-5 MOE gapmers with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672650	CTTGCTACAGGCTGCGGT	sooossssssssssooss	1413	1430	27	813
672651	GCTTGCTACAGGCTGCGG	sooossssssssssooss	1414	1431	53	814
672652	AGCTTGCTACAGGCTGCG	sooossssssssssooss	1415	1432	38	815
672653	GAGCTTGCTACAGGCTGC	sooossssssssssooss	1416	1433	7	816
672654	AGAGCTTGCTACAGGCTG	sooossssssssssooss	1417	1434	7	817
672655	CAGAGCTTGCTACAGGCT	sooossssssssssooss	1418	1435	15	818
672656	CCAGAGCTTGCTACAGGC	sooossssssssssooss	1419	1436	22	819
672657	TCCAGAGCTTGCTACAGG	sooossssssssssooss	1420	1437	43	820
672658	TTCCAGAGCTTGCTACAG	sooossssssssssooss	1421	1438	20	821
672659	GTTCCAGAGCTTGCTACA	sooossssssssssooss	1422	1439	12	822
672660	AGTTCCAGAGCTTGCTAC	sooossssssssssooss	1423	1440	11	823
672661	GAGTTCCAGAGCTTGCTA	sooossssssssssooss	1424	1441	39	824
672662	TGAGTTCCAGAGCTTGCT	sooossssssssssooss	1425	1442	18	825
672663	CTGAGTTCCAGAGCTTGC	sooossssssssssooss	1426	1443	26	826
672664	CCTGAGTTCCAGAGCTTG	sooossssssssssooss	1427	1444	69	827
672665	TCCTGAGTTCCAGAGCTT	sooossssssssssooss	1428	1445	56	828
672666	CTCCTGAGTTCCAGAGCT	sooossssssssssooss	1429	1446	28	829
672667	ACTCCTGAGTTCCAGAGC	sooossssssssssooss	1430	1447	51	830
672668	GACTCCTGAGTTCCAGAG	sooossssssssssooss	1431	1448	39	831
672669	CGACTCCTGAGTTCCAGA	sooossssssssssooss	1432	1449	32	832
672670	GCGACTCCTGAGTTCCAG	sooossssssssssooss	1433	1450	66	833
672671	CGCGACTCCTGAGTTCCA	sooossssssssssooss	1434	1451	67	834
672672	GCGCGACTCCTGAGTTCC	sooossssssssssooss	1435	1452	48	835
672673	CGCGCGACTCCTGAGTTC	sooossssssssssooss	1436	1453	38	836
672674	GCGCGCGACTCCTGAGTT	sooossssssssssooss	1437	1454	53	837
672675	AGCGCGCGACTCCTGAGT	sooossssssssssooss	1438	1455	58	838
672676	TAGCGCGCGACTCCTGAG	sooossssssssssooss	1439	1456	55	839
672677	CTAGCGCGCGACTCCTGA	sooossssssssssooss	1440	1457	47	840
672678	CCTAGCGCGCGACTCCTG	sooossssssssssooss	1441	1458	56	841
672679	CCCTAGCGCGCGACTCCT	sooossssssssssooss	1442	1459	74	842
672680	CCCCTAGCGCGCGACTCC	sooossssssssssooss	1443	1460	43	843
672681	GCCCCTAGCGCGCGACTC	sooossssssssssooss	1444	1461	53	844
672682	GGCCCCTAGCGCGCGACT	sooossssssssssooss	1445	1462	42	845
672683	CGGCCCCTAGCGCGCGAC	sooossssssssssooss	1446	1463	69	846
672684	CCGGCCCCTAGCGCGCGA	sooossssssssssooss	1447	1464	29	847

TABLE 38-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by 5-8-5 MOE gapmers with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672685	CCCGGCCCCTAGCGCGCG	sooooooooooooooss	1448	1465	21	848
672686	CCCCGGCCCCTAGCGCGC	sooooooooooooooss	1449	1466	35	849
672687	GCCCCGGCCCCTAGCGCG	sooooooooooooooss	1450	1467	41	850
672688	GGCCCCGGCCCCTAGCGC	sooooooooooooooss	1451	1468	46	851
672689	CGGCCCCGGCCCCTAGCG	sooooooooooooooss	1452	1469	28	852
672690	CCGGCCCCGGCCCCTAGC	sooooooooooooooss	1453	1470	33	853
672691	CCCGGCCCCGGCCCCTAG	sooooooooooooooss	1454	1471	10	854
672692	CCCCGGCCCCGGCCCCTA	sooooooooooooooss	1455	1472	35	855
672693	ACGCCCCGGCCCCGGCCC	sooooooooooooooss	1464	1481	57	856
672694	CACGCCCCGGCCCCGGCC	sooooooooooooooss	1465	1482	39	857
672695	CCACGCCCCGGCCCCGGC	sooooooooooooooss	1466	1483	48	858
672696	ACCACGCCCCGGCCCCGG	sooooooooooooooss	1467	1484	39	859
672697	GACCACGCCCCGGCCCCG	sooooooooooooooss	1468	1485	54	860
672698	CGACCACGCCCCGGCCCC	sooooooooooooooss	1469	1486	48	861
672699	CCGACCACGCCCCGGCCC	sooooooooooooooss	1470	1487	52	862
672700	CCCGACCACGCCCCGGCC	sooooooooooooooss	1471	1488	67	863
672701	CCCCGACCACGCCCCGGC	sooooooooooooooss	1472	1489	42	864
672702	GCCCCGACCACGCCCCGG	sooooooooooooooss	1473	1490	11	865
672703	CGCCCCGACCACGCCCCG	sooooooooooooooss	1474	1491	23	866
672704	CCGCCCCGACCACGCCCC	sooooooooooooooss	1475	1492	50	867
672705	CCCGCCCCGACCACGCCC	sooooooooooooooss	1476	1493	23	868
672706	GCCCGCCCCGACCACGCC	sooooooooooooooss	1477	1494	24	869
672707	GGCCCGCCCCGACCACGC	sooooooooooooooss	1478	1495	44	870
672708	GGGCCCGCCCCGACCACG	sooooooooooooooss	1479	1496	29	871
672709	CGGGCCCGCCCCGACCAC	sooooooooooooooss	1480	1497	7	872
672710	CCGGGCCCGCCCCGACCA	sooooooooooooooss	1481	1498	30	873
672711	CCCGGGCCCGCCCCGACC	sooooooooooooooss	1482	1499	16	874
672712	CCCCGGGCCCGCCCCGAC	sooooooooooooooss	1483	1500	14	875
672713	AGCCCCGCCCCGGGCCCG	sooooooooooooooss	1502	1519	32	876
672714	CAGCCCCGCCCCGGGCCC	sooooooooooooooss	1503	1520	22	877
672715	GCAGCCCCGCCCCGGGCC	sooooooooooooooss	1504	1521	1	878
672716	CGCAGCCCCGCCCCGGGC	sooooooooooooooss	1505	1522	29	879
672717	CCGCAGCCCCGCCCCGGG	sooooooooooooooss	1506	1523	51	880
672718	ACCGCAGCCCCGCCCCGG	sooooooooooooooss	1507	1524	45	881

TABLE 38-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by 5-8-5 MOE gapmers with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672719	AACCGCAGCCCCGCCCCG	soooossssssssooss	1508	1525	12	882
672720	CAACCGCAGCCCCGCCCC	soooossssssssooss	1509	1526	7	883
672721	GCAACCGCAGCCCCGCCC	soooossssssssooss	1510	1527	38	884
672722	CGCAACCGCAGCCCCGCC	soooossssssssooss	1511	1528	34	885
672723	CCGCAACCGCAGCCCCGC	soooossssssssooss	1512	1529	58	886
672724	ACCGCAACCGCAGCCCCG	soooossssssssooss	1513	1530	39	887
672725	CACCGCAACCGCAGCCCC	soooossssssssooss	1514	1531	43	888
672726	GCACCGCAACCGCAGCCC	soooossssssssooss	1515	1532	41	889
672727	GGCACCGCAACCGCAGCC	soooossssssssooss	1516	1533	18	890
672728	AGGCACCGCAACCGCAGC	soooossssssssooss	1517	1534	53	891
672729	CAGGCACCGCAACCGCAG	soooossssssssooss	1518	1535	26	892
672730	GCAGGCACCGCAACCGCA	soooossssssssooss	1519	1536	54	893
672731	CGCAGGCACCGCAACCGC	soooossssssssooss	1520	1537	41	894
672732	GCGCAGGCACCGCAACCG	soooossssssssooss	1521	1538	46	895
672733	GGCGCAGGCACCGCAACC	soooossssssssooss	1522	1539	7	896
672734	GGGCGCAGGCACCGCAAC	soooossssssssooss	1523	1540	26	897

TABLE 39

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
576816	GCCTTACTCTAGGACCAAGA	eeeee-d10-eeeee	sssssssssssssssssss	7990	8009	71	20
672735	TGAGAGCAAGTAGTGGG	eeekk-d7-kkeee	soossssssssooss	1326	1342	5	898
672736	GTGAGAGCAAGTAGTGG	eeekk-d7-kkeee	soossssssssooss	1327	1343	35	899
672737	TGTGAGAGCAAGTAGTG	eeekk-d7-kkeee	soossssssssooss	1328	1344	0	900
672738	CTGTGAGAGCAAGTAGT	eeekk-d7-kkeee	soossssssssooss	1329	1345	63	901
672739	ACTGTGAGAGCAAGTAG	eeekk-d7-kkeee	soossssssssooss	1330	1346	0	902
672740	TACTGTGAGAGCAAGTA	eeekk-d7-kkeee	soossssssssooss	1331	1347	65	903
672741	GTACTGTGAGAGCAAGT	eeekk-d7-kkeee	soossssssssooss	1332	1348	13	904
672742	AGTACTGTGAGAGCAAG	eeekk-d7-kkeee	soossssssssooss	1333	1349	46	905
672743	GAGTACTGTGAGAGCAA	eeekk-d7-kkeee	soossssssssooss	1334	1350	55	906
672744	CGAGTACTGTGAGAGCA	eeekk-d7-kkeee	soossssssssooss	1335	1351	76	907
672745	GCGAGTACTGTGAGAGC	eeekk-d7-kkeee	soossssssssooss	1336	1352	11	908

TABLE 39-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672746	AGCGAGTACTGTGAGAG	eeekk-d7-kkeee	soossssssssooss	1337	1353	11	909
672747	CAGCGAGTACTGTGAGA	eeekk-d7-kkeee	soossssssssooss	1338	1354	51	910
672748	TCAGCGAGTACTGTGAG	eeekk-d7-kkeee	soossssssssooss	1339	1355	20	911
672749	CTCAGCGAGTACTGTGA	eeekk-d7-kkeee	soossssssssooss	1340	1356	12	912
672750	CCTCAGCGAGTACTGTG	eeekk-d7-kkeee	soossssssssooss	1341	1357	2	913
672751	CCCTCAGCGAGTACTGT	eeekk-d7-kkeee	soossssssssooss	1342	1358	39	914
672752	ACCCTCAGCGAGTACTG	eeekk-d7-kkeee	soossssssssooss	1343	1359	27	915
672753	CACCCTCAGCGAGTACT	eeekk-d7-kkeee	soossssssssooss	1344	1360	33	916
672754	TCACCCTCAGCGAGTAC	eeekk-d7-kkeee	soossssssssooss	1345	1361	0	917
672755	TTCACCCTCAGCGAGTA	eeekk-d7-kkeee	soossssssssooss	1346	1362	13	918
672756	GTTCACCCTCAGCGAGT	eeekk-d7-kkeee	soossssssssooss	1347	1363	45	919
672757	TGTTCACCCTCAGCGAG	eeekk-d7-kkeee	soossssssssooss	1348	1364	11	920
672758	TTGTTCACCCTCAGCGA	eeekk-d7-kkeee	soossssssssooss	1349	1365	0	921
672759	CTTGTTCACCCTCAGCG	eeekk-d7-kkeee	soossssssssooss	1350	1366	12	922
672760	TCTTGTTCACCCTCAGC	eeekk-d7-kkeee	soossssssssooss	1351	1367	21	923
672761	TTCTTGTTCACCCTCAG	eeekk-d7-kkeee	soossssssssooss	1352	1368	0	924
672762	TTTCTTGTTCACCCTCA	eeekk-d7-kkeee	soossssssssooss	1353	1369	34	925
672763	TTTTCTTGTTCACCCTC	eeekk-d7-kkeee	soossssssssooss	1354	1370	16	926
672764	CTTTTCTTGTTCACCCT	eeekk-d7-kkeee	soossssssssooss	1355	1371	2	927
672765	TCTTTTCTTGTTCACCC	eeekk-d7-kkeee	soossssssssooss	1356	1372	24	928
672766	GTCTTTTCTTGTTCACC	eeekk-d7-kkeee	soossssssssooss	1357	1373	28	929
672767	GGTCTTTTCTTGTTCAC	eeekk-d7-kkeee	soossssssssooss	1358	1374	30	930
672768	AGGTCTTTTCTTGTTCA	eeekk-d7-kkeee	soossssssssooss	1359	1375	14	931
672769	CAGGTCTTTTCTTGTTC	eeekk-d7-kkeee	soossssssssooss	1360	1376	20	932
672770	TCAGGTCTTTTCTTGTT	eeekk-d7-kkeee	soossssssssooss	1361	1377	2	933
672771	ATCAGGTCTTTTCTTGT	eeekk-d7-kkeee	soossssssssooss	1362	1378	0	934
672772	TATCAGGTCTTTTCTTG	eeekk-d7-kkeee	soossssssssooss	1363	1379	23	935
672773	TTATCAGGTCTTTTCTT	eeekk-d7-kkeee	soossssssssooss	1364	1380	39	936
672774	ATCTTTATCAGGTCTTT	eeekk-d7-kkeee	soossssssssooss	1368	1384	92	937
672775	AATCTTTATCAGGTCTT	eeekk-d7-kkeee	soossssssssooss	1369	1385	61	938
672776	TAATCTTTATCAGGTCT	eeekk-d7-kkeee	soossssssssooss	1370	1386	2	939
672777	TTAATCTTTATCAGGTC	eeekk-d7-kkeee	soossssssssooss	1371	1387	33	940
672778	GTTAATCTTTATCAGGT	eeekk-d7-kkeee	soossssssssooss	1372	1388	53	941
672779	GGTTAATCTTTATCAGG	eeekk-d7-kkeee	soossssssssooss	1373	1389	40	942

TABLE 39-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672780	TGGTTAATCTTTATCAG	eeekk-d7-kkeee	soossssssssooss	1374	1390	26	943
672781	CTGGTTAATCTTTATCA	eeekk-d7-kkeee	soossssssssooss	1375	1391	25	944
672782	TCTGGTTAATCTTTATC	eeekk-d7-kkeee	soossssssssooss	1376	1392	44	945
672783	CCCTCCTTGTTTTCTTC	eeekk-d7-kkeee	soossssssssooss	1391	1407	37	946
672784	TCCCTCCTTGTTTTCTT	eeekk-d7-kkeee	soossssssssooss	1392	1408	10	947
672785	TTCCCTCCTTGTTTTCT	eeekk-d7-kkeee	soossssssssooss	1393	1409	0	948
672786	TTTCCCTCCTTGTTTTC	eeekk-d7-kkeee	soossssssssooss	1394	1410	0	949
672787	GTTTCCCTCCTTGTTTT	eeekk-d7-kkeee	soossssssssooss	1395	1411	0	950
672788	TGTTTCCCTCCTTGTTT	eeekk-d7-kkeee	soossssssssooss	1396	1412	0	951
672789	TTGTTTCCCTCCTTGTT	eeekk-d7-kkeee	soossssssssooss	1397	1413	0	952
672790	GGTTGTTTCCCTCCTTG	eeekk-d7-kkeee	soossssssssooss	1399	1415	33	953
672791	CGGTTGTTTCCCTCCTT	eeekk-d7-kkeee	soossssssssooss	1400	1416	0	954
672792	GCGGTTGTTTCCCTCCT	eeekk-d7-kkeee	soossssssssooss	1401	1417	23	955
672793	TGCGGTTGTTTCCCTCC	eeekk-d7-kkeee	soossssssssooss	1402	1418	0	956
672794	CTGCGGTTGTTTCCCTC	eeekk-d7-kkeee	soossssssssooss	1403	1419	25	957
672795	GCTGCGGTTGTTTCCCT	eeekk-d7-kkeee	soossssssssooss	1404	1420	5	958
672796	GGCTGCGGTTGTTTCCC	eeekk-d7-kkeee	soossssssssooss	1405	1421	16	959
672797	AGGCTGCGGTTGTTTCC	eeekk-d7-kkeee	soossssssssooss	1406	1422	7	960
672798	CAGGCTGCGGTTGTTTC	eeekk-d7-kkeee	soossssssssooss	1407	1423	0	961
672799	ACAGGCTGCGGTTGTTT	eeekk-d7-kkeee	soossssssssooss	1408	1424	28	962
672800	TACAGGCTGCGGTTGTT	eeekk-d7-kkeee	soossssssssooss	1409	1425	33	963
672801	CTACAGGCTGCGGTTGT	eeekk-d7-kkeee	soossssssssooss	1410	1426	53	964
672802	GCTACAGGCTGCGGTTG	eeekk-d7-kkeee	soossssssssooss	1411	1427	0	965
672803	TGCTACAGGCTGCGGTT	eeekk-d7-kkeee	soossssssssooss	1412	1428	0	966
672804	TTGCTACAGGCTGCGGT	eeekk-d7-kkeee	soossssssssooss	1413	1429	0	967
672805	CTTGCTACAGGCTGCGG	eeekk-d7-kkeee	soossssssssooss	1414	1430	1	968
672806	GCTTGCTACAGGCTGCG	eeekk-d7-kkeee	soossssssssooss	1415	1431	58	969
672807	AGCTTGCTACAGGCTGC	eeekk-d7-kkeee	soossssssssooss	1416	1432	0	970
672808	GAGCTTGCTACAGGCTG	eeekk-d7-kkeee	soossssssssooss	1417	1433	0	971
672809	AGAGCTTGCTACAGGCT	eeekk-d7-kkeee	soossssssssooss	1418	1434	71	972
672810	CAGAGCTTGCTACAGGC	eeekk-d7-kkeee	soossssssssooss	1419	1435	18	973
672811	CCAGAGCTTGCTACAGG	eeekk-d7-kkeee	soossssssssooss	1420	1436	0	974
672812	TCCAGAGCTTGCTACAG	eeekk-d7-kkeee	soossssssssooss	1421	1437	19	975
672813	TTCCAGAGCTTGCTACA	eeekk-d7-kkeee	soossssssssooss	1422	1438	19	976

TABLE 39-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672814	GTTCCAGAGCTTGCTAC	eeekk-d7-kkeee	soossssssssooss	1423	1439	0	977
672815	AGTTCCAGAGCTTGCTA	eeekk-d7-kkeee	soossssssssooss	1424	1440	0	978
672816	GAGTTCCAGAGCTTGCT	eeekk-d7-kkeee	soossssssssooss	1425	1441	14	979
672817	TGAGTTCCAGAGCTTGC	eeekk-d7-kkeee	soossssssssooss	1426	1442	31	980
672818	CTGAGTTCCAGAGCTTG	eeekk-d7-kkeee	soossssssssooss	1427	1443	21	981
672819	CCTGAGTTCCAGAGCTT	eeekk-d7-kkeee	soossssssssooss	1428	1444	19	982
672820	TCCTGAGTTCCAGAGCT	eeekk-d7-kkeee	soossssssssooss	1429	1445	46	983
672821	CTCCTGAGTTCCAGAGC	eeekk-d7-kkeee	soossssssssooss	1430	1446	7	984
672822	ACTCCTGAGTTCCAGAG	eeekk-d7-kkeee	soossssssssooss	1431	1447	13	985
672823	GACTCCTGAGTTCCAGA	eeekk-d7-kkeee	soossssssssooss	1432	1448	19	986
672824	CGACTCCTGAGTTCCAG	eeekk-d7-kkeee	soossssssssooss	1433	1449	23	987
672825	GCGACTCCTGAGTTCCA	eeekk-d7-kkeee	soossssssssooss	1434	1450	0	988
672826	CGCGACTCCTGAGTTCC	eeekk-d7-kkeee	soossssssssooss	1435	1451	19	989
672827	GCGCGACTCCTGAGTTC	eeekk-d7-kkeee	soossssssssooss	1436	1452	31	990
672828	CGCGCGACTCCTGAGTT	eeekk-d7-kkeee	soossssssssooss	1437	1453	63	991
672829	GCGCGCGACTCCTGAGT	eeekk-d7-kkeee	soossssssssooss	1438	1454	28	992
672830	AGCGCGCGACTCCTGAG	eeekk-d7-kkeee	soossssssssooss	1439	1455	58	993
672831	TAGCGCGCGACTCCTGA	eeekk-d7-kkeee	soossssssssooss	1440	1456	42	994
672832	CTAGCGCGCGACTCCTG	eeekk-d7-kkeee	soossssssssooss	1441	1457	42	995
672833	CCTAGCGCGCGACTCCT	eeekk-d7-kkeee	soossssssssooss	1442	1458	24	996
672834	CCCTAGCGCGCGACTCC	eeekk-d7-kkeee	soossssssssooss	1443	1459	58	997
672835	CCCCTAGCGCGCGACTC	eeekk-d7-kkeee	soossssssssooss	1444	1460	40	998
672836	GCCCCTAGCGCGCGACT	eeekk-d7-kkeee	soossssssssooss	1445	1461	0	999
672837	GGCCCCTAGCGCGCGAC	eeekk-d7-kkeee	soossssssssooss	1446	1462	2	1000
672838	CGGCCCCTAGCGCGCGA	eeekk-d7-kkeee	soossssssssooss	1447	1463	72	1001
672839	CCGGCCCCTAGCGCGCG	eeekk-d7-kkeee	soossssssssooss	1448	1464	0	1002
672840	CCCGGCCCCTAGCGCGC	eeekk-d7-kkeee	soossssssssooss	1449	1465	28	1003
672841	CCCCGGCCCCTAGCGCG	eeekk-d7-kkeee	soossssssssooss	1450	1466	28	1004
672842	GCCCCGGCCCCTAGCGC	eeekk-d7-kkeee	soossssssssooss	1451	1467	0	1005
672843	GGCCCCGGCCCCTAGCG	eeekk-d7-kkeee	soossssssssooss	1452	1468	23	1006
672844	CGGCCCCGGCCCCTAGC	eeekk-d7-kkeee	soossssssssooss	1453	1469	26	1007
672845	CCGGCCCCGGCCCCTAG	eeekk-d7-kkeee	soossssssssooss	1454	1470	24	1008
672846	CCCGGCCCCGGCCCCTA	eeekk-d7-kkeee	soossssssssooss	1455	1471	9	1009
672847	ACGCCCCGGCCCCGGCC	eeekk-d7-kkeee	soossssssssooss	1465	1481	3	1010

TABLE 39-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672848	CACGCCCCGGCCCCGGC	eeekk-d7-kkeee	soossssssssooss	1466	1482	19	1011
672849	CCACGCCCCGGCCCCGG	eeekk-d7-kkeee	soossssssssooss	1467	1483	50	1012
672850	ACCACGCCCCGGCCCCG	eeekk-d7-kkeee	soossssssssooss	1468	1484	0	1013
672851	GACCACGCCCCGGCCCC	eeekk-d7-kkeee	soossssssssooss	1469	1485	3	1014
672852	CGACCACGCCCCGGCCC	eeekk-d7-kkeee	soossssssssooss	1470	1486	9	1015
672853	CCGACCACGCCCCGGCC	eeekk-d7-kkeee	soossssssssooss	1471	1487	24	1016
672854	CCCGACCACGCCCCGGC	eeekk-d7-kkeee	soossssssssooss	1472	1488	9	1017
672855	CCCCGACCACGCCCCGG	eeekk-d7-kkeee	soossssssssooss	1473	1489	18	1018
672856	GCCCCGACCACGCCCCG	eeekk-d7-kkeee	soossssssssooss	1474	1490	8	1019
672857	CGCCCCGACCACGCCCC	eeekk-d7-kkeee	soossssssssooss	1475	1491	0	1020
672858	CCGCCCCGACCACGCCC	eeekk-d7-kkeee	soossssssssooss	1476	1492	48	1021
672859	CCCGCCCCGACCACGCC	eeekk-d7-kkeee	soossssssssooss	1477	1493	28	1022
672860	GCCCGCCCCGACCACGC	eeekk-d7-kkeee	soossssssssooss	1478	1494	0	1023
672861	GGCCCGCCCCGACCACG	eeekk-d7-kkeee	soossssssssooss	1479	1495	33	1024
672862	GGGCCCGCCCCGACCAC	eeekk-d7-kkeee	soossssssssooss	1480	1496	32	1025
672863	CGGGCCCGCCCCGACCA	eeekk-d7-kkeee	soossssssssooss	1481	1497	0	1026
672864	CCGGGCCCGCCCCGACC	eeekk-d7-kkeee	soossssssssooss	1482	1498	0	1027
672865	CCCGGGCCCGCCCCGAC	eeekk-d7-kkeee	soossssssssooss	1483	1499	11	1028
672866	GCAGCCCCGCCCCGGGC	eeekk-d7-kkeee	soossssssssooss	1505	1521	23	1029
672867	CGCAGCCCCGCCCCGGG	eeekk-d7-kkeee	soossssssssooss	1506	1522	26	1030
672868	CCGCAGCCCCGCCCCGG	eeekk-d7-kkeee	soossssssssooss	1507	1523	2	1031
672869	ACCGCAGCCCCGCCCCG	eeekk-d7-kkeee	soossssssssooss	1508	1524	8	1032
672870	AACCGCAGCCCCGCCCC	eeekk-d7-kkeee	soossssssssooss	1509	1525	7	1033
672871	CAACCGCAGCCCCGCCC	eeekk-d7-kkeee	soossssssssooss	1510	1526	1	1034
672872	GCAACCGCAGCCCCGCC	eeekk-d7-kkeee	soossssssssooss	1511	1527	37	1035
672873	CGCAACCGCAGCCCCGC	eeekk-d7-kkeee	soossssssssooss	1512	1528	20	1036
672874	CCGCAACCGCAGCCCCG	eeekk-d7-kkeee	soossssssssooss	1513	1529	23	1037
672875	ACCGCAACCGCAGCCCC	eeekk-d7-kkeee	soossssssssooss	1514	1530	8	1038
672876	CACCGCAACCGCAGCCC	eeekk-d7-kkeee	soossssssssooss	1515	1531	22	1039
672877	GCACCGCAACCGCAGCC	eeekk-d7-kkeee	soossssssssooss	1516	1532	19	1040
672878	GGCACCGCAACCGCAGC	eeekk-d7-kkeee	soossssssssooss	1517	1533	25	1041
672879	AGGCACCGCAACCGCAG	eeekk-d7-kkeee	soossssssssooss	1518	1534	21	1042
672880	CAGGCACCGCAACCGCA	eeekk-d7-kkeee	soossssssssooss	1519	1535	12	1043
672881	GCAGGCACCGCAACCGC	eeekk-d7-kkeee	soossssssssooss	1520	1536	18	1044

TABLE 39-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672882	CGCAGGCACCGCAACCG	eeekk-d7-kkeee	soossssssssooss	1521	1537	15	1045
672883	GCGCAGGCACCGCAACC	eeekk-d7-kkeee	soossssssssooss	1522	1538	0	1046
672884	GGCGCAGGCACCGCAAC	eeekk-d7-kkeee	soossssssssooss	1523	1539	0	1047

Example 9

Dose-Dependent Antisense Inhibition of Human C9ORF72 mRNA in HepG2 Cells

[0471] Antisense oligonucleotides from the study described in Example 8 hereinabove exhibiting significant in vitro inhibition of C9ORF72 mRNA were selected and tested at various doses in HepG2 cells. ISIS 576816, previously tested in PCT/US2013/065073 (claiming priority to U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.11 μM, 0.33 μM, 1.00 μM, or 3.00 μM concentrations of antisense oligonucleotide. After a treatment period of approximately 16 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant, which is the product of a pre-mRNA containing a hexanucleotide repeat. C9ORF72 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72 levels, relative to untreated control cells.

[0472] As shown in Tables 39 and 40, total C9ORF72 mRNA levels were reduced in a dose-dependent manner in some of the antisense oligonucleotide treated cells.

TABLE 39

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.11 μM	0.33 μM	1.00 μM	3.00 μM
672651	12	39	63	88
672611	13	50	59	79
672602	16	36	62	79
672624	25	54	80	95
672657	19	30	53	75
672582	0	0	20	57
672683	33	57	73	84
672595	39	39	66	88
576816	23	53	78	87
672640	17	26	56	84
672599	28	59	69	87
672637	36	50	66	88
672592	16	38	37	65
672636	24	39	69	88
672652	26	48	63	94
672619	8	12	6	0

TABLE 39-continued

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.11 μM	0.33 μM	1.00 μM	3.00 μM
672608	12	45	37	59
672598	21	9	55	69
672642	28	35	59	72

TABLE 40

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.11 μM	0.33 μM	1.00 μM	3.00 μM
672679	2	52	81	95
672693	14	41	53	83
672681	27	42	63	87
672683	10	35	64	82
672678	24	56	77	91
672699	17	31	46	83
672664	7	28	58	87
672665	35	46	62	75
576816	15	55	71	84
672671	36	66	79	87
672676	33	53	71	77
672700	25	45	68	81
672730	24	41	59	80
672670	25	40	60	75
672697	11	41	64	80
672723	30	48	68	88
672728	11	44	49	68
672675	41	48	74	88
672674	19	34	55	61

Example 10

Antisense Inhibition of C9ORF72 by Deoxy, MOE and cEt Antisense Oligonucleotides with Mixed Backbones

[0473] Antisense oligonucleotides described in Example 6 hereinabove (see Tables 25-28 hereinabove) were tested in HepG2 cells in a series of experiments that had similar culture conditions. ISIS 576816, which was previously tested in PCT/US2013/065073 (claiming priority to U.S. Application No. 61/714,132, filed Oct. 15, 2012) was used as a benchmark oligonucleotide for study with deoxy, MOE, and cEt anti-

sense oligonucleotides. The results for each experiment are presented in tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 700 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant, which is the product of a pre-mRNA containing a hexanucleotide repeat. The levels of the C9ORF72 pathogenic associated mRNA variant were normalized to the total RNA content of the cell, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72, relative to untreated control cells.

TABLE 41

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO:	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
576816	GCCTTACTCTAGGACCAAGA	eeeee-d10-eeeee	sssssssssssssssssss	7990	8009	52	20
672885	TGAGAGCAAGTAGTGGG	eekk-d8-kkeee	soossssssssooss	1326	1342	10	898
672886	GTGAGAGCAAGTAGTGG	eekk-d8-kkeee	soossssssssooss	1327	1343	24	899
672887	TGTGAGAGCAAGTAGTG	eekk-d8-kkeee	soossssssssooss	1328	1344	38	900
672888	CTGTGAGAGCAAGTAGT	eekk-d8-kkeee	soossssssssooss	1329	1345	37	901
672889	ACTGTGAGAGCAAGTAG	eekk-d8-kkeee	soossssssssooss	1330	1346	16	902
672890	TACTGTGAGAGCAAGTA	eekk-d8-kkeee	soossssssssooss	1331	1347	26	903
672891	GTACTGTGAGAGCAAGT	eekk-d8-kkeee	soossssssssooss	1332	1348	38	904
672892	AGTACTGTGAGAGCAAG	eekk-d8-kkeee	soossssssssooss	1333	1349	44	905
672893	GAGTACTGTGAGAGCAA	eekk-d8-kkeee	soossssssssooss	1334	1350	57	906
672894	CGAGTACTGTGAGAGCA	eekk-d8-kkeee	soossssssssooss	1335	1351	62	907
672895	GCGAGTACTGTGAGAGC	eekk-d8-kkeee	soossssssssooss	1336	1352	39	908
672896	AGCGAGTACTGTGAGAG	eekk-d8-kkeee	soossssssssooss	1337	1353	54	909
672897	CAGCGAGTACTGTGAGA	eekk-d8-kkeee	soossssssssooss	1338	1354	57	910
672898	TCAGCGAGTACTGTGAG	eekk-d8-kkeee	soossssssssooss	1339	1355	23	911
672899	CTCAGCGAGTACTGTGA	eekk-d8-kkeee	soossssssssooss	1340	1356	34	912
672900	CCTCAGCGAGTACTGTG	eekk-d8-kkeee	soossssssssooss	1341	1357	33	913
672901	CCCTCAGCGAGTACTGT	eekk-d8-kkeee	soossssssssooss	1342	1358	32	914
672902	ACCCTCAGCGAGTACTG	eekk-d8-kkeee	soossssssssooss	1343	1359	62	915
672903	CACCCTCAGCGAGTACT	eekk-d8-kkeee	soossssssssooss	1344	1360	59	916
672904	TCACCCTCAGCGAGTAC	eekk-d8-kkeee	soossssssssooss	1345	1361	52	917
672905	TTCACCCTCAGCGAGTA	eekk-d8-kkeee	soossssssssooss	1346	1362	50	918
672906	GTTCACCCTCAGCGAGT	eekk-d8-kkeee	soossssssssooss	1347	1363	38	919
672907	TGTTCACCCTCAGCGAG	eekk-d8-kkeee	soossssssssooss	1348	1364	20	920
672908	TTGTTCACCCTCAGCGA	eekk-d8-kkeee	soossssssssooss	1349	1365	57	921
672909	CTTGTTCACCCTCAGCG	eekk-d8-kkeee	soossssssssooss	1350	1366	66	922
672910	TCTTGTTCACCCTCAGC	eekk-d8-kkeee	soossssssssooss	1351	1367	47	923
672911	TTCTTGTTCACCCTCAG	eekk-d8-kkeee	soossssssssooss	1352	1368	37	924
672912	TTTCTTGTTCACCCTCA	eekk-d8-kkeee	soossssssssooss	1353	1369	36	925

TABLE 41-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO:	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672913	TTTTCTTGTTCACCCTC	eekk-d8-kkeee	soossssssssooss	1354	1370	34	926
672914	CTTTTCTTGTTCACCCT	eekk-d8-kkeee	soossssssssooss	1355	1371	35	927
672915	TCTTTTCTTGTTCACCC	eekk-d8-kkeee	soossssssssooss	1356	1372	41	928
672916	GTCTTTTCTTGTTCACC	eekk-d8-kkeee	soossssssssooss	1357	1373	34	929
672917	GGTCTTTTCTTGTTCAC	eekk-d8-kkeee	soossssssssooss	1358	1374	23	930
672918	AGGTCTTTTCTTGTTCA	eekk-d8-kkeee	soossssssssooss	1359	1375	31	931
672919	CAGGTCTTTTCTTGTTC	eekk-d8-kkeee	soossssssssooss	1360	1376	51	932
672920	TCAGGTCTTTTCTTGTT	eekk-d8-kkeee	soossssssssooss	1361	1377	15	933
672921	ATCAGGTCTTTTCTTGT	eekk-d8-kkeee	soossssssssooss	1362	1378	0	934
672922	TATCAGGTCTTTTCTTG	eekk-d8-kkeee	soossssssssooss	1363	1379	31	935
672923	TTATCAGGTCTTTTCTT	eekk-d8-kkeee	soossssssssooss	1364	1380	14	936
672924	ATCTTTATCAGGTCTTT	eekk-d8-kkeee	soossssssssooss	1368	1384	71	937
672925	AATCTTTATCAGGTCTT	eekk-d8-kkeee	soossssssssooss	1369	1385	72	938
672926	TAATCTTTATCAGGTCT	eekk-d8-kkeee	soossssssssooss	1370	1386	40	939
672927	TTAATCTTTATCAGGTC	eekk-d8-kkeee	soossssssssooss	1371	1387	66	940
672928	GTTAATCTTTATCAGGT	eekk-d8-kkeee	soossssssssooss	1372	1388	56	941
672929	GGTTAATCTTTATCAGG	eekk-d8-kkeee	soossssssssooss	1373	1389	80	942
672930	TGGTTAATCTTTATCAG	eekk-d8-kkeee	soossssssssooss	1374	1390	48	943
672931	CTGGTTAATCTTTATCA	eekk-d8-kkeee	soossssssssooss	1375	1391	48	944
672932	TCTGGTTAATCTTTATC	eekk-d8-kkeee	soossssssssooss	1376	1392	54	945
672933	CCCTCCTTGTTTTCTTC	eekk-d8-kkeee	soossssssssooss	1391	1407	18	946
672934	TCCCTCCTTGTTTTCTT	eekk-d8-kkeee	soossssssssooss	1392	1408	7	947
672935	TTCCCTCCTTGTTTTCT	eekk-d8-kkeee	soossssssssooss	1393	1409	10	948
672936	TTTCCCTCCTTGTTTTC	eekk-d8-kkeee	soossssssssooss	1394	1410	0	949
672937	GTTTCCCTCCTTGTTTT	eekk-d8-kkeee	soossssssssooss	1395	1411	0	950
672938	TGTTTCCCTCCTTGTTT	eekk-d8-kkeee	soossssssssooss	1396	1412	9	951
672939	TTGTTTCCCTCCTTGTT	eekk-d8-kkeee	soossssssssooss	1397	1413	27	952
672940	GGTTGTTTCCCTCCTTG	eekk-d8-kkeee	soossssssssooss	1399	1415	49	953
672941	CGGTTGTTTCCCTCCTT	eekk-d8-kkeee	soossssssssooss	1400	1416	17	954
672942	GCGGTTGTTTCCCTCCT	eekk-d8-kkeee	soossssssssooss	1401	1417	10	955
672943	TGCGGTTGTTTCCCTCC	eekk-d8-kkeee	soossssssssooss	1402	1418	33	956
672944	CTGCGGTTGTTTCCCTC	eekk-d8-kkeee	soossssssssooss	1403	1419	29	957
672945	GCTGCGGTTGTTTCCCT	eekk-d8-kkeee	soossssssssooss	1404	1420	23	958

TABLE 41-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO:	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672946	GGCTGCGGTTGTTTCCC	eekk-d8-kkeee	soossssssssooss	1405	1421	15	959
672947	AGGCTGCGGTTGTTTCC	eekk-d8-kkeee	soossssssssooss	1406	1422	24	960
672948	CAGGCTGCGGTTGTTTC	eekk-d8-kkeee	soossssssssooss	1407	1423	49	961
672949	ACAGGCTGCGGTTGTTT	eekk-d8-kkeee	soossssssssooss	1408	1424	35	962
672950	TACAGGCTGCGGTTGTT	eekk-d8-kkeee	soossssssssooss	1409	1425	37	963
672951	CTACAGGCTGCGGTTGT	eekk-d8-kkeee	soossssssssooss	1410	1426	4	964
672952	GCTACAGGCTGCGGTTG	eekk-d8-kkeee	soossssssssooss	1411	1427	4	965
672953	TGCTACAGGCTGCGGTT	eekk-d8-kkeee	soossssssssooss	1412	1428	24	966
672954	TTGCTACAGGCTGCGGT	eekk-d8-kkeee	soossssssssooss	1413	1429	8	967
672955	CTTGCTACAGGCTGCGG	eekk-d8-kkeee	soossssssssooss	1414	1430	28	968
672956	GCTTGCTACAGGCTGCG	eekk-d8-kkeee	soossssssssooss	1415	1431	8	969
672957	AGCTTGCTACAGGCTGC	eekk-d8-kkeee	soossssssssooss	1416	1432	5	970
672958	GAGCTTGCTACAGGCTG	eekk-d8-kkeee	soossssssssooss	1417	1433	4	971
672959	AGAGCTTGCTACAGGCT	eekk-d8-kkeee	soossssssssooss	1418	1434	0	972
672960	CAGAGCTTGCTACAGGC	eekk-d8-kkeee	soossssssssooss	1419	1435	12	973
672961	CCAGAGCTTGCTACAGG	eekk-d8-kkeee	soossssssssooss	1420	1436	36	974
672962	TCCAGAGCTTGCTACAG	eekk-d8-kkeee	soossssssssooss	1421	1437	0	975
672963	TTCCAGAGCTTGCTACA	eekk-d8-kkeee	soossssssssooss	1422	1438	11	976
672964	GTTCCAGAGCTTGCTAC	eekk-d8-kkeee	soossssssssooss	1423	1439	8	977
672965	AGTTCCAGAGCTTGCTA	eekk-d8-kkeee	soossssssssooss	1424	1440	19	978
672966	GAGTTCCAGAGCTTGCT	eekk-d8-kkeee	soossssssssooss	1425	1441	48	979
672967	TGAGTTCCAGAGCTTGC	eekk-d8-kkeee	soossssssssooss	1426	1442	41	980
672968	CTGAGTTCCAGAGCTTG	eekk-d8-kkeee	soossssssssooss	1427	1443	54	981
672969	CCTGAGTTCCAGAGCTT	eekk-d8-kkeee	soossssssssooss	1428	1444	58	982
672970	TCCTGAGTTCCAGAGCT	eekk-d8-kkeee	soossssssssooss	1429	1445	12	983
672971	CTCCTGAGTTCCAGAGC	eekk-d8-kkeee	soossssssssooss	1430	1446	23	984
672972	ACTCCTGAGTTCCAGAG	eekk-d8-kkeee	soossssssssooss	1431	1447	30	985
672973	GACTCCTGAGTTCCAGA	eekk-d8-kkeee	soossssssssooss	1432	1448	39	986
672974	CGACTCCTGAGTTCCAG	eekk-d8-kkeee	soossssssssooss	1433	1449	41	987
672975	GCGACTCCTGAGTTCCA	eekk-d8-kkeee	soossssssssooss	1434	1450	31	988
672976	CGCGACTCCTGAGTTCC	eekk-d8-kkeee	soossssssssooss	1435	1451	56	989
672977	GCGCGACTCCTGAGTTC	eekk-d8-kkeee	soossssssssooss	1436	1452	31	990
672978	CGCGCGACTCCTGAGTT	eekk-d8-kkeee	soossssssssooss	1437	1453	45	991
672979	GCGCGCGACTCCTGAGT	eekk-d8-kkeee	soossssssssooss	1438	1454	29	992

TABLE 41-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO:	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
672980	AGCGCGCGACTCCTGAG	eekk-d8-kkeee	soossssssssooss	1439	1455	48	993
672981	TAGCGCGCGACTCCTGA	eekk-d8-kkeee	soossssssssooss	1440	1456	68	994
672982	CTAGCGCGCGACTCCTG	eekk-d8-kkeee	soossssssssooss	1441	1457	59	995
672983	CCTAGCGCGCGACTCCT	eekk-d8-kkeee	soossssssssooss	1442	1458	62	996
672984	CCCTAGCGCGCGACTCC	eekk-d8-kkeee	soossssssssooss	1443	1459	69	997
672985	CCCCTAGCGCGCGACTC	eekk-d8-kkeee	soossssssssooss	1444	1460	65	998
672986	GCCCCTAGCGCGCGACT	eekk-d8-kkeee	soossssssssooss	1445	1461	34	999
672987	GGCCCCTAGCGCGCGAC	eekk-d8-kkeee	soossssssssooss	1446	1462	22	1000
672988	CGGCCCCTAGCGCGCGA	eekk-d8-kkeee	soossssssssooss	1447	1463	16	1001
672989	CCGGCCCCTAGCGCGCG	eekk-d8-kkeee	soossssssssooss	1448	1464	24	1002
672990	CCCGGCCCCTAGCGCGC	eekk-d8-kkeee	soossssssssooss	1449	1465	10	1003
672991	CCCCGGCCCCTAGCGCG	eekk-d8-kkeee	soossssssssooss	1450	1466	24	1004
672992	GCCCCGGCCCCTAGCGC	eekk-d8-kkeee	soossssssssooss	1451	1467	29	1005
672993	GGCCCCGGCCCCTAGCG	eekk-d8-kkeee	soossssssssooss	1452	1468	24	1006
672994	CGGCCCCGGCCCCTAGC	eekk-d8-kkeee	soossssssssooss	1453	1469	25	1007
672995	CCGGCCCCGGCCCCTAG	eekk-d8-kkeee	soossssssssooss	1454	1470	28	1008
672996	CCCGGCCCCGGCCCCTA	eekk-d8-kkeee	soossssssssooss	1455	1471	25	1009
672997	ACGCCCCGGCCCCGGCC	eekk-d8-kkeee	soossssssssooss	1465	1481	21	1010
672998	CACGCCCCGGCCCCGGC	eekk-d8-kkeee	soossssssssooss	1466	1482	17	1011
672999	CCACGCCCCGGCCCCGG	eekk-d8-kkeee	soossssssssooss	1467	1483	30	1012
673000	ACCACGCCCCGGCCCCG	eekk-d8-kkeee	soossssssssooss	1468	1484	29	1013
673001	GACCACGCCCCGGCCCC	eekk-d8-kkeee	soossssssssooss	1469	1485	25	1014
673002	CGACCACGCCCCGGCCC	eekk-d8-kkeee	soossssssssooss	1470	1486	37	1015
673003	CCGACCACGCCCCGGCC	eekk-d8-kkeee	soossssssssooss	1471	1487	23	1016
673004	CCCGACCACGCCCCGGC	eekk-d8-kkeee	soossssssssooss	1472	1488	21	1017
673005	CCCCGACCACGCCCCGG	eekk-d8-kkeee	soossssssssooss	1473	1489	9	1018
673006	GCCCCGACCACGCCCCG	eekk-d8-kkeee	soossssssssooss	1474	1490	13	1019
673007	CGCCCCGACCACGCCCC	eekk-d8-kkeee	soossssssssooss	1475	1491	17	1020
673008	CCGCCCCGACCACGCCC	eekk-d8-kkeee	soossssssssooss	1476	1492	20	1021
673009	CCCGCCCCGACCACGCC	eekk-d8-kkeee	soossssssssooss	1477	1493	36	1022
673010	GCCCGCCCCGACCACGC	eekk-d8-kkeee	soossssssssooss	1478	1494	16	1023
673011	GGCCCGCCCCGACCACG	eekk-d8-kkeee	soossssssssooss	1479	1495	3	1024
673012	GGGCCCGCCCCGACCAC	eekk-d8-kkeee	soossssssssooss	1480	1496	21	1025

TABLE 41-continued

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO:	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673013	CGGGCCCGCCCCGACCA	eekk-d8-kkeee	soossssssssooss	1481	1497	4	1026
673014	CCGGGCCCGCCCCGACC	eekk-d8-kkeee	soossssssssooss	1482	1498	21	1027
673015	CCCGGGCCCGCCCCGAC	eekk-d8-kkeee	soossssssssooss	1483	1499	15	1028
673016	GCAGCCCCGCCCCGGGC	eekk-d8-kkeee	soossssssssooss	1505	1521	3	1029
673017	CGCAGCCCCGCCCCGGG	eekk-d8-kkeee	soossssssssooss	1506	1522	7	1030
673018	CCGCAGCCCCGCCCCGG	eekk-d8-kkeee	soossssssssooss	1507	1523	7	1031
673019	ACCGCAGCCCCGCCCCG	eekk-d8-kkeee	soossssssssooss	1508	1524	8	1032
673020	AACCGCAGCCCCGCCCC	eekk-d8-kkeee	soossssssssooss	1509	1525	42	1033
673021	CAACCGCAGCCCCGCCC	eekk-d8-kkeee	soossssssssooss	1510	1526	58	1034
673022	GCAACCGCAGCCCCGCC	eekk-d8-kkeee	soossssssssooss	1511	1527	44	1035
673023	CGCAACCGCAGCCCCGC	eekk-d8-kkeee	soossssssssooss	1512	1528	46	1036
673024	CCGCAACCGCAGCCCCG	eekk-d8-kkeee	soossssssssooss	1513	1529	26	1037
673025	ACCGCAACCGCAGCCCC	eekk-d8-kkeee	soossssssssooss	1514	1530	20	1038
673026	CACCGCAACCGCAGCCC	eekk-d8-kkeee	soossssssssooss	1515	1531	52	1039
673027	GCACCGCAACCGCAGCC	eekk-d8-kkeee	soossssssssooss	1516	1532	22	1040
673028	GGCACCGCAACCGCAGC	eekk-d8-kkeee	soossssssssooss	1517	1533	32	1041
673029	AGGCACCGCAACCGCAG	eekk-d8-kkeee	soossssssssooss	1518	1534	27	1042
673030	CAGGCACCGCAACCGCA	eekk-d8-kkeee	soossssssssooss	1519	1535	32	1043
673031	GCAGGCACCGCAACCGC	eekk-d8-kkeee	soossssssssooss	1520	1536	38	1044
673032	CGCAGGCACCGCAACCG	eekk-d8-kkeee	soossssssssooss	1521	1537	54	1045
673033	GCGCAGGCACCGCAACC	eekk-d8-kkeee	soossssssssooss	1522	1538	24	1046
673034	GGCGCAGGCACCGCAAC	eekk-d8-kkeee	soossssssssooss	1523	1539	17	1047

TABLE 42

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
576816	GCCTTACTCTAGGACCAAGA	eeeee-d10-eeeee	sssssssssssssssssss	7990	8009	75	20
673035	TGAGAGCAAGTAGTGGG	ek-d8-ekekeee	sosssssssssoooss	1326	1342	47	898
673036	GTGAGAGCAAGTAGTGG	ek-d8-ekekeee	sosssssssssoooss	1327	1343	57	899
673037	TGTGAGAGCAAGTAGTG	ek-d8-ekekeee	sosssssssssoooss	1328	1344	39	900
673038	CTGTGAGAGCAAGTAGT	ek-d8-ekekeee	sosssssssssoooss	1329	1345	48	901

TABLE 42-continued

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673039	ACTGTGAGAGCAAGTAG	ek-d8-ekekeee	sosssssssssooooss	1330	1346	35	902
673040	TACTGTGAGAGCAAGTA	ek-d8-ekekeee	sosssssssssooooss	1331	1347	40	903
673041	GTACTGTGAGAGCAAGT	ek-d8-ekekeee	sosssssssssooooss	1332	1348	35	904
673042	AGTACTGTGAGAGCAAG	ek-d8-ekekeee	sosssssssssooooss	1333	1349	26	905
673043	GAGTACTGTGAGAGCAA	ek-d8-ekekeee	sosssssssssooooss	1334	1350	44	906
673044	CGAGTACTGTGAGAGCA	ek-d8-ekekeee	sosssssssssooooss	1335	1351	44	907
673045	GCGAGTACTGTGAGAGC	ek-d8-ekekeee	sosssssssssooooss	1336	1352	36	908
673046	AGCGAGTACTGTGAGAG	ek-d8-ekekeee	sosssssssssooooss	1337	1353	37	909
673047	CAGCGAGTACTGTGAGA	ek-d8-ekekeee	sosssssssssooooss	1338	1354	66	910
673048	TCAGCGAGTACTGTGAG	ek-d8-ekekeee	sosssssssssooooss	1339	1355	22	911
673049	CTCAGCGAGTACTGTGA	ek-d8-ekekeee	sosssssssssooooss	1340	1356	49	912
673050	CCTCAGCGAGTACTGTG	ek-d8-ekekeee	sosssssssssooooss	1341	1357	53	913
673051	CCCTCAGCGAGTACTGT	ek-d8-ekekeee	sosssssssssooooss	1342	1358	52	914
673052	ACCCTCAGCGAGTACTG	ek-d8-ekekeee	sosssssssssooooss	1343	1359	20	915
673053	CACCCTCAGCGAGTACT	ek-d8-ekekeee	sosssssssssooooss	1344	1360	66	916
673054	TCACCCTCAGCGAGTAC	ek-d8-ekekeee	sosssssssssooooss	1345	1361	50	917
673055	TTCACCCTCAGCGAGTA	ek-d8-ekekeee	sosssssssssooooss	1346	1362	35	918
673056	GTTCACCCTCAGCGAGT	ek-d8-ekekeee	sosssssssssooooss	1347	1363	46	919
673057	TGTTCACCCTCAGCGAG	ek-d8-ekekeee	sosssssssssooooss	1348	1364	51	920
673058	TTGTTCACCCTCAGCGA	ek-d8-ekekeee	sosssssssssooooss	1349	1365	57	921
673059	CTTGTTCACCCTCAGCG	ek-d8-ekekeee	sosssssssssooooss	1350	1366	50	922
673060	TCTTGTTCACCCTCAGC	ek-d8-ekekeee	sosssssssssooooss	1351	1367	41	923
673061	TTCTTGTTCACCCTCAG	ek-d8-ekekeee	sosssssssssooooss	1352	1368	14	924
673062	TTTCTTGTTCACCCTCA	ek-d8-ekekeee	sosssssssssooooss	1353	1369	28	925
673063	TTTTCTTGTTCACCCTC	ek-d8-ekekeee	sosssssssssooooss	1354	1370	38	926
673064	CTTTTCTTGTTCACCCT	ek-d8-ekekeee	sosssssssssooooss	1355	1371	38	927
673065	TCTTTTCTTGTTCACCC	ek-d8-ekekeee	sosssssssssooooss	1356	1372	30	928
673066	GTCTTTTCTTGTTCACC	ek-d8-ekekeee	sosssssssssooooss	1357	1373	47	929
673067	GGTCTTTTCTTGTTCAC	ek-d8-ekekeee	sosssssssssooooss	1358	1374	63	930
673068	AGGTCTTTTCTTGTTCA	ek-d8-ekekeee	sosssssssssooooss	1359	1375	71	931
673069	CAGGTCTTTTCTTGTTC	ek-d8-ekekeee	sosssssssssooooss	1360	1376	41	932
673070	TCAGGTCTTTTCTTGTT	ek-d8-ekekeee	sosssssssssooooss	1361	1377	24	933
673071	ATCAGGTCTTTTCTTGT	ek-d8-ekekeee	sosssssssssooooss	1362	1378	53	934
673072	TATCAGGTCTTTTCTTG	ek-d8-ekekeee	sosssssssssooooss	1363	1379	44	935
673073	TTATCAGGTCTTTTCTT	ek-d8-ekekeee	sosssssssssooooss	1364	1380	28	936

TABLE 42-continued

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673074	ATCTTTATCAGGTCTTT	ek-d8-ekekeee	sosssssssssooooss	1368	1384	57	937
673075	AATCTTTATCAGGTCTT	ek-d8-ekekeee	sosssssssssooooss	1369	1385	49	938
673076	TAATCTTTATCAGGTCT	ek-d8-ekekeee	sosssssssssooooss	1370	1386	40	939
673077	TTAATCTTTATCAGGTC	ek-d8-ekekeee	sosssssssssooooss	1371	1387	33	940
673078	GTTAATCTTTATCAGGT	ek-d8-ekekeee	sosssssssssooooss	1372	1388	46	941
673079	GGTTAATCTTTATCAGG	ek-d8-ekekeee	sosssssssssooooss	1373	1389	87	942
673080	TGGTTAATCTTTATCAG	ek-d8-ekekeee	sosssssssssooooss	1374	1390	35	943
673081	CTGGTTAATCTTTATCA	ek-d8-ekekeee	sosssssssssooooss	1375	1391	56	944
673082	TCTGGTTAATCTTTATC	ek-d8-ekekeee	sosssssssssooooss	1376	1392	59	945
673083	CCCTCCTTGTTTTCTTC	ek-d8-ekekeee	sosssssssssooooss	1391	1407	11	946
673084	TCCCTCCTTGTTTTCTT	ek-d8-ekekeee	sosssssssssooooss	1392	1408	10	947
673085	TTCCCTCCTTGTTTTCT	ek-d8-ekekeee	sosssssssssooooss	1393	1409	8	948
673086	TTTCCCTCCTTGTTTTC	ek-d8-ekekeee	sosssssssssooooss	1394	1410	26	949
673087	GTTTCCCTCCTTGTTTT	ek-d8-ekekeee	sosssssssssooooss	1395	1411	25	950
673088	TGTTTCCCTCCTTGTTT	ek-d8-ekekeee	sosssssssssooooss	1396	1412	62	951
673089	TTGTTTCCCTCCTTGTT	ek-d8-ekekeee	sosssssssssooooss	1397	1413	51	952
673090	GGTTGTTTCCCTCCTTG	ek-d8-ekekeee	sosssssssssooooss	1399	1415	34	953
673091	CGGTTGTTTCCCTCCTT	ek-d8-ekekeee	sosssssssssooooss	1400	1416	13	954
673092	GCGGTTGTTTCCCTCCT	ek-d8-ekekeee	sosssssssssooooss	1401	1417	37	955
673093	TGCGGTTGTTTCCCTCC	ek-d8-ekekeee	sosssssssssooooss	1402	1418	49	956
673094	CTGCGGTTGTTTCCCTC	ek-d8-ekekeee	sosssssssssooooss	1403	1419	15	957
673095	GCTGCGGTTGTTTCCCT	ek-d8-ekekeee	sosssssssssooooss	1404	1420	17	958
673096	GGCTGCGGTTGTTTCCC	ek-d8-ekekeee	sosssssssssooooss	1405	1421	33	959
673097	AGGCTGCGGTTGTTTCC	ek-d8-ekekeee	sosssssssssooooss	1406	1422	43	960
673098	CAGGCTGCGGTTGTTTC	ek-d8-ekekeee	sosssssssssooooss	1407	1423	53	961
673099	ACAGGCTGCGGTTGTTT	ek-d8-ekekeee	sosssssssssooooss	1408	1424	21	962
673100	TACAGGCTGCGGTTGTT	ek-d8-ekekeee	sosssssssssooooss	1409	1425	23	963
673101	CTACAGGCTGCGGTTGT	ek-d8-ekekeee	sosssssssssooooss	1410	1426	16	964
673102	GCTACAGGCTGCGGTTG	ek-d8-ekekeee	sosssssssssooooss	1411	1427	24	965
673103	TGCTACAGGCTGCGGTT	ek-d8-ekekeee	sosssssssssooooss	1412	1428	41	966
673104	TTGCTACAGGCTGCGGT	ek-d8-ekekeee	sosssssssssooooss	1413	1429	30	967
673105	CTTGCTACAGGCTGCGG	ek-d8-ekekeee	sosssssssssooooss	1414	1430	13	968
673106	GCTTGCTACAGGCTGCG	ek-d8-ekekeee	sosssssssssooooss	1415	1431	7	969
673107	AGCTTGCTACAGGCTGC	ek-d8-ekekeee	sosssssssssooooss	1416	1432	7	970

TABLE 42-continued

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673108	GAGCTTGCTACAGGCTG	ek-d8-ekekeee	sosssssssssooooss	1417	1433	23	971
673109	AGAGCTTGCTACAGGCT	ek-d8-ekekeee	sosssssssssooooss	1418	1434	38	972
673110	CAGAGCTTGCTACAGGC	ek-d8-ekekeee	sosssssssssooooss	1419	1435	22	973
673111	CCAGAGCTTGCTACAGG	ek-d8-ekekeee	sosssssssssooooss	1420	1436	14	974
673112	TCCAGAGCTTGCTACAG	ek-d8-ekekeee	sosssssssssooooss	1421	1437	11	975
673113	TTCCAGAGCTTGCTACA	ek-d8-ekekeee	sosssssssssooooss	1422	1438	24	976
673114	GTTCCAGAGCTTGCTAC	ek-d8-ekekeee	sosssssssssooooss	1423	1439	37	977
673115	AGTTCCAGAGCTTGCTA	ek-d8-ekekeee	sosssssssssooooss	1424	1440	34	978
673116	GAGTTCCAGAGCTTGCT	ek-d8-ekekeee	sosssssssssooooss	1425	1441	21	979
673117	TGAGTTCCAGAGCTTGC	ek-d8-ekekeee	sosssssssssooooss	1426	1442	47	980
673118	CTGAGTTCCAGAGCTTG	ek-d8-ekekeee	sosssssssssooooss	1427	1443	27	981
673119	CCTGAGTTCCAGAGCTT	ek-d8-ekekeee	sosssssssssooooss	1428	1444	44	982
673120	TCCTGAGTTCCAGAGCT	ek-d8-ekekeee	sosssssssssooooss	1429	1445	39	983
673121	CTCCTGAGTTCCAGAGC	ek-d8-ekekeee	sosssssssssooooss	1430	1446	37	984
673122	ACTCCTGAGTTCCAGAG	ek-d8-ekekeee	sosssssssssooooss	1431	1447	40	985
673123	GACTCCTGAGTTCCAGA	ek-d8-ekekeee	sosssssssssooooss	1432	1448	26	986
673124	CGACTCCTGAGTTCCAG	ek-d8-ekekeee	sosssssssssooooss	1433	1449	36	987
673125	GCGACTCCTGAGTTCCA	ek-d8-ekekeee	sosssssssssooooss	1434	1450	55	988
673126	CGCGACTCCTGAGTTCC	ek-d8-ekekeee	sosssssssssooooss	1435	1451	55	989
673127	GCGCGACTCCTGAGTTC	ek-d8-ekekeee	sosssssssssooooss	1436	1452	64	990
673128	CGCGCGACTCCTGAGTT	ek-d8-ekekeee	sosssssssssooooss	1437	1453	59	991
673129	GCGCGCGACTCCTGAGT	ek-d8-ekekeee	sosssssssssooooss	1438	1454	42	992
673130	AGCGCGCGACTCCTGAG	ek-d8-ekekeee	sosssssssssooooss	1439	1455	60	993
673131	TAGCGCGCGACTCCTGA	ek-d8-ekekeee	sosssssssssooooss	1440	1456	59	994
673132	CTAGCGCGCGACTCCTG	ek-d8-ekekeee	sosssssssssooooss	1441	1457	49	995
673133	CCTAGCGCGCGACTCCT	ek-d8-ekekeee	sosssssssssooooss	1442	1458	62	996
673134	CCCTAGCGCGCGACTCC	ek-d8-ekekeee	sosssssssssooooss	1443	1459	62	997
673135	CCCCTAGCGCGCGACTC	ek-d8-ekekeee	sosssssssssooooss	1444	1460	65	998
673136	GCCCCTAGCGCGCGACT	ek-d8-ekekeee	sosssssssssooooss	1445	1461	27	999
673137	GGCCCCTAGCGCGCGAC	ek-d8-ekekeee	sosssssssssooooss	1446	1462	6	1000
673138	CGGCCCCTAGCGCGCGA	ek-d8-ekekeee	sosssssssssooooss	1447	1463	26	1001
673139	CCGGCCCCTAGCGCGCG	ek-d8-ekekeee	sosssssssssooooss	1448	1464	15	1002
673140	CCCGGCCCCTAGCGCGC	ek-d8-ekekeee	sosssssssssooooss	1449	1465	24	1003
673141	CCCCGGCCCCTAGCGCG	ek-d8-ekekeee	sosssssssssooooss	1450	1466	27	1004
673142	GCCCCGGCCCCTAGCGC	ek-d8-ekekeee	sosssssssssooooss	1451	1467	28	1005

TABLE 42-continued

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673143	GGCCCCGGCCCCTAGCG	ek-d8-ekekeee	sosssssssssooooss	1452	1468	28	1006
673144	CGGCCCCGGCCCCTAGC	ek-d8-ekekeee	sosssssssssooooss	1453	1469	49	1007
673145	CCGGCCCCGGCCCCTAG	ek-d8-ekekeee	sosssssssssooooss	1454	1470	24	1008
673146	CCCGGCCCCGGCCCCTA	ek-d8-ekekeee	sosssssssssooooss	1455	1471	32	1009
673147	ACGCCCCGGCCCCGGCC	ek-d8-ekekeee	sosssssssssooooss	1465	1481	12	1010
673148	CACGCCCCGGCCCCGGC	ek-d8-ekekeee	sosssssssssooooss	1466	1482	4	1011
673149	CCACGCCCCGGCCCCGG	ek-d8-ekekeee	sosssssssssooooss	1467	1483	18	1012
673150	ACCACGCCCCGGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1468	1484	5	1013
673151	GACCACGCCCCGGCCCC	ek-d8-ekekeee	sosssssssssooooss	1469	1485	20	1014
673152	CGACCACGCCCCGGCCC	ek-d8-ekekeee	sosssssssssooooss	1470	1486	52	1015
673153	CCGACCACGCCCCGGCC	ek-d8-ekekeee	sosssssssssooooss	1471	1487	4	1016
673154	CCCGACCACGCCCCGGC	ek-d8-ekekeee	sosssssssssooooss	1472	1488	9	1017
673155	CCCCGACCACGCCCCGG	ek-d8-ekekeee	sosssssssssooooss	1473	1489	23	1018
673156	GCCCCGACCACGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1474	1490	9	1019
673157	CGCCCCGACCACGCCCC	ek-d8-ekekeee	sosssssssssooooss	1475	1491	22	1020
673158	CCGCCCCGACCACGCCC	ek-d8-ekekeee	sosssssssssooooss	1476	1492	27	1021
673159	CCCGCCCCGACCACGCC	ek-d8-ekekeee	sosssssssssooooss	1477	1493	49	1022
673160	GCCCGCCCCGACCACGC	ek-d8-ekekeee	sosssssssssooooss	1478	1494	33	1023
673161	GGCCCGCCCCGACCACG	ek-d8-ekekeee	sosssssssssooooss	1479	1495	7	1024
673162	GGGCCCGCCCCGACCAC	ek-d8-ekekeee	sosssssssssooooss	1480	1496	8	1025
673163	CGGGCCCGCCCCGACCA	ek-d8-ekekeee	sosssssssssooooss	1481	1497	12	1026
673164	CCGGGCCCGCCCCGACC	ek-d8-ekekeee	sosssssssssooooss	1482	1498	2	1027
673165	CCCGGGCCCGCCCCGAC	ek-d8-ekekeee	sosssssssssooooss	1483	1499	13	1028
673166	GCAGCCCCGCCCCGGGC	ek-d8-ekekeee	sosssssssssooooss	1505	1521	7	1029
673167	CGCAGCCCCGCCCCGGG	ek-d8-ekekeee	sosssssssssooooss	1506	1522	14	1030
673168	CCGCAGCCCCGCCCCGG	ek-d8-ekekeee	sosssssssssooooss	1507	1523	17	1031
673169	ACCGCAGCCCCGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1508	1524	44	1032
673170	AACCGCAGCCCCGCCCC	ek-d8-ekekeee	sosssssssssooooss	1509	1525	40	1033
673171	CAACCGCAGCCCCGCCC	ek-d8-ekekeee	sosssssssssooooss	1510	1526	45	1034
673172	GCAACCGCAGCCCCGCC	ek-d8-ekekeee	sosssssssssooooss	1511	1527	28	1035
673173	CGCAACCGCAGCCCCGC	ek-d8-ekekeee	sosssssssssooooss	1512	1528	31	1036
673174	CCGCAACCGCAGCCCCG	ek-d8-ekekeee	sosssssssssooooss	1513	1529	38	1037
673175	ACCGCAACCGCAGCCCC	ek-d8-ekekeee	sosssssssssooooss	1514	1530	47	1038
673176	CACCGCAACCGCAGCCC	ek-d8-ekekeee	sosssssssssooooss	1515	1531	37	1039

TABLE 42-continued

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673177	GCACCGCAACCGCAGCC	ek-d8-ekekeee	sosssssssssooooss	1516	1532	41	1040
673178	GGCACCGCAACCGCAGC	ek-d8-ekekeee	sosssssssssooooss	1517	1533	34	1041
673179	AGGCACCGCAACCGCAG	ek-d8-ekekeee	sosssssssssooooss	1518	1534	19	1042
673180	CAGGCACCGCAACCGCA	ek-d8-ekekeee	sosssssssssooooss	1519	1535	36	1043
673181	GCAGGCACCGCAACCGC	ek-d8-ekekeee	sosssssssssooooss	1520	1536	33	1044
673182	CGCAGGCACCGCAACCG	ek-d8-ekekeee	sosssssssssooooss	1521	1537	37	1045
673183	GCGCAGGCACCGCAACC	ek-d8-ekekeee	sosssssssssooooss	1522	1538	6	1046
673184	GGCGCAGGCACCGCAAC	ek-d8-ekekeee	sosssssssssooooss	1523	1539	11	1047

TABLE 43

Percent inhibition of the C90RF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
576816	GCCTTACTCTAGGACCAAGA	eeeee-d10-eeeee	sssssssssssssssssss	7990	8009	79	20
673185	TGAGAGCAAGTAGTGGG	keke-d8-ekeke	soosssssssssooss	1326	1342	37	898
673186	GTGAGAGCAAGTAGTGG	keke-d8-ekeke	soosssssssssooss	1327	1343	39	899
673187	TGTGAGAGCAAGTAGTG	keke-d8-ekeke	soosssssssssooss	1328	1344	33	900
673188	CTGTGAGAGCAAGTAGT	keke-d8-ekeke	soosssssssssooss	1329	1345	40	901
673189	ACTGTGAGAGCAAGTAG	keke-d8-ekeke	soosssssssssooss	1330	1346	26	902
673190	TACTGTGAGAGCAAGTA	keke-d8-ekeke	soosssssssssooss	1331	1347	23	903
673191	GTACTGTGAGAGCAAGT	keke-d8-ekeke	soosssssssssooss	1332	1348	50	904
673192	AGTACTGTGAGAGCAAG	keke-d8-ekeke	soosssssssssooss	1333	1349	39	905
673193	GAGTACTGTGAGAGCAA	keke-d8-ekeke	soosssssssssooss	1334	1350	69	906
673194	CGAGTACTGTGAGAGCA	keke-d8-ekeke	soosssssssssooss	1335	1351	72	907
673195	GCGAGTACTGTGAGAGC	keke-d8-ekeke	soosssssssssooss	1336	1352	51	908
673196	AGCGAGTACTGTGAGAG	keke-d8-ekeke	soosssssssssooss	1337	1353	51	909
673197	CAGCGAGTACTGTGAGA	keke-d8-ekeke	soosssssssssooss	1338	1354	59	910
673198	TCAGCGAGTACTGTGAG	keke-d8-ekeke	soosssssssssooss	1339	1355	33	911
673199	CTCAGCGAGTACTGTGA	keke-d8-ekeke	soosssssssssooss	1340	1356	32	912
673200	CCTCAGCGAGTACTGTG	keke-d8-ekeke	soosssssssssooss	1341	1357	46	913
673201	CCCTCAGCGAGTACTGT	keke-d8-ekeke	soosssssssssooss	1342	1358	53	914
673202	ACCCTCAGCGAGTACTG	keke-d8-ekeke	soosssssssssooss	1343	1359	58	915

TABLE 43-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673203	CACCCTCAGCGAGTACT	keke-d8-ekeke	soossssssssooss	1344	1360	68	916
673204	TCACCCTCAGCGAGTAC	keke-d8-ekeke	soossssssssooss	1345	1361	70	917
673205	TTCACCCTCAGCGAGTA	keke-d8-ekeke	soossssssssooss	1346	1362	47	918
673206	GTTCACCCTCAGCGAGT	keke-d8-ekeke	soossssssssooss	1347	1363	65	919
673207	TGTTCACCCTCAGCGAG	keke-d8-ekeke	soossssssssooss	1348	1364	31	920
673208	TTGTTCACCCTCAGCGA	keke-d8-ekeke	soossssssssooss	1349	1365	51	921
673209	CTTGTTCACCCTCAGCG	keke-d8-ekeke	soossssssssooss	1350	1366	49	922
673210	TCTTGTTCACCCTCAGC	keke-d8-ekeke	soossssssssooss	1351	1367	61	923
673211	TTCTTGTTCACCCTCAG	keke-d8-ekeke	soossssssssooss	1352	1368	49	924
673212	TTTCTTGTTCACCCTCA	keke-d8-ekeke	soossssssssooss	1353	1369	42	925
673213	TTTTCTTGTTCACCCTC	keke-d8-ekeke	soossssssssooss	1354	1370	36	926
673214	CTTTTCTTGTTCACCCT	keke-d8-ekeke	soossssssssooss	1355	1371	41	927
673215	TCTTTTCTTGTTCACCC	keke-d8-ekeke	soossssssssooss	1356	1372	41	928
673216	GTCTTTTCTTGTTCACC	keke-d8-ekeke	soossssssssooss	1357	1373	36	929
673217	GGTCTTTTCTTGTTCAC	keke-d8-ekeke	soossssssssooss	1358	1374	29	930
673218	AGGTCTTTTCTTGTTCA	keke-d8-ekeke	soossssssssooss	1359	1375	50	931
673219	CAGGTCTTTTCTTGTTC	keke-d8-ekeke	soossssssssooss	1360	1376	60	932
673220	TCAGGTCTTTTCTTGTT	keke-d8-ekeke	soossssssssooss	1361	1377	34	933
673221	ATCAGGTCTTTTCTTGT	keke-d8-ekeke	soossssssssooss	1362	1378	33	934
673222	TATCAGGTCTTTTCTTG	keke-d8-ekeke	soossssssssooss	1363	1379	31	935
673223	TTATCAGGTCTTTTCTT	keke-d8-ekeke	soossssssssooss	1364	1380	10	936
673224	ATCTTTATCAGGTCTTT	keke-d8-ekeke	soossssssssooss	1368	1384	61	937
673225	AATCTTTATCAGGTCTT	keke-d8-ekeke	soossssssssooss	1369	1385	74	938
673226	TAATCTTTATCAGGTCT	keke-d8-ekeke	soossssssssooss	1370	1386	62	939
673227	TTAATCTTTATCAGGTC	keke-d8-ekeke	soossssssssooss	1371	1387	51	940
673228	GTTAATCTTTATCAGGT	keke-d8-ekeke	soossssssssooss	1372	1388	73	941
673229	GGTTAATCTTTATCAGG	keke-d8-ekeke	soossssssssooss	1373	1389	66	942
673230	TGGTTAATCTTTATCAG	keke-d8-ekeke	soossssssssooss	1374	1390	38	943
673231	CTGGTTAATCTTTATCA	keke-d8-ekeke	soossssssssooss	1375	1391	41	944
673232	TCTGGTTAATCTTTATC	keke-d8-ekeke	soossssssssooss	1376	1392	37	945
673233	CCCTCCTTGTTTTCTTC	keke-d8-ekeke	soossssssssooss	1391	1407	26	946
673234	TCCCTCCTTGTTTTCTT	keke-d8-ekeke	soossssssssooss	1392	1408	10	947
673235	TTCCCTCCTTGTTTTCT	keke-d8-ekeke	soossssssssooss	1393	1409	20	948
673236	TTTCCCTCCTTGTTTTC	keke-d8-ekeke	soossssssssooss	1394	1410	1	949

TABLE 43-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673237	GTTTCCCTCCTTGTTTT	keke-d8-ekeke	soossssssssooss	1395	1411	7	950
673238	TGTTTCCCTCCTTGTTT	keke-d8-ekeke	soossssssssooss	1396	1412	26	951
673239	TTGTTTCCCTCCTTGTT	keke-d8-ekeke	soossssssssooss	1397	1413	28	952
673240	GGTTGTTTCCCTCCTTG	keke-d8-ekeke	soossssssssooss	1399	1415	70	953
673241	CGGTTGTTTCCCTCCTT	keke-d8-ekeke	soossssssssooss	1400	1416	36	954
673242	GCGGTTGTTTCCCTCCT	keke-d8-ekeke	soossssssssooss	1401	1417	21	955
673243	TGCGGTTGTTTCCCTCC	keke-d8-ekeke	soossssssssooss	1402	1418	24	956
673244	CTGCGGTTGTTTCCCTC	keke-d8-ekeke	soossssssssooss	1403	1419	41	957
673245	GCTGCGGTTGTTTCCCT	keke-d8-ekeke	soossssssssooss	1404	1420	23	958
673246	GGCTGCGGTTGTTTCCC	keke-d8-ekeke	soossssssssooss	1405	1421	39	959
673247	AGGCTGCGGTTGTTTCC	keke-d8-ekeke	soossssssssooss	1406	1422	50	960
673248	CAGGCTGCGGTTGTTTC	keke-d8-ekeke	soossssssssooss	1407	1423	32	961
673249	ACAGGCTGCGGTTGTTT	keke-d8-ekeke	soossssssssooss	1408	1424	28	962
673250	TACAGGCTGCGGTTGTT	keke-d8-ekeke	soossssssssooss	1409	1425	41	963
673251	CTACAGGCTGCGGTTGT	keke-d8-ekeke	soossssssssooss	1410	1426	18	964
673252	GCTACAGGCTGCGGTTG	keke-d8-ekeke	soossssssssooss	1411	1427	28	965
673253	TGCTACAGGCTGCGGTT	keke-d8-ekeke	soossssssssooss	1412	1428	28	966
673254	TTGCTACAGGCTGCGGT	keke-d8-ekeke	soossssssssooss	1413	1429	9	967
673255	CTTGCTACAGGCTGCGG	keke-d8-ekeke	soossssssssooss	1414	1430	34	968
673256	GCTTGCTACAGGCTGCG	keke-d8-ekeke	soossssssssooss	1415	1431	28	969
673257	AGCTTGCTACAGGCTGC	keke-d8-ekeke	soossssssssooss	1416	1432	25	970
673258	GAGCTTGCTACAGGCTG	keke-d8-ekeke	soossssssssooss	1417	1433	23	971
673259	AGAGCTTGCTACAGGCT	keke-d8-ekeke	soossssssssooss	1418	1434	0	972
673260	CAGAGCTTGCTACAGGC	keke-d8-ekeke	soossssssssooss	1419	1435	4	973
673261	CCAGAGCTTGCTACAGG	keke-d8-ekeke	soossssssssooss	1420	1436	40	974
673262	TCCAGAGCTTGCTACAG	keke-d8-ekeke	soossssssssooss	1421	1437	35	975
673263	TTCCAGAGCTTGCTACA	keke-d8-ekeke	soossssssssooss	1422	1438	23	976
673264	GTTCCAGAGCTTGCTAC	keke-d8-ekeke	soossssssssooss	1423	1439	36	977
673265	AGTTCCAGAGCTTGCTA	keke-d8-ekeke	soossssssssooss	1424	1440	54	978
673266	GAGTTCCAGAGCTTGCT	keke-d8-ekeke	soossssssssooss	1425	1441	32	979
673267	TGAGTTCCAGAGCTTGC	keke-d8-ekeke	soossssssssooss	1426	1442	32	980
673268	CTGAGTTCCAGAGCTTG	keke-d8-ekeke	soossssssssooss	1427	1443	44	981
673269	CCTGAGTTCCAGAGCTT	keke-d8-ekeke	soossssssssooss	1428	1444	70	982

TABLE 43-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673270	TCCTGAGTTCCAGAGCT	keke-d8-ekeke	soossssssssooss	1429	1445	43	983
673271	CTCCTGAGTTCCAGAGC	keke-d8-ekeke	soossssssssooss	1430	1446	41	984
673272	ACTCCTGAGTTCCAGAG	keke-d8-ekeke	soossssssssooss	1431	1447	23	985
673273	GACTCCTGAGTTCCAGA	keke-d8-ekeke	soossssssssooss	1432	1448	48	986
673274	CGACTCCTGAGTTCCAG	keke-d8-ekeke	soossssssssooss	1433	1449	34	987
673275	GCGACTCCTGAGTTCCA	keke-d8-ekeke	soossssssssooss	1434	1450	64	988
673276	CGCGACTCCTGAGTTCC	keke-d8-ekeke	soossssssssooss	1435	1451	65	989
673277	GCGCGACTCCTGAGTTC	keke-d8-ekeke	soossssssssooss	1436	1452	62	990
673278	CGCGCGACTCCTGAGTT	keke-d8-ekeke	soossssssssooss	1437	1453	45	991
673279	GCGCGCGACTCCTGAGT	keke-d8-ekeke	soossssssssooss	1438	1454	52	992
673280	AGCGCGCGACTCCTGAG	keke-d8-ekeke	soossssssssooss	1439	1455	65	993
673281	TAGCGCGCGACTCCTGA	keke-d8-ekeke	soossssssssooss	1440	1456	89	994
673282	CTAGCGCGCGACTCCTG	keke-d8-ekeke	soossssssssooss	1441	1457	69	995
673283	CCTAGCGCGCGACTCCT	keke-d8-ekeke	soossssssssooss	1442	1458	68	996
673284	CCCTAGCGCGCGACTCC	keke-d8-ekeke	soossssssssooss	1443	1459	73	997
673285	CCCCTAGCGCGCGACTC	keke-d8-ekeke	soossssssssooss	1444	1460	70	998
673286	GCCCCTAGCGCGCGACT	keke-d8-ekeke	soossssssssooss	1445	1461	45	999
673287	GGCCCCTAGCGCGCGAC	keke-d8-ekeke	soossssssssooss	1446	1462	33	1000
673288	CGGCCCCTAGCGCGCGA	keke-d8-ekeke	soossssssssooss	1447	1463	29	1001
673289	CCGGCCCCTAGCGCGCG	keke-d8-ekeke	soossssssssooss	1448	1464	0	1002
673290	CCCGGCCCCTAGCGCGC	keke-d8-ekeke	soossssssssooss	1449	1465	31	1003
673291	CCCCGGCCCCTAGCGCG	keke-d8-ekeke	soossssssssooss	1450	1466	28	1004
673292	GCCCCGGCCCCTAGCGC	keke-d8-ekeke	soossssssssooss	1451	1467	12	1005
673293	GGCCCCGGCCCCTAGCG	keke-d8-ekeke	soossssssssooss	1452	1468	29	1006
673294	CGGCCCCGGCCCCTAGC	keke-d8-ekeke	soossssssssooss	1453	1469	39	1007
673295	CCGGCCCCGGCCCCTAG	keke-d8-ekeke	soossssssssooss	1454	1470	28	1008
673296	CCCGGCCCCGGCCCCTA	keke-d8-ekeke	soossssssssooss	1455	1471	4	1009
673297	ACGCCCCGGCCCCGGCC	keke-d8-ekeke	soossssssssooss	1465	1481	17	1010
673298	CACGCCCCGGCCCCGGC	keke-d8-ekeke	soossssssssooss	1466	1482	35	1011
673299	CCACGCCCCGGCCCCGG	keke-d8-ekeke	soossssssssooss	1467	1483	28	1012
673300	ACCACGCCCCGGCCCCG	keke-d8-ekeke	soossssssssooss	1468	1484	21	1013
673301	GACCACGCCCCGGCCCC	keke-d8-ekeke	soossssssssooss	1469	1485	28	1014
673302	CGACCACGCCCCGGCCC	keke-d8-ekeke	soossssssssooss	1470	1486	46	1015
673303	CCGACCACGCCCCGGCC	keke-d8-ekeke	soossssssssooss	1471	1487	40	1016

TABLE 43-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673304	CCCGACCACGCCCCGGC	keke-d8-ekeke	soossssssssooss	1472	1488	16	1017
673305	CCCCGACCACGCCCCGG	keke-d8-ekeke	soossssssssooss	1473	1489	11	1018
673306	GCCCCGACCACGCCCCG	keke-d8-ekeke	soossssssssooss	1474	1490	13	1019
673307	CGCCCCGACCACGCCCC	keke-d8-ekeke	soossssssssooss	1475	1491	43	1020
673308	CCGCCCCGACCACGCCC	keke-d8-ekeke	soossssssssooss	1476	1492	20	1021
673309	CCCGCCCCGACCACGCC	keke-d8-ekeke	soossssssssooss	1477	1493	16	1022
673310	GCCCGCCCCGACCACGC	keke-d8-ekeke	soossssssssooss	1478	1494	44	1023
673311	GGCCCGCCCCGACCACG	keke-d8-ekeke	soossssssssooss	1479	1495	33	1024
673312	GGGCCCGCCCCGACCAC	keke-d8-ekeke	soossssssssooss	1480	1496	1	1025
673313	CGGGCCCGCCCCGACCA	keke-d8-ekeke	soossssssssooss	1481	1497	0	1026
673314	CCGGGCCCGCCCCGACC	keke-d8-ekeke	soossssssssooss	1482	1498	0	1027
673315	CCCGGGCCCGCCCCGAC	keke-d8-ekeke	soossssssssooss	1483	1499	8	1028
673316	GCAGCCCCGCCCCGGGC	keke-d8-ekeke	soossssssssooss	1505	1521	31	1029
673317	CGCAGCCCCGCCCCGGG	keke-d8-ekeke	soossssssssooss	1506	1522	4	1030
673318	CCGCAGCCCCGCCCCGG	keke-d8-ekeke	soossssssssooss	1507	1523	18	1031
673319	ACCGCAGCCCCGCCCCG	keke-d8-ekeke	soossssssssooss	1508	1524	16	1032
673320	AACCGCAGCCCCGCCCC	keke-d8-ekeke	soossssssssooss	1509	1525	39	1033
673321	CAACCGCAGCCCCGCCC	keke-d8-ekeke	soossssssssooss	1510	1526	50	1034
673322	GCAACCGCAGCCCCGCC	keke-d8-ekeke	soossssssssooss	1511	1527	45	1035
673323	CGCAACCGCAGCCCCGC	keke-d8-ekeke	soossssssssooss	1512	1528	56	1036
673324	CCGCAACCGCAGCCCCG	keke-d8-ekeke	soossssssssooss	1513	1529	10	1037
673325	ACCGCAACCGCAGCCCC	keke-d8-ekeke	soossssssssooss	1514	1530	43	1038
673326	CACCGCAACCGCAGCCC	keke-d8-ekeke	soossssssssooss	1515	1531	49	1039
673327	GCACCGCAACCGCAGCC	keke-d8-ekeke	soossssssssooss	1516	1532	35	1040
673328	GGCACCGCAACCGCAGC	keke-d8-ekeke	soossssssssooss	1517	1533	24	1041
673329	AGGCACCGCAACCGCAG	keke-d8-ekeke	soossssssssooss	1518	1534	52	1042
673330	CAGGCACCGCAACCGCA	keke-d8-ekeke	soossssssssooss	1519	1535	38	1043
673331	GCAGGCACCGCAACCGC	keke-d8-ekeke	soossssssssooss	1520	1536	51	1044
673332	CGCAGGCACCGCAACCG	keke-d8-ekeke	soossssssssooss	1521	1537	59	1045
673333	GCGCAGGCACCGCAACC	keke-d8-ekeke	soossssssssooss	1522	1538	24	1046
673334	GGCGCAGGCACCGCAAC	keke-d8-ekeke	soossssssssooss	1523	1539	18	1047

TABLE 44

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
576816	GCCTTACTCTAGGACCAAGA	eeeee-d10-eeeee	sssssssssssssssssss	7990	8009	60	20
673335	TGAGAGCAAGTAGTGGG	ekek-d8-kekee	soossssssssooss	1326	1342	28	898
673336	GTGAGAGCAAGTAGTGG	ekek-d8-kekee	soossssssssooss	1327	1343	27	899
673337	TGTGAGAGCAAGTAGTG	ekek-d8-kekee	soossssssssooss	1328	1344	32	900
673338	CTGTGAGAGCAAGTAGT	ekek-d8-kekee	soossssssssooss	1329	1345	43	901
673339	ACTGTGAGAGCAAGTAG	ekek-d8-kekee	soossssssssooss	1330	1346	22	902
673340	TACTGTGAGAGCAAGTA	ekek-d8-kekee	soossssssssooss	1331	1347	20	903
673341	GTACTGTGAGAGCAAGT	ekek-d8-kekee	soossssssssooss	1332	1348	53	904
673342	AGTACTGTGAGAGCAAG	ekek-d8-kekee	soossssssssooss	1333	1349	20	905
673343	GAGTACTGTGAGAGCAA	ekek-d8-kekee	soossssssssooss	1334	1350	50	906
673344	CGAGTACTGTGAGAGCA	ekek-d8-kekee	soossssssssooss	1335	1351	45	907
673345	GCGAGTACTGTGAGAGC	ekek-d8-kekee	soossssssssooss	1336	1352	45	908
673346	AGCGAGTACTGTGAGAG	ekek-d8-kekee	soossssssssooss	1337	1353	53	909
673347	CAGCGAGTACTGTGAGA	ekek-d8-kekee	soossssssssooss	1338	1354	35	910
673348	TCAGCGAGTACTGTGAG	ekek-d8-kekee	soossssssssooss	1339	1355	36	911
673349	CTCAGCGAGTACTGTGA	ekek-d8-kekee	soossssssssooss	1340	1356	19	912
673350	CCTCAGCGAGTACTGTG	ekek-d8-kekee	soossssssssooss	1341	1357	21	913
673351	CCCTCAGCGAGTACTGT	ekek-d8-kekee	soossssssssooss	1342	1358	46	914
673352	ACCCTCAGCGAGTACTG	ekek-d8-kekee	soossssssssooss	1343	1359	43	915
673353	CACCCTCAGCGAGTACT	ekek-d8-kekee	soossssssssooss	1344	1360	46	916
673354	TCACCCTCAGCGAGTAC	ekek-d8-kekee	soossssssssooss	1345	1361	40	917
673355	TTCACCCTCAGCGAGTA	ekek-d8-kekee	soossssssssooss	1346	1362	33	918
673356	GTTCACCCTCAGCGAGT	ekek-d8-kekee	soossssssssooss	1347	1363	11	919
673357	TGTTCACCCTCAGCGAG	ekek-d8-kekee	soossssssssooss	1348	1364	34	920
673358	TTGTTCACCCTCAGCGA	ekek-d8-kekee	soossssssssooss	1349	1365	47	921
673359	CTTGTTCACCCTCAGCG	ekek-d8-kekee	soossssssssooss	1350	1366	54	922
673360	TCTTGTTCACCCTCAGC	ekek-d8-kekee	soossssssssooss	1351	1367	26	923
673361	TTCTTGTTCACCCTCAG	ekek-d8-kekee	soossssssssooss	1352	1368	36	924
673362	TTTCTTGTTCACCCTCA	ekek-d8-kekee	soossssssssooss	1353	1369	29	925
673363	TTTTCTTGTTCACCCTC	ekek-d8-kekee	soossssssssooss	1354	1370	29	926
673364	CTTTTCTTGTTCACCCT	ekek-d8-kekee	soossssssssooss	1355	1371	23	927
673365	TCTTTTCTTGTTCACCC	ekek-d8-kekee	soossssssssooss	1356	1372	36	928
673366	GTCTTTTCTTGTTCACC	ekek-d8-kekee	soossssssssooss	1357	1373	27	929
673367	GGTCTTTTCTTGTTCAC	ekek-d8-kekee	soossssssssooss	1358	1374	21	930

TABLE 44-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673368	AGGTCTTTTCTTGTTCA	ekek-d8-kekee	soossssssssooss	1359	1375	29	931
673369	CAGGTCTTTTCTTGTTC	ekek-d8-kekee	soossssssssooss	1360	1376	65	932
673370	TCAGGTCTTTTCTTGTT	ekek-d8-kekee	soossssssssooss	1361	1377	2	933
673371	ATCAGGTCTTTTCTTGT	ekek-d8-kekee	soossssssssooss	1362	1378	23	934
673372	TATCAGGTCTTTTCTTG	ekek-d8-kekee	soossssssssooss	1363	1379	40	935
673373	TTATCAGGTCTTTTCTT	ekek-d8-kekee	soossssssssooss	1364	1380	13	936
673374	ATCTTTATCAGGTCTTT	ekek-d8-kekee	soossssssssooss	1368	1384	76	937
673375	AATCTTTATCAGGTCTT	ekek-d8-kekee	soossssssssooss	1369	1385	62	938
673376	TAATCTTTATCAGGTCT	ekek-d8-kekee	soossssssssooss	1370	1386	39	939
673377	TTAATCTTTATCAGGTC	ekek-d8-kekee	soossssssssooss	1371	1387	71	940
673378	GTTAATCTTTATCAGGT	ekek-d8-kekee	soossssssssooss	1372	1388	61	941
673379	GGTTAATCTTTATCAGG	ekek-d8-kekee	soossssssssooss	1373	1389	74	942
673380	TGGTTAATCTTTATCAG	ekek-d8-kekee	soossssssssooss	1374	1390	24	943
673381	CTGGTTAATCTTTATCA	ekek-d8-kekee	soossssssssooss	1375	1391	32	944
673382	TCTGGTTAATCTTTATC	ekek-d8-kekee	soossssssssooss	1376	1392	38	945
673383	CCCTCCTTGTTTTCTTC	ekek-d8-kekee	soossssssssooss	1391	1407	21	946
673384	TCCCTCCTTGTTTTCTT	ekek-d8-kekee	soossssssssooss	1392	1408	0	947
673385	TTCCCTCCTTGTTTTCT	ekek-d8-kekee	soossssssssooss	1393	1409	0	948
673386	TTTCCCTCCTTGTTTTC	ekek-d8-kekee	soossssssssooss	1394	1410	5	949
673387	GTTTCCCTCCTTGTTTT	ekek-d8-kekee	soossssssssooss	1395	1411	0	950
673388	TGTTTCCCTCCTTGTTT	ekek-d8-kekee	soossssssssooss	1396	1412	0	951
673389	TTGTTTCCCTCCTTGTT	ekek-d8-kekee	soossssssssooss	1397	1413	22	952
673390	GGTTGTTTCCCTCCTTG	ekek-d8-kekee	soossssssssooss	1399	1415	55	953
673391	CGGTTGTTTCCCTCCTT	ekek-d8-kekee	soossssssssooss	1400	1416	25	954
673392	GCGGTTGTTTCCCTCCT	ekek-d8-kekee	soossssssssooss	1401	1417	19	955
673393	TGCGGTTGTTTCCCTCC	ekek-d8-kekee	soossssssssooss	1402	1418	0	956
673394	CTGCGGTTGTTTCCCTC	ekek-d8-kekee	soossssssssooss	1403	1419	13	957
673395	GCTGCGGTTGTTTCCCT	ekek-d8-kekee	soossssssssooss	1404	1420	19	958
673396	GGCTGCGGTTGTTTCCC	ekek-d8-kekee	soossssssssooss	1405	1421	27	959
673397	AGGCTGCGGTTGTTTCC	ekek-d8-kekee	soossssssssooss	1406	1422	13	960
673398	CAGGCTGCGGTTGTTTC	ekek-d8-kekee	soossssssssooss	1407	1423	22	961
673399	ACAGGCTGCGGTTGTTT	ekek-d8-kekee	soossssssssooss	1408	1424	5	962
673400	TACAGGCTGCGGTTGTT	ekek-d8-kekee	soossssssssooss	1409	1425	0	963

TABLE 44-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673401	CTACAGGCTGCGGTTGT	ekek-d8-kekee	soossssssssooss	1410	1426	0	964
673402	GCTACAGGCTGCGGTTG	ekek-d8-kekee	soossssssssooss	1411	1427	39	965
673403	TGCTACAGGCTGCGGTT	ekek-d8-kekee	soossssssssooss	1412	1428	20	966
673404	TTGCTACAGGCTGCGGT	ekek-d8-kekee	soossssssssooss	1413	1429	24	967
673405	CTTGCTACAGGCTGCGG	ekek-d8-kekee	soossssssssooss	1414	1430	0	968
673406	GCTTGCTACAGGCTGCG	ekek-d8-kekee	soossssssssooss	1415	1431	18	969
673407	AGCTTGCTACAGGCTGC	ekek-d8-kekee	soossssssssooss	1416	1432	3	970
673408	GAGCTTGCTACAGGCTG	ekek-d8-kekee	soossssssssooss	1417	1433	13	971
673409	AGAGCTTGCTACAGGCT	ekek-d8-kekee	soossssssssooss	1418	1434	29	972
673410	CAGAGCTTGCTACAGGC	ekek-d8-kekee	soossssssssooss	1419	1435	22	973
673411	CCAGAGCTTGCTACAGG	ekek-d8-kekee	soossssssssooss	1420	1436	24	974
673412	TCCAGAGCTTGCTACAG	ekek-d8-kekee	soossssssssooss	1421	1437	4	975
673413	TTCCAGAGCTTGCTACA	ekek-d8-kekee	soossssssssooss	1422	1438	0	976
673414	GTTCCAGAGCTTGCTAC	ekek-d8-kekee	soossssssssooss	1423	1439	19	977
673415	AGTTCCAGAGCTTGCTA	ekek-d8-kekee	soossssssssooss	1424	1440	0	978
673416	GAGTTCCAGAGCTTGCT	ekek-d8-kekee	soossssssssooss	1425	1441	48	979
673417	TGAGTTCCAGAGCTTGC	ekek-d8-kekee	soossssssssooss	1426	1442	14	980
673418	CTGAGTTCCAGAGCTTG	ekek-d8-kekee	soossssssssooss	1427	1443	37	981
673419	CCTGAGTTCCAGAGCTT	ekek-d8-kekee	soossssssssooss	1428	1444	80	982
673420	TCCTGAGTTCCAGAGCT	ekek-d8-kekee	soossssssssooss	1429	1445	26	983
673421	CTCCTGAGTTCCAGAGC	ekek-d8-kekee	soossssssssooss	1430	1446	5	984
673422	ACTCCTGAGTTCCAGAG	ekek-d8-kekee	soossssssssooss	1431	1447	23	985
673423	GACTCCTGAGTTCCAGA	ekek-d8-kekee	soossssssssooss	1432	1448	37	986
673424	CGACTCCTGAGTTCCAG	ekek-d8-kekee	soossssssssooss	1433	1449	5	987
673425	GCGACTCCTGAGTTCCA	ekek-d8-kekee	soossssssssooss	1434	1450	39	988
673426	CGCGACTCCTGAGTTCC	ekek-d8-kekee	soossssssssooss	1435	1451	46	989
673427	GCGCGACTCCTGAGTTC	ekek-d8-kekee	soossssssssooss	1436	1452	50	990
673428	CGCGCGACTCCTGAGTT	ekek-d8-kekee	soossssssssooss	1437	1453	19	991
673429	GCGCGCGACTCCTGAGT	ekek-d8-kekee	soossssssssooss	1438	1454	13	992
673430	AGCGCGCGACTCCTGAG	ekek-d8-kekee	soossssssssooss	1439	1455	51	993
673431	TAGCGCGCGACTCCTGA	ekek-d8-kekee	soossssssssooss	1440	1456	83	994
673432	CTAGCGCGCGACTCCTG	ekek-d8-kekee	soossssssssooss	1441	1457	60	995
673433	CCTAGCGCGCGACTCCT	ekek-d8-kekee	soossssssssooss	1442	1458	37	996
673434	CCCTAGCGCGCGACTCC	ekek-d8-kekee	soossssssssooss	1443	1459	60	997

TABLE 44-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673435	CCCCTAGCGCGCGACTC	ekek-d8-kekee	soossssssssooss	1444	1460	62	998
673436	GCCCCTAGCGCGCGACT	ekek-d8-kekee	soossssssssooss	1445	1461	41	999
673437	GGCCCCTAGCGCGCGAC	ekek-d8-kekee	soossssssssooss	1446	1462	8	1000
673438	CGGCCCCTAGCGCGCGA	ekek-d8-kekee	soossssssssooss	1447	1463	31	1001
673439	CCGGCCCCTAGCGCGCG	ekek-d8-kekee	soossssssssooss	1448	1464	18	1002
673440	CCCGGCCCCTAGCGCGC	ekek-d8-kekee	soossssssssooss	1449	1465	6	1003
673441	CCCCGGCCCCTAGCGCG	ekek-d8-kekee	soossssssssooss	1450	1466	23	1004
673442	GCCCCGGCCCCTAGCGC	ekek-d8-kekee	soossssssssooss	1451	1467	8	1005
673443	GGCCCCGGCCCCTAGCG	ekek-d8-kekee	soossssssssooss	1452	1468	18	1006
673444	CGGCCCCGGCCCCTAGC	ekek-d8-kekee	soossssssssooss	1453	1469	28	1007
673445	CCGGCCCCGGCCCCTAG	ekek-d8-kekee	soossssssssooss	1454	1470	9	1008
673446	CCCGGCCCCGGCCCCTA	ekek-d8-kekee	soossssssssooss	1455	1471	5	1009
673447	ACGCCCCGGCCCCGGCC	ekek-d8-kekee	soossssssssooss	1465	1481	23	1010
673448	CACGCCCCGGCCCCGGC	ekek-d8-kekee	soossssssssooss	1466	1482	14	1011
673449	CCACGCCCCGGCCCCGG	ekek-d8-kekee	soossssssssooss	1467	1483	35	1012
673450	ACCACGCCCCGGCCCCG	ekek-d8-kekee	soossssssssooss	1468	1484	30	1013
673451	GACCACGCCCCGGCCCC	ekek-d8-kekee	soossssssssooss	1469	1485	0	1014
673452	CGACCACGCCCCGGCCC	ekek-d8-kekee	soossssssssooss	1470	1486	15	1015
673453	CCGACCACGCCCCGGCC	ekek-d8-kekee	soossssssssooss	1471	1487	42	1016
673454	CCCGACCACGCCCCGGC	ekek-d8-kekee	soossssssssooss	1472	1488	19	1017
673455	CCCCGACCACGCCCCGG	ekek-d8-kekee	soossssssssooss	1473	1489	21	1018
673456	GCCCCGACCACGCCCCG	ekek-d8-kekee	soossssssssooss	1474	1490	9	1019
673457	CGCCCCGACCACGCCCC	ekek-d8-kekee	soossssssssooss	1475	1491	45	1020
673458	CCGCCCCGACCACGCCC	ekek-d8-kekee	soossssssssooss	1476	1492	14	1021
673459	CCCGCCCCGACCACGCC	ekek-d8-kekee	soossssssssooss	1477	1493	2	1022
673460	GCCCGCCCCGACCACGC	ekek-d8-kekee	soossssssssooss	1478	1494	28	1023
673461	GGCCCGCCCCGACCACG	ekek-d8-kekee	soossssssssooss	1479	1495	19	1024
673462	GGGCCCGCCCCGACCAC	ekek-d8-kekee	soossssssssooss	1480	1496	26	1025
673463	CGGGCCCGCCCCGACCA	ekek-d8-kekee	soossssssssooss	1481	1497	12	1026
673464	CCGGGCCCGCCCCGACC	ekek-d8-kekee	soossssssssooss	1482	1498	18	1027
673465	CCCGGGCCCGCCCCGAC	ekek-d8-kekee	soossssssssooss	1483	1499	19	1028
673466	GCAGCCCCGCCCCGGGC	ekek-d8-kekee	soossssssssooss	1505	1521	11	1029
673467	CGCAGCCCCGCCCCGGG	ekek-d8-kekee	soossssssssooss	1506	1522	40	1030

TABLE 44-continued

Percent inhibition of the C9ORF72 pathogenic associated mRNA variant compared to PBS control by Deoxy, MOE and cEt antisense oligonucleotides with mixed backbones targeting SEQ ID NO: 2

ISIS NO	Sequence	Chemistry	Linkage	SEQ ID NO: 2 Start Site	SEQ ID NO: 2 Stop Site	% inhibition	SEQ ID NO
673468	CCGCAGCCCCGCCCCGG	ekek-d8-kekee	soossssssssooss	1507	1523	12	1031
673469	ACCGCAGCCCCGCCCCG	ekek-d8-kekee	soossssssssooss	1508	1524	26	1032
673470	AACCGCAGCCCCGCCCC	ekek-d8-kekee	soossssssssooss	1509	1525	36	1033
673471	CAACCGCAGCCCCGCCC	ekek-d8-kekee	soossssssssooss	1510	1526	63	1034
673472	GCAACCGCAGCCCCGCC	ekek-d8-kekee	soossssssssooss	1511	1527	35	1035
673473	CGCAACCGCAGCCCCGC	ekek-d8-kekee	soossssssssooss	1512	1528	51	1036
673474	CCGCAACCGCAGCCCCG	ekek-d8-kekee	soossssssssooss	1513	1529	27	1037
673475	ACCGCAACCGCAGCCCC	ekek-d8-kekee	soossssssssooss	1514	1530	49	1038
673476	CACCGCAACCGCAGCCC	ekek-d8-kekee	soossssssssooss	1515	1531	34	1039
673477	GCACCGCAACCGCAGCC	ekek-d8-kekee	soossssssssooss	1516	1532	36	1040
673478	GGCACCGCAACCGCAGC	ekek-d8-kekee	soossssssssooss	1517	1533	22	1041
673479	AGGCACCGCAACCGCAG	ekek-d8-kekee	soossssssssooss	1518	1534	23	1042
673480	CAGGCACCGCAACCGCA	ekek-d8-kekee	soossssssssooss	1519	1535	27	1043
673481	GCAGGCACCGCAACCGC	ekek-d8-kekee	soossssssssooss	1520	1536	41	1044
673482	CGCAGGCACCGCAACCG	ekek-d8-kekee	soossssssssooss	1521	1537	60	1045
673483	GCGCAGGCACCGCAACC	ekek-d8-kekee	soossssssssooss	1522	1538	22	1046
673484	GGCGCAGGCACCGCAAC	ekek-d8-kekee	soossssssssooss	1523	1539	11	1047

Example 11

Dose-Dependent Antisense Inhibition of Human C9ORF72 mRNA in HepG2 Cells

[0474] Antisense oligonucleotides from the study described in Example 10 hereinabove exhibiting significant in vitro inhibition of C9ORF72 mRNA were selected and tested at various doses in HepG2 cells. ISIS 576816, which was previously tested in PCT/US2013/065073 (claiming priority to U.S. Application No. 61/714,132, filed Oct. 15, 2012) was used as a benchmark oligonucleotide. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.185 μM, 0.56 μM, 1.67 μM, or 5.00 μM concentrations of antisense oligonucleotide. After a treatment period of approximately 16 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3905 was used to measure the C9ORF72 pathogenic associated mRNA variant, which is the product of a pre-mRNA containing a hexanucleotide repeat. C9ORF72 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72 levels, relative to untreated control cells.

[0475] As shown in Tables 45-52, total C9ORF72 mRNA levels were reduced in a dose-dependent manner in some of the antisense oligonucleotide treated cells.

TABLE 45

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	21	66	82	91	0.5
672893	4	50	83	82	0.8
672894	13	55	70	88	0.7
672896	13	57	81	89	0.6
672897	2	38	72	79	1.1
672902	20	40	83	88	0.7
672903	19	44	73	80	0.8
672904	16	35	49	85	1.2
672905	15	30	67	82	1.0
672908	41	49	79	83	0.4
672909	20	54	72	90	0.6
672919	31	58	69	92	0.5
672924	34	60	89	97	0.4
672925	41	58	88	94	0.3
672927	31	78	81	92	0.3
672928	30	62	79	92	0.4
672929	51	71	89	94	0.1
672932	10	54	83	88	0.7
672940	14	36	58	87	1.0

TABLE 46

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	24	51	78	88	0.6
672948	17	27	52	65	1.8
672966	1	36	77	73	1.1
672967	24	36	44	75	1.4
672968	15	46	69	83	0.8
672969	1	39	66	93	1.0
672976	47	65	74	80	0.2
672978	36	32	52	76	1.0
672980	24	45	77	86	0.6
672981	48	74	86	93	0.1
672982	42	63	91	90	0.2
672983	38	56	83	92	0.4
672984	33	53	72	88	0.5
672985	38	46	66	78	0.6
673021	43	48	76	79	0.4
673022	2	52	58	89	1.0
673023	44	36	76	78	0.5
673026	22	77	70	76	0.4
673032	19	37	55	80	1.1

TABLE 47

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	48	49	80	95	0.3
673036	38	54	73	88	0.4
673047	27	75	93	87	0.3
673050	6	66	83	82	0.6
673051	31	57	69	85	0.5
673053	17	59	76	92	0.6
673054	15	45	76	90	0.7
673057	37	67	64	81	0.3
673058	35	62	79	87	0.4
673067	59	74	98	97	<0.2
673068	37	71	85	95	0.3
673071	43	5	59	64	1.9
673074	37	44	89	89	0.4
673079	41	71	89	95	0.2
673081	21	37	80	76	0.8
673082	27	58	87	92	0.4
673088	63	79	96	97	<0.2
673089	11	41	71	83	0.9
673098	15	61	68	93	0.6

TABLE 48

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	43	71	80	93	0.2
673098	18	53	81	88	0.6
673117	22	45	76	80	0.7
673119	17	47	65	90	0.8
673125	33	64	79	79	0.4
673126	41	56	70	82	0.4
673127	46	85	92	96	<0.2
673128	32	71	88	99	0.3
673130	42	69	91	91	0.2
673131	34	62	74	99	0.4

TABLE 48-continued

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
673132	47	44	75	89	0.4
673133	54	61	78	84	<0.2
673134	41	62	77	77	0.3
673135	28	60	77	82	0.5
673144	24	58	64	92	0.6
673152	18	59	70	72	0.7
673159	4	50	75	80	0.9
673171	17	43	58	86	0.9
673175	30	45	76	78	0.6

TABLE 49

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	22	69	78	93	0.4
673193	60	74	90	89	<0.2
673194	15	54	75	77	0.7
673196	36	38	71	73	0.7
673197	28	39	68	78	0.8
673201	0	40	69	91	1.0
673202	9	50	77	89	0.7
673203	40	52	84	98	0.4
673204	44	67	91	92	0.2
673206	27	40	70	90	0.7
673210	22	43	79	94	0.6
673211	0	45	53	85	1.2
673219	27	36	67	88	0.8
673224	41	65	86	95	0.3
673225	34	73	78	97	0.3
673226	19	59	83	94	0.5
673228	8	67	79	94	0.6
673229	46	76	89	86	<0.2
673240	18	58	75	93	0.6

TABLE 50

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	31	65	81	89	0.4
673265	24	43	73	83	0.7
673269	12	58	81	89	0.6
673275	31	57	63	94	0.5
673276	33	56	69	91	0.5
673277	37	51	65	66	0.6
673279	11	57	68	86	0.8
673280	38	60	80	95	0.3
673281	51	83	92	86	<0.2
673282	60	73	93	95	<0.2
673283	59	66	94	96	<0.2
673284	45	59	78	91	0.3
673285	30	59	78	86	0.4
673321	10	44	72	79	0.9
673323	43	54	76	86	0.3
673326	0	46	72	81	0.8
673329	15	30	64	76	1.2
673331	47	40	66	79	0.5
673332	58	49	71	78	<0.2

TABLE 51

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	38	61	75	92	0.3
673338	0	13	48	61	2.6
673341	29	39	66	82	0.8
673343	38	33	58	81	0.8
673344	29	40	69	72	0.8
673345	24	11	51	66	2.2
673346	19	27	52	74	1.4
673351	17	40	61	82	1.0
673352	18	36	62	71	1.2
673353	29	38	47	74	1.2
673358	11	39	63	71	1.2
673359	15	46	51	65	1.4
673369	17	33	55	70	1.4
673374	42	62	77	87	0.3
673375	28	66	79	94	0.4
673377	32	51	77	87	0.5
673378	32	47	76	89	0.5
673379	33	58	76	83	0.4
673390	21	40	57	74	1.1

TABLE 52

Dose-dependent inhibition of the C9ORF72 pathogenic associated mRNA variant transcript levels in HepG2 cells

ISIS No	0.185 μM	0.56 μM	1.67 μM	5.00 μM	IC_{50} (μM)
576816	6	54	75	88	0.7
673416	31	51	61	67	0.7
673419	16	34	41	67	2.0
673426	31	62	53	68	0.7
673427	41	52	52	59	0.8
673430	27	46	76	83	0.6
673431	49	68	83	96	0.2
673432	43	72	72	86	0.2
673434	41	70	80	90	0.2
673435	8	48	71	69	1.0
673436	15	20	65	66	1.6
673453	18	49	57	72	1.0
673457	0	19	43	63	2.5
673467	12	25	35	42	>5.0
673471	13	45	57	79	1.0
673473	13	48	62	92	0.8
673475	23	30	65	61	1.4
673481	26	33	52	41	>5.0
673482	14	45	56	75	1.1

Example 12

Antisense Inhibition of C9ORF72 by Human-Rhesus Cross-Reactive Antisense Oligonucleotides in LLC-MK2 Cells by Mixed Backbone 5-8-5 MOE and 5-10-5 MOE Gapmers

[0476] Antisense oligonucleotides targeting a human C9ORF72 nucleic acid and cross-reactive with a rhesus C9ORF72 nucleic acid were designed and tested for their effects on rhesus C9ORF72 mRNA expression in vitro. ISIS 576816, previously tested in U.S. Application No. 61/714,132, filed Oct. 15, 2012, was used as a benchmark oligonucleotide. ISIS 619420, which is the mixed backbone version of ISIS 576816, described in Example 2 hereinabove was also tested. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cultured LLC-MK2 cells at a density of 20,000 cells per well were transfected using electroporation with 3,500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3750 (a TAQ-man primer probe set) was used to measure total C9ORF72 mRNA levels. RTS3750 targets exon 2 of the mRNA transcripts and, therefore, measures total mRNA transcripts. In cases where the oligonucleotide overlapped the amplicon of the primer probe set RTS3750 (see, e.g., Table 53), an alternative primer probe set, RTS3760_MGB (forward sequence TTCCAATGCTTACTGGAGAAGTGA, designated herein as SEQ ID NO: 1546; reverse sequence GGAACACTGTGTGATTTCATAGATGA, designated herein as SEQ ID NO: 1547; probe sequence TCCTGTAATGGAACTGC, designated herein as SEQ ID NO: 1548—a TAQ-man primer probe set) was used to measure total mRNA transcripts. The levels of the C9ORF72 mRNA were normalized to the total RNA content of the cell, as measured by RIBOGREEN®. Results are presented as percent inhibition of rhesus C9ORF72 mRNA expression, relative to untreated control cells. The oligonucleotides marked with as asterisk (*) target the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of these oligonucleotides. 'n.d.' indicates that there was no signal reading in the assay for that particular oligonucleotide.

[0477] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as 5-8-5 MOE gapmers and 5-10-5 MOE gapmers.

[0478] The 5-8-5 MOE gapmers are 18 nucleosides in length, wherein the central gap segment comprises eight 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment has a 2'-MOE group. All cytosine residues throughout each gapmer are 5-methylcytosines. The internucleoside linkages for the gapmers are mixed phosphorothioate and phosphodiester linkages. The internucleoside linkages for each gapmer are presented in the Linkage column, where 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage.

[0479] The 5-10-5 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises ten 2'-deoxynucleosides and is flanked by wing segments on both the 5' end and the 3' end comprising five nucleosides each. Each nucleoside in the 5' wing segment and each nucleoside in the 3' wing segment has a 2'-MOE group. All cytosine residues throughout each gapmer are 5-methylcytosines. The internucleoside linkages for the gapmers are mixed phosphorothioate and phosphodiester linkages. The internucleoside linkages for each gapmer are presented in the Linkage column, where 'o' indicates a phosphodiester linkage and 's' indicates a phosphorothioate linkage.

[0480] "Start site" indicates the 5'-most nucleoside to which the gapmer is targeted in the gene sequence. "Stop site" indicates the 3'-most nucleoside to which the gapmer is targeted in the gene sequence. Each gapmer listed in the Tables below is targeted to one or more of human C9ORF72 mRNA sequence, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_001256054.1), human C9ORF72 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NT_008413.18

truncated from nucleosides 27535000 to 27565000), and rhesus C9ORF72 genomic sequence designated herein as SEQ ID NO: 19 (GENBANK Accession No. NW_001101662.1 truncated from nucleosides 8522000 to 8552000). The 'Mismatches' column indicates the number of mismatches the human antisense oligonucleotide has with the rhesus genomic sequence. 'n/a' in the rhesus sequence columns indicates that the human oligonucleotide has more than 3 mismatches with the rhesus genomic sequence. Where the 'Mismtaches' column is not provided in a Table, it is understood that the human oligonucleotides of the Table are fully cross-reactive with the rhesus genomic sequence.

TABLE 53

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	% inhibition (RTS3760_MGB)	SEQ ID NO
576816	310	7990	8043	GCCTTACTCTAGGACCAAGA	sssssssssssssssss	5-10-5	60	54	20
619420	310	7990	8043	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	5-10-5	69	68	20
688005*	221	7901	7954	AACAGCTGGAGATGGCGG	sooossssssssssooss	5-8-5	12	36	1057
688006*	222	7902	7955	CAACAGCTGGAGATGGCG	sooossssssssssooss	5-8-5	29	20	1058
688007*	223	7903	7956	GCAACAGCTGGAGATGGC	sooossssssssssooss	5-8-5	46	33	1059
688008*	224	7904	7957	GGCAACAGCTGGAGATGG	sooossssssssssooss	5-8-5	59	41	1060
688009*	225	7905	7958	TGGCAACAGCTGGAGATG	sooossssssssssooss	5-8-5	43	24	1061
688010*	226	7906	7959	TTGGCAACAGCTGGAGAT	sooossssssssssooss	5-8-5	31	0	1062
688011*	227	7907	7960	CTTGGCAACAGCTGGAGA	sooossssssssssooss	5-8-5	45	18	1063
688012*	228	7908	7961	TCTTGGCAACAGCTGGAG	sooossssssssssooss	5-8-5	67	0	1064
688013*	229	7909	7962	GTCTTGGCAACAGCTGGA	sooossssssssssooss	5-8-5	48	13	1065
688014*	230	7910	7963	TGTCTTGGCAACAGCTGG	sooossssssssssooss	5-8-5	16	16	1066
688015*	231	7911	7964	CTGTCTTGGCAACAGCTG	sooossssssssssooss	5-8-5	75	33	1067
688016*	232	7912	7965	TCTGTCTTGGCAACAGCT	sooossssssssssooss	5-8-5	37	36	1068
688017*	233	7913	7966	CTCTGTCTTGGCAACAGC	sooossssssssssooss	5-8-5	59	28	1069
688018*	234	7914	7967	TCTCTGTCTTGGCAACAG	sooossssssssssooss	5-8-5	40	36	1070
688019*	235	7915	7968	ATCTCTGTCTTGGCAACA	sooossssssssssooss	5-8-5	34	31	1071
688020*	236	7916	7969	AATCTCTGTCTTGGCAAC	sooossssssssssooss	5-8-5	30	26	1072
688021*	237	7917	7970	CAATCTCTGTCTTGGCAA	sooossssssssssooss	5-8-5	58	0	1073
688022*	238	7918	7971	GCAATCTCTGTCTTGGCA	sooossssssssssooss	5-8-5	90	52	1074
688023*	239	7919	7972	AGCAATCTCTGTCTTGGC	sooossssssssssooss	5-8-5	92	73	1075
688024*	240	7920	7973	AAGCAATCTCTGTCTTGG	sooossssssssssooss	5-8-5	74	24	1076
688025*	241	7921	7974	AAAGCAATCTCTGTCTTG	sooossssssssssooss	5-8-5	76	0	1077
688026*	242	7922	7975	TAAAGCAATCTCTGTCTT	sooossssssssssooss	5-8-5	28	6	1078
688027*	243	7923	7976	TTAAAGCAATCTCTGTCT	sooossssssssssooss	5-8-5	32	18	1079
688028*	244	7924	7977	CTTAAAGCAATCTCTGTC	sooossssssssssooss	5-8-5	41	4	1080
688029*	245	7925	7978	ACTTAAAGCAATCTCTGT	sooossssssssssooss	5-8-5	63	0	1081
688030*	246	7926	7979	CACTTAAAGCAATCTCTG	sooossssssssssooss	5-8-5	26	31	1082
688031*	247	7927	7980	CCACTTAAAGCAATCTCT	sooossssssssssooss	5-8-5	73	33	1083
688032	267	7947	8000	CTGCTAATAAAGGTGATT	sooossssssssssooss	5-8-5	24	34	1084

TABLE 53-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	% inhibition (RTS3760_MGB)	SEQ ID NO
688033	268	7948	8001	GCTGCTAATAAAGGTGAT	soooossssssssooss	5-8-5	11	18	1085
688034	269	7949	8002	AGCTGCTAATAAAGGTGA	soooossssssssooss	5-8-5	40	34	1086
688035	270	7950	8003	TAGCTGCTAATAAAGGTG	soooossssssssooss	5-8-5	46	1	1087
688036	271	7951	8004	GTAGCTGCTAATAAAGGT	soooossssssssooss	5-8-5	13	17	1088
688037	272	7952	8005	AGTAGCTGCTAATAAAGG	soooossssssssooss	5-8-5	0	0	1089
688038	273	7953	8006	AAGTAGCTGCTAATAAAG	soooossssssssooss	5-8-5	0	0	1090
688039	274	7954	8007	AAAGTAGCTGCTAATAAA	soooossssssssooss	5-8-5	0	3	1091
688040	275	7955	8008	AAAAGTAGCTGCTAATAA	soooossssssssooss	5-8-5	0	0	1092
688041	276	7956	8009	CAAAAGTAGCTGCTAATA	soooossssssssooss	5-8-5	0	0	1093
688042	277	7957	8010	GCAAAAGTAGCTGCTAAT	soooossssssssooss	5-8-5	32	31	1094
688043	278	7958	8011	AGCAAAAGTAGCTGCTAA	soooossssssssooss	5-8-5	18	14	1095
688044	279	7959	8012	AAGCAAAAGTAGCTGCTA	soooossssssssooss	5-8-5	17	24	1096
688045	280	7960	8013	TAAGCAAAAGTAGCTGCT	soooossssssssooss	5-8-5	38	7	1097
688046	281	7961	8014	GTAAGCAAAAGTAGCTGC	soooossssssssooss	5-8-5	37	38	1098
688047	282	7962	8015	AGTAAGCAAAAGTAGCTG	soooossssssssooss	5-8-5	12	33	1099
688048	283	7963	8016	CAGTAAGCAAAAGTAGCT	soooossssssssooss	5-8-5	17	29	1100
688049	284	7964	8017	CCAGTAAGCAAAAGTAGC	soooossssssssooss	5-8-5	17	0	1101
688050	285	7965	8018	CCCAGTAAGCAAAAGTAG	soooossssssssooss	5-8-5	7	27	1102
688051	286	7966	8019	TCCCAGTAAGCAAAAGTA	soooossssssssooss	5-8-5	0	0	1103
688052	287	7967	8020	GTCCCAGTAAGCAAAAGT	soooossssssssooss	5-8-5	16	24	1104
688053	288	7968	8021	TGTCCCAGTAAGCAAAAG	soooossssssssooss	5-8-5	0	0	1105
688054	289	7969	8022	TTGTCCCAGTAAGCAAAA	soooossssssssooss	5-8-5	9	10	1106
688055	290	7970	8023	ATTGTCCCAGTAAGCAAA	soooossssssssooss	5-8-5	6	15	1107
688056	291	7971	8024	TATTGTCCCAGTAAGCAA	soooossssssssooss	5-8-5	21	0	1108
688057	292	7972	8025	ATATTGTCCCAGTAAGCA	soooossssssssooss	5-8-5	11	4	1109
688058	293	7973	8026	AATATTGTCCCAGTAAGC	soooossssssssooss	5-8-5	23	29	1110
688059	294	7974	8027	GAATATTGTCCCAGTAAG	soooossssssssooss	5-8-5	5	13	1111
688060	295	7975	8028	AGAATATTGTCCCAGTAA	soooossssssssooss	5-8-5	0	31	1112
688061	296	7976	8029	AAGAATATTGTCCCAGTA	soooossssssssooss	5-8-5	0	16	1113
688062	297	7977	8030	CAAGAATATTGTCCCAGT	soooossssssssooss	5-8-5	29	0	1114
688063	298	7978	8031	CCAAGAATATTGTCCCAG	soooossssssssooss	5-8-5	26	42	1115
688064	299	7979	8032	ACCAAGAATATTGTCCCA	soooossssssssooss	5-8-5	19	11	1116
688065	300	7980	8033	GACCAAGAATATTGTCCC	soooossssssssooss	5-8-5	31	26	1117

TABLE 53-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	% inhibition (RTS3760_MGB)	SEQ ID NO
688066	301	7981	8034	GGACCAAGAATATTGTCC	sooossssssssssss	5-8-5	0	0	1118
688067	302	7982	8035	AGGACCAAGAATATTGTC	sooossssssssssss	5-8-5	5	10	1119
688068	303	7983	8036	TAGGACCAAGAATATTGT	sooossssssssssss	5-8-5	20	13	1120
688069	304	7984	8037	CTAGGACCAAGAATATTG	sooossssssssssss	5-8-5	0	22	1121
688070	305	7985	8038	TCTAGGACCAAGAATATT	sooossssssssssss	5-8-5	0	0	1122
688071	306	7986	8039	CTCTAGGACCAAGAATAT	sooossssssssssss	5-8-5	5	18	1123
688072	307	7987	8040	ACTCTAGGACCAAGAATA	sooossssssssssss	5-8-5	17	27	1124
688073	308	7988	8041	TACTCTAGGACCAAGAAT	sooossssssssssss	5-8-5	13	8	1125
688074	309	7989	8042	TTACTCTAGGACCAAGAA	sooossssssssssss	5-8-5	0	0	1126
688075	310	7990	8043	CTTACTCTAGGACCAAGA	sooossssssssssss	5-8-5	0	0	1127
688076	311	7991	8044	CCTTACTCTAGGACCAAG	sooossssssssssss	5-8-5	45	71	1128
688077	312	7992	8045	GCCTTACTCTAGGACCAA	sooossssssssssss	5-8-5	62	67	1129
688078	313	7993	8046	TGCCTTACTCTAGGACCA	sooossssssssssss	5-8-5	33	30	1130
688079	314	7994	8047	GTGCCTTACTCTAGGACC	sooossssssssssss	5-8-5	59	49	1131
688080	315	7995	8048	TGTGCCTTACTCTAGGAC	sooossssssssssss	5-8-5	47	41	1132
688081	316	7996	8049	ATGTGCCTTACTCTAGGA	sooossssssssssss	5-8-5	44	13	1133
688082	317	7997	8050	AATGTGCCTTACTCTAGG	sooossssssssssss	5-8-5	6	0	1134

TABLE 54

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
576816	310	7990	8043	GCCTTACTCTAGGACCAAGA	ssssssssssssssssss	5-10-5	58	20
619420	310	7990	8043	GCCTTACTCTAGGACCAAGA	soooossssssssssoss	5-10-5	70	20
688078	313	7993	8046	TGCCTTACTCTAGGACCA	sooossssssssssss	5-8-5	50	1130
688083	318	7998	8051	AAATGTGCCTTACTCTAG	sooossssssssssss	5-8-5	7	1135
688084	319	7999	8052	CAAATGTGCCTTACTCTA	sooossssssssssss	5-8-5	11	1136
688085	320	8000	8053	CCAAATGTGCCTTACTCT	sooossssssssssss	5-8-5	29	1137
688086	321	8001	8054	CCCAAATGTGCCTTACTC	sooossssssssssss	5-8-5	43	1138
688087	322	8002	8055	GCCCAAATGTGCCTTACT	sooossssssssssss	5-8-5	58	1139
688088	323	8003	8056	AGCCCAAATGTGCCTTAC	sooossssssssssss	5-8-5	66	1140

TABLE 54-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688089	324	8004	8057	GAGCCCAAATGTGCCTTA	soooossssssssooss	5-8-5	61	1141
688090	325	8005	8058	GGAGCCCAAATGTGCCTT	soooossssssssooss	5-8-5	52	1142
688091	326	8006	8059	TGGAGCCCAAATGTGCCT	soooossssssssooss	5-8-5	34	1143
688092	327	8007	8060	TTGGAGCCCAAATGTGCC	soooossssssssooss	5-8-5	15	1144
688093	328	8008	8061	TTTGGAGCCCAAATGTGC	soooossssssssooss	5-8-5	31	1145
688094	329	8009	8062	CTTTGGAGCCCAAATGTG	soooossssssssooss	5-8-5	23	1146
688095	330	8010	8063	TCTTTGGAGCCCAAATGT	soooossssssssooss	5-8-5	13	1147
688096	331	8011	8064	GTCTTTGGAGCCCAAATG	soooossssssssooss	5-8-5	30	1148
688097	332	8012	8065	TGTCTTTGGAGCCCAAAT	soooossssssssooss	5-8-5	32	1149
688098	333	8013	8066	CTGTCTTTGGAGCCCAAA	soooossssssssooss	5-8-5	48	1150
688099	334	8014	8067	TCTGTCTTTGGAGCCCAA	soooossssssssooss	5-8-5	62	1151
688100	335	8015	8068	TTCTGTCTTTGGAGCCCA	soooossssssssooss	5-8-5	61	1152
688101	336	8016	8069	GTTCTGTCTTTGGAGCCC	soooossssssssooss	5-8-5	77	1153
688102	337	8017	8070	TGTTCTGTCTTTGGAGCC	soooossssssssooss	5-8-5	58	1154
688103	338	8018	8071	CTGTTCTGTCTTTGGAGC	soooossssssssooss	5-8-5	57	1155
688104	339	8019	8072	CCTGTTCTGTCTTTGGAG	soooossssssssooss	5-8-5	55	1156
688105	340	8020	8073	ACCTGTTCTGTCTTTGGA	soooossssssssooss	5-8-5	51	1157
688106	341	8021	8074	TACCTGTTCTGTCTTTGG	soooossssssssooss	5-8-5	42	1158
688107	342	8022	8075	GTACCTGTTCTGTCTTTG	soooossssssssooss	5-8-5	49	1159
688108	343	8023	8076	AGTACCTGTTCTGTCTTT	soooossssssssooss	5-8-5	25	1160
688109	344	8024	8077	AAGTACCTGTTCTGTCTT	soooossssssssooss	5-8-5	22	1161
688110	345	8025	8078	GAAGTACCTGTTCTGTCT	soooossssssssooss	5-8-5	42	1162
688111	346	8026	8079	AGAAGTACCTGTTCTGTC	soooossssssssooss	5-8-5	21	1163
688112	347	8027	8080	GAGAAGTACCTGTTCTGT	soooossssssssooss	5-8-5	22	1164
688113	348	8028	8081	TGAGAAGTACCTGTTCTG	soooossssssssooss	5-8-5	13	1165
688114	349	8029	8082	CTGAGAAGTACCTGTTCT	soooossssssssooss	5-8-5	25	1166
688115	350	8030	8083	ACTGAGAAGTACCTGTTC	soooossssssssooss	5-8-5	14	1167
688116	351	8031	8084	CACTGAGAAGTACCTGTT	soooossssssssooss	5-8-5	36	1168
688117	352	8032	8085	TCACTGAGAAGTACCTGT	soooossssssssooss	5-8-5	28	1169
688118	353	8033	8086	ATCACTGAGAAGTACCTG	soooossssssssooss	5-8-5	36	1170
688119	354	8034	8087	CATCACTGAGAAGTACCT	soooossssssssooss	5-8-5	32	1171
688120	355	8035	8088	CCATCACTGAGAAGTACC	soooossssssssooss	5-8-5	44	1172
688121	356	8036	8089	TCCATCACTGAGAAGTAC	soooossssssssooss	5-8-5	31	1173
688122	357	8037	8090	CTCCATCACTGAGAAGTA	soooossssssssooss	5-8-5	56	1174

TABLE 54-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688123	358	8038	8091	TCTCCATCACTGAGAAGT	sooossssssssssooss	5-8-5	53	1175
688124	359	8039	8092	TTCTCCATCACTGAGAAG	sooossssssssssooss	5-8-5	14	1176
688125	360	8040	8093	TTTCTCCATCACTGAGAA	sooossssssssssooss	5-8-5	12	1177
688126	361	8041	8094	ATTTCTCCATCACTGAGA	sooossssssssssooss	5-8-5	18	1178
688127	362	8042	8095	TATTTCTCCATCACTGAG	sooossssssssssooss	5-8-5	11	1179
688128	364	8044	8097	GTTATTTCTCCATCACTG	sooossssssssssooss	5-8-5	40	1180
688129	365	8045	8098	AGTTATTTCTCCATCACT	sooossssssssssooss	5-8-5	37	1181
688130	366	8046	8099	AAGTTATTTCTCCATCAC	sooossssssssssooss	5-8-5	20	1182
688131	367	8047	8100	AAAGTTATTTCTCCATCA	sooossssssssssooss	5-8-5	22	1183
688132	368	8048	8101	AAAAGTTATTTCTCCATC	sooossssssssssooss	5-8-5	31	1184
688133	369	8049	8102	GAAAAGTTATTTCTCCAT	sooossssssssssooss	5-8-5	19	1185
688134	371	8051	8104	AAGAAAAGTTATTTCTCC	sooossssssssssooss	5-8-5	34	1186
688135	372	8052	8105	CAAGAAAAGTTATTTCTC	sooossssssssssooss	5-8-5	0	1187
688136	373	8053	8106	GCAAGAAAAGTTATTTCT	sooossssssssssooss	5-8-5	21	1188
688137	374	8054	8107	GGCAAGAAAAGTTATTTC	sooossssssssssooss	5-8-5	51	1189
688138	375	8055	8108	TGGCAAGAAAAGTTATTT	sooossssssssssooss	5-8-5	42	1190
688139	376	8056	8109	TTGGCAAGAAAAGTTATT	sooossssssssssooss	5-8-5	13	1191
688140	377	8057	8110	GTTGGCAAGAAAAGTTAT	sooossssssssssooss	5-8-5	16	1192
688141	378	8058	8111	GGTTGGCAAGAAAAGTTA	sooossssssssssooss	5-8-5	23	1193
688142	379	8059	8112	TGGTTGGCAAGAAAAGTT	sooossssssssssooss	5-8-5	7	1194
688143	380	8060	8113	GTGGTTGGCAAGAAAAGT	sooossssssssssooss	5-8-5	30	1195
688144	381	8061	8114	TGTGGTTGGCAAGAAAAG	sooossssssssssooss	5-8-5	12	1196
688145	382	8062	8115	GTGTGGTTGGCAAGAAAA	sooossssssssssooss	5-8-5	7	1197
688146	383	8063	8116	AGTGTGGTTGGCAAGAAA	sooossssssssssooss	5-8-5	0	1198
688147	384	8064	8117	GAGTGTGGTTGGCAAGAA	sooossssssssssooss	5-8-5	27	1199
688148	385	8065	8118	AGAGTGTGGTTGGCAAGA	sooossssssssssooss	5-8-5	17	1200
688149	386	8066	8119	TAGAGTGTGGTTGGCAAG	sooossssssssssooss	5-8-5	17	1201
688150	387	8067	8120	TTAGAGTGTGGTTGGCAA	sooossssssssssooss	5-8-5	20	1202
688151	388	8068	8121	TTTAGAGTGTGGTTGGCA	sooossssssssssooss	5-8-5	22	1203
688152	389	8069	8122	ATTTAGAGTGTGGTTGGC	sooossssssssssooss	5-8-5	47	1204
688153	390	8070	8123	CATTTAGAGTGTGGTTGG	sooossssssssssooss	5-8-5	21	1205
688154	391	8071	8124	CCATTTAGAGTGTGGTTG	sooossssssssssooss	5-8-5	23	1206
688155	392	8072	8125	TCCATTTAGAGTGTGGTT	sooossssssssssooss	5-8-5	9	1207

TABLE 54-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688156	393	8073	8126	CTCCATTTAGAGTGTGGT	sooosssssssssooss	5-8-5	25	1208
688157	394	8074	8127	TCTCCATTTAGAGTGTGG	sooossssssssooss	5-8-5	41	1209
688158	395	8075	8128	TTCTCCATTTAGAGTGTG	sooossssssssooss	5-8-5	19	1210
688159	396	8076	8129	TTTCTCCATTTAGAGTGT	sooossssssssooss	5-8-5	0	1211

TABLE 55

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
576816	310	7990	8043	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	5-10-5	57	20
619420	310	7990	8043	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	5-10-5	64	20
688078	313	7993	8046	TGCCTTACTCTAGGACCA	sooossssssssooss	5-8-5	37	1130
688160	397	8077	8130	ATTTCTCCATTTAGAGTG	sooossssssssooss	5-8-5	0	1212
688161	398	8078	8131	GATTTCTCCATTTAGAGT	sooossssssssooss	5-8-5	9	1213
688162	399	8079	8132	GGATTTCTCCATTTAGAG	sooossssssssooss	5-8-5	2	1214
688163	400	8080	8133	AGGATTTCTCCATTTAGA	sooossssssssooss	5-8-5	0	1215
688164	401	8081	8134	AAGGATTTCTCCATTTAG	sooossssssssooss	5-8-5	12	1216
688165	402	8082	8135	GAAGGATTTCTCCATTTA	sooossssssssooss	5-8-5	0	1217
688166	403	8083	8136	CGAAGGATTTCTCCATTT	sooossssssssooss	5-8-5	18	1218
688167	404	8084	8137	TCGAAGGATTTCTCCATT	sooossssssssooss	5-8-5	12	1219
688168	405	8085	8138	TTCGAAGGATTTCTCCAT	sooossssssssooss	5-8-5	6	1220
688169	406	8086	8139	TTTCGAAGGATTTCTCCA	sooossssssssooss	5-8-5	0	1221
688170	407	8087	8140	ATTTCGAAGGATTTCTCC	sooossssssssooss	5-8-5	8	1222
688171	408	8088	8141	CATTTCGAAGGATTTCTC	sooossssssssooss	5-8-5	16	1223
688172	409	8089	8142	GCATTTCGAAGGATTTCT	sooossssssssooss	5-8-5	55	1224
688173	410	8090	8143	TGCATTTCGAAGGATTTC	sooossssssssooss	5-8-5	0	1225
688174	411	8091	8144	CTGCATTTCGAAGGATTT	sooossssssssooss	5-8-5	25	1226
688175	412	8092	8145	TCTGCATTTCGAAGGATT	sooossssssssooss	5-8-5	33	1227
688176	413	8093	8146	CTCTGCATTTCGAAGGAT	sooossssssssooss	5-8-5	12	1228
688177	414	8094	8147	TCTCTGCATTTCGAAGGA	sooossssssssooss	5-8-5	14	1229
688178	415	8095	8148	CTCTCTGCATTTCGAAGG	sooossssssssooss	5-8-5	5	1230

TABLE 55-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688179	416	8096	8149	ACTCTCTGCATTTCGAAG	soooossssssssooss	5-8-5	0	1231
688180	417	8097	8150	CACTCTCTGCATTTCGAA	soooossssssssooss	5-8-5	12	1232
688181	418	8098	8151	CCACTCTCTGCATTTCGA	soooossssssssooss	5-8-5	40	1233
688182	419	8099	8152	ACCACTCTCTGCATTTCG	soooossssssssooss	5-8-5	40	1234
688183	420	8100	8153	CACCACTCTCTGCATTTC	soooossssssssooss	5-8-5	45	1235
688184	421	8101	8154	GCACCACTCTCTGCATTT	soooossssssssooss	5-8-5	24	1236
688185	422	8102	8155	AGCACCACTCTCTGCATT	soooossssssssooss	5-8-5	12	1237
688186	423	8103	8156	TAGCACCACTCTCTGCAT	soooossssssssooss	5-8-5	21	1238
688187	424	8104	8157	ATAGCACCACTCTCTGCA	soooossssssssooss	5-8-5	25	1239
688188	425	8105	8158	TATAGCACCACTCTCTGC	soooossssssssooss	5-8-5	12	1240
688189	426	8106	8159	CTATAGCACCACTCTCTG	soooossssssssooss	5-8-5	0	1241
688190	427	8107	8160	TCTATAGCACCACTCTCT	soooossssssssooss	5-8-5	0	1242
688191	428	8108	8161	ATCTATAGCACCACTCTC	soooossssssssooss	5-8-5	20	1243
688192	429	8109	8162	CATCTATAGCACCACTCT	soooossssssssooss	5-8-5	0	1244
688193	430	8110	8163	ACATCTATAGCACCACTC	soooossssssssooss	5-8-5	31	1245
688194	431	8111	8164	TACATCTATAGCACCACT	soooossssssssooss	5-8-5	4	1246
688195	432	8112	8165	TTACATCTATAGCACCAC	soooossssssssooss	5-8-5	32	1247
688196	433	8113	8166	TTTACATCTATAGCACCA	soooossssssssooss	5-8-5	37	1248
688197	434	8114	8167	CTTTACATCTATAGCACC	soooossssssssooss	5-8-5	25	1249
688198	435	8115	8168	ACTTTACATCTATAGCAC	soooossssssssooss	5-8-5	0	1250
688199	436	8116	8169	AACTTTACATCTATAGCA	soooossssssssooss	5-8-5	9	1251
688200	437	8117	8170	AAACTTTACATCTATAGC	soooossssssssooss	5-8-5	8	1252
688201	438	8118	8171	AAAACTTTACATCTATAG	soooossssssssooss	5-8-5	4	1253
688202	440	8120	8173	AAAAAACTTTACATCTAT	soooossssssssooss	5-8-5	8	1254
688203	441	8121	8174	CAAAAAACTTTACATCTA	soooossssssssooss	5-8-5	4	1255
688204	442	8122	8175	ACAAAAAACTTTACATCT	soooossssssssooss	5-8-5	0	1256
688205	443	8123	8176	GACAAAAAACTTTACATC	soooossssssssooss	5-8-5	0	1257
688206	446	8126	8179	CAAGACAAAAAACTTTAC	soooossssssssooss	5-8-5	5	1258
688207	448	8128	8181	GACAAGACAAAAAACTTT	soooossssssssooss	5-8-5	27	1259
688208	449	8129	8182	AGACAAGACAAAAAACTT	soooossssssssooss	5-8-5	9	1260
688209	450	8130	8183	CAGACAAGACAAAAAACT	soooossssssssooss	5-8-5	0	1261
688210	451	8131	8184	TCAGACAAGACAAAAAAC	soooossssssssooss	5-8-5	11	1262
688211	452	8132	8185	TTCAGACAAGACAAAAAA	soooossssssssooss	5-8-5	0	1263
688212	454	8134	8187	TTTTCAGACAAGACAAAA	soooossssssssooss	5-8-5	0	1264

TABLE 55-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688213	455	8135	8188	CTTTTCAGACAAGACAAA	sooossssssssooss	5-8-5	0	1265
688214	456	8136	8189	CCTTTTCAGACAAGACAA	sooossssssssooss	5-8-5	30	1266
688215	457	8137	8190	CCCTTTTCAGACAAGACA	sooossssssssooss	5-8-5	31	1267
688216	458	8138	8191	TCCCTTTTCAGACAAGAC	sooossssssssooss	5-8-5	24	1268
688217	459	8139	8192	CTCCCTTTTCAGACAAGA	sooossssssssooss	5-8-5	47	1269
688218	460	8140	8193	ACTCCCTTTTCAGACAAG	sooossssssssooss	5-8-5	34	1270
688219	461	8141	8194	CACTCCCTTTTCAGACAA	sooossssssssooss	5-8-5	5	1271
688220	462	8142	8195	TCACTCCCTTTTCAGACA	sooossssssssooss	5-8-5	15	1272
688221	463	8143	8196	ATCACTCCCTTTTCAGAC	sooossssssssooss	5-8-5	14	1273
688222	464	8144	8197	AATCACTCCCTTTTCAGA	sooossssssssooss	5-8-5	18	1274
688223	465	8145	8198	TAATCACTCCCTTTTCAG	sooossssssssooss	5-8-5	14	1275
688224	466	8146	8199	ATAATCACTCCCTTTTCA	sooossssssssooss	5-8-5	15	1276
688225	467	8147	8200	AATAATCACTCCCTTTTC	sooossssssssooss	5-8-5	8	1277
688226	468	8148	8201	CAATAATCACTCCCTTTT	sooossssssssooss	5-8-5	9	1278
688227	469	8149	8202	ACAATAATCACTCCCTTT	sooossssssssooss	5-8-5	24	1279
688228	470	8150	8203	AACAATAATCACTCCCTT	sooossssssssooss	5-8-5	21	1280
688229	471	8151	8204	AAACAATAATCACTCCCT	sooossssssssooss	5-8-5	21	1281
688230	472	8152	8205	GAAACAATAATCACTCCC	sooossssssssooss	5-8-5	36	1282
688231	473	8153	8206	TGAAACAATAATCACTCC	sooossssssssooss	5-8-5	7	1283
688232	474	8154	8207	ATGAAACAATAATCACTC	sooossssssssooss	5-8-5	20	1284
688233	475	8155	8208	AATGAAACAATAATCACT	sooossssssssooss	5-8-5	0	1285
688234	476	8156	8209	TAATGAAACAATAATCAC	sooossssssssooss	5-8-5	16	1286
688235	477	8157	8210	TTAATGAAACAATAATCA	sooossssssssooss	5-8-5	0	1287
688236	478	8158	8211	ATTAATGAAACAATAATC	sooossssssssooss	5-8-5	0	1288

TABLE 56

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
576816	310	7990	8043	GCCTTACTCTAGGACCAAGA	sssssssssssssssss	5-10-5	61	20
619420	310	7990	8043	GCCTTACTCTAGGACCAAGA	soooossssssssooss	5-10-5	75	20

TABLE 56-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688078	313	7993	8046	TGCCTTACTCTAGGACCA	sooossssssssssooss	5-8-5	67	1130
688237	485	8165	8218	ATCAAAGATTAATGAAAC	sooossssssssssooss	5-8-5	43	1289
688238	486	8166	8219	CATCAAAGATTAATGAAA	sooossssssssssooss	5-8-5	0	1290
688239	487	8167	8220	CCATCAAAGATTAATGAA	sooossssssssssooss	5-8-5	42	1291
688240	488	8168	8221	TCCATCAAAGATTAATGA	sooossssssssssooss	5-8-5	46	1292
688241	489	8169	8222	TTCCATCAAAGATTAATG	sooossssssssssooss	5-8-5	0	1293
688242	490	8170	8223	TTTCCATCAAAGATTAAT	sooossssssssssooss	5-8-5	55	1294
688243	491	8171	8224	GTTTCCATCAAAGATTAA	sooossssssssssooss	5-8-5	49	1295
688244	492	8172	8225	AGTTTCCATCAAAGATTA	sooossssssssssooss	5-8-5	44	1296
688245	493	8173	8226	CAGTTTCCATCAAAGATT	sooossssssssssooss	5-8-5	0	1297
688246	494	8174	8227	CCAGTTTCCATCAAAGAT	sooossssssssssooss	5-8-5	46	1298
688247	495	8175	8228	TCCAGTTTCCATCAAAGA	sooossssssssssooss	5-8-5	46	1299
688248	496	8176	8229	TTCCAGTTTCCATCAAAG	sooossssssssssooss	5-8-5	59	1300
688249	497	8177	8230	ATTCCAGTTTCCATCAAA	sooossssssssssooss	5-8-5	48	1301
688250	498	8178	8231	CATTCCAGTTTCCATCAA	sooossssssssssooss	5-8-5	54	1302
688251	499	8179	8232	CCATTCCAGTTTCCATCA	sooossssssssssooss	5-8-5	59	1303
688252	500	8180	8233	CCCATTCCAGTTTCCATC	sooossssssssssooss	5-8-5	67	1304
688253	501	8181	8234	CCCCATTCCAGTTTCCAT	sooossssssssssooss	5-8-5	58	1305
688254	502	8182	8235	TCCCCATTCCAGTTTCCA	sooossssssssssooss	5-8-5	55	1306
688255	503	8183	8236	ATCCCCATTCCAGTTTCC	sooossssssssssooss	5-8-5	61	1307
688256	504	8184	8237	GATCCCCATTCCAGTTTC	sooossssssssssooss	5-8-5	51	1308
688257	505	8185	8238	CGATCCCCATTCCAGTTT	sooossssssssssooss	5-8-5	49	1309
688258	506	8186	8239	GCGATCCCCATTCCAGTT	sooossssssssssooss	5-8-5	43	1310
688259	507	8187	8240	TGCGATCCCCATTCCAGT	sooossssssssssooss	5-8-5	51	1311
688260	508	8188	8241	CTGCGATCCCCATTCCAG	sooossssssssssooss	5-8-5	70	1312
688261	509	8189	8242	GCTGCGATCCCCATTCCA	sooossssssssssooss	5-8-5	72	1313
688262	510	8190	8243	TGCTGCGATCCCCATTCC	sooossssssssssooss	5-8-5	49	1314
688263	511	8191	8244	GTGCTGCGATCCCCATTC	sooossssssssssooss	5-8-5	0	1315
688264	512	8192	8245	TGTGCTGCGATCCCCATT	sooossssssssssooss	5-8-5	58	1316
688265	513	8193	8246	ATGTGCTGCGATCCCCAT	sooossssssssssooss	5-8-5	66	1317
688266	514	8194	8247	TATGTGCTGCGATCCCCA	sooossssssssssooss	5-8-5	63	1318
688267	533	8213	8266	AAGTATAATTGATAGTCC	sooossssssssssooss	5-8-5	49	1319
688268	534	8214	8267	GAAGTATAATTGATAGTC	sooossssssssssooss	5-8-5	50	1320

TABLE 56-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688269	535	8215	8268	GGAAGTATAATTGATAGT	sooossssssssssooss	5-8-5	39	1321
688270	536	8216	8269	TGGAAGTATAATTGATAG	sooossssssssssooss	5-8-5	43	1322
688271	537	8217	8270	GTGGAAGTATAATTGATA	sooossssssssssooss	5-8-5	49	1323
688272	538	8218	8271	TGTGGAAGTATAATTGAT	sooossssssssssooss	5-8-5	40	1324
688273	539	8219	8272	CTGTGGAAGTATAATTGA	sooossssssssssooss	5-8-5	34	1325
688274	540	8220	8273	TCTGTGGAAGTATAATTG	sooossssssssssooss	5-8-5	0	1326
688275	541	8221	8274	GTCTGTGGAAGTATAATT	sooossssssssssooss	5-8-5	0	1327
688276	542	8222	8275	TGTCTGTGGAAGTATAAT	sooossssssssssooss	5-8-5	66	1328
688277	543	8223	8276	CTGTCTGTGGAAGTATAA	sooossssssssssooss	5-8-5	0	1329
688278	544	8224	8277	TCTGTCTGTGGAAGTATA	sooossssssssssooss	5-8-5	63	1330
688279	545	8225	8278	TTCTGTCTGTGGAAGTAT	sooossssssssssooss	5-8-5	55	1331
688280	546	8226	8279	GTTCTGTCTGTGGAAGTA	sooossssssssssooss	5-8-5	78	1332
688281	547	8227	8280	AGTTCTGTCTGTGGAAGT	sooossssssssssooss	5-8-5	63	1333
688282	548	8228	8281	AAGTTCTGTCTGTGGAAG	sooossssssssssooss	5-8-5	50	1334
688283	549	8229	8282	TAAGTTCTGTCTGTGGAA	sooossssssssssooss	5-8-5	0	1335
688284	550	8230	8283	CTAAGTTCTGTCTGTGGA	sooossssssssssooss	5-8-5	55	1336
688285	551	8231	8284	ACTAAGTTCTGTCTGTGG	sooossssssssssooss	5-8-5	69	1337
688286	552	8232	8285	AACTAAGTTCTGTCTGTG	sooossssssssssooss	5-8-5	66	1338
688287	553	8233	8286	AAACTAAGTTCTGTCTGT	sooossssssssssooss	5-8-5	43	1339
688288	554	8234	8287	GAAACTAAGTTCTGTCTG	sooossssssssssooss	5-8-5	37	1340
688289	555	8235	8288	AGAAACTAAGTTCTGTCT	sooossssssssssooss	5-8-5	47	1341
688290	556	8236	8289	TAGAAACTAAGTTCTGTC	sooossssssssssooss	5-8-5	50	1342
688291	557	8237	8290	GTAGAAACTAAGTTCTGT	sooossssssssssooss	5-8-5	47	1343
688292	558	8238	8291	GGTAGAAACTAAGTTCTG	sooossssssssssooss	5-8-5	46	1344
688293	559	8239	8292	AGGTAGAAACTAAGTTCT	sooossssssssssooss	5-8-5	59	1345
688294	560	8240	8293	GAGGTAGAAACTAAGTTC	sooossssssssssooss	5-8-5	48	1346
688295	561	8241	8294	GGAGGTAGAAACTAAGTT	sooossssssssssooss	5-8-5	47	1347
688296	562	8242	8295	GGGAGGTAGAAACTAAGT	sooossssssssssooss	5-8-5	43	1348
688297	563	8243	8296	TGGGAGGTAGAAACTAAG	sooossssssssssooss	5-8-5	49	1349
688298	564	8244	8297	GTGGGAGGTAGAAACTAA	sooossssssssssooss	5-8-5	51	1350
688299	565	8245	8298	AGTGGGAGGTAGAAACTA	sooossssssssssooss	5-8-5	45	1351
688300	566	8246	8299	AAGTGGGAGGTAGAAACT	sooossssssssssooss	5-8-5	40	1352
688301	567	8247	8300	GAAGTGGGAGGTAGAAAC	sooossssssssssooss	5-8-5	39	1353
688302	568	8248	8301	TGAAGTGGGAGGTAGAAA	sooossssssssssooss	5-8-5	0	1354

TABLE 56-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688303	569	8249	8302	ATGAAGTGGGAGGTAGAA	sooossssssssssooss	5-8-5	42	1355
688304	570	8250	8303	TATGAAGTGGGAGGTAGA	sooossssssssssooss	5-8-5	32	1356
688305	571	8251	8304	CTATGAAGTGGGAGGTAG	sooossssssssssooss	5-8-5	47	1357
688306	572	8252	8305	TCTATGAAGTGGGAGGTA	sooossssssssssooss	5-8-5	33	1358
688307	573	8253	8306	CTCTATGAAGTGGGAGGT	sooossssssssssooss	5-8-5	55	1359
688308	574	8254	8307	ACTCTATGAAGTGGGAGG	sooossssssssssooss	5-8-5	51	1360
688309	575	8255	8308	CACTCTATGAAGTGGGAG	sooossssssssssooss	5-8-5	59	1361
688310	576	8256	8309	ACACTCTATGAAGTGGGA	sooossssssssssooss	5-8-5	58	1362
688311	577	8257	8310	CACACTCTATGAAGTGGG	sooossssssssssooss	5-8-5	59	1363
688312	578	8258	8311	ACACACTCTATGAAGTGG	sooossssssssssooss	5-8-5	42	1364
688313	579	8259	8312	CACACACTCTATGAAGTG	sooossssssssssooss	5-8-5	40	1365

TABLE 57

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Mismatches with SEQ ID NO: 19	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
576816	310	7990	8043	0	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	5-10-5	47	20
619420	310	7990	8043	0	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	5-10-5	51	20
688078	313	7993	8046	0	TGCCTTACTCTAGGACCA	soooossssssssooss	5-8-5	32	1130
688314	580	8260	8313	0	ACACACACTCTATGAAGT	soooossssssssooss	5-8-5	9	1366
688315	691	n/a	9505	1	CCCTGATCTTCCATTCTC	soooossssssssooss	5-8-5	21	1367
688316	692	n/a	9506	2	ACCCTGATCTTCCATTCT	soooossssssssooss	5-8-5	2	1368
688317	693	n/a	9507	3	GACCCTGATCTTCCATTC	soooossssssssooss	5-8-5	16	1369
688318	694	n/a	9508	3	TGACCCTGATCTTCCATT	soooossssssssooss	5-8-5	18	1370
688319	695	n/a	n/a	n/a	CTGACCCTGATCTTCCAT	soooossssssssooss	5-8-5	5	1371
688320	696	n/a	n/a	n/a	TCTGACCCTGATCTTCCA	soooossssssssooss	5-8-5	11	1372
688321	697	n/a	n/a	n/a	CTCTGACCCTGATCTTCC	soooossssssssooss	5-8-5	21	1373
688322	698	n/a	n/a	n/a	ACTCTGACCCTGATCTTC	soooossssssssooss	5-8-5	26	1374
688323	699	n/a	n/a	n/a	TACTCTGACCCTGATCTT	soooossssssssooss	5-8-5	6	1375
688324	700	n/a	12554	3	ATACTCTGACCCTGATCT	soooossssssssooss	5-8-5	7	1376
688325	701	n/a	12555	2	AATACTCTGACCCTGATC	soooossssssssooss	5-8-5	0	1377

TABLE 57-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Mismatches with SEQ ID NO: 19	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
687955	n/a	13641	13680	0	ATGATTTCTTGTCTGGGA	sooossssssssooss	5-8-5	31	1378
687956	n/a	13642	13681	0	CATGATTTCTTGTCTGGG	sooossssssssooss	5-8-5	38	1379
687957	n/a	13643	13682	0	CCATGATTTCTTGTCTGG	sooossssssssooss	5-8-5	12	1380
687958	n/a	13644	13683	0	GCCATGATTTCTTGTCTG	sooossssssssooss	5-8-5	27	1381
687959	n/a	13645	13684	0	GGCCATGATTTCTTGTCT	sooossssssssooss	5-8-5	26	1382
687960	n/a	13646	13685	0	GGGCCATGATTTCTTGTC	sooossssssssooss	5-8-5	18	1383
687961	n/a	14089	14136	0	AACTAACATGTAGGCACT	sooossssssssooss	5-8-5	38	1384
687962	n/a	14090	14137	0	GAACTAACATGTAGGCAC	sooossssssssooss	5-8-5	57	1385
687963	n/a	14091	14138	0	GGAACTAACATGTAGGCA	sooossssssssooss	5-8-5	63	1386
687964	n/a	14092	14139	0	AGGAACTAACATGTAGGC	sooossssssssooss	5-8-5	29	1387
687965	n/a	14302	14349	0	CTTCTGATTCAAGCCATT	sooossssssssooss	5-8-5	25	1388
687966	n/a	14303	14350	0	GCTTCTGATTCAAGCCAT	sooossssssssooss	5-8-5	51	1389
687967	n/a	14304	14351	0	TGCTTCTGATTCAAGCCA	sooossssssssooss	5-8-5	46	1390
687968	n/a	14305	14352	0	GTGCTTCTGATTCAAGCC	sooossssssssooss	5-8-5	21	1391
687969	n/a	14306	14353	0	AGTGCTTCTGATTCAAGC	sooossssssssooss	5-8-5	36	1392
687970	n/a	14307	14354	0	AAGTGCTTCTGATTCAAG	sooossssssssooss	5-8-5	25	1393
687971	n/a	14308	14355	0	AAAGTGCTTCTGATTCAA	sooossssssssooss	5-8-5	28	1394
687972	n/a	14309	14356	0	TAAAGTGCTTCTGATTCA	sooossssssssooss	5-8-5	0	1395
687973	n/a	14310	14357	0	CTAAAGTGCTTCTGATTC	sooossssssssooss	5-8-5	25	1396
687974	n/a	14311	14358	0	ACTAAAGTGCTTCTGATT	sooossssssssooss	5-8-5	17	1397
687975	n/a	14312	14359	0	GACTAAAGTGCTTCTGAT	sooossssssssooss	5-8-5	33	1398
687976	n/a	14313	14360	0	GGACTAAAGTGCTTCTGA	sooossssssssooss	5-8-5	47	1399
687977	n/a	14314	14361	0	AGGACTAAAGTGCTTCTG	sooossssssssooss	5-8-5	44	1400
687978	n/a	14315	14362	0	CAGGACTAAAGTGCTTCT	sooossssssssooss	5-8-5	57	1401
687979	n/a	14316	14363	0	ACAGGACTAAAGTGCTTC	sooossssssssooss	5-8-5	31	1402
687980	n/a	14317	14364	0	TACAGGACTAAAGTGCTT	sooossssssssooss	5-8-5	24	1403
687981	n/a	14318	14365	0	ATACAGGACTAAAGTGCT	sooossssssssooss	5-8-5	21	1404
687982	n/a	14319	14366	0	GATACAGGACTAAAGTGC	sooossssssssooss	5-8-5	15	1405
687983	n/a	14320	14367	0	AGATACAGGACTAAAGTG	sooossssssssooss	5-8-5	8	1406
687984	n/a	14321	14368	0	CAGATACAGGACTAAAGT	sooossssssssooss	5-8-5	1	1407
687985	n/a	14322	14369	0	ACAGATACAGGACTAAAG	sooossssssssooss	5-8-5	10	1408
687986	n/a	14323	14370	0	AACAGATACAGGACTAAA	sooossssssssooss	5-8-5	11	1409
687987	n/a	14324	14371	0	GAACAGATACAGGACTAA	sooossssssssooss	5-8-5	20	1410
687988	n/a	14325	14372	0	TGAACAGATACAGGACTA	sooossssssssooss	5-8-5	25	1411

TABLE 57-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Mismatches with SEQ ID NO: 19	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
687989	n/a	14326	14373	0	CTGAACAGATACAGGACT	sooossssssssssooss	5-8-5	12	1412
687990	n/a	14327	14374	0	ACTGAACAGATACAGGAC	sooossssssssssooss	5-8-5	25	1413
687991	n/a	14328	14375	0	CACTGAACAGATACAGGA	sooossssssssssooss	5-8-5	8	1414
687992	n/a	14329	14376	0	ACACTGAACAGATACAGG	sooossssssssssooss	5-8-5	10	1415
687993	n/a	14330	14377	0	GACACTGAACAGATACAG	sooossssssssssooss	5-8-5	13	1416
687994	n/a	14331	14378	0	TGACACTGAACAGATACA	sooossssssssssooss	5-8-5	25	1417
687995	n/a	14332	14379	0	CTGACACTGAACAGATAC	sooossssssssssooss	5-8-5	35	1418
687996	n/a	14333	14380	0	GCTGACACTGAACAGATA	sooossssssssssooss	5-8-5	24	1419
687997	n/a	14334	14381	0	GGCTGACACTGAACAGAT	sooossssssssssooss	5-8-5	40	1420
687998	n/a	14335	14382	0	AGGCTGACACTGAACAGA	sooossssssssssooss	5-8-5	10	1421
687999	n/a	14336	14383	0	AAGGCTGACACTGAACAG	sooossssssssssooss	5-8-5	3	1422
688000	n/a	14337	14384	0	AAAGGCTGACACTGAACA	sooossssssssssooss	5-8-5	18	1423
688001	n/a	14338	14385	0	GAAAGGCTGACACTGAAC	sooossssssssssooss	5-8-5	17	1424
688002	n/a	14339	14386	0	TGAAAGGCTGACACTGAA	sooossssssssssooss	5-8-5	12	1425
688003	n/a	14358	14405	0	TGGGATTTAAAATGATGT	sooossssssssssooss	5-8-5	17	1426
688004	n/a	14359	14406	0	ATGGGATTTAAAATGATG	sooossssssssssooss	5-8-5	11	1427
688326	n/a	13402	13443	0	CTTGAGAAGAAAGCCTTC	sooossssssssssooss	5-8-5	6	1428
688327	n/a	14287	14334	0	ATTAAGGCTCTTAGGTTA	sooossssssssssooss	5-8-5	0	1429
688328	n/a	13499	13530	0	GTAGACAGTCTGTTATTT	sooossssssssssooss	5-8-5	27	1430
688329	n/a	14397	14444	0	TGACATGTAGAGAGATTA	sooossssssssssooss	5-8-5	43	1431
688330	n/a	13827	13866	0	TGGTTTAAGGGCACAAAC	sooossssssssssooss	5-8-5	0	1432
688331	n/a	13403	13444	0	ACTTGAGAAGAAAGCCTT	sooossssssssssooss	5-8-5	27	1433
688332	n/a	14257	14304	0	CCTCTGATACTCCATCAT	sooossssssssssooss	5-8-5	28	1434
688333	n/a	13471	13502	0	AAATCTTGTCATAGGTGA	sooossssssssssooss	5-8-5	21	1435
688334	n/a	13410	13451	0	AATTCTTACTTGAGAAGA	sooossssssssssooss	5-8-5	7	1436
688335	n/a	13885	13924	0	GGTGTATAGAGAATTCAG	sooossssssssssooss	5-8-5	41	1437
688336	n/a	14250	14297	0	TACTCCATCATGAGCCTA	sooossssssssssooss	5-8-5	35	1438
688337	n/a	13788	13827	0	GCTGGATGGAAAAAGATC	sooossssssssssooss	5-8-5	12	1439
688338	n/a	13517	13548	0	GTCCCTAGAACAATCTAA	sooossssssssssooss	5-8-5	28	1440
688339	n/a	14405	14452	0	GAAGAAATTGACATGTAG	sooossssssssssooss	5-8-5	12	1441
688340	n/a	13724	13763	0	CATCTACAGTACAACTTA	sooossssssssssooss	5-8-5	4	1442

TABLE 58

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688341	233	7913	7966	ATCTCTGTCTTGGCAACAGC	soooossssssssssooss	5-10-5	57	1443
688342	234	7914	7967	AATCTCTGTCTTGGCAACAG	soooossssssssssooss	5-10-5	35	1444
688343	235	7915	7968	CAATCTCTGTCTTGGCAACA	soooossssssssssooss	5-10-5	36	1445
655153	236	7916	7969	GCAATCTCTGTCTTGGCAAC	soooossssssssssooss	5-10-5	89	463
655154	237	7917	7970	AGCAATCTCTGTCTTGGCAA	soooossssssssssooss	5-10-5	81	464
688344	238	7918	7971	AAGCAATCTCTGTCTTGGCA	soooossssssssssooss	5-10-5	83	1446
655172	306	7986	8039	TACTCTAGGACCAAGAATAT	soooossssssssssooss	5-10-5	10	483
688345	307	7987	8040	TTACTCTAGGACCAAGAATA	soooossssssssssooss	5-10-5	1	1447
688346	308	7988	8041	CTTACTCTAGGACCAAGAAT	soooossssssssssooss	5-10-5	10	1448
625183	309	7989	8042	CCTTACTCTAGGACCAAGAA	soooossssssssssooss	5-10-5	44	484
576816	310	7990	8043	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	5-10-5	56	20
619420	310	7990	8043	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	5-10-5	73	20
688347	311	7991	8044	TGCCTTACTCTAGGACCAAG	soooossssssssssooss	5-10-5	62	1449
655173	312	7992	8045	GTGCCTTACTCTAGGACCAA	soooossssssssssooss	5-10-5	59	485
688348	313	7993	8046	TGTGCCTTACTCTAGGACCA	soooossssssssssooss	5-10-5	62	1450
688349	314	7994	8047	ATGTGCCTTACTCTAGGACC	soooossssssssssooss	5-10-5	62	1451
655174	315	7995	8048	AATGTGCCTTACTCTAGGAC	soooossssssssssooss	5-10-5	60	486
688350	319	7999	8052	CCCAAATGTGCCTTACTCTA	soooossssssssssooss	5-10-5	41	1452
688351	320	8000	8053	GCCCAAATGTGCCTTACTCT	soooossssssssssooss	5-10-5	62	1453
627833	321	8001	8054	AGCCCAAATGTGCCTTACTC	soooossssssssssooss	5-10-5	51	487
688352	322	8002	8055	GAGCCCAAATGTGCCTTACT	soooossssssssssooss	5-10-5	66	1454
688353	323	8003	8056	GGAGCCCAAATGTGCCTTAC	soooossssssssssooss	5-10-5	48	1455
619411	324	8004	8057	TGGAGCCCAAATGTGCCTTA	soooossssssssssooss	5-10-5	67	51
688354	325	8005	8058	TTGGAGCCCAAATGTGCCTT	soooossssssssssooss	5-10-5	53	1456
627608	326	8006	8059	TTTGGAGCCCAAATGTGCCT	soooossssssssssooss	5-10-5	42	1457
655175	327	8007	8060	CTTTGGAGCCCAAATGTGCC	soooossssssssssooss	5-10-5	26	488
655176	330	8010	8063	TGTCTTTGGAGCCCAAATGT	soooossssssssssooss	5-10-5	33	489
688355	331	8011	8064	CTGTCTTTGGAGCCCAAATG	soooossssssssssooss	5-10-5	56	1458
619412	332	8012	8065	TCTGTCTTTGGAGCCCAAAT	soooossssssssssooss	5-10-5	60	53
655177	333	8013	8066	TTCTGTCTTTGGAGCCCAAA	soooossssssssssooss	5-10-5	47	490
655178	334	8014	8067	GTTCTGTCTTTGGAGCCCAA	soooossssssssssooss	5-10-5	67	491
688356	335	8015	8068	TGTTCTGTCTTTGGAGCCCA	soooossssssssssooss	5-10-5	60	1459
655179	336	8016	8069	CTGTTCTGTCTTTGGAGCCC	soooossssssssssooss	5-10-5	59	492
688357	337	8017	8070	CCTGTTCTGTCTTTGGAGCC	soooossssssssssooss	5-10-5	50	1460

TABLE 58-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688358	338	8018	8071	ACCTGTTCTGTCTTTGGAGC	soooossssssssssooss	5-10-5	44	1461
655180	339	8019	8072	TACCTGTTCTGTCTTTGGAG	soooossssssssssooss	5-10-5	42	493
619413	340	8020	8073	GTACCTGTTCTGTCTTTGGA	soooossssssssssooss	5-10-5	56	135
688359	341	8021	8074	AGTACCTGTTCTGTCTTTGG	soooossssssssssooss	5-10-5	38	1462
655181	342	8022	8075	AAGTACCTGTTCTGTCTTTG	soooossssssssssooss	5-10-5	31	494
655185	351	8031	8084	ATCACTGAGAAGTACCTGTT	soooossssssssssooss	5-10-5	32	498
688360	352	8032	8085	CATCACTGAGAAGTACCTGT	soooossssssssssooss	5-10-5	53	1463
619414	353	8033	8086	CCATCACTGAGAAGTACCTG	soooossssssssssooss	5-10-5	48	136
655186	354	8034	8087	TCCATCACTGAGAAGTACCT	soooossssssssssooss	5-10-5	56	499
688361	355	8035	8088	CTCCATCACTGAGAAGTACC	soooossssssssssooss	5-10-5	54	1464
655201	415	8095	8148	CACTCTCTGCATTTCGAAGG	soooossssssssssooss	5-10-5	32	521
688362	416	8096	8149	CCACTCTCTGCATTTCGAAG	soooossssssssssooss	5-10-5	39	1465
688363	417	8097	8150	ACCACTCTCTGCATTTCGAA	soooossssssssssooss	5-10-5	37	1466
655202	418	8098	8151	CACCACTCTCTGCATTTCGA	soooossssssssssooss	5-10-5	50	522
688364	419	8099	8152	GCACCACTCTCTGCATTTCG	soooossssssssssooss	5-10-5	56	1467
655203	420	8100	8153	AGCACCACTCTCTGCATTTC	soooossssssssssooss	5-10-5	56	523
655204	421	8101	8154	TAGCACCACTCTCTGCATTT	soooossssssssssooss	5-10-5	25	524
688365	422	8102	8155	ATAGCACCACTCTCTGCATT	soooossssssssssooss	5-10-5	29	1468
688366	423	8103	8156	TATAGCACCACTCTCTGCAT	soooossssssssssooss	5-10-5	31	1469
655206	427	8107	8160	CATCTATAGCACCACTCTCT	soooossssssssssooss	5-10-5	20	526
688367	428	8108	8161	ACATCTATAGCACCACTCTC	soooossssssssssooss	5-10-5	28	1470
688368	429	8109	8162	TACATCTATAGCACCACTCT	soooossssssssssooss	5-10-5	24	1471
671081	430	8110	8163	TTACATCTATAGCACCACTC	soooossssssssssooss	5-10-5	26	1052
688369	431	8111	8164	TTTACATCTATAGCACCACT	soooossssssssssooss	5-10-5	40	1472
688370	432	8112	8165	CTTTACATCTATAGCACCAC	soooossssssssssooss	5-10-5	43	1473
655207	433	8113	8166	ACTTTACATCTATAGCACCA	soooossssssssssooss	5-10-5	33	527
688371	434	8114	8167	AACTTTACATCTATAGCACC	soooossssssssssooss	5-10-5	15	1474
688372	435	8115	8168	AAACTTTACATCTATAGCAC	soooossssssssssooss	5-10-5	24	1475
655208	436	8116	8169	AAAACTTTACATCTATAGCA	soooossssssssssooss	5-10-5	28	528
655215	456	8136	8189	TCCCTTTTCAGACAAGACAA	soooossssssssssooss	5-10-5	35	535
688373	457	8137	8190	CTCCCTTTTCAGACAAGACA	soooossssssssssooss	5-10-5	42	1476
688374	458	8138	8191	ACTCCCTTTTCAGACAAGAC	soooossssssssssooss	5-10-5	57	1477
655216	459	8139	8192	CACTCCCTTTTCAGACAAGA	soooossssssssssooss	5-10-5	51	536

TABLE 58-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
671082	460	8140	8193	TCACTCCCTTTTCAGACAAG	soooossssssssssooss	5-10-5	45	1053
688375	461	8141	8194	ATCACTCCCTTTTCAGACAA	soooossssssssssooss	5-10-5	39	1478
655217	462	8142	8195	AATCACTCCCTTTTCAGACA	soooossssssssssooss	5-10-5	45	537
688376	463	8143	8196	TAATCACTCCCTTTTCAGAC	soooossssssssssooss	5-10-5	7	1479
688377	464	8144	8197	ATAATCACTCCCTTTTCAGA	soooossssssssssooss	5-10-5	1	1480
655218	465	8145	8198	AATAATCACTCCCTTTTCAG	soooossssssssssooss	5-10-5	23	538
655230	500	8180	8233	TCCCCATTCCAGTTTCCATC	soooossssssssssooss	5-10-5	57	550
688378	501	8181	8234	ATCCCCATTCCAGTTTCCAT	soooossssssssssooss	5-10-5	60	1481
688379	502	8182	8235	GATCCCCATTCCAGTTTCCA	soooossssssssssooss	5-10-5	55	1482
655231	503	8183	8236	CGATCCCCATTCCAGTTTCC	soooossssssssssooss	5-10-5	58	551
688380	504	8184	8237	GCGATCCCCATTCCAGTTTC	soooossssssssssooss	5-10-5	56	1483
688381	505	8185	8238	TGCGATCCCCATTCCAGTTT	soooossssssssssooss	5-10-5	46	1484

TABLE 59

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Mismatches with SEQ ID NO: 19	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
576816	310	7990	8043	0	GCCTTACTCTAGGACCAAGA	sssssssssssssssssss	5-10-5	53	20
619420	310	7990	8043	0	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	5-10-5	54	20
655173	312	7992	8045	0	GTGCCTTACTCTAGGACCAA	soooossssssssssooss	5-10-5	77	485
655232	506	8186	8239	0	CTGCGATCCCCATTCCAGTT	soooossssssssssooss	5-10-5	46	552
688382	507	8187	8240	0	GCTGCGATCCCCATTCCAGT	soooossssssssssooss	5-10-5	65	1485
688383	508	8188	8241	0	TGCTGCGATCCCCATTCCAG	soooossssssssssooss	5-10-5	60	1486
655233	509	8189	8242	0	GTGCTGCGATCCCCATTCCA	soooossssssssssooss	5-10-5	57	553
688384	510	8190	8243	0	TGTGCTGCGATCCCCATTCC	soooossssssssssooss	5-10-5	45	1487
688385	511	8191	8244	0	ATGTGCTGCGATCCCCATTC	soooossssssssssooss	5-10-5	68	1488
655234	512	8192	8245	0	TATGTGCTGCGATCCCCATT	soooossssssssssooss	5-10-5	32	554
655240	548	8228	8281	0	CTAAGTTCTGTCTGTGGAAG	soooossssssssssooss	5-10-5	23	560
688386	549	8229	8282	0	ACTAAGTTCTGTCTGTGGAA	soooossssssssssooss	5-10-5	45	1489
671083	550	8230	8283	0	AACTAAGTTCTGTCTGTGGA	soooossssssssssooss	5-10-5	41	1054
655241	551	8231	8284	0	AAACTAAGTTCTGTCTGTGG	soooossssssssssooss	5-10-5	28	561

TABLE 59-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Mismatches with SEQ ID NO: 19	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688387	552	8232	8285	0	GAAACTAAGTTCTGTCTGTG	soooossssssssssooss	5-10-5	49	1490
688388	553	8233	8286	0	AGAAACTAAGTTCTGTCTGT	soooossssssssssooss	5-10-5	46	1491
655242	554	8234	8287	0	TAGAAACTAAGTTCTGTCTG	soooossssssssssooss	5-10-5	25	562
655289	691	n/a	9505	3	GACCCTGATCTTCCATTCTC	soooossssssssssooss	5-10-5	34	627
688389	692	n/a	9506	3	TGACCCTGATCTTCCATTCT	soooossssssssssooss	5-10-5	19	1492
688390	693	n/a	n/a	n/a	CTGACCCTGATCTTCCATTC	soooossssssssssooss	5-10-5	21	1493
655290	694	n/a	n/a	n/a	TCTGACCCTGATCTTCCATT	soooossssssssssooss	5-10-5	67	628
672561	695	n/a	n/a	n/a	CTCTGACCCTGATCTTCCAT	soooossssssssssooss	5-10-5	25	1056
625345	696	n/a	n/a	n/a	ACTCTGACCCTGATCTTCCA	soooossssssssssooss	5-10-5	15	1494
655291	697	n/a	n/a	n/a	TACTCTGACCCTGATCTTCC	soooossssssssssooss	5-10-5	18	629
688391	698	n/a	n/a	n/a	ATACTCTGACCCTGATCTTC	soooossssssssssooss	5-10-5	40	1495
688392	699	n/a	n/a	n/a	AATACTCTGACCCTGATCTT	soooossssssssssooss	5-10-5	8	1496
619423	n/a	13642	13681	0	GCCATGATTTCTTGTCTGGG	soooossssssssssooss	5-10-5	77	383
655417	n/a	14089	14136	0	GGAACTAACATGTAGGCACT	soooossssssssssooss	5-10-5	83	738
655420	n/a	14331	14378	0	GCTGACACTGAACAGATACA	soooossssssssssooss	5-10-5	51	741
655422	n/a	14452	14499	0	ATCATTTAATTAATGTATTT	soooossssssssssooss	5-10-5	33	743
671084	n/a	14316	14363	0	ATACAGGACTAAAGTGCTTC	soooossssssssssooss	5-10-5	74	1055
688393	n/a	13641	13680	0	CCATGATTTCTTGTCTGGGA	soooossssssssssooss	5-10-5	54	1497
688394	n/a	13643	13682	0	GGCCATGATTTCTTGTCTGG	soooossssssssssooss	5-10-5	52	1498
688395	n/a	13644	13683	0	GGGCCATGATTTCTTGTCTG	soooossssssssssooss	5-10-5	46	1499
688396	n/a	13731	13770	0	ACTTAAGTTCATCTACAGTA	soooossssssssssooss	5-10-5	18	1500
688397	n/a	13792	13831	0	TCCACTGCTGGATGGAAAAA	soooossssssssssooss	5-10-5	20	1501
688398	n/a	13968	14007	0	ATATTATTTATCTTACTCAA	soooossssssssssooss	5-10-5	32	1502
688399	n/a	13982	14021	0	GTTCTAAGTGCTTTATATTA	soooossssssssssooss	5-10-5	30	1503
688400	n/a	14090	14137	0	AGGAACTAACATGTAGGCAC	soooossssssssssooss	5-10-5	87	1504
688401	n/a	14122	14169	0	ATATAAGATAATACATGTAA	soooossssssssssooss	5-10-5	6	1505
688402	n/a	14243	14290	0	CATCATGAGCCTAAAGGAAA	soooossssssssssooss	5-10-5	22	1506
688403	n/a	14244	14291	0	CCATCATGAGCCTAAAGGAA	soooossssssssssooss	5-10-5	37	1507
688404	n/a	14300	14347	0	CTTCTGATTCAAGCCATTAA	soooossssssssssooss	5-10-5	76	1508
688405	n/a	14302	14349	0	TGCTTCTGATTCAAGCCATT	soooossssssssssooss	5-10-5	55	1509
688406	n/a	14303	14350	0	GTGCTTCTGATTCAAGCCAT	soooossssssssssooss	5-10-5	50	1510
688407	n/a	14304	14351	0	AGTGCTTCTGATTCAAGCCA	soooossssssssssooss	5-10-5	68	1511
688408	n/a	14305	14352	0	AAGTGCTTCTGATTCAAGCC	soooossssssssssooss	5-10-5	41	1512
688409	n/a	14306	14353	0	AAAGTGCTTCTGATTCAAGC	soooossssssssssooss	5-10-5	18	1513

TABLE 59-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Mismatches with SEQ ID NO: 19	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688410	n/a	14307	14354	0	TAAAGTGCTTCTGATTCAAG	soooossssssssssooss	5-10-5	29	1514
688411	n/a	14308	14355	0	CTAAAGTGCTTCTGATTCAA	soooossssssssssooss	5-10-5	43	1515
688412	n/a	14309	14356	0	ACTAAAGTGCTTCTGATTCA	soooossssssssssooss	5-10-5	33	1516
688413	n/a	14310	14357	0	GACTAAAGTGCTTCTGATTC	soooossssssssssooss	5-10-5	49	1517
688414	n/a	14311	14358	0	GGACTAAAGTGCTTCTGATT	soooossssssssssooss	5-10-5	49	1518
688415	n/a	14312	14359	0	AGGACTAAAGTGCTTCTGAT	soooossssssssssooss	5-10-5	55	1519
688416	n/a	14313	14360	0	CAGGACTAAAGTGCTTCTGA	soooossssssssssooss	5-10-5	71	1520
688417	n/a	14314	14361	0	ACAGGACTAAAGTGCTTCTG	soooossssssssssooss	5-10-5	66	1521
688418	n/a	14315	14362	0	TACAGGACTAAAGTGCTTCT	soooossssssssssooss	5-10-5	61	1522
688419	n/a	14317	14364	0	GATACAGGACTAAAGTGCTT	soooossssssssssooss	5-10-5	76	1523
688420	n/a	14318	14365	0	AGATACAGGACTAAAGTGCT	soooossssssssssooss	5-10-5	46	1524
688421	n/a	14319	14366	0	CAGATACAGGACTAAAGTGC	soooossssssssssooss	5-10-5	53	1525
688422	n/a	14320	14367	0	ACAGATACAGGACTAAAGTG	soooossssssssssooss	5-10-5	23	1526
688423	n/a	14321	14368	0	AACAGATACAGGACTAAAGT	soooossssssssssooss	5-10-5	28	1527
688424	n/a	14322	14369	0	GAACAGATACAGGACTAAAG	soooossssssssssooss	5-10-5	26	1528
688425	n/a	14323	14370	0	TGAACAGATACAGGACTAAA	soooossssssssssooss	5-10-5	13	1529
688426	n/a	14324	14371	0	CTGAACAGATACAGGACTAA	soooossssssssssooss	5-10-5	27	1530
688427	n/a	14325	14372	0	ACTGAACAGATACAGGACTA	soooossssssssssooss	5-10-5	37	1531
688428	n/a	14326	14373	0	CACTGAACAGATACAGGACT	soooossssssssssooss	5-10-5	35	1532
688429	n/a	14327	14374	0	ACACTGAACAGATACAGGAC	soooossssssssssooss	5-10-5	28	1533
688430	n/a	14328	14375	0	GACACTGAACAGATACAGGA	soooossssssssssooss	5-10-5	39	1534
688431	n/a	14329	14376	0	TGACACTGAACAGATACAGG	soooossssssssssooss	5-10-5	38	1535
688432	n/a	14330	14377	0	CTGACACTGAACAGATACAG	soooossssssssssooss	5-10-5	64	1536
688433	n/a	14332	14379	0	GGCTGACACTGAACAGATAC	soooossssssssssooss	5-10-5	50	1537
688434	n/a	14333	14380	0	AGGCTGACACTGAACAGATA	soooossssssssssooss	5-10-5	18	1538
688435	n/a	14334	14381	0	AAGGCTGACACTGAACAGAT	soooossssssssssooss	5-10-5	45	1539
688436	n/a	14335	14382	0	AAAGGCTGACACTGAACAGA	soooossssssssssooss	5-10-5	28	1540
688437	n/a	14336	14383	0	GAAAGGCTGACACTGAACAG	soooossssssssssooss	5-10-5	18	1541
688438	n/a	14337	14384	0	TGAAAGGCTGACACTGAACA	soooossssssssssooss	5-10-5	23	1542
688439	n/a	14358	14405	0	AATGGGATTTAAAATGATGT	soooossssssssssooss	5-10-5	26	1543
688440	n/a	14359	14406	0	AAATGGGATTTAAAATGATG	soooossssssssssooss	5-10-5	30	1544
688441	n/a	14360	14407	0	CAAATGGGATTTAAAATGAT	soooossssssssssooss	5-10-5	16	1545

TABLE 60

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
576816	310	7990	8043	GCCTTACTCTAGGACCAAGA	sssssssssssssssss	5-10-5	62	20
619420	310	7990	8043	GCCTTACTCTAGGACCAAGA	soooossssssssssooss	5-10-5	75	20
688078	313	7993	8046	TGCCTTACTCTAGGACCA	soooossssssssooss	5-8-5	67	1130
688237	485	8165	8218	ATCAAAGATTAATGAAAC	soooossssssssooss	5-8-5	42	1289
688238	486	8166	8219	CATCAAAGATTAATGAAA	soooossssssssooss	5-8-5	0	1290
688239	487	8167	8220	CCATCAAAGATTAATGAA	soooossssssssooss	5-8-5	39	1291
688240	488	8168	8221	TCCATCAAAGATTAATGA	soooossssssssooss	5-8-5	45	1292
688241	489	8169	8222	TTCCATCAAAGATTAATG	soooossssssssooss	5-8-5	0	1293
688242	490	8170	8223	TTTCCATCAAAGATTAAT	soooossssssssooss	5-8-5	53	1294
688243	491	8171	8224	GTTTCCATCAAAGATTAA	soooossssssssooss	5-8-5	46	1295
688244	492	8172	8225	AGTTTCCATCAAAGATTA	soooossssssssooss	5-8-5	43	1296
688245	493	8173	8226	CAGTTTCCATCAAAGATT	soooossssssssooss	5-8-5	0	1297
688246	494	8174	8227	CCAGTTTCCATCAAAGAT	soooossssssssooss	5-8-5	44	1298
688247	495	8175	8228	TCCAGTTTCCATCAAAGA	soooossssssssooss	5-8-5	46	1299
688248	496	8176	8229	TTCCAGTTTCCATCAAAG	soooossssssssooss	5-8-5	58	1300
688249	497	8177	8230	ATTCCAGTTTCCATCAAA	soooossssssssooss	5-8-5	46	1301
688250	498	8178	8231	CATTCCAGTTTCCATCAA	soooossssssssooss	5-8-5	50	1302
688251	499	8179	8232	CCATTCCAGTTTCCATCA	soooossssssssooss	5-8-5	58	1303
688252	500	8180	8233	CCCATTCCAGTTTCCATC	soooossssssssooss	5-8-5	66	1304
688253	501	8181	8234	CCCCATTCCAGTTTCCAT	soooossssssssooss	5-8-5	59	1305
688254	502	8182	8235	TCCCCATTCCAGTTTCCA	soooossssssssooss	5-8-5	52	1306
688255	503	8183	8236	ATCCCCATTCCAGTTTCC	soooossssssssooss	5-8-5	61	1307
688256	504	8184	8237	GATCCCCATTCCAGTTTC	soooossssssssooss	5-8-5	50	1308
688257	505	8185	8238	CGATCCCCATTCCAGTTT	soooossssssssooss	5-8-5	48	1309
688258	506	8186	8239	GCGATCCCCATTCCAGTT	soooossssssssooss	5-8-5	44	1310
688259	507	8187	8240	TGCGATCCCCATTCCAGT	soooossssssssooss	5-8-5	49	1311
688260	508	8188	8241	CTGCGATCCCCATTCCAG	soooossssssssooss	5-8-5	70	1312
688261	509	8189	8242	GCTGCGATCCCCATTCCA	soooossssssssooss	5-8-5	72	1313
688262	510	8190	8243	TGCTGCGATCCCCATTCC	soooossssssssooss	5-8-5	47	1314
688263	511	8191	8244	GTGCTGCGATCCCCATTC	soooossssssssooss	5-8-5	0	1315
688264	512	8192	8245	TGTGCTGCGATCCCCATT	soooossssssssooss	5-8-5	56	1316
688265	513	8193	8246	ATGTGCTGCGATCCCCAT	soooossssssssooss	5-8-5	65	1317
688266	514	8194	8247	TATGTGCTGCGATCCCCA	soooossssssssooss	5-8-5	61	1318
688267	533	8213	8266	AAGTATAATTGATAGTCC	soooossssssssooss	5-8-5	47	1319

TABLE 60-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688268	534	8214	8267	GAAGTATAATTGATAGTC	soooossssssssooss	5-8-5	48	1320
688269	535	8215	8268	GGAAGTATAATTGATAGT	soooossssssssooss	5-8-5	35	1321
688270	536	8216	8269	TGGAAGTATAATTGATAG	soooossssssssooss	5-8-5	41	1322
688271	537	8217	8270	GTGGAAGTATAATTGATA	soooossssssssooss	5-8-5	48	1323
688272	538	8218	8271	TGTGGAAGTATAATTGAT	soooossssssssooss	5-8-5	42	1324
688273	539	8219	8272	CTGTGGAAGTATAATTGA	soooossssssssooss	5-8-5	32	1325
688274	540	8220	8273	TCTGTGGAAGTATAATTG	soooossssssssooss	5-8-5	0	1326
688275	541	8221	8274	GTCTGTGGAAGTATAATT	soooossssssssooss	5-8-5	0	1327
688276	542	8222	8275	TGTCTGTGGAAGTATAAT	soooossssssssooss	5-8-5	65	1328
688277	543	8223	8276	CTGTCTGTGGAAGTATAA	soooossssssssooss	5-8-5	0	1329
688278	544	8224	8277	TCTGTCTGTGGAAGTATA	soooossssssssooss	5-8-5	62	1330
688279	545	8225	8278	TTCTGTCTGTGGAAGTAT	soooossssssssooss	5-8-5	55	1331
688280	546	8226	8279	GTTCTGTCTGTGGAAGTA	soooossssssssooss	5-8-5	77	1332
688281	547	8227	8280	AGTTCTGTCTGTGGAAGT	soooossssssssooss	5-8-5	63	1333
688282	548	8228	8281	AAGTTCTGTCTGTGGAAG	soooossssssssooss	5-8-5	49	1334
688283	549	8229	8282	TAAGTTCTGTCTGTGGAA	soooossssssssooss	5-8-5	0	1335
688284	550	8230	8283	CTAAGTTCTGTCTGTGGA	soooossssssssooss	5-8-5	54	1336
688285	551	8231	8284	ACTAAGTTCTGTCTGTGG	soooossssssssooss	5-8-5	69	1337
688286	552	8232	8285	AACTAAGTTCTGTCTGTG	soooossssssssooss	5-8-5	65	1338
688287	553	8233	8286	AAACTAAGTTCTGTCTGT	soooossssssssooss	5-8-5	40	1339
688288	554	8234	8287	GAAACTAAGTTCTGTCTG	soooossssssssooss	5-8-5	36	1340
688289	555	8235	8288	AGAAACTAAGTTCTGTCT	soooossssssssooss	5-8-5	47	1341
688290	556	8236	8289	TAGAAACTAAGTTCTGTC	soooossssssssooss	5-8-5	48	1342
688291	557	8237	8290	GTAGAAACTAAGTTCTGT	soooossssssssooss	5-8-5	45	1343
688292	558	8238	8291	GGTAGAAACTAAGTTCTG	soooossssssssooss	5-8-5	44	1344
688293	559	8239	8292	AGGTAGAAACTAAGTTCT	soooossssssssooss	5-8-5	58	1345
688294	560	8240	8293	GAGGTAGAAACTAAGTTC	soooossssssssooss	5-8-5	45	1346
688295	561	8241	8294	GGAGGTAGAAACTAAGTT	soooossssssssooss	5-8-5	47	1347
688296	562	8242	8295	GGGAGGTAGAAACTAAGT	soooossssssssooss	5-8-5	44	1348
688297	563	8243	8296	TGGGAGGTAGAAACTAAG	soooossssssssooss	5-8-5	47	1349
688298	564	8244	8297	GTGGGAGGTAGAAACTAA	soooossssssssooss	5-8-5	53	1350
688299	565	8245	8298	AGTGGGAGGTAGAAACTA	soooossssssssooss	5-8-5	43	1351
688300	566	8246	8299	AAGTGGGAGGTAGAAACT	soooossssssssooss	5-8-5	36	1352

TABLE 60-continued

Percent inhibition of the C9ORF72 mRNA levels compared to PBS control by antisense oligonucleotides targeting SEQ ID NOs: 1, 2 and 19

ISIS NO	SEQ ID NO: 1 Start Site	SEQ ID NO: 2 Start Site	SEQ ID NO: 19 Start Site	Sequence	Linkage	Motif	% inhibition (RTS3750)	SEQ ID NO
688301	567	8247	8300	GAAGTGGGAGGTAGAAAC	sooossssssssooss	5-8-5	41	1353
688302	568	8248	8301	TGAAGTGGGAGGTAGAAA	sooossssssssooss	5-8-5	0	1354
688303	569	8249	8302	ATGAAGTGGGAGGTAGAA	sooossssssssooss	5-8-5	41	1355
688304	570	8250	8303	TATGAAGTGGGAGGTAGA	sooossssssssooss	5-8-5	31	1356
688305	571	8251	8304	CTATGAAGTGGGAGGTAG	sooossssssssooss	5-8-5	46	1357
688306	572	8252	8305	TCTATGAAGTGGGAGGTA	sooossssssssooss	5-8-5	30	1358
688307	573	8253	8306	CTCTATGAAGTGGGAGGT	sooossssssssooss	5-8-5	54	1359
688308	574	8254	8307	ACTCTATGAAGTGGGAGG	sooossssssssooss	5-8-5	50	1360
688309	575	8255	8308	CACTCTATGAAGTGGGAG	sooossssssssooss	5-8-5	60	1361
688310	576	8256	8309	ACACTCTATGAAGTGGGA	sooossssssssooss	5-8-5	56	1362
688311	577	8257	8310	CACACTCTATGAAGTGGG	sooossssssssooss	5-8-5	57	1363
688312	578	8258	8311	ACACACTCTATGAAGTGG	sooossssssssooss	5-8-5	41	1364
688313	579	8259	8312	CACACACTCTATGAAGTG	sooossssssssooss	5-8-5	38	1365

Example 13

Dose-Dependent Antisense Inhibition of Human C9ORF72 mRNA in HepG2 Cells

[0481] Antisense oligonucleotides from the study described above exhibiting significant in vitro inhibition of C9ORF72 mRNA were selected and tested at various doses in HepG2 cells. ISIS 619420 described in Example 2 hereinabove was also tested. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.33 μM, 1.00 μM, 3.00 μM, or 9.00 μM concentrations of antisense oligonucleotide. After a treatment period of approximately 16 hours, RNA was isolated from the cells and C9ORF72 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3750 was used to measure the total C9ORF72 mRNA levels. C9ORF72 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of C9ORF72 levels, relative to untreated control cells.

[0482] The half maximal inhibitory concentration (IC^{50}) of each oligonucleotide is also presented in the Tables below. As shown in the Tables below, total C9ORF72 mRNA levels were reduced in a dose-dependent manner in some of the antisense oligonucleotide treated cells.

TABLE 61

Dose-dependent inhibition of total C9ORF72 mRNA levels in HepG2 cells

ISIS No	0.33 μM	1.00 μM	3.00 μM	9.00 μM	IC_{50} (μM)
619411	55	68	89	90	<0.3
619420	41	74	87	95	0.4
655178	45	63	84	92	0.4
687962	48	61	83	86	0.4
687963	34	53	77	76	0.8
687978	50	60	70	65	<0.3
688077	49	63	70	86	0.3
688088	51	72	86	92	<0.3
688089	51	66	76	85	<0.3
688099	41	66	73	85	0.5
688100	44	64	79	86	0.4
688101	53	64	77	74	<0.3
688102	29	43	52	62	2.4
688172	31	47	77	88	1.0
688261	35	41	50	49	6.7
688347	41	60	79	89	0.5
688348	57	61	85	92	<0.3
688352	51	64	71	75	<0.3

TABLE 62

Dose-dependent inhibition of total C9ORF72 mRNA levels in HepG2 cells

ISIS No	0.33 μM	1.00 μM	3.00 μM	9.00 μM	IC_{50} (μM)
619412	48	69	83	87	0.3
619413	34	33	47	85	1.8

TABLE 62-continued

Dose-dependent inhibition of total C9ORF72 mRNA levels in HepG2 cells

ISIS No	0.33 μM	1.00 μM	3.00 μM	9.00 μM	IC_{50} (μM)
619420	53	72	80	84	<0.3
655173	48	62	76	89	0.4
655174	38	59	82	79	0.6
655179	47	68	83	80	0.3
655186	40	53	82	85	0.6
655203	28	61	87	79	0.7
655230	36	59	77	89	0.7
655231	49	69	78	81	<0.3
688349	40	63	84	79	0.5
688351	47	74	81	88	0.3
688355	50	56	79	83	0.4
688356	47	64	78	85	0.3
688364	29	44	57	78	1.5
688374	38	50	79	70	0.8
688378	41	67	85	87	0.4
688379	39	42	83	83	0.8
688380	50	62	68	82	0.3

TABLE 63

Dose-dependent inhibition of total C9ORF72 mRNA levels in HepG2 cells

ISIS No	0.33 μM	1.00 μM	3.00 μM	9.00 μM	IC_{50} (μM)
619420	37	39	87	92	0.3
619423	55	73	84	82	<0.3
655173	53	83	82	94	<0.3
655233	34	28	41	79	2.4
655290	16	27	68	78	2.0
655417	73	42	87	84	<0.3
688022	51	71	80	76	<0.3
688360	32	63	85	72	0.6
688361	46	70	34	85	0.5
688382	42	58	79	81	0.5
688383	43	63	80	87	0.5
688385	41	60	82	84	0.5
688400	61	81	85	85	<0.3
688404	38	62	78	81	0.6
688407	37	53	71	85	0.8
688415	43	18	79	73	1.3
688416	48	74	73	81	<0.3
688417	33	61	67	74	0.8
688418	34	55	77	82	0.8

Example 14

Tolerability of Antisense Oligonucleotides Targeting Human C9ORF72 in Mice

[0483] Antisense oligonucleotides from the Examples above were tested in a standard mouse model to assess tolerability of the oligonucleotides. The rodents were assessed by standard FOB assays and measurement of GFAP and/or AIF expression levels.

[0484] Groups of mice were administered a single ICV dose of 700 μg of ISIS oligonucleotides.

Mouse FOB Assay

[0485] At 3 hours, one week, 2 weeks, 4 weeks, 6 weeks, and 8 weeks post injection, each mouse was evaluated according to 7 different criteria. The 7 criteria are (1) the mouse was bright, alert, and responsive; (2) the mouse was standing or hunched without stimuli; (3) the mouse showed any movement without stimuli (4) the mouse demonstrated forward movement after it was lifted; (5) the mouse demonstrated any movement after it was lifted; (6) the mouse responded to a tail pinch; (7) regular breathing. For each of the 7 different criteria, each mouse was given a sub-score of 0 if it met the criteria or 1 if it did not. After all of the 7 criteria were evaluated, the sub-scores were summed for each mouse and then averaged for each group. For example, if a mouse was bright, alert, and responsive 3 hours after the 700 μg ICV dose, and met all other criteria, it would get a summed score of 0. If another mouse was not bright, alert, and responsive 3 hours after the 700 μg ICV dose but met all other criteria, it would receive a score of 1.

[0486] The results are presented as individual scores for each mouse in each group. The expression levels of GFAP and AIF1 in the thoracic spinal cord of the mice were assessed by qRT-PCR after 8 weeks.

Study 1 with cEt Oligonucleotides

[0487] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 64

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site
672744	1335
672747	1338
672774	1368
672775	1369
672778	1372
672779	1373
672831	1440
672908	1349
672909	1350
672919	1360
672924	1368
672925	1369
672927	1371
672928	1372
672929	1373
672976	1435
672980	1439
672981	1440

TABLE 65

FOB scores in C57/Bl6 mice

	3 hr
PBS	0, 0, 0, 0
672744	7, 7, 7, 7
672747	7, 7, 7, 7
672774	5, 5, 5, 5
672775	5, 7, 5, 7
672778	1, 7, 1, 1
672779	7, 2, 5, 7
672831	7, 7, 7, 7
672908	6, 1, 1, 1
672909	6, 6, 6, 4
672919	5, 5, 5, 5
672924	4, 6, 7, 6
672925	4, 4, 4, 4
672927	0, 0, 0, 0
672928	0, 0, 0, 0
672929	7, 7, 7, 7
672976	5, 7, 7, 7
672980	2, 2, 2, 2
672981	6, 6, 6, 6

Study 2 with cEt Oligonucleotides

[0488] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 66

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
672982	1441
672983	1442
672984	1443
672985	1444
673021	1510
673023	1512
673026	1515
673036	1327
673047	1338
673057	1348
673058	1349
673067	1358
673068	1359
673074	1368
673079	1373
673082	1376
673088	1396
673125	1434
673126	1435
673127	1436
672832	1441

TABLE 67

FOB scores in C57/Bl6 mice

	3 hr
PBS	0, 0, 0, 0
672982	7, 7, 7, 7
672983	5, 7, 5, 7
672984	4, 6, 6, 6
672985	5, 4, 4, 4
673021	5, 5, 5, 5
673023	7, 7, 7, 7
673026	0, 0, 0, 0
673036	7, 7, 7, 7
673047	5, 5, 5, 6
673057	5, 0, 5, 7
673058	3, 3, 3, 3
673067	7, 0, 0, 6
673068	6, 6, 6, 6
673074	1, 1, 1, 1
673079	6, 6, 7, 6
673082	1, 3, 0, 3
673088	6, 6, 6, 6
673125	0, 0, 0, 0
673126	0, 0, 0, 0
673127	7, 7, 7, 7
672832	6, 6, 6, 7

Study 3 with cEt Oligonucleotides

[0489] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 68

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
673131	1440
673132	1441

TABLE 68-continued

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
673133	1442
673134	1443
673135	1444
673193	1334
673203	1344
673204	1345
673224	1368
673225	1369
673226	1370
673228	1372
673229	1373
673275	1434
673276	1435
673280	1439

TABLE 69

FOB scores in C57/Bl6 mice

	3 hr
PBS	0, 0, 0, 0
673131	4, 4, 4, 5
673132	4, 4, 4, 4
673133	4, 3, 4, 3
673134	5, 6, 3, 6
673135	5, 3, 4, 5
673193	6, 6, 6, 6
673203	4, 3, 4, 4
673204	2, 3, 2, 2
673224	5, 5, 5, 5
673225	1, 2, 2, 2
673226	1, 1, 1, 1
673228	2, 2, 2, 2
673229	6, 6, 6, 6
673275	1, 1, 1, 1
673276	3, 3, 3, 3
673280	1, 1, 1, 1

Study 4 with cEt Oligonucleotides

[0490] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 70

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
673128	1437
673130	1439
673281	1440
673282	1441
673283	1442
673284	1443
673285	1444
673323	1512
673331	1520
673332	1521
673374	1368
673375	1369
673377	1371
673378	1372
673379	1373
673430	1439
673431	1440

TABLE 70-continued

Antisense oligonucleotides targeted to SEQ ID NO: 2	
ISIS No	Start Site of SEQ ID NO: 2
673432	1441
673434	1443

TABLE 71

FOB scores in C57/Bl6 mice	
	3 hr
PBS	0, 0, 0, 0
673128	0, 0, 0, 0
673130	0, 0, 0, 0
673281	7, 7, 7, 7
673282	7, 7, 7, 7
673283	5, 6, 6, 7
673284	5, 0, 0, 0
673285	3, 3, 3, 3
673323	7, 7, 7, 7
673331	7, 7, 7, 7
673332	7, 7, 7, 7
673374	7, 7, 7, 7
673375	2, 3, 2, 3
673377	0, 0, 0, 0
673378	1, 1, 1, 1
673379	4, 6, 6, 7
673430	7, 7, 7, 7
673431	7, 7, 7, 7
673432	7, 6, 6, 7
673434	7, 7, 7, 7

Study 1 with MOE Oligonucleotides

[0491] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 72

Antisense oligonucleotides targeted to SEQ ID NO: 2	
ISIS No	Start Site of SEQ ID NO: 2
619253	1406
619293	1446
619322	1481
619352	1519
619353	1520
619410	7840
619414	8033
619420	7990
619421	28251
619422	3452
619423	13642
653222	1553
653223	5325
655016	1458
655017	1459

TABLE 73

FOB scores in C57/Bl6 mice	
	3 hrs
PBS	0, 0, 0, 0
619253	3, 3, 7, 7

TABLE 73-continued

FOB scores in C57/Bl6 mice	
	3 hrs
619293	6, 4, 2, 4
619322	2, 2, 2, 4
619352	0, 0, 7, 0
619353	6, 4, 7, 6
619410	7, 3, 3, 3
619414	6, 6, 6, 6
619420	7, 7, 7, 7
619421	1, 1, 1, 1
619422	5, 5, 5, 5
619423	6, 6, 6, 6
653222	3, 3, 3, 3
653223	1, 1, 7, 1
655016	5, 5, 3, 3
655017	0, 0, 0, 0

Study 2 with MOE Oligonucleotides

[0492] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 74

Antisense oligonucleotides targeted to SEQ ID NO: 2	
ISIS No	Start Site of SEQ ID NO: 2
619173	1326
619174	1327
619178	1331
619179	1332
619180	1333
619181	1334
619182	1335
619185	1338
619186	1339
619187	1340
619191	1344
619201	1354
619202	1355
619203	1356
619216	1369
619217	1370

TABLE 75

FOB scores in C57/Bl6 mice	
	3 hrs
PBS	0, 0, 0, 0
619173	7, 7, 7, 7
619174	7, 7, 7, 7
619178	7, 7, 7, 7
619179	7, 7, 7, 7
619180	7, 7, 7, 7
619181	7, 7, 7, 7
619182	7, 7, 7, 7
619185	7, 6, 6, 7
619186	6, 6, 6, 7
619187	5, 7, 5, 7
619191	5, 5, 3, 5
619201	3, 3, 3, 1
619202	6, 6, 6, 6
619203	5, 1, 5, 5
619216	4, 4, 4, 4
619217	2, 2, 2, 2

Study 3 with MOE Oligonucleotides

[0493] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 76

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
619218	1371
619245	1398
619246	1399
619247	1400
619251	1404
619252	1405
619259	1412
619260	1413
619276	1429
619278	1431
619280	1433
619284	1437
619292	1445
619297	1450
619298	1451
619307	1466

TABLE 77

FOB scores in C57/Bl6 mice

	3 hrs
PBS	0, 0, 0, 0
619218	4, 2, 7, 7
619245	4, 4, 4, 4
619246	4, 5, 5, 4
619247	1, 1, 1, 1
619251	4, 7, 4, 4
619252	6, 6, 6, 0
619259	6, 6, 4, 5
619260	4, 4, 4, 4
619276	3, 3, 4, 4
619278	4, 7, 2, 7
619280	7, 7, 7, 7
619284	0, 0, 0, 0
619292	6, 5, 5, 5
619297	7, 4, 4, 4
619298	0, 0, 0, 0
619307	0, 0, 0, 0

Study 4 with MOE Oligonucleotides

[0494] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 78

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
619266	1419
619268	1421
619316	1475
619317	1476
619336	1503
619337	1504
619338	1505
619339	1506

TABLE 78-continued

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
619341	1508
619342	1509
619343	1510
619344	1511
619346	1513
619350	1517
619351	1518
619413	8020

TABLE 79

FOB scores in C57/Bl6 mice

	3 hrs
PBS	0, 0, 0, 0
619266	2, 2, 2, 2
619268	6, 6, 7, 7
619316	5, 5, 5, 7
619317	2, 3, 3, 4
619336	1, 1, 1, 7
619338	0, 0, 7, 7
619339	0, 5, 7, 7
619341	7, 7, 7, 7
619342	6, 6, 6, 6
619343	6, 6, 6, 6
619344	7, 7, 7, 7
619346	5, 5, 7, 7
619350	6, 6, 7, 7
619351	6, 6, 7, 7
619413	7, 6, 7, 6

Example 15

Tolerability of Antisense Oligonucleotides Targeting Human C9ORF72 in Rats

[0495] Antisense oligonucleotides from the Examples above were tested in a standard rat model to assess tolerability of the oligonucleotides. The rodents were assessed by standard FOB assays and measurement of GFAP and/or AIF expression levels. Groups of Sprague-Dawley rats were administered intrathecally with 2,000 μg or 3,000 μg of ISIS oligonucleotide as specified in the Tables below.

Rat FOB Assay

[0496] At 3 hours, one week, 2 weeks, 4 weeks, 6 weeks, and 8 weeks post injection the movement of 7 different parts of the body was evaluated for each rat. The 7 body parts were (1) the rat's tail; (2) the rat's posterior posture; (3) the rat's hind limbs; (4) the rat's hind paws; (5) the rat's forepaws; (6) the rat's anterior posture; and (7) the rat's head. For each of the 7 different body parts, each rat was given a sub-score of 0 if the body part was moving or 1 if the body part was paralyzed. After each of the 7 body parts was evaluated, the sub-scores were summed for each rat and then averaged for each group. Saline treated rats generally receive a score of 0.

Study 1 with 5-10-5 MOE Oligonucleotides at 3,000 μg

[0497] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 80

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
619185	1338
619186	1339
619187	1340
619191	1344
619201	1354
619203	1356
619217	1370
619218	1371
619251	1404
619252	1405
619259	1412
619266	1419
619268	1421
619278	1431
619284	1437
619346	1513
619350	1517
619351	1518

TABLE 81

FOB scores in Sprague-Dawley rats

	3 hrs
PBS	0, 0, 0
619185	7, 7, 6
619186	0, 6, 6
619187	7, 0, 6
619191	6, 0, 0
619201	1, 4, 4
619203	5, 1, 5
619217	4, 4, 4
619218	4, 2, 4
619251	6, 6, 6
619252	6, 6, 6
619259	0, 6, 6
619266	4, 4, 3
619268	6, 6, 6
619278	6, 6, 6
619284	1
619346	0, 0, 6
619350	4, 0, 3
619351	3, 3, 3

Study 2 with 5-10-5 MOE Oligonucleotides at 3,000 μg

[0498] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 82

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
619176	1329
619183	1336
619253	1406
619260	1413
619276	1429
619292	1445

TABLE 82-continued

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
619293	1446
619307	1466
619317	1476
619338	1505
619339	1506
619342	1509

TABLE 83

FOB scores in Sprague-Dawley rats

	3 hrs
PBS	0, 1, 1
619176	7, 7, 6
619183	6, 6, 6
619253	7, 6, 6
619260	6, 5, 5
619276	4, 5, 5
619292	6, 6, 0
619293	ND
619307	ND
619317	6, 6, 7
619338	7, 7, 7
619339	7, 6, 3
619342	6, 6, 6

Study 3 with 5-8-5 MOE Oligonucleotides at 3,000 μg

[0499] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 84

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
672595	1340
672599	1344
672602	1347
672624	1372
672636	1399
672637	1400
672640	1403
672642	1405
672651	1414
672652	1415

TABLE 85

FOB scores in Sprague-Dawley rats

	3 hr
PBS	0, 0, 0, 0
672595	4, 0, 3, 0, 4, 4, 4
672599	3, 0, 3, 0, 1, 3
672602	3, 3, 1, 0, 0, 3
672624	3, 0, 0, 7, 3, 3, 3
672636	5, 5, 5, 5
672637	1, 1, 2, 1
672640	6, 6, 5, 1, 6, 0

TABLE 85-continued

FOB scores in Sprague-Dawley rats	
	3 hr
672642	6, 7, 0, 7, 6, 6
672651	6, 1, 1, 6, 3, 2, 3
672652	6, 1, 1, 5, 4, 4

Study 4 with 5-8-5 MOE Oligonucleotides at 3,000 μg

[0500] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below.

TABLE 86

Antisense oligonucleotides targeted to SEQ ID NO: 2	
ISIS No	Start Site of SEQ ID NO: 2
672664	1427
672665	1428
672670	1433
672671	1434
672675	1438
672676	1439
672678	1441
672679	1442
672681	1444
672683	1446
672693	1464
672697	1468
672699	1470
672700	1471
672723	1512
672730	1519

TABLE 87

FOB scores in Sprague-Dawley rats	
ISIS No	3 hr
PBS	1, 1, 1, 0
672664	6, 6, 6, 5
672665	4, 0, 4, 0, 4, 1
672670	4, 2, 4, 4
672671	4, 0, 0, 2, 5, 3
672675	3, 3, 0, 2, 1, 1
672676	1, 5, 0, 1, 1, 4
672678	0, 6, 6, 4, 6, 6
672679	3, 0, 0, 2, 3, 4
672681	1, 2, 2, 2
672683	0, 2, 5, 5, 0, 4
672693	ND
672697	6, 6, 6, 6
672699	0, 0, 3, 4, 4, 1
672700	4, 7, 4, 4
672723	4, 4, 4, 4
672730	2, 2, 3, 3

Study 5 with 5-10-5 MOE Oligonucleotides at 3,000 μg

[0501] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below and indicate that several oligonucleotides were tolerable in this model.

TABLE 88

Antisense oligonucleotides targeted to SEQ ID NO: 2	
ISIS No	Start Site of SEQ ID NO: 2
619411	8004
619412	8012
627833	8001
655153	7916
655173	7992
655178	8014
655179	8016
655202	8098
655231	8183
655232	8186
655233	8189
655417	14089
655420	14331
671081	8110
671082	8140
671083	8230
671084	14316
672561	mRNA

TABLE 89

FOB scores in Sprague-Dawley rats	
	3 hr
PBS	0, 0, 0, 0
619411	6, 6, 6, 6
619412	4, 6, 6, 0
627833	6, 6, 6, 1
655153	6, 6, 6, 1
655173	6, 6, 6, 0
655178	5, 5, 6, 6
655179	6, 6, 6, 6
655202	6, 0, 6, 6
655231	4, 4, 4, 4
655232	0, 0, 0, 0
655233	6, 6, 6, 6
655417	1, 6, 6, 7
655420	5, 7, 5, 5
671081	5, 5, 5, 5
671082	5, 6, 5, 7
671083	7, 6, 7, 6
671084	6, 0, 4, 3
672561	4, 4, 4, 1

Study 6 with 5-8-5 MOE Oligonucleotides at 2,000 μg

[0502] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below and indicate that several oligonucleotides were tolerable in this model.

TABLE 90

Antisense oligonucleotides targeted to SEQ ID NO: 2	
ISIS No	Start Site of SEQ ID NO: 2
687978	14315
688022	7918
688077	7992
688088	8003
688089	8004
688099	8014
688100	8015

TABLE 91

FOB scores in Sprague-Dawley rats

	3 hr
PBS	0, 0, 0, 0
687978	3, 3, 3, 3
688022	3, 3, 3, 3
688077	0, 3, 1, 3, 3, 3
688088	3, 0, 3, 3, 3
688089	1, 3, 3, 3, 3
688099	3, 4, 0, 3, 3
688100	0, 3, 4, 4

Study 7 with 5-8-5 and 5-10-5 MOE Oligonucleotides at 2,000 μg

[0503] The oligonucleotides tested in this study are presented in the Table below. The FOB scores are given below and indicate that several oligonucleotides were tolerable in this model.

TABLE 92

Antisense oligonucleotides targeted to SEQ ID NO: 2

ISIS No	Start Site of SEQ ID NO: 2
688101	8016
688348	7993
688356	8015
688380	8184
688382	8187
688400	14090
688407	14304
688415	14312
688416	14313
688417	14314

TABLE 93

FOB scores in Sprague-Dawley rats

	3 hr
PBS	0, 0, 0, 0
688101	3, 2, 2, 2
688348	2, 1, 2, 2
688356	3, 3, 3, 3
688380	0, 0, 1, 0
688382	2, 2, 2, 2
688400	0, 3, 3, 3
688407	2, 2, 2, 2
688415	3, 0, 3, 3
688416	3, 3, 3, 3
688417	4, 4, 3, 3

Example 16

Acute Tolerability of Oligonucleotides from WO 2014/062691

[0504] Oligonucleotides described in WO 2014/062691 were tested in an acute tolerability study in mice. The tested oligonucleotides include ISIS 576816, ISIS 576974, ISIS 577061, ISIS 577065, and ISIS 577083, which are 5-10-5 MOE gapmers with a full phosphorothioate backbone and each cytosine is a 5-methylcytosine. The sequences are provided in the Table below. Mice were separated into groups of 3 or 4 mice. Each mouse in each group of mice was administered a single ICV dose of either 700 ug of the oligonucleotides in the table below. At 3 hours post injection, each mouse was evaluated according to 7 different criteria. The 7 criteria are (1) the mouse was bright, alert, and responsive; (2) the mouse was standing or hunched without stimuli; (3) the mouse shows any movement without stimuli (4) the mouse demonstrates forward movement after its lifted; (5) the mouse demonstrates any movement after its lifted; (6) the mouse responds to a tail pinch; (7) regular breathing. For each of the 7 different criteria, each mouse was given a sub-score of 0 if it met the criteria or 1 if it did not. After all of the 7 criteria were evaluated, the sub-scores were summed for each mouse and then averaged for each group. For example, if a mouse was bright, alert, and responsive 3 hours after the ICV dose, and met all other criteria, it would get a summed score of 0. If another mouse was not bright, alert, and responsive 3 hours after the dose but met all other criteria, it would receive a score of 1. Saline treated mice generally receive a score of 0. Results are presented as the average score for each treatment group in the Table below. These results demonstrate that ISIS 576816, ISIS 576974, ISIS 577061, ISIS 577065, and ISIS 577083 were poorly tolerated.

TABLE 94

Antisense oligonucleotides from WO 2014/062691

ISIS No	Sequence	SEQ ID NO
576816	GCCTTACTCTAGGACCAAGA	20
576974	GGGACACTACAAGGTAGTAT	401
577061	TACAGGCTGCGGTTGTTTCC	97
577065	CCCGGCCCCTAGCGCGCGAC	98
577083	GGTAACTTCAAACTCTTGGG	382

TABLE 95

FOB scores in mice

	3 hrs	5 hrs
576816	7, 7, 7	7, 7, 7
576974	6, 5, 6	7, 5, 7
577061	7, 7, 7	7, 7, 7
577065	6, 6, 6	7, 7, 7
577083	7, 7, 7	7, 7, 7

[0505] Gapmers from the studies described above, including compounds ISIS 576816, ISIS 576974, ISIS 577061, ISIS 577065, and ISIS 577083 which were previously disclosed in WO 2014/062691, are tested for tolerability in Sprague-Dawley rats. Rats are injected intrathecally with 3 mg of a single dose of ISIS oligonucleotide. A control group of rats is injected intrathecally with PBS. Acute tolerability is assessed 3 hours post-dose using a functional observational battery (FOB). This score is used to evaluate the acute tolerability of a compound with lower scores denoting better tolerated compounds. Control animals usually have a score of '0' or '1'. At 3 hours post injection, the rats are observed by placing each rat on the cage top and evaluating certain functions, assigning a number of '0' or '1' depending on whether the rat exhibits normal function in the region of interest (0) or does not (1) for each function, and then adding the total scores. Seven regions

are assessed, including tail, hind paws, hind legs, hind end, front posture, fore paws, and head.

[0506] Poor acute tolerability in mice is generally predictive of poor acute tolerability in rats. For example, ISIS 619185 (see Example 14 hereinabove) had an acute tolerability score of 7,6,6,6 (4 animals) in mouse, and an acute tolerability score of 7,7,6 (3 animals) in rats (See Example 15 hereinabove). ISIS 619342 (see Example 14 hereinabove) had an acute tolerability score of 6,6,6,6 (4 animals) in mouse, and an acute tolerability score of 6,6,6 (3 animals) in rats (see Example 15 hereinabove). Both compounds were deemed to be poorly tolerated acutely. It is therefore expected that the compounds ISIS 576816, ISIS 576974, ISIS 577061, ISIS 577065, and ISIS 577083, which were previously disclosed in WO 2014/062691, will show similarly high FOB scores in rats as they did in mice.

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ttgtcgctgt ttgacgcacc 20

<210> SEQ ID NO 53
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 53

tctgtctttg gagcccaaat 20

<210> SEQ ID NO 54
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 54

cttgtcgctg tttgacgcac 20

<210> SEQ ID NO 55
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 55

acttgtcgct gtttgacgca 20

<210> SEQ ID NO 56
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 56

aacttgtcgc tgtttgacgc 20

<210> SEQ ID NO 57
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 57

gaacttgtcg ctgtttgacg 20

<210> SEQ ID NO 58
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 58

ggaacttgtc gctgtttgac 20

<210> SEQ ID NO 59
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 59

cggaacttgt cgctgtttga 20

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<210> SEQ ID NO 60
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 60

tagcgggaca ccgtaggtta 20

<210> SEQ ID NO 61
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 61

gcggaacttg tcgctgtttg 20

<210> SEQ ID NO 62
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 62

ggcggaactt gtcgctgttt 20

<210> SEQ ID NO 63
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 63

gggcggaact tgtcgctgtt 20

<210> SEQ ID NO 64
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 64

tgggcggaac ttgtcgctgt 20

<210> SEQ ID NO 65
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 65

gtgggcggaa cttgtcgctg 20

<210> SEQ ID NO 66
<211> LENGTH: 20

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 66

cgtgggcgga acttgtcgct 20

<210> SEQ ID NO 67
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 67

ctagcgggac accgtaggtt 20

<210> SEQ ID NO 68
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 68

cctagcggga caccgtaggt 20

<210> SEQ ID NO 69
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 69

tcctagcggg acaccgtagg 20

<210> SEQ ID NO 70
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 70

gacggctgac acaccaagcg 20

<210> SEQ ID NO 71
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 71

ggacggctga cacaccaagc 20

<210> SEQ ID NO 72
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 72

gggacggctg acacaccaag 20

<210> SEQ ID NO 73
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 73

agggacggct gacacaccaa 20

<210> SEQ ID NO 74
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 74

cagggacggc tgacacacca 20

<210> SEQ ID NO 75
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 75

ttcctagcgg gacaccgtag 20

<210> SEQ ID NO 76
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 76

gcagggacgg ctgacacacc 20

<210> SEQ ID NO 77
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 77

agcagggacg gctgacacac 20

<210> SEQ ID NO 78
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 78

cagcagggac ggctgacaca 20

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<210> SEQ ID NO 79
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 79

gcagcaggga cggctgacac 20

<210> SEQ ID NO 80
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 80

ggcagcaggg acggctgaca 20

<210> SEQ ID NO 81
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 81

tttcctagcg ggacaccgta 20

<210> SEQ ID NO 82
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 82

agaagcaacc gggcagcagg 20

<210> SEQ ID NO 83
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 83

gagaagcaac cgggcagcag 20

<210> SEQ ID NO 84
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 84

agagaagcaa ccgggcagca 20

<210> SEQ ID NO 85
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 85

aagagaagca accgggcagc 20

<210> SEQ ID NO 86
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 86

aaagagaagc aaccgggcag 20

<210> SEQ ID NO 87
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 87

gttttctatg tgcgatgacg 20

<210> SEQ ID NO 88
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 88

tgttttctat gtgcgatgac 20

<210> SEQ ID NO 89
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 89

ctgttttcta tgtgcgatga 20

<210> SEQ ID NO 90
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 90

tctgttttct atgtgcgatg 20

<210> SEQ ID NO 91
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 91

gtctgttttc tatgtgcgat 20

<210> SEQ ID NO 92
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 92

tgtctgtttt ctatgtgcga 20

<210> SEQ ID NO 93
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 93

ctgtctgttt tctatgtgcg 20

<210> SEQ ID NO 94
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 94

tctgtctgtt ttctatgtgc 20

<210> SEQ ID NO 95
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 95

gtctgtctgt tttctatgtg 20

<210> SEQ ID NO 96
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 96

cgtctgtctg ttttctatgt 20

<210> SEQ ID NO 97
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 97

tacaggctgc ggttgtttcc 20

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<210> SEQ ID NO 98
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 98

cccggcccct agcgcgcgac 20

<210> SEQ ID NO 99
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 99

ccccaaaaga gaagcaaccg 20

<210> SEQ ID NO 100
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 100

cccccaaaag agaagcaacc 20

<210> SEQ ID NO 101
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 101

gcccccaaaa gagaagcaac 20

<210> SEQ ID NO 102
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 102

cgcccccaaa agagaagcaa 20

<210> SEQ ID NO 103
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 103

ccgcccccaa aagagaagca 20

<210> SEQ ID NO 104
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 104

cccgccccca aaagagaagc 20

<210> SEQ ID NO 105
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 105

ccccgccccc aaaagagaag 20

<210> SEQ ID NO 106
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 106

accccgcccc caaaagagaa 20

<210> SEQ ID NO 107
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 107

gaccccgccc ccaaaagaga 20

<210> SEQ ID NO 108
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 108

agaccccgcc cccaaaagag 20

<210> SEQ ID NO 109
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 109

tagaccccgc ccccaaaaga 20

<210> SEQ ID NO 110
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 110

ctagaccccg cccccaaaag 20

<210> SEQ ID NO 111
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 111

gctagacccc gcccccaaaa 20

<210> SEQ ID NO 112
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 112

tgctagaccc cgcccccaaa 20

<210> SEQ ID NO 113
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 113

ttgctagacc ccgccccaa 20

<210> SEQ ID NO 114
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 114

cttgctagac cccgccccca 20

<210> SEQ ID NO 115
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 115

tcttgctaga ccccgccccc 20

<210> SEQ ID NO 116
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 116

ctcttgctag accccgcccc 20

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<210> SEQ ID NO 117
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 117

gctcttgcta gaccccgccc 20

<210> SEQ ID NO 118
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 118

tgctcttgct agaccccgcc 20

<210> SEQ ID NO 119
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 119

ctgctcttgc tagaccccgc 20

<210> SEQ ID NO 120
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 120

cctgctcttg ctagaccccg 20

<210> SEQ ID NO 121
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 121

acctgctctt gctagacccc 20

<210> SEQ ID NO 122
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 122

cacctgctct tgctagaccc 20

<210> SEQ ID NO 123
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 123

acacctgctc ttgctagacc 20

<210> SEQ ID NO 124
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 124

cacacctgct cttgctagac 20

<210> SEQ ID NO 125
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 125

ccacacctgc tcttgctaga 20

<210> SEQ ID NO 126
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 126

cccacacctg ctcttgctag 20

<210> SEQ ID NO 127
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 127

acccacacct gctcttgcta 20

<210> SEQ ID NO 128
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 128

aacccacacc tgctcttgct 20

<210> SEQ ID NO 129
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 129

aaacccacac ctgctcttgc 20

<210> SEQ ID NO 130
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 130

taaacccaca cctgctcttg 20

<210> SEQ ID NO 131
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 131

ctaaacccac acctgctctt 20

<210> SEQ ID NO 132
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 132

cctaaaccca cacctgctct 20

<210> SEQ ID NO 133
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 133

tcctaaaccc acacctgctc 20

<210> SEQ ID NO 134
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 134

ctcctaaacc cacacctgct 20

<210> SEQ ID NO 135
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 135

gtacctgttc tgtctttgga 20

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<210> SEQ ID NO 136
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 136

ccatcactga gaagtacctg 20

<210> SEQ ID NO 137
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 137

ggcataatgt tctgactatc 20

<210> SEQ ID NO 138
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 138

cctcctaaac ccacacctgc 20

<210> SEQ ID NO 139
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 139

acctcctaaa cccacacctg 20

<210> SEQ ID NO 140
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 140

cacctcctaa acccacacct 20

<210> SEQ ID NO 141
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 141

acacctccta aacccacacc 20

<210> SEQ ID NO 142
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 142

cacacctcct aaacccacac 20

<210> SEQ ID NO 143
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 143

acacacctcc taaacccaca 20

<210> SEQ ID NO 144
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 144

cacacacctc ctaaacccac 20

<210> SEQ ID NO 145
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 145

acacacacct cctaaaccca 20

<210> SEQ ID NO 146
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 146

aacacacacc tcctaaaccc 20

<210> SEQ ID NO 147
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 147

aaacacacac ctcctaaacc 20

<210> SEQ ID NO 148
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 148

aaaacacaca cctcctaaac 20

<210> SEQ ID NO 149
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 149

aaaaacacac acctcctaaa 20

<210> SEQ ID NO 150
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 150

caaaaacaca cacctcctaa 20

<210> SEQ ID NO 151
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 151

acaaaaacac acacctccta 20

<210> SEQ ID NO 152
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 152

aacaaaaaca cacacctcct 20

<210> SEQ ID NO 153
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 153

aaacaaaaac acacacctcc 20

<210> SEQ ID NO 154
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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aaaacaaaaa cacacacctc 20

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gaaaaacaaa aacacacacc 20

<210> SEQ ID NO 156
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 156

ggaaaaacaa aaacacacac 20

<210> SEQ ID NO 157
<211> LENGTH: 20
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tgggaaaaac aaaaacacac 20

<210> SEQ ID NO 158
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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gtgggaaaaa caaaaacaca 20

<210> SEQ ID NO 159
<211> LENGTH: 20
<212> TYPE: DNA
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ggtgggaaaa acaaaaacac 20

<210> SEQ ID NO 160
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ctgtgagagc aagtagtggg 20

<210> SEQ ID NO 161
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<212> TYPE: DNA
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actgtgagag caagtagtgg 20

<210> SEQ ID NO 162
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tactgtgaga gcaagtagtg 20

<210> SEQ ID NO 163
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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gtactgtgag agcaagtagt 20

<210> SEQ ID NO 164
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agtactgtga gagcaagtag 20

<210> SEQ ID NO 165
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gagtactgtg agagcaagta 20

<210> SEQ ID NO 166
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cgagtactgt gagagcaagt 20

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gcgagtactg tgagagcaag 20

<210> SEQ ID NO 168
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agcgagtact gtgagagcaa 20

<210> SEQ ID NO 169
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<212> TYPE: DNA
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cagcgagtac tgtgagagca 20

<210> SEQ ID NO 170
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<212> TYPE: DNA
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tcagcgagta ctgtgagagc 20

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<212> TYPE: DNA
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ctcagcgagt actgtgagag 20

<210> SEQ ID NO 172
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<212> TYPE: DNA
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<400> SEQUENCE: 172

cctcagcgag tactgtgaga 20

<210> SEQ ID NO 173
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<212> TYPE: DNA
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ccctcagcga gtactgtgag 20

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accctcagcg agtactgtga 20

<210> SEQ ID NO 175
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<212> TYPE: DNA
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<400> SEQUENCE: 175

caccctcagc gagtactgtg 20

<210> SEQ ID NO 176
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<212> TYPE: DNA
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<400> SEQUENCE: 176

tcaccctcag cgagtactgt 20

<210> SEQ ID NO 177
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 177

ttcaccctca gcgagtactg 20

<210> SEQ ID NO 178
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 178

gttcaccctc agcgagtact 20

<210> SEQ ID NO 179
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 179

tgttcaccct cagcgagtac 20

<210> SEQ ID NO 180
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 180

ttgttcaccc tcagcgagta 20

<210> SEQ ID NO 181
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 181

cttgttcacc ctcagcgagt 20

<210> SEQ ID NO 182
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 182

tcttgttcac cctcagcgag 20

<210> SEQ ID NO 183
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 183

ttcttgttca ccctcagcga 20

<210> SEQ ID NO 184
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 184

tttcttgttc accctcagcg 20

<210> SEQ ID NO 185
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 185

ttttcttgtt caccctcagc 20

<210> SEQ ID NO 186
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 186

cttttcttgt tcaccctcag 20

<210> SEQ ID NO 187
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 187

tcttttcttg ttcaccctca 20

<210> SEQ ID NO 188
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 188

gtcttttctt gttcaccctc 20

<210> SEQ ID NO 189
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 189

ggtcttttct tgttcaccct 20

<210> SEQ ID NO 190
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 190

aggtcttttc ttgttcaccc 20

<210> SEQ ID NO 191
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 191

caggtctttt cttgttcacc 20

<210> SEQ ID NO 192
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 192

tcaggtcttt tcttgttcac 20

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<210> SEQ ID NO 193
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 193

atcaggtctt ttcttgttca 20

<210> SEQ ID NO 194
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 194

tatcaggtct tttcttgttc 20

<210> SEQ ID NO 195
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 195

ttatcaggtc ttttcttgtt 20

<210> SEQ ID NO 196
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 196

tttatcaggt cttttcttgt 20

<210> SEQ ID NO 197
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 197

ctttatcagg tcttttcttg 20

<210> SEQ ID NO 198
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 198

tctttatcag gtcttttctt 20

<210> SEQ ID NO 199
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 199

atctttatca ggtcttttct 20

<210> SEQ ID NO 200
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 200

aatctttatc aggtcttttc 20

<210> SEQ ID NO 201
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 201

taatctttat caggtctttt 20

<210> SEQ ID NO 202
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 202

ttaatcttta tcaggtcttt 20

<210> SEQ ID NO 203
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 203

gttaatcttt atcaggtctt 20

<210> SEQ ID NO 204
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 204

ggttaatctt tatcaggtct 20

<210> SEQ ID NO 205
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 205

tggttaatct ttatcaggtc 20

<210> SEQ ID NO 206
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 206

ctggttaatc tttatcaggt 20

<210> SEQ ID NO 207
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 207

tctggttaat ctttatcagg 20

<210> SEQ ID NO 208
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 208

ttctggttaa tctttatcag 20

<210> SEQ ID NO 209
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 209

cttctggtta atctttatca 20

<210> SEQ ID NO 210
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 210

tcttctggtt aatctttatc 20

<210> SEQ ID NO 211
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 211

ttcttctggt taatctttat 20

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<210> SEQ ID NO 212
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 212

tttcttctgg ttaatcttta 20

<210> SEQ ID NO 213
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 213

ttttcttctg gttaatcttt 20

<210> SEQ ID NO 214
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 214

gttttcttct ggttaatctt 20

<210> SEQ ID NO 215
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 215

tgttttcttc tggttaatct 20

<210> SEQ ID NO 216
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 216

ttgttttctt ctggttaatc 20

<210> SEQ ID NO 217
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 217

cttgttttct tctggttaat 20

<210> SEQ ID NO 218
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 218

ccttgttttc ttctggttaa 20

<210> SEQ ID NO 219
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 219

tccttgtttt cttctggtta 20

<210> SEQ ID NO 220
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 220

ctccttgttt tcttctggtt 20

<210> SEQ ID NO 221
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 221

cctccttgtt ttcttctggt 20

<210> SEQ ID NO 222
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 222

ccctccttgt tttcttctgg 20

<210> SEQ ID NO 223
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 223

tccctccttg ttttcttctg 20

<210> SEQ ID NO 224
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 224

ttccctcctt gttttcttct 20

<210> SEQ ID NO 225
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 225

tttccctcct tgttttcttc 20

<210> SEQ ID NO 226
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 226

gtttccctcc ttgttttctt 20

<210> SEQ ID NO 227
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 227

tgtttccctc cttgttttct 20

<210> SEQ ID NO 228
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 228

ttgtttccct ccttgttttc 20

<210> SEQ ID NO 229
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 229

gttgtttccc tccttgtttt 20

<210> SEQ ID NO 230
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 230

ggttgtttcc ctccttgttt 20

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<210> SEQ ID NO 231
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 231

cggttgtttc cctccttgtt 20

<210> SEQ ID NO 232
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 232

gcggttgttt ccctccttgt 20

<210> SEQ ID NO 233
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 233

tgcggttgtt tccctccttg 20

<210> SEQ ID NO 234
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 234

ctgcggttgt ttccctcctt 20

<210> SEQ ID NO 235
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 235

gctgcggttg tttccctcct 20

<210> SEQ ID NO 236
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 236

ggctgcggtt gtttccctcc 20

<210> SEQ ID NO 237
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 237

aggctgcggt tgtttccctc 20

<210> SEQ ID NO 238
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 238

caggctgcgg ttgtttccct 20

<210> SEQ ID NO 239
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 239

acaggctgcg gttgtttccc 20

<210> SEQ ID NO 240
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 240

ctacaggctg cggttgtttc 20

<210> SEQ ID NO 241
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 241

gctacaggct gcggttgttt 20

<210> SEQ ID NO 242
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 242

tgctacaggc tgcggttgtt 20

<210> SEQ ID NO 243
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 243

ttgctacagg ctgcggttgt 20

<210> SEQ ID NO 244
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 244

cttgctacag gctgcggttg 20

<210> SEQ ID NO 245
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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gcttgctaca ggctgcggtt 20

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<400> SEQUENCE: 246

agcttgctac aggctgcggt 20

<210> SEQ ID NO 247
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<212> TYPE: DNA
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gagcttgcta caggctgcgg 20

<210> SEQ ID NO 248
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agagcttgct acaggctgcg 20

<210> SEQ ID NO 249
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gccaactcag atttcacctt 20

<210> SEQ ID NO 251
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ccagagcttg ctacaggctg 20

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tccagagctt gctacaggct 20

<210> SEQ ID NO 253
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ttccagagct tgctacaggc 20

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<212> TYPE: DNA
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gttccagagc ttgctacagg 20

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agttccagag cttgctacag 20

<210> SEQ ID NO 256
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gagttccaga gcttgctaca 20

<210> SEQ ID NO 257
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tgagttccag agcttgctac 20

<210> SEQ ID NO 258
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ctgagttcca gagcttgcta 20

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tcctgagttc cagagcttgc 20

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ctcctgagtt ccagagcttg 20

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actcctgagt tccagagctt 20

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gactcctgag ttccagagct 20

<210> SEQ ID NO 264
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cgactcctga gttccagagc 20

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cgcgactcct gagttccaga 20

<210> SEQ ID NO 267
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<400> SEQUENCE: 267

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<210> SEQ ID NO 268
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cgcgcgactc ctgagttcca 20

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gcgcgcgact cctgagttcc 20

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tagcgcgcga ctcctgagtt 20

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<400> SEQUENCE: 273

cctagcgcgc gactcctgag 20

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 274

ccctagcgcg cgactcctga 20

<210> SEQ ID NO 275
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 275

cccctagcgc gcgactcctg 20

<210> SEQ ID NO 276
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<212> TYPE: DNA
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<400> SEQUENCE: 276

gcccctagcg cgcgactcct 20

<210> SEQ ID NO 277
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<212> TYPE: DNA
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<400> SEQUENCE: 277

ggcccctagc gcgcgactcc 20

<210> SEQ ID NO 278
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<212> TYPE: DNA
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<400> SEQUENCE: 278

cggcccctag cgcgcgactc 20

<210> SEQ ID NO 279
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 279

ccggccccta gcgcgcgact 20

<210> SEQ ID NO 280
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<212> TYPE: DNA
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<400> SEQUENCE: 280

ccccggcccc tagcgcgcga 20

<210> SEQ ID NO 281
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<212> TYPE: DNA
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gccccggccc ctagcgcgcg 20

<210> SEQ ID NO 282
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ggccccggcc cctagcgcgc 20

<210> SEQ ID NO 283
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 283

cggccccggc ccctagcgcg 20

<210> SEQ ID NO 284
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<212> TYPE: DNA
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<400> SEQUENCE: 284

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<210> SEQ ID NO 285
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<212> TYPE: DNA
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<400> SEQUENCE: 285

cccggccccg gcccctagcg 20

<210> SEQ ID NO 286
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<400> SEQUENCE: 286

ccccggcccc ggcccctagc 20

<210> SEQ ID NO 287
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<400> SEQUENCE: 287

gccccggccc cggcccctag 20

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<210> SEQ ID NO 288
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<400> SEQUENCE: 288

ggccccggcc ccggccccta 20

<210> SEQ ID NO 289
<211> LENGTH: 20
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<400> SEQUENCE: 289

acgccccggc cccggccccg 20

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 290

cacgccccgg ccccggcccc 20

<210> SEQ ID NO 291
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 291

ccacgccccg gccccggccc 20

<210> SEQ ID NO 292
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 292

accacgcccc ggccccggcc 20

<210> SEQ ID NO 293
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 293

gaccacgccc cggccccggc 20

<210> SEQ ID NO 294
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<212> TYPE: DNA
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<400> SEQUENCE: 294

cgaccacgcc ccggccccgg 20

<210> SEQ ID NO 295
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 295

ccgaccacgc cccggccccg 20

<210> SEQ ID NO 296
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 296

cccgaccacg ccccggcccc 20

<210> SEQ ID NO 297
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 297

ccccgaccac gccccggccc 20

<210> SEQ ID NO 298
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 298

gccccgacca cgccccggcc 20

<210> SEQ ID NO 299
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<212> TYPE: DNA
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<400> SEQUENCE: 299

cgccccgacc acgccccggc 20

<210> SEQ ID NO 300
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<212> TYPE: DNA
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<400> SEQUENCE: 300

ccgccccgac cacgccccgg 20

<210> SEQ ID NO 301
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 301

cccgccccga ccacgccccg 20

<210> SEQ ID NO 302
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 302

gcccgccccg accacgcccc 20

<210> SEQ ID NO 303
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 303

ggcccgcccc gaccacgccc 20

<210> SEQ ID NO 304
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 304

gggcccgccc cgaccacgcc 20

<210> SEQ ID NO 305
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 305

cgggcccgcc ccgaccacgc 20

<210> SEQ ID NO 306
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 306

ccgggcccgc cccgaccacg 20

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<210> SEQ ID NO 307
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 307

cccgggcccg ccccgaccac 20

<210> SEQ ID NO 308
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 308

ccccgggccc gccccgacca 20

<210> SEQ ID NO 309
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 309

cccccgggcc cgccccgacc 20

<210> SEQ ID NO 310
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 310

gcccccgggc ccgccccgac 20

<210> SEQ ID NO 311
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 311

cgcccccggg cccgccccga 20

<210> SEQ ID NO 312
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 312

cccgcccccg ggcccgcccc 20

<210> SEQ ID NO 313
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 313

gcccgccccc gggcccgccc 20

<210> SEQ ID NO 314
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 314

ggcccgcccc cgggcccgcc 20

<210> SEQ ID NO 315
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 315

cgccccgggc ccgccccgg 20

<210> SEQ ID NO 316
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 316

cccgccccgg gcccgccccc 20

<210> SEQ ID NO 317
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 317

ccccgccccg ggcccgcccc 20

<210> SEQ ID NO 318
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 318

gccccgcccc gggcccgccc 20

<210> SEQ ID NO 319
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 319

agccccgccc cgggcccgcc 20

<210> SEQ ID NO 320
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 320

cagccccgcc ccgggcccgc 20

<210> SEQ ID NO 321
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 321

gcagccccgc cccgggcccg 20

<210> SEQ ID NO 322
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 322

cgcagccccg ccccgggccc 20

<210> SEQ ID NO 323
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 323

ctacacacca aagaatgcca 20

<210> SEQ ID NO 324
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 324

ggaataaggt cactagttcg 20

<210> SEQ ID NO 325
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 325

cagagcttgc tacaggctgc 20

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<210> SEQ ID NO 326
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 326

ccgcagcccc gccccgggcc 20

<210> SEQ ID NO 327
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 327

accgcagccc cgccccgggc 20

<210> SEQ ID NO 328
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 328

aaccgcagcc ccgccccggg 20

<210> SEQ ID NO 329
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 329

caaccgcagc cccgccccgg 20

<210> SEQ ID NO 330
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 330

gcaaccgcag ccccgccccg 20

<210> SEQ ID NO 331
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 331

cgcaaccgca gccccgcccc 20

<210> SEQ ID NO 332
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 332

ccgcaaccgc agccccgccc 20

<210> SEQ ID NO 333
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 333

accgcaaccg cagccccgcc 20

<210> SEQ ID NO 334
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 334

caccgcaacc gcagccccgc 20

<210> SEQ ID NO 335
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 335

gcaccgcaac cgcagccccg 20

<210> SEQ ID NO 336
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 336

ggcaccgcaa ccgcagcccc 20

<210> SEQ ID NO 337
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 337

aggcaccgca accgcagccc 20

<210> SEQ ID NO 338
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 338

caggcaccgc aaccgcagcc 20

<210> SEQ ID NO 339
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 339

gcaggcaccg caaccgcagc 20

<210> SEQ ID NO 340
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 340

cgcaggcacc gcaaccgcag 20

<210> SEQ ID NO 341
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 341

gcgcaggcac cgcaaccgca 20

<210> SEQ ID NO 342
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 342

ggcgcaggca ccgcaaccgc 20

<210> SEQ ID NO 343
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<212> TYPE: DNA
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<400> SEQUENCE: 343

gggcgcaggc accgcaaccg 20

<210> SEQ ID NO 344
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 344

tctctctttc ctagcgggac 20

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<210> SEQ ID NO 345
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 345

atctctcttt cctagcggga 20

<210> SEQ ID NO 346
<211> LENGTH: 20
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<400> SEQUENCE: 346

tatctctctt tcctagcggg 20

<210> SEQ ID NO 347
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 347

atatctctct ttcctagcgg 20

<210> SEQ ID NO 348
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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gatatctctc tttcctagcg 20

<210> SEQ ID NO 349
<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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agatatctct ctttcctagc 20

<210> SEQ ID NO 350
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gagatatctc tctttcctag 20

<210> SEQ ID NO 351
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ggagatatct ctctttccta 20

<210> SEQ ID NO 352
<211> LENGTH: 20
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cggagatatc tctctttcct 20

<210> SEQ ID NO 353
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ccggagatat ctctctttcc 20

<210> SEQ ID NO 354
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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tccggagata tctctctttc 20

<210> SEQ ID NO 355
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctccggagat atctctcttt 20

<210> SEQ ID NO 356
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 356

gctccggaga tatctctctt 20

<210> SEQ ID NO 357
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<212> TYPE: DNA
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tgctccggag atatctctct 20

<210> SEQ ID NO 358
<211> LENGTH: 20
<212> TYPE: DNA
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atgctccgga gatatctctc 20

<210> SEQ ID NO 359
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aatgctccgg agatatctct 20

<210> SEQ ID NO 360
<211> LENGTH: 20
<212> TYPE: DNA
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tctgctcttg ctagaccccg 20

<210> SEQ ID NO 361
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 361

atctgctctt gctagacccc 20

<210> SEQ ID NO 362
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tatctgctct tgctagaccc 20

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atatctgctc ttgctagacc 20

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gatatctgct cttgctagac 20

<210> SEQ ID NO 365
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 365

agatatctgc tcttgctaga 20

<210> SEQ ID NO 366
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 366

gagatatctg ctcttgctag 20

<210> SEQ ID NO 367
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 367

ggagatatct gctcttgcta 20

<210> SEQ ID NO 368
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 368

cggagatatc tgctcttgct 20

<210> SEQ ID NO 369
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 369

ccggagatat ctgctcttgc 20

<210> SEQ ID NO 370
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<212> TYPE: DNA
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tccggagata tctgctcttg 20

<210> SEQ ID NO 371
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 371

ctccggagat atctgctctt 20

<210> SEQ ID NO 372
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 372

gctccggaga tatctgctct 20

<210> SEQ ID NO 373
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 373

tgctccggag atatctgctc 20

<210> SEQ ID NO 374
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 374

atgctccgga gatatctgct 20

<210> SEQ ID NO 375
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 375

aatgctccgg agatatctgc 20

<210> SEQ ID NO 376
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 376

aaatgctccg gagatatctg 20

<210> SEQ ID NO 377
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<212> TYPE: DNA
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<400> SEQUENCE: 377

tgctccggag atatcaagcg 20

<210> SEQ ID NO 378
<211> LENGTH: 20
<212> TYPE: DNA
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atgctccgga gatatcaagc 20

<210> SEQ ID NO 379
<211> LENGTH: 20
<212> TYPE: DNA
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aatgctccgg agatatcaag 20

<210> SEQ ID NO 380
<211> LENGTH: 20
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<400> SEQUENCE: 380

aaatgctccg gagatatcaa 20

<210> SEQ ID NO 381
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 381

caaatgctcc ggagatatca 20

<210> SEQ ID NO 382
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 382

ggtaacttca aactcttggg 20

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<210> SEQ ID NO 383
<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 383

gccatgattt cttgtctggg 20

<210> SEQ ID NO 384
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 384

tctcctaaac ccacacctgc 20

<210> SEQ ID NO 385
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 385

atctcctaaa cccacacctg 20

<210> SEQ ID NO 386
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 386

tatctcctaa acccacacct 20

<210> SEQ ID NO 387
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 387

atatctccta aacccacacc 20

<210> SEQ ID NO 388
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 388

gatatctcct aaacccacac 20

<210> SEQ ID NO 389
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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agatatctcc taaacccaca 20

<210> SEQ ID NO 390
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 390

gagatatctc ctaaacccac 20

<210> SEQ ID NO 391
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 391

ggagatatct cctaaaccca 20

<210> SEQ ID NO 392
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 392

cggagatatc tcctaaaccc 20

<210> SEQ ID NO 393
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 393

ccggagatat ctcctaaacc 20

<210> SEQ ID NO 394
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 394

tccggagata tctcctaaac 20

<210> SEQ ID NO 395
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 395

ctccggagat atctcctaaa 20

<210> SEQ ID NO 396
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 396

gctccggaga tatctcctaa 20

<210> SEQ ID NO 397
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 397

tgctccggag atatctccta 20

<210> SEQ ID NO 398
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 398

atgctccgga gatatctcct 20

<210> SEQ ID NO 399
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 399

aatgctccgg agatatctcc 20

<210> SEQ ID NO 400
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 400

aaatgctccg gagatatctc 20

<210> SEQ ID NO 401
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 401

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 441

tcacattatc caaatgctcc 20

<210> SEQ ID NO 442
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 442

ctgtcacatt atccaaatgc 20

<210> SEQ ID NO 443
<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 443

caactgtcac attatccaaa 20

<210> SEQ ID NO 444
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 444

ttccaactgt cacattatcc 20

<210> SEQ ID NO 445
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 445

tgcattccaa ctgtcacatt 20

<210> SEQ ID NO 446
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 446

cactgcattc caactgtcac 20

<210> SEQ ID NO 447
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 447

catcactgca ttccaactgt 20

<210> SEQ ID NO 448
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 448

cgacatcact gcattccaac 20

<210> SEQ ID NO 449
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 449

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agtcgacatc actgcattcc 20

<210> SEQ ID NO 450
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 450

aagagtcgac atcactgcat 20

<210> SEQ ID NO 451
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 451

gcaaagagtc gacatcactg 20

<210> SEQ ID NO 452
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 452

tgggcaaaga gtcgacatca 20

<210> SEQ ID NO 453
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 453

cggtgggcaa agagtcgaca 20

<210> SEQ ID NO 454
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 454

tggcggtggg caaagagtcg 20

<210> SEQ ID NO 455
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 455

gagatggcgg tgggcaaaga 20

<210> SEQ ID NO 456

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 456

ctggagatgg cggtgggcaa 20

<210> SEQ ID NO 457
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 457

acagctggag atggcggtgg 20

<210> SEQ ID NO 458
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 458

gcaacagctg gagatggcgg 20

<210> SEQ ID NO 459
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 459

ttggcaacag ctggagatgg 20

<210> SEQ ID NO 460
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 460

gtcttggcaa cagctggaga 20

<210> SEQ ID NO 461
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<212> TYPE: DNA
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<400> SEQUENCE: 461

tctgtcttgg caacagctgg 20

<210> SEQ ID NO 462
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 462

tctctgtctt ggcaacagct 20

<210> SEQ ID NO 463
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 463

gcaatctctg tcttggcaac 20

<210> SEQ ID NO 464
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 464

agcaatctct gtcttggcaa 20

<210> SEQ ID NO 465
<211> LENGTH: 20
<212> TYPE: DNA
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aaagcaatct ctgtcttggc 20

<210> SEQ ID NO 466
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 466

cttaaagcaa tctctgtctt 20

<210> SEQ ID NO 467
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 467

ccacttaaag caatctctgt 20

<210> SEQ ID NO 468
<211> LENGTH: 20
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agctgctaat aaaggtgatt 20

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agtagctgct aataaaggtg 20

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aaaagtagct gctaataaag 20

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agcaaaagta gctgctaata 20

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gtaagcaaaa gtagctgcta 20

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ccagtaagca aaagtagctg 20

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gtcccagtaa gcaaaagtag 20

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attgtcccag taagcaaaag 20

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atattgtccc agtaagcaaa 20

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aatattgtcc cagtaagcaa 20

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aagaatattg tcccagtaag 20

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accaagaata ttgtcccagt 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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aggaccaaga atattgtccc 20

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tactctagga ccaagaatat 20

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ccttactcta ggaccaagaa 20

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gtgccttact ctaggaccaa 20

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aatgtgcctt actctaggac 20

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agcccaaatg tgccttactc 20

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ctttggagcc caaatgtgcc 20

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tgtctttgga gcccaaatgt 20

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ttctgtcttt ggagcccaaa 20

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gttctgtctt tggagcccaa 20

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tacctgttct gtctttggag 20

<210> SEQ ID NO 494

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aagtacctgt tctgtctttg 20

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gagaagtacc tgttctgtct 20

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<400> SEQUENCE: 496

actgagaagt acctgttctg 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 497

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<212> TYPE: DNA
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<400> SEQUENCE: 498

atcactgaga agtacctgtt 20

<210> SEQ ID NO 499
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<400> SEQUENCE: 499

tccatcactg agaagtacct 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 500

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 501

ttatttctcc atcactgaga 20

<210> SEQ ID NO 502
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 502

aagttatttc tccatcactg 20

<210> SEQ ID NO 503
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 503

gaaaagttat ttctccatca 20

<210> SEQ ID NO 504
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 504

ggttggcaag aaaagttatt 20

<210> SEQ ID NO 505
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 505

tgtggttggc aagaaaagtt 20

<210> SEQ ID NO 506
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 506

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gagtgtggtt ggcaagaaaa 20

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<400> SEQUENCE: 507

ttagagtgtg gttggcaaga 20

<210> SEQ ID NO 508
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 508

catttagagt gtggttggca 20

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<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 509

ccatttagag tgtggttggc 20

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<212> TYPE: DNA
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<400> SEQUENCE: 510

tttctccatt tagagtgtgg 20

<210> SEQ ID NO 511
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 511

ggatttctcc atttagagtg 20

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 512

cgccaccgcc tgcgcctccg 20

<210> SEQ ID NO 513

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<211> LENGTH: 20
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<220> FEATURE:
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<400> SEQUENCE: 513

actcgccacc gcctgcgcct 20

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 514

tccactcgcc accgcctgcg 20

<210> SEQ ID NO 515
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<212> TYPE: DNA
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<400> SEQUENCE: 515

atatccactc gccaccgcct 20

<210> SEQ ID NO 516
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<212> TYPE: DNA
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<400> SEQUENCE: 516

gagatatcca ctcgccaccg 20

<210> SEQ ID NO 517
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 517

ttcgaaggat ttctccattt 20

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<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 518

catttcgaag gatttctcca 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<212> TYPE: DNA
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<400> SEQUENCE: 521

cactctctgc atttcgaagg 20

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<212> TYPE: DNA
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<400> SEQUENCE: 522

caccactctc tgcatttcga 20

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<212> TYPE: DNA
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agcaccactc tctgcatttc 20

<210> SEQ ID NO 524
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<212> TYPE: DNA
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<400> SEQUENCE: 524

tagcaccact ctctgcattt 20

<210> SEQ ID NO 525
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctatagcacc actctctgca 20

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catctatagc accactctct 20

<210> SEQ ID NO 527
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<212> TYPE: DNA
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actttacatc tatagcacca 20

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<212> TYPE: DNA
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aaaactttac atctatagca 20

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 529

aagacaaaaa actttacatc 20

<210> SEQ ID NO 530
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 530

caagacaaaa aactttacat 20

<210> SEQ ID NO 531
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<212> TYPE: DNA
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<220> FEATURE:
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gacaagacaa aaaactttac 20

<210> SEQ ID NO 532

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<211> LENGTH: 20
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 532

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<210> SEQ ID NO 533
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 533

tttcagacaa gacaaaaaac 20

<210> SEQ ID NO 534
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 534

cttttcagac aagacaaaaa 20

<210> SEQ ID NO 535
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 535

tccctttca gacaagacaa 20

<210> SEQ ID NO 536
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 536

cactcccttt tcagacaaga 20

<210> SEQ ID NO 537
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 537

aatcactccc ttttcagaca 20

<210> SEQ ID NO 538
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 538

aataatcact ccctttcag 20

<210> SEQ ID NO 539
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 539

acaataatca ctccctttc 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 540

aacaataatc actccctttt 20

<210> SEQ ID NO 541
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 541

tgaaacaata atcactccct 20

<210> SEQ ID NO 542
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 542

taatgaaaca ataatcactc 20

<210> SEQ ID NO 543
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 543

gattaatgaa acaataatca 20

<210> SEQ ID NO 544
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 544

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tcaaagatta atgaaacaat 20

<210> SEQ ID NO 545
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 545

ccatcaaaga ttaatgaaac 20

<210> SEQ ID NO 546
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 546

tttccatcaa agattaatga 20

<210> SEQ ID NO 547
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 547

cagtttccat caaagattaa 20

<210> SEQ ID NO 548
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 548

ttccagtttc catcaaagat 20

<210> SEQ ID NO 549
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 549

ccattccagt ttccatcaaa 20

<210> SEQ ID NO 550
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 550

tccccattcc agtttccatc 20

<210> SEQ ID NO 551

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<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 551

cgatccccat tccagtttcc 20

<210> SEQ ID NO 552
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 552

ctgcgatccc cattccagtt 20

<210> SEQ ID NO 553
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 553

gtgctgcgat ccccattcca 20

<210> SEQ ID NO 554
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 554

tatgtgctgc gatccccatt 20

<210> SEQ ID NO 555
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 555

ggaagtataa ttgatagtcc 20

<210> SEQ ID NO 556
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 556

tgtggaagta taattgatag 20

<210> SEQ ID NO 557
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 557

gtctgtggaa gtataattga 20

<210> SEQ ID NO 558
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 558

tctgtctgtg gaagtataat 20

<210> SEQ ID NO 559
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 559

agttctgtct gtggaagtat 20

<210> SEQ ID NO 560
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 560

ctaagttctg tctgtggaag 20

<210> SEQ ID NO 561
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 561

aaactaagtt ctgtctgtgg 20

<210> SEQ ID NO 562
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 562

tagaaactaa gttctgtctg 20

<210> SEQ ID NO 563
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 563

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aggtagaaac taagttctgt 20

<210> SEQ ID NO 564
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 564

gggaggtaga aactaagttc 20

<210> SEQ ID NO 565
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 565

agtgggaggt agaaactaag 20

<210> SEQ ID NO 566
<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 566

tgaagtggga ggtagaaact 20

<210> SEQ ID NO 567
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 567

ctatgaagtg ggaggtagaa 20

<210> SEQ ID NO 568
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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actctatgaa gtgggaggta 20

<210> SEQ ID NO 569
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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acactctatg aagtgggagg 20

<210> SEQ ID NO 570

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<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
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cacactctat gaagtgggag 20

<210> SEQ ID NO 571
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 571

acacactcta tgaagtggga 20

<210> SEQ ID NO 572
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 572

acacacactc tatgaagtgg 20

<210> SEQ ID NO 573
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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tcaacacaca ctctatgaag 20

<210> SEQ ID NO 574
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 574

ctatcaacac acactctatg 20

<210> SEQ ID NO 575
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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aatctatcaa cacacactct 20

<210> SEQ ID NO 576
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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gttaatctat caacacacac 20

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 577

gtgttaatct atcaacacac 20

<210> SEQ ID NO 578
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 578

tgtgttaatc tatcaacaca 20

<210> SEQ ID NO 579
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 579

tatgtgttaa tctatcaaca 20

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 580

atatgtgtta atctatcaac 20

<210> SEQ ID NO 581
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 581

attatatgtg ttaatctatc 20

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<211> LENGTH: 20
<212> TYPE: DNA
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cggattatat gtgttaatct 20

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 583

ttccggatta tatgtgttaa 20

<210> SEQ ID NO 584
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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cctttccgga ttatatgtgt 20

<210> SEQ ID NO 585
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 585

cttcctttcc ggattatatg 20

<210> SEQ ID NO 586
<211> LENGTH: 20
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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attcttcctt tccggattat 20

<210> SEQ ID NO 587
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 587

catattcttc ctttccggat 20

<210> SEQ ID NO 588
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 588

atccatattc ttcctttccg 20

<210> SEQ ID NO 589

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<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 589

atgcatccat attcttcctt 20

<210> SEQ ID NO 590
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 590

ttatgcatcc atattcttcc 20

<210> SEQ ID NO 591
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 591

tccttatgca tccatattct 20

<210> SEQ ID NO 592
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 592

ctttccttat gcatccatat 20

<210> SEQ ID NO 593
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 593

tgtctttcct tatgcatcca 20

<210> SEQ ID NO 594
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 594

aggacctccc tcctgtttct 20

<210> SEQ ID NO 595
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 595

agaagtaatg ccagacagat 20

<210> SEQ ID NO 596
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 596

ctttgtttct ctgaaagcaa 20

<210> SEQ ID NO 597
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 597

gtggttggtc cactgctatt 20

<210> SEQ ID NO 598
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 598

ttgagggaag ccaagattca 20

<210> SEQ ID NO 599
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 599

agagctgtac aattatttta 20

<210> SEQ ID NO 600
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 600

ggtaatgaca ctactgctgt 20

<210> SEQ ID NO 601
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 601

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gatcctaatc ctgtctatgc 20

<210> SEQ ID NO 602
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 602

acttgtgggt tgaattgtgt 20

<210> SEQ ID NO 603
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 603

taccttatgc atccatattc 20

<210> SEQ ID NO 604
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 604

gatgttcact gcatataatt 20

<210> SEQ ID NO 605
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 605

ccagatgtat ttgtatctaa 20

<210> SEQ ID NO 606
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 606

taatgtggag ctaccatttc 20

<210> SEQ ID NO 607
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 607

gctcccaaga agaatccagg 20

<210> SEQ ID NO 608

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 608

acttacacat agtagtaagc 20

<210> SEQ ID NO 609
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 609

aaagagacca aaggctacat 20

<210> SEQ ID NO 610
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 610

ggaattctct tgggaaccat 20

<210> SEQ ID NO 611
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 611

tcttgtcttt ccttatgcat 20

<210> SEQ ID NO 612
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 612

ttttcttgtc tttccttatg 20

<210> SEQ ID NO 613
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 613

tggacatttt cttgtctttc 20

<210> SEQ ID NO 614
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 614

ttctggacat tttcttgtct 20

<210> SEQ ID NO 615
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 615

atcttctgga cattttcttg 20

<210> SEQ ID NO 616
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 616

taatcttctg gacattttct 20

<210> SEQ ID NO 617
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 617

aagataatct tctggacatt 20

<210> SEQ ID NO 618
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 618

tctaagataa tcttctggac 20

<210> SEQ ID NO 619
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 619

ccttctaaga taatcttctg 20

<210> SEQ ID NO 620
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 620

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gtgccttcta agataatctt 20

<210> SEQ ID NO 621
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 621

tctgtgcctt ctaagataat 20

<210> SEQ ID NO 622
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 622

ctctctgtgc cttctaagat 20

<210> SEQ ID NO 623
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 623

attctctctg tgccttctaa 20

<210> SEQ ID NO 624
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 624

tccattctct ctgtgccttc 20

<210> SEQ ID NO 625
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 625

tcttccattc tctctgtgcc 20

<210> SEQ ID NO 626
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 626

tgatcttcca ttctctctgt 20

<210> SEQ ID NO 627

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 627

gaccctgatc ttccattctc 20

<210> SEQ ID NO 628
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 628

tctgaccctg atcttccatt 20

<210> SEQ ID NO 629
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 629

tactctgacc ctgatcttcc 20

<210> SEQ ID NO 630
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 630

taatactctg accctgatct 20

<210> SEQ ID NO 631
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 631

gaataatact ctgaccctga 20

<210> SEQ ID NO 632
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 632

gcattggaat aatactctga 20

<210> SEQ ID NO 633
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 633

taagcattgg aataatactc 20

<210> SEQ ID NO 634
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 634

cagtaagcat tggaataata 20

<210> SEQ ID NO 635
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 635

ctccagtaag cattggaata 20

<210> SEQ ID NO 636
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 636

cttctccagt aagcattgga 20

<210> SEQ ID NO 637
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 637

tcacttctcc agtaagcatt 20

<210> SEQ ID NO 638
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 638

gaatcacttc tccagtaagc 20

<210> SEQ ID NO 639
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 639

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caggaatcac ttctccagta 20

<210> SEQ ID NO 640
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 640

ttacaggaat cacttctcca 20

<210> SEQ ID NO 641
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 641

ccattacagg aatcacttct 20

<210> SEQ ID NO 642
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 642

aagcagttcc attacaggaa 20

<210> SEQ ID NO 643
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 643

tgaaagcagt tccattacag 20

<210> SEQ ID NO 644
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 644

agatgaaagc agttccatta 20

<210> SEQ ID NO 645
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 645

catagatgaa agcagttcca 20

<210> SEQ ID NO 646

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 646

tttcatagat gaaagcagtt 20

<210> SEQ ID NO 647
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 647

gtgtgatttc atagatgaaa 20

<210> SEQ ID NO 648
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 648

cactgtgtga tttcatagat 20

<210> SEQ ID NO 649
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 649

gaacactgtg tgatttcata 20

<210> SEQ ID NO 650
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 650

caggaacact gtgtgatttc 20

<210> SEQ ID NO 651
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 651

tttcttcagg aacactgtgt 20

<210> SEQ ID NO 652
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 652

ctatttcttc aggaacactg 20

<210> SEQ ID NO 653
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 653

tatctatttc ttcaggaaca 20

<210> SEQ ID NO 654
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 654

ctatatctat ttcttcagga 20

<210> SEQ ID NO 655
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 655

cagctatatc tatttcttca 20

<210> SEQ ID NO 656
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 656

tatcagctat atctatttct 20

<210> SEQ ID NO 657
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 657

ctgtatcagc tatatctatt 20

<210> SEQ ID NO 658
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 658

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gtactgtatc agctatatct 20

<210> SEQ ID NO 659
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 659

tgagtactgt atcagctata 20

<210> SEQ ID NO 660
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 660

cattgagtac tgtatcagct 20

<210> SEQ ID NO 661
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 661

catcattgag tactgtatca 20

<210> SEQ ID NO 662
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 662

catcatcatt gagtactgta 20

<210> SEQ ID NO 663
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 663

tatcatcatc attgagtact 20

<210> SEQ ID NO 664
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 664

caatatcatc atcattgagt 20

<210> SEQ ID NO 665

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<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 665

gtcaccaata tcatcatcat 20

<210> SEQ ID NO 666
<211> LENGTH: 20
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 666

ttgagaagaa agccttcatg 20

<210> SEQ ID NO 667
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 667

actatactga aatgtaaata 20

<210> SEQ ID NO 668
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 668

tatcaaactg gaacacagga 20

<210> SEQ ID NO 669
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 669

tgggcaaaag ccttttaaaa 20

<210> SEQ ID NO 670
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 670

gcaaacatag taaaaaatta 20

<210> SEQ ID NO 671
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 671

ttctcctgat tttaagagtt 20

<210> SEQ ID NO 672
<211> LENGTH: 20
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 672

aagaatgact tgcacttttc 20

<210> SEQ ID NO 673
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 673

aaagataact tcacagaaaa 20

<210> SEQ ID NO 674
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 674

ctttctactt tagggaaaaa 20

<210> SEQ ID NO 675
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 675

tttttcaata gacatgttct 20

<210> SEQ ID NO 676
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 676

cttgtctttc ctgagcaaga 20

<210> SEQ ID NO 677
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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acctgatctt ccattctctc 20

<210> SEQ ID NO 678
<211> LENGTH: 20
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 678

ctccataaaa gctccattaa 20

<210> SEQ ID NO 679
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgtttactga tttaactctt 20

<210> SEQ ID NO 680
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 680

aacagaaaaa aaaagggagc 20

<210> SEQ ID NO 681
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 681

gtaccttaaa gaacatatca 20

<210> SEQ ID NO 682
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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aaatgtaaat tgcatgagtc 20

<210> SEQ ID NO 683
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 683

gggtaagaaa tatcactgac 20

<210> SEQ ID NO 684

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<211> LENGTH: 20
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<400> SEQUENCE: 684

aaccatgctt ctcaaactct 20

<210> SEQ ID NO 685
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 685

aagaacttct ctgctttaca 20

<210> SEQ ID NO 686
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 686

aatggaagta aaagtgaaga 20

<210> SEQ ID NO 687
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 687

aacagccatg tttaaaatat 20

<210> SEQ ID NO 688
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 688

ttaaagtatc atctgtctca 20

<210> SEQ ID NO 689
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 689

caatttggta aaggagatca 20

<210> SEQ ID NO 690
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 690

acacagaata actgtctctg 20

<210> SEQ ID NO 691
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 691

gcttattgac cagcaaataa 20

<210> SEQ ID NO 692
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 692

cccagtaaaa gcagaatttt 20

<210> SEQ ID NO 693
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 693

attaatagta gtcaacttaa 20

<210> SEQ ID NO 694
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 694

acttgaactt ctcagcagta 20

<210> SEQ ID NO 695
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 695

agaagaggct ctaaaagaaa 20

<210> SEQ ID NO 696
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 696

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aaaggcaact cctccttttc 20

<210> SEQ ID NO 697
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 697

acagtattgt tcaaaataaa 20

<210> SEQ ID NO 698
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 698

cagatgacag ctacaactga 20

<210> SEQ ID NO 699
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 699

gaataatgac tagatccgtg 20

<210> SEQ ID NO 700
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 700

ataattatca tgcctgttta 20

<210> SEQ ID NO 701
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 701

ctttagtaac ctccacaact 20

<210> SEQ ID NO 702
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 702

gaatttaaat gtgatgctac 20

<210> SEQ ID NO 703

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 703

tgtcagaccc agggccattt 20

<210> SEQ ID NO 704
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 704

acttatttta tgaaatgatt 20

<210> SEQ ID NO 705
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 705

ttttagctaa acatattttt 20

<210> SEQ ID NO 706
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 706

cagtctcatc agttttgtga 20

<210> SEQ ID NO 707
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 707

tgtaaagtgt ctcaaatatg 20

<210> SEQ ID NO 708
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 708

cttgaaattg taattttgaa 20

<210> SEQ ID NO 709
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 709

aatcaaaatc agcacatata 20

<210> SEQ ID NO 710
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 710

accaaatagg taaggaaaac 20

<210> SEQ ID NO 711
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 711

aaagatctcc tttaaaattt 20

<210> SEQ ID NO 712
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 712

ctttagagag tatggaatca 20

<210> SEQ ID NO 713
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 713

caaagctcac ttttattctt 20

<210> SEQ ID NO 714
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 714

acacagtatc aaacaagtct 20

<210> SEQ ID NO 715
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 715

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aagctgggca ataaaaaata 20

<210> SEQ ID NO 716
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 716

ctgaccctgc acaataaagt 20

<210> SEQ ID NO 717
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 717

catctatttc ttcaggaaca 20

<210> SEQ ID NO 718
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 718

gaatattaat aatatacata 20

<210> SEQ ID NO 719
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 719

aggattttgt gtgtgcttat 20

<210> SEQ ID NO 720
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 720

ttttaggaat tataaaagta 20

<210> SEQ ID NO 721
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 721

acacagtttt gtttcaaaag 20

<210> SEQ ID NO 722

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 722

ggaaactaaa tttgtgacta 20

<210> SEQ ID NO 723
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 723

ctcttaacac tcatagtgtg 20

<210> SEQ ID NO 724
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 724

gagactaacc taaatgacaa 20

<210> SEQ ID NO 725
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 725

caaatgtgaa agctggtcaa 20

<210> SEQ ID NO 726
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 726

taacacactg ccttcatttc 20

<210> SEQ ID NO 727
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 727

tatctaaaat gcatcaaaaa 20

<210> SEQ ID NO 728
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 728

cttgagaaga aagccttcat 20

<210> SEQ ID NO 729
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 729

ccaaatcttg tcataggtga 20

<210> SEQ ID NO 730
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 730

taacacaaat ttaagcaaca 20

<210> SEQ ID NO 731
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 731

aaatagcaaa tggaataaca 20

<210> SEQ ID NO 732
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 732

aaaccagaat caagcaaggg 20

<210> SEQ ID NO 733
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 733

catctacagt acaacttaat 20

<210> SEQ ID NO 734
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 734

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agatcagtat aaatatgaat 20

<210> SEQ ID NO 735
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 735

gtttaagggc acaaactctt 20

<210> SEQ ID NO 736
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 736

aggtgtatag agaattcagg 20

<210> SEQ ID NO 737
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 737

tactcaatgc ttataacaac 20

<210> SEQ ID NO 738
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 738

ggaactaaca tgtaggcact 20

<210> SEQ ID NO 739
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 739

cataaaagtg aatactttat 20

<210> SEQ ID NO 740
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 740

aggctcttag gttaaacaca 20

<210> SEQ ID NO 741

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 741

gctgacactg aacagataca 20

<210> SEQ ID NO 742
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 742

catgtagaga gattaagtga 20

<210> SEQ ID NO 743
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 743

atcatttaat taatgtattt 20

<210> SEQ ID NO 744
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 744

gtgagagcaa gtagtggg 18

<210> SEQ ID NO 745
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 745

tgtgagagca agtagtgg 18

<210> SEQ ID NO 746
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 746

ctgtgagagc aagtagtg 18

<210> SEQ ID NO 747
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 747

actgtgagag caagtagt 18

<210> SEQ ID NO 748
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 748

tactgtgaga gcaagtag 18

<210> SEQ ID NO 749
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 749

gtactgtgag agcaagta 18

<210> SEQ ID NO 750
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 750

agtactgtga gagcaagt 18

<210> SEQ ID NO 751
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 751

gagtactgtg agagcaag 18

<210> SEQ ID NO 752
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 752

cgagtactgt gagagcaa 18

<210> SEQ ID NO 753
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 753

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gcgagtactg tgagagca 18

<210> SEQ ID NO 754
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 754

agcgagtact gtgagagc 18

<210> SEQ ID NO 755
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 755

cagcgagtac tgtgagag 18

<210> SEQ ID NO 756
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 756

tcagcgagta ctgtgaga 18

<210> SEQ ID NO 757
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 757

ctcagcgagt actgtgag 18

<210> SEQ ID NO 758
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 758

cctcagcgag tactgtga 18

<210> SEQ ID NO 759
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 759

ccctcagcga gtactgtg 18

<210> SEQ ID NO 760

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<211> LENGTH: 18
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 760

accctcagcg agtactgt 18

<210> SEQ ID NO 761
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 761

caccctcagc gagtactg 18

<210> SEQ ID NO 762
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 762

tcaccctcag cgagtact 18

<210> SEQ ID NO 763
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 763

ttcaccctca gcgagtac 18

<210> SEQ ID NO 764
<211> LENGTH: 18
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<400> SEQUENCE: 764

gttcaccctc agcgagta 18

<210> SEQ ID NO 765
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 765

tgttcaccct cagcgagt 18

<210> SEQ ID NO 766
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<212> TYPE: DNA
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<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 766

ttgttcaccc tcagcgag 18

<210> SEQ ID NO 767
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 767

cttgttcacc ctcagcga 18

<210> SEQ ID NO 768
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 768

tcttgttcac cctcagcg 18

<210> SEQ ID NO 769
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 769

ttcttgttca ccctcagc 18

<210> SEQ ID NO 770
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 770

tttcttgttc accctcag 18

<210> SEQ ID NO 771
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 771

ttttcttgtt caccctca 18

<210> SEQ ID NO 772
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 772

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cttttcttgt tcaccctc 18

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<220> FEATURE:
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<400> SEQUENCE: 773

tcttttcttg ttcaccct 18

<210> SEQ ID NO 774
<211> LENGTH: 18
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 774

gtcttttctt gttcaccc 18

<210> SEQ ID NO 775
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 775

ggtcttttct tgttcacc 18

<210> SEQ ID NO 776
<211> LENGTH: 18
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 776

aggtcttttc ttgttcac 18

<210> SEQ ID NO 777
<211> LENGTH: 18
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 777

caggtctttt cttgttca 18

<210> SEQ ID NO 778
<211> LENGTH: 18
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 778

tcaggtcttt tcttgttc 18

<210> SEQ ID NO 779

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 779

atcaggtctt ttcttgtt 18

<210> SEQ ID NO 780
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 780

tatcaggtct tttcttgt 18

<210> SEQ ID NO 781
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 781

ttatcaggtc ttttcttg 18

<210> SEQ ID NO 782
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 782

tttatcaggt cttttctt 18

<210> SEQ ID NO 783
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 783

aatctttatc aggtcttt 18

<210> SEQ ID NO 784
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 784

taatctttat caggtctt 18

<210> SEQ ID NO 785
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 785

ttaatcttta tcaggtct 18

<210> SEQ ID NO 786
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 786

gttaatcttt atcaggtc 18

<210> SEQ ID NO 787
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 787

ggttaatctt tatcaggt 18

<210> SEQ ID NO 788
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 788

tggttaatct ttatcagg 18

<210> SEQ ID NO 789
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 789

ctggttaatc tttatcag 18

<210> SEQ ID NO 790
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 790

tctggttaat ctttatca 18

<210> SEQ ID NO 791
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 791

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ttctggttaa tctttatc 18

<210> SEQ ID NO 792
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 792

tccctccttg ttttcttc 18

<210> SEQ ID NO 793
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 793

tttccctcct tgttttct 18

<210> SEQ ID NO 794
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 794

gtttccctcc ttgttttc 18

<210> SEQ ID NO 795
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 795

tgtttccctc cttgtttt 18

<210> SEQ ID NO 796
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 796

ttgtttccct ccttgttt 18

<210> SEQ ID NO 797
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 797

gttgtttccc tccttgtt 18

<210> SEQ ID NO 798

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 798

ggttgtttcc ctccttgt 18

<210> SEQ ID NO 799
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 799

cggttgtttc cctccttg 18

<210> SEQ ID NO 800
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 800

gcggttgttt ccctcctt 18

<210> SEQ ID NO 801
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 801

tgcggttgtt tccctcct 18

<210> SEQ ID NO 802
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 802

ctgcggttgt ttccctcc 18

<210> SEQ ID NO 803
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 803

gctgcggttg tttccctc 18

<210> SEQ ID NO 804
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 804

ggctgcggtt gtttccct 18

<210> SEQ ID NO 805
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 805

aggctgcggt tgtttccc 18

<210> SEQ ID NO 806
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 806

caggctgcgg ttgtttcc 18

<210> SEQ ID NO 807
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 807

acaggctgcg gttgtttc 18

<210> SEQ ID NO 808
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 808

tacaggctgc ggttgttt 18

<210> SEQ ID NO 809
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 809

ctacaggctg cggttgtt 18

<210> SEQ ID NO 810
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 810

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gctacaggct gcggttgt 18

<210> SEQ ID NO 811
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 811

tgctacaggc tgcggttg 18

<210> SEQ ID NO 812
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 812

ttgctacagg ctgcggtt 18

<210> SEQ ID NO 813
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 813

cttgctacag gctgcggt 18

<210> SEQ ID NO 814
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 814

gcttgctaca ggctgcgg 18

<210> SEQ ID NO 815
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 815

agcttgctac aggctgcg 18

<210> SEQ ID NO 816
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 816

gagcttgcta caggctgc 18

<210> SEQ ID NO 817

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 817

agagcttgct acaggctg 18

<210> SEQ ID NO 818
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 818

cagagcttgc tacaggct 18

<210> SEQ ID NO 819
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 819

ccagagcttg ctacaggc 18

<210> SEQ ID NO 820
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 820

tccagagctt gctacagg 18

<210> SEQ ID NO 821
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 821

ttccagagct tgctacag 18

<210> SEQ ID NO 822
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 822

gttccagagc ttgctaca 18

<210> SEQ ID NO 823
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 823

agttccagag cttgctac 18

<210> SEQ ID NO 824
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 824

gagttccaga gcttgcta 18

<210> SEQ ID NO 825
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 825

tgagttccag agcttgct 18

<210> SEQ ID NO 826
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 826

ctgagttcca gagcttgc 18

<210> SEQ ID NO 827
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 827

cctgagttcc agagcttg 18

<210> SEQ ID NO 828
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 828

tcctgagttc cagagctt 18

<210> SEQ ID NO 829
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 829

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ctcctgagtt ccagagct 18

<210> SEQ ID NO 830
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 830

actcctgagt tccagagc 18

<210> SEQ ID NO 831
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 831

gactcctgag ttccagag 18

<210> SEQ ID NO 832
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 832

cgactcctga gttccaga 18

<210> SEQ ID NO 833
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 833

gcgactcctg agttccag 18

<210> SEQ ID NO 834
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 834

cgcgactcct gagttcca 18

<210> SEQ ID NO 835
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 835

gcgcgactcc tgagttcc 18

<210> SEQ ID NO 836

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 836

cgcgcgactc ctgagttc 18

<210> SEQ ID NO 837
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 837

gcgcgcgact cctgagtt 18

<210> SEQ ID NO 838
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 838

agcgcgcgac tcctgagt 18

<210> SEQ ID NO 839
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 839

tagcgcgcga ctcctgag 18

<210> SEQ ID NO 840
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 840

ctagcgcgcg actcctga 18

<210> SEQ ID NO 841
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 841

cctagcgcgc gactcctg 18

<210> SEQ ID NO 842
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 842

ccctagcgcg cgactcct 18

<210> SEQ ID NO 843
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 843

cccctagcgc gcgactcc 18

<210> SEQ ID NO 844
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 844

gcccctagcg cgcgactc 18

<210> SEQ ID NO 845
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 845

ggcccctagc gcgcgact 18

<210> SEQ ID NO 846
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 846

cggcccctag cgcgcgac 18

<210> SEQ ID NO 847
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 847

ccggccccta gcgcgcga 18

<210> SEQ ID NO 848
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 848

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cccggcccct agcgcgcg 18

<210> SEQ ID NO 849
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 849

ccccggcccc tagcgcgc 18

<210> SEQ ID NO 850
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 850

gccccggccc ctagcgcg 18

<210> SEQ ID NO 851
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 851

ggccccggcc cctagcgc 18

<210> SEQ ID NO 852
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 852

cggccccggc ccctagcg 18

<210> SEQ ID NO 853
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<400> SEQUENCE: 853

ccggccccgg cccctagc 18

<210> SEQ ID NO 854
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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cccggccccg gcccctag 18

<210> SEQ ID NO 855

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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 855

ccccggcccc ggccccta 18

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<213> ORGANISM: Artificial sequence
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acgccccggc cccggccc 18

<210> SEQ ID NO 857
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<400> SEQUENCE: 857

cacgccccgg ccccggcc 18

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<211> LENGTH: 18
<212> TYPE: DNA
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ccacgccccg gccccggc 18

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accacgcccc ggccccgg 18

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gaccacgccc cggccccg 18

<210> SEQ ID NO 861
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 861

cgaccacgcc ccggcccc 18

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ccgaccacgc cccggccc 18

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<400> SEQUENCE: 863

cccgaccacg ccccggcc 18

<210> SEQ ID NO 864
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<400> SEQUENCE: 864

ccccgaccac gccccggc 18

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gccccgacca cgccccgg 18

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<400> SEQUENCE: 866

cgccccgacc acgccccg 18

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ccgccccgac cacgcccc 18

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cccgccccga ccacgccc 18

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gcccgccccg accacgcc 18

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ggcccgcccc gaccacgc 18

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gggcccgccc cgaccacg 18

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<400> SEQUENCE: 872

cgggcccgcc ccgaccac 18

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<400> SEQUENCE: 873

ccgggcccgc cccgacca 18

<210> SEQ ID NO 874

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<400> SEQUENCE: 874

cccgggcccg ccccgacc 18

<210> SEQ ID NO 875
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<400> SEQUENCE: 875

ccccgggccc gccccgac 18

<210> SEQ ID NO 876
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<400> SEQUENCE: 876

agccccgccc cgggcccg 18

<210> SEQ ID NO 877
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<400> SEQUENCE: 877

cagccccgcc ccgggccc 18

<210> SEQ ID NO 878
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<400> SEQUENCE: 878

gcagccccgc cccgggcc 18

<210> SEQ ID NO 879
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 879

cgcagccccg ccccgggc 18

<210> SEQ ID NO 880
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 880

ccgcagcccc gccccggg 18

<210> SEQ ID NO 881
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 881

accgcagccc cgccccgg 18

<210> SEQ ID NO 882
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 882

aaccgcagcc ccgccccg 18

<210> SEQ ID NO 883
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 883

caaccgcagc cccgcccc 18

<210> SEQ ID NO 884
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 884

gcaaccgcag ccccgccc 18

<210> SEQ ID NO 885
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 885

cgcaaccgca gccccgcc 18

<210> SEQ ID NO 886
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 886

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ccgcaaccgc agccccgc 18

<210> SEQ ID NO 887
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 887

accgcaaccg cagccccg 18

<210> SEQ ID NO 888
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 888

caccgcaacc gcagcccc 18

<210> SEQ ID NO 889
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 889

gcaccgcaac cgcagccc 18

<210> SEQ ID NO 890
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 890

ggcaccgcaa ccgcagcc 18

<210> SEQ ID NO 891
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 891

aggcaccgca accgcagc 18

<210> SEQ ID NO 892
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 892

caggcaccgc aaccgcag 18

<210> SEQ ID NO 893

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 893

gcaggcaccg caaccgca 18

<210> SEQ ID NO 894
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 894

cgcaggcacc gcaaccgc 18

<210> SEQ ID NO 895
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 895

gcgcaggcac cgcaaccg 18

<210> SEQ ID NO 896
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 896

ggcgcaggca ccgcaacc 18

<210> SEQ ID NO 897
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 897

gggcgcaggc accgcaac 18

<210> SEQ ID NO 898
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 898

tgagagcaag tagtggg 17

<210> SEQ ID NO 899
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 899

gtgagagcaa gtagtgg 17

<210> SEQ ID NO 900
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 900

tgtgagagca agtagtg 17

<210> SEQ ID NO 901
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 901

ctgtgagagc aagtagt 17

<210> SEQ ID NO 902
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 902

actgtgagag caagtag 17

<210> SEQ ID NO 903
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 903

tactgtgaga gcaagta 17

<210> SEQ ID NO 904
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 904

gtactgtgag agcaagt 17

<210> SEQ ID NO 905
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 905

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agtactgtga gagcaag 17

<210> SEQ ID NO 906
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 906

gagtactgtg agagcaa 17

<210> SEQ ID NO 907
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 907

cgagtactgt gagagca 17

<210> SEQ ID NO 908
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 908

gcgagtactg tgagagc 17

<210> SEQ ID NO 909
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 909

agcgagtact gtgagag 17

<210> SEQ ID NO 910
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 910

cagcgagtac tgtgaga 17

<210> SEQ ID NO 911
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 911

tcagcgagta ctgtgag 17

<210> SEQ ID NO 912

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<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 912

ctcagcgagt actgtga 17

<210> SEQ ID NO 913
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 913

cctcagcgag tactgtg 17

<210> SEQ ID NO 914
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 914

ccctcagcga gtactgt 17

<210> SEQ ID NO 915
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 915

accctcagcg agtactg 17

<210> SEQ ID NO 916
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 916

caccctcagc gagtact 17

<210> SEQ ID NO 917
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 917

tcaccctcag cgagtac 17

<210> SEQ ID NO 918
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 918

ttcaccctca gcgagta 17

<210> SEQ ID NO 919
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 919

gttcaccctc agcgagt 17

<210> SEQ ID NO 920
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 920

tgttcaccct cagcgag 17

<210> SEQ ID NO 921
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 921

ttgttcaccc tcagcga 17

<210> SEQ ID NO 922
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 922

cttgttcacc ctcagcg 17

<210> SEQ ID NO 923
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 923

tcttgttcac cctcagc 17

<210> SEQ ID NO 924
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 924

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ttcttgttca ccctcag 17

<210> SEQ ID NO 925
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 925

tttcttgttc accctca 17

<210> SEQ ID NO 926
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 926

ttttcttgtt caccctc 17

<210> SEQ ID NO 927
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 927

cttttcttgt tcaccct 17

<210> SEQ ID NO 928
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 928

tcttttcttg ttcaccc 17

<210> SEQ ID NO 929
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 929

gtcttttctt gttcacc 17

<210> SEQ ID NO 930
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 930

ggtcttttct tgttcac 17

<210> SEQ ID NO 931

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<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 931

aggtcttttc ttgttca 17

<210> SEQ ID NO 932
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 932

caggtctttt cttgttc 17

<210> SEQ ID NO 933
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 933

tcaggtcttt tcttgtt 17

<210> SEQ ID NO 934
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 934

atcaggtctt ttcttgt 17

<210> SEQ ID NO 935
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 935

tatcaggtct tttcttg 17

<210> SEQ ID NO 936
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 936

ttatcaggtc ttttctt 17

<210> SEQ ID NO 937
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 937

atctttatca ggtcttt 17

<210> SEQ ID NO 938
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 938

aatctttatc aggtctt 17

<210> SEQ ID NO 939
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 939

taatctttat caggtct 17

<210> SEQ ID NO 940
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 940

ttaatcttta tcaggtc 17

<210> SEQ ID NO 941
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 941

gttaatcttt atcaggt 17

<210> SEQ ID NO 942
<211> LENGTH: 17
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 942

ggttaatctt tatcagg 17

<210> SEQ ID NO 943
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 943

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tggttaatct ttatcag 17

<210> SEQ ID NO 944
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 944

ctggttaatc tttatca 17

<210> SEQ ID NO 945
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 945

tctggttaat ctttatc 17

<210> SEQ ID NO 946
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 946

ccctccttgt tttcttc 17

<210> SEQ ID NO 947
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 947

tccctccttg ttttctt 17

<210> SEQ ID NO 948
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 948

ttccctcctt gttttct 17

<210> SEQ ID NO 949
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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tttccctcct tgttttc 17

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<220> FEATURE:
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gtttccctcc ttgtttt 17

<210> SEQ ID NO 951
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 951

tgtttccctc cttgttt 17

<210> SEQ ID NO 952
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<212> TYPE: DNA
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<400> SEQUENCE: 952

ttgtttccct ccttgtt 17

<210> SEQ ID NO 953
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 953

ggttgtttcc ctccttg 17

<210> SEQ ID NO 954
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cggttgtttc cctcctt 17

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gcggttgttt ccctcct 17

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgcggttgtt tccctcc 17

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<400> SEQUENCE: 957

ctgcggttgt ttccctc 17

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gctgcggttg tttccct 17

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ggctgcggtt gtttccc 17

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aggctgcggt tgtttcc 17

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caggctgcgg ttgtttc 17

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acaggctgcg gttgttt 17

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tacaggctgc ggttgtt 17

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ctacaggctg cggttgt 17

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gctacaggct gcggttg 17

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<400> SEQUENCE: 966

tgctacaggc tgcggtt 17

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ttgctacagg ctgcggt 17

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<400> SEQUENCE: 968

cttgctacag gctgcgg 17

<210> SEQ ID NO 969

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<400> SEQUENCE: 969

gcttgctaca ggctgcg 17

<210> SEQ ID NO 970
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 970

agcttgctac aggctgc 17

<210> SEQ ID NO 971
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 971

gagcttgcta caggctg 17

<210> SEQ ID NO 972
<211> LENGTH: 17
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 972

agagcttgct acaggct 17

<210> SEQ ID NO 973
<211> LENGTH: 17
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 973

cagagcttgc tacaggc 17

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<211> LENGTH: 17
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 974

ccagagcttg ctacagg 17

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 975

tccagagctt gctacag 17

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 976

ttccagagct tgctaca 17

<210> SEQ ID NO 977
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 977

gttccagagc ttgctac 17

<210> SEQ ID NO 978
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 978

agttccagag cttgcta 17

<210> SEQ ID NO 979
<211> LENGTH: 17
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 979

gagttccaga gcttgct 17

<210> SEQ ID NO 980
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 980

tgagttccag agcttgc 17

<210> SEQ ID NO 981
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 981

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ctgagttcca gagcttg 17

<210> SEQ ID NO 982
<211> LENGTH: 17
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 982

cctgagttcc agagctt 17

<210> SEQ ID NO 983
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 983

tcctgagttc cagagct 17

<210> SEQ ID NO 984
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 984

ctcctgagtt ccagagc 17

<210> SEQ ID NO 985
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 985

actcctgagt tccagag 17

<210> SEQ ID NO 986
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 986

gactcctgag ttccaga 17

<210> SEQ ID NO 987
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 987

cgactcctga gttccag 17

<210> SEQ ID NO 988

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<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 988

gcgactcctg agttcca 17

<210> SEQ ID NO 989
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 989

cgcgactcct gagttcc 17

<210> SEQ ID NO 990
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 990

gcgcgactcc tgagttc 17

<210> SEQ ID NO 991
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 991

cgcgcgactc ctgagtt 17

<210> SEQ ID NO 992
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 992

gcgcgcgact cctgagt 17

<210> SEQ ID NO 993
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 993

agcgcgcgac tcctgag 17

<210> SEQ ID NO 994
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 994

tagcgcgcga ctcctga 17

<210> SEQ ID NO 995
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 995

ctagcgcgcg actcctg 17

<210> SEQ ID NO 996
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 996

cctagcgcgc gactcct 17

<210> SEQ ID NO 997
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 997

ccctagcgcg cgactcc 17

<210> SEQ ID NO 998
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 998

cccctagcgc gcgactc 17

<210> SEQ ID NO 999
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 999

gcccctagcg cgcgact 17

<210> SEQ ID NO 1000
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1000

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ggcccctagc gcgcgac 17

<210> SEQ ID NO 1001
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1001

cggcccctag cgcgcga 17

<210> SEQ ID NO 1002
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1002

ccggccccta gcgcgcg 17

<210> SEQ ID NO 1003
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1003

cccggccccc agcgcgc 17

<210> SEQ ID NO 1004
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1004

ccccggcccc tagcgcg 17

<210> SEQ ID NO 1005
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1005

gccccggccc ctagcgc 17

<210> SEQ ID NO 1006
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1006

ggccccggcc cctagcg 17

<210> SEQ ID NO 1007

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<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1007

cggccccggc ccctagc 17

<210> SEQ ID NO 1008
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1008

ccggccccgg cccctag 17

<210> SEQ ID NO 1009
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1009

cccggccccg gccccta 17

<210> SEQ ID NO 1010
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1010

acgccccggc cccggcc 17

<210> SEQ ID NO 1011
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1011

cacgccccgg ccccggc 17

<210> SEQ ID NO 1012
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1012

ccacgccccg gccccgg 17

<210> SEQ ID NO 1013
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1013

accacgcccc ggccccg 17

<210> SEQ ID NO 1014
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1014

gaccacgccc cggcccc 17

<210> SEQ ID NO 1015
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1015

cgaccacgcc ccggccc 17

<210> SEQ ID NO 1016
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1016

ccgaccacgc cccggcc 17

<210> SEQ ID NO 1017
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1017

cccgaccacg ccccggc 17

<210> SEQ ID NO 1018
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1018

ccccgaccac gccccgg 17

<210> SEQ ID NO 1019
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1019

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gccccgacca cgccccg 17

<210> SEQ ID NO 1020
<211> LENGTH: 17
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1020

cgccccgacc acgcccc 17

<210> SEQ ID NO 1021
<211> LENGTH: 17
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1021

ccgccccgac cacgccc 17

<210> SEQ ID NO 1022
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1022

cccgccccga ccacgcc 17

<210> SEQ ID NO 1023
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1023

gcccgccccg accacgc 17

<210> SEQ ID NO 1024
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1024

ggcccgcccc gaccacg 17

<210> SEQ ID NO 1025
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1025

gggcccgccc cgaccac 17

<210> SEQ ID NO 1026

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<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1026

cgggcccgcc ccgacca 17

<210> SEQ ID NO 1027
<211> LENGTH: 17
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1027

ccgggcccgc cccgacc 17

<210> SEQ ID NO 1028
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1028

cccgggcccg ccccgac 17

<210> SEQ ID NO 1029
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1029

gcagccccgc cccgggc 17

<210> SEQ ID NO 1030
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1030

cgcagccccg ccccggg 17

<210> SEQ ID NO 1031
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1031

ccgcagcccc gccccgg 17

<210> SEQ ID NO 1032
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1032

accgcagccc cgccccg 17

<210> SEQ ID NO 1033
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1033

aaccgcagcc ccgcccc 17

<210> SEQ ID NO 1034
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1034

caaccgcagc cccgccc 17

<210> SEQ ID NO 1035
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1035

gcaaccgcag ccccgcc 17

<210> SEQ ID NO 1036
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1036

cgcaaccgca gccccgc 17

<210> SEQ ID NO 1037
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1037

ccgcaaccgc agccccg 17

<210> SEQ ID NO 1038
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1038

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accgcaaccg cagcccc 17

<210> SEQ ID NO 1039
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1039

caccgcaacc gcagccc 17

<210> SEQ ID NO 1040
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1040

gcaccgcaac cgcagcc 17

<210> SEQ ID NO 1041
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1041

ggcaccgcaa ccgcagc 17

<210> SEQ ID NO 1042
<211> LENGTH: 17
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1042

aggcaccgca accgcag 17

<210> SEQ ID NO 1043
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1043

caggcaccgc aaccgca 17

<210> SEQ ID NO 1044
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1044

gcaggcaccg caaccgc 17

<210> SEQ ID NO 1045

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<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1045

cgcaggcacc gcaaccg 17

<210> SEQ ID NO 1046
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1046

gcgcaggcac cgcaacc 17

<210> SEQ ID NO 1047
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1047

ggcgcaggca ccgcaac 17

<210> SEQ ID NO 1048
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1048

ccactcgcca ccgcctgcgc 20

<210> SEQ ID NO 1049
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1049

tgcattccta agcaatgtgt 20

<210> SEQ ID NO 1050
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1050

cccggccccg gccccggccc 20

<210> SEQ ID NO 1051
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1051

ccccggcccc ggccccggcc 20

<210> SEQ ID NO 1052
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1052

ttacatctat agcaccactc 20

<210> SEQ ID NO 1053
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1053

tcactccctt ttcagacaag 20

<210> SEQ ID NO 1054
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1054

aactaagttc tgtctgtgga 20

<210> SEQ ID NO 1055
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1055

atacaggact aaagtgcttc 20

<210> SEQ ID NO 1056
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1056

ctctgaccct gatcttccat 20

<210> SEQ ID NO 1057
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1057

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aacagctgga gatggcgg 18

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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1058

caacagctgg agatggcg 18

<210> SEQ ID NO 1059
<211> LENGTH: 18
<212> TYPE: DNA
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gcaacagctg gagatggc 18

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1060

ggcaacagct ggagatgg 18

<210> SEQ ID NO 1061
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tggcaacagc tggagatg 18

<210> SEQ ID NO 1062
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1062

ttggcaacag ctggagat 18

<210> SEQ ID NO 1063
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cttggcaaca gctggaga 18

<210> SEQ ID NO 1064

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<211> LENGTH: 18
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<220> FEATURE:
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<400> SEQUENCE: 1064

tcttggcaac agctggag 18

<210> SEQ ID NO 1065
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1065

gtcttggcaa cagctgga 18

<210> SEQ ID NO 1066
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1066

tgtcttggca acagctgg 18

<210> SEQ ID NO 1067
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1067

ctgtcttggc aacagctg 18

<210> SEQ ID NO 1068
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1068

tctgtcttgg caacagct 18

<210> SEQ ID NO 1069
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1069

ctctgtcttg gcaacagc 18

<210> SEQ ID NO 1070
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1070

tctctgtctt ggcaacag 18

<210> SEQ ID NO 1071
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1071

atctctgtct tggcaaca 18

<210> SEQ ID NO 1072
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1072

aatctctgtc ttggcaac 18

<210> SEQ ID NO 1073
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1073

caatctctgt cttggcaa 18

<210> SEQ ID NO 1074
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1074

gcaatctctg tcttggca 18

<210> SEQ ID NO 1075
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1075

agcaatctct gtcttggc 18

<210> SEQ ID NO 1076
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1076

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aagcaatctc tgtcttgg 18

<210> SEQ ID NO 1077
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1077

aaagcaatct ctgtcttg 18

<210> SEQ ID NO 1078
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1078

taaagcaatc tctgtctt 18

<210> SEQ ID NO 1079
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1079

ttaaagcaat ctctgtct 18

<210> SEQ ID NO 1080
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1080

cttaaagcaa tctctgtc 18

<210> SEQ ID NO 1081
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1081

acttaaagca atctctgt 18

<210> SEQ ID NO 1082
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1082

cacttaaagc aatctctg 18

<210> SEQ ID NO 1083

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1083

ccacttaaag caatctct 18

<210> SEQ ID NO 1084
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1084

ctgctaataa aggtgatt 18

<210> SEQ ID NO 1085
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1085

gctgctaata aaggtgat 18

<210> SEQ ID NO 1086
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1086

agctgctaat aaaggtga 18

<210> SEQ ID NO 1087
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1087

tagctgctaa taaaggtg 18

<210> SEQ ID NO 1088
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1088

gtagctgcta ataaaggt 18

<210> SEQ ID NO 1089
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1089

agtagctgct aataaagg 18

<210> SEQ ID NO 1090
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1090

aagtagctgc taataaag 18

<210> SEQ ID NO 1091
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1091

aaagtagctg ctaataaa 18

<210> SEQ ID NO 1092
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1092

aaaagtagct gctaataa 18

<210> SEQ ID NO 1093
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1093

caaaagtagc tgctaata 18

<210> SEQ ID NO 1094
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1094

gcaaaagtag ctgctaat 18

<210> SEQ ID NO 1095
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1095

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agcaaaagta gctgctaa 18

<210> SEQ ID NO 1096
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1096

aagcaaaagt agctgcta 18

<210> SEQ ID NO 1097
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1097

taagcaaaag tagctgct 18

<210> SEQ ID NO 1098
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1098

gtaagcaaaa gtagctgc 18

<210> SEQ ID NO 1099
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1099

agtaagcaaa agtagctg 18

<210> SEQ ID NO 1100
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1100

cagtaagcaa aagtagct 18

<210> SEQ ID NO 1101
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1101

ccagtaagca aaagtagc 18

<210> SEQ ID NO 1102

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1102

cccagtaagc aaaagtag 18

<210> SEQ ID NO 1103
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1103

tcccagtaag caaaagta 18

<210> SEQ ID NO 1104
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1104

gtcccagtaa gcaaaagt 18

<210> SEQ ID NO 1105
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1105

tgtcccagta agcaaaag 18

<210> SEQ ID NO 1106
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1106

ttgtcccagt aagcaaaa 18

<210> SEQ ID NO 1107
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1107

attgtcccag taagcaaa 18

<210> SEQ ID NO 1108
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1108

tattgtccca gtaagcaa 18

<210> SEQ ID NO 1109
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1109

atattgtccc agtaagca 18

<210> SEQ ID NO 1110
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1110

aatattgtcc cagtaagc 18

<210> SEQ ID NO 1111
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1111

gaatattgtc ccagtaag 18

<210> SEQ ID NO 1112
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1112

agaatattgt cccagtaa 18

<210> SEQ ID NO 1113
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1113

aagaatattg tcccagta 18

<210> SEQ ID NO 1114
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1114

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caagaatatt gtcccagt 18

<210> SEQ ID NO 1115
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1115

ccaagaatat tgtcccag 18

<210> SEQ ID NO 1116
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1116

accaagaata ttgtccca 18

<210> SEQ ID NO 1117
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1117

gaccaagaat attgtccc 18

<210> SEQ ID NO 1118
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1118

ggaccaagaa tattgtcc 18

<210> SEQ ID NO 1119
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1119

aggaccaaga atattgtc 18

<210> SEQ ID NO 1120
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1120

taggaccaag aatattgt 18

<210> SEQ ID NO 1121

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1121

ctaggaccaa gaatattg 18

<210> SEQ ID NO 1122
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1122

tctaggacca agaatatt 18

<210> SEQ ID NO 1123
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1123

ctctaggacc aagaatat 18

<210> SEQ ID NO 1124
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1124

actctaggac caagaata 18

<210> SEQ ID NO 1125
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1125

tactctagga ccaagaat 18

<210> SEQ ID NO 1126
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1126

ttactctagg accaagaa 18

<210> SEQ ID NO 1127
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1127

cttactctag gaccaaga 18

<210> SEQ ID NO 1128
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1128

ccttactcta ggaccaag 18

<210> SEQ ID NO 1129
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1129

gccttactct aggaccaa 18

<210> SEQ ID NO 1130
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1130

tgccttactc taggacca 18

<210> SEQ ID NO 1131
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1131

gtgccttact ctaggacc 18

<210> SEQ ID NO 1132
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1132

tgtgccttac tctaggac 18

<210> SEQ ID NO 1133
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1133

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atgtgcctta ctctagga 18

<210> SEQ ID NO 1134
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1134

aatgtgcctt actctagg 18

<210> SEQ ID NO 1135
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1135

aaatgtgcct tactctag 18

<210> SEQ ID NO 1136
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1136

caaatgtgcc ttactcta 18

<210> SEQ ID NO 1137
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1137

ccaaatgtgc cttactct 18

<210> SEQ ID NO 1138
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1138

cccaaatgtg ccttactc 18

<210> SEQ ID NO 1139
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1139

gcccaaatgt gccttact 18

<210> SEQ ID NO 1140

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1140

agcccaaatg tgccttac 18

<210> SEQ ID NO 1141
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1141

gagcccaaat gtgcctta 18

<210> SEQ ID NO 1142
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ggagcccaaa tgtgcctt 18

<210> SEQ ID NO 1143
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<213> ORGANISM: Artificial sequence
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tggagcccaa atgtgcct 18

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ttggagccca aatgtgcc 18

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<212> TYPE: DNA
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<400> SEQUENCE: 1145

tttggagccc aaatgtgc 18

<210> SEQ ID NO 1146
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctttggagcc caaatgtg 18

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tctttggagc ccaaatgt 18

<210> SEQ ID NO 1148
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<400> SEQUENCE: 1148

gtctttggag cccaaatg 18

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<212> TYPE: DNA
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<400> SEQUENCE: 1149

tgtctttgga gcccaaat 18

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<400> SEQUENCE: 1150

ctgtctttgg agcccaaa 18

<210> SEQ ID NO 1151
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1151

tctgtctttg gagcccaa 18

<210> SEQ ID NO 1152
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ttctgtcttt ggagccca 18

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gttctgtctt tggagccc 18

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tgttctgtct ttggagcc 18

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ctgttctgtc tttggagc 18

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cctgttctgt ctttggag 18

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acctgttctg tctttgga 18

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tacctgttct gtctttgg 18

<210> SEQ ID NO 1159

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gtacctgttc tgtctttg 18

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agtacctgtt ctgtcttt 18

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aagtacctgt tctgtctt 18

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gaagtacctg ttctgtct 18

<210> SEQ ID NO 1163
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agaagtacct gttctgtc 18

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gagaagtacc tgttctgt 18

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgagaagtac ctgttctg 18

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ctgagaagta cctgttct 18

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actgagaagt acctgttc 18

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cactgagaag tacctgtt 18

<210> SEQ ID NO 1169
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<220> FEATURE:
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tcactgagaa gtacctgt 18

<210> SEQ ID NO 1170
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1170

atcactgaga agtacctg 18

<210> SEQ ID NO 1171
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1171

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catcactgag aagtacct 18

<210> SEQ ID NO 1172
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<400> SEQUENCE: 1172

ccatcactga gaagtacc 18

<210> SEQ ID NO 1173
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1173

tccatcactg agaagtac 18

<210> SEQ ID NO 1174
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1174

ctccatcact gagaagta 18

<210> SEQ ID NO 1175
<211> LENGTH: 18
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<400> SEQUENCE: 1175

tctccatcac tgagaagt 18

<210> SEQ ID NO 1176
<211> LENGTH: 18
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1176

ttctccatca ctgagaag 18

<210> SEQ ID NO 1177
<211> LENGTH: 18
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1177

tttctccatc actgagaa 18

<210> SEQ ID NO 1178

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1178

atttctccat cactgaga 18

<210> SEQ ID NO 1179
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1179

tatttctcca tcactgag 18

<210> SEQ ID NO 1180
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1180

gttatttctc catcactg 18

<210> SEQ ID NO 1181
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1181

agttatttct ccatcact 18

<210> SEQ ID NO 1182
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1182

aagttatttc tccatcac 18

<210> SEQ ID NO 1183
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1183

aaagttattt ctccatca 18

<210> SEQ ID NO 1184
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1184

aaaagttatt tctccatc 18

<210> SEQ ID NO 1185
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1185

gaaaagttat ttctccat 18

<210> SEQ ID NO 1186
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1186

aagaaaagtt atttctcc 18

<210> SEQ ID NO 1187
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1187

caagaaaagt tatttctc 18

<210> SEQ ID NO 1188
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1188

gcaagaaaag ttatttct 18

<210> SEQ ID NO 1189
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1189

ggcaagaaaa gttatttc 18

<210> SEQ ID NO 1190
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1190

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tggcaagaaa agttattt 18

<210> SEQ ID NO 1191
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1191

ttggcaagaa aagttatt 18

<210> SEQ ID NO 1192
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1192

gttggcaaga aaagttat 18

<210> SEQ ID NO 1193
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1193

ggttggcaag aaaagtta 18

<210> SEQ ID NO 1194
<211> LENGTH: 18
<212> TYPE: DNA
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<400> SEQUENCE: 1194

tggttggcaa gaaaagtt 18

<210> SEQ ID NO 1195
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1195

gtggttggca agaaaagt 18

<210> SEQ ID NO 1196
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1196

tgtggttggc aagaaaag 18

<210> SEQ ID NO 1197

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1197

gtgtggttgg caagaaaa 18

<210> SEQ ID NO 1198
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1198

agtgtggttg gcaagaaa 18

<210> SEQ ID NO 1199
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1199

gagtgtggtt ggcaagaa 18

<210> SEQ ID NO 1200
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1200

agagtgtggt tggcaaga 18

<210> SEQ ID NO 1201
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1201

tagagtgtgg ttggcaag 18

<210> SEQ ID NO 1202
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1202

ttagagtgtg gttggcaa 18

<210> SEQ ID NO 1203
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1203

tttagagtgt ggttggca 18

<210> SEQ ID NO 1204
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1204

atttagagtg tggttggc 18

<210> SEQ ID NO 1205
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1205

catttagagt gtggttgg 18

<210> SEQ ID NO 1206
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1206

ccatttagag tgtggttg 18

<210> SEQ ID NO 1207
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1207

tccatttaga gtgtggtt 18

<210> SEQ ID NO 1208
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1208

ctccatttag agtgtggt 18

<210> SEQ ID NO 1209
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1209

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tctccattta gagtgtgg 18

<210> SEQ ID NO 1210
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1210

ttctccattt agagtgtg 18

<210> SEQ ID NO 1211
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1211

tttctccatt tagagtgt 18

<210> SEQ ID NO 1212
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1212

atttctccat ttagagtg 18

<210> SEQ ID NO 1213
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1213

gatttctcca tttagagt 18

<210> SEQ ID NO 1214
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1214

ggatttctcc atttagag 18

<210> SEQ ID NO 1215
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1215

aggatttctc catttaga 18

<210> SEQ ID NO 1216

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1216

aaggatttct ccatttag 18

<210> SEQ ID NO 1217
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1217

gaaggatttc tccattta 18

<210> SEQ ID NO 1218
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1218

cgaaggattt ctccattt 18

<210> SEQ ID NO 1219
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1219

tcgaaggatt tctccatt 18

<210> SEQ ID NO 1220
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1220

ttcgaaggat ttctccat 18

<210> SEQ ID NO 1221
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1221

tttcgaagga tttctcca 18

<210> SEQ ID NO 1222
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1222

atttcgaagg atttctcc 18

<210> SEQ ID NO 1223
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1223

catttcgaag gatttctc 18

<210> SEQ ID NO 1224
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1224

gcatttcgaa ggatttct 18

<210> SEQ ID NO 1225
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1225

tgcatttcga aggatttc 18

<210> SEQ ID NO 1226
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1226

ctgcatttcg aaggattt 18

<210> SEQ ID NO 1227
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1227

tctgcatttc gaaggatt 18

<210> SEQ ID NO 1228
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1228

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ctctgcattt cgaaggat 18

<210> SEQ ID NO 1229
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1229

tctctgcatt tcgaagga 18

<210> SEQ ID NO 1230
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1230

ctctctgcat ttcgaagg 18

<210> SEQ ID NO 1231
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1231

actctctgca tttcgaag 18

<210> SEQ ID NO 1232
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1232

cactctctgc atttcgaa 18

<210> SEQ ID NO 1233
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1233

ccactctctg catttcga 18

<210> SEQ ID NO 1234
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1234

accactctct gcatttcg 18

<210> SEQ ID NO 1235

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1235

caccactctc tgcatttc 18

<210> SEQ ID NO 1236
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1236

gcaccactct ctgcattt 18

<210> SEQ ID NO 1237
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1237

agcaccactc tctgcatt 18

<210> SEQ ID NO 1238
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1238

tagcaccact ctctgcat 18

<210> SEQ ID NO 1239
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1239

atagcaccac tctctgca 18

<210> SEQ ID NO 1240
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1240

tatagcacca ctctctgc 18

<210> SEQ ID NO 1241
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1241

ctatagcacc actctctg 18

<210> SEQ ID NO 1242
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<400> SEQUENCE: 1242

tctatagcac cactctct 18

<210> SEQ ID NO 1243
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atctatagca ccactctc 18

<210> SEQ ID NO 1244
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1244

catctatagc accactct 18

<210> SEQ ID NO 1245
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1245

acatctatag caccactc 18

<210> SEQ ID NO 1246
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1246

tacatctata gcaccact 18

<210> SEQ ID NO 1247
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1247

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ttacatctat agcaccac 18

<210> SEQ ID NO 1248
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1248

tttacatcta tagcacca 18

<210> SEQ ID NO 1249
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctttacatct atagcacc 18

<210> SEQ ID NO 1250
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1250

actttacatc tatagcac 18

<210> SEQ ID NO 1251
<211> LENGTH: 18
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1251

aactttacat ctatagca 18

<210> SEQ ID NO 1252
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<212> TYPE: DNA
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<400> SEQUENCE: 1252

aaactttaca tctatagc 18

<210> SEQ ID NO 1253
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aaaactttac atctatag 18

<210> SEQ ID NO 1254

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<211> LENGTH: 18
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1254

aaaaaacttt acatctat 18

<210> SEQ ID NO 1255
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1255

caaaaaactt tacatcta 18

<210> SEQ ID NO 1256
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1256

acaaaaaact ttacatct 18

<210> SEQ ID NO 1257
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1257

gacaaaaaac tttacatc 18

<210> SEQ ID NO 1258
<211> LENGTH: 18
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1258

caagacaaaa aactttac 18

<210> SEQ ID NO 1259
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1259

gacaagacaa aaaacttt 18

<210> SEQ ID NO 1260
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1260

agacaagaca aaaaactt 18

<210> SEQ ID NO 1261
<211> LENGTH: 18
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1261

cagacaagac aaaaaact 18

<210> SEQ ID NO 1262
<211> LENGTH: 18
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<400> SEQUENCE: 1262

tcagacaaga caaaaaac 18

<210> SEQ ID NO 1263
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1263

ttcagacaag acaaaaaa 18

<210> SEQ ID NO 1264
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1264

ttttcagaca agacaaaa 18

<210> SEQ ID NO 1265
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1265

cttttcagac aagacaaa 18

<210> SEQ ID NO 1266
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1266

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cctttttcaga caagacaa 18

<210> SEQ ID NO 1267
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1267

ccctttttcag acaagaca 18

<210> SEQ ID NO 1268
<211> LENGTH: 18
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1268

tccctttttca gacaagac 18

<210> SEQ ID NO 1269
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1269

ctccctttttc agacaaga 18

<210> SEQ ID NO 1270
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1270

actccctttt cagacaag 18

<210> SEQ ID NO 1271
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1271

cactcccttt tcagacaa 18

<210> SEQ ID NO 1272
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1272

tcactccctt ttcagaca 18

<210> SEQ ID NO 1273

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1273

atcactccct tttcagac 18

<210> SEQ ID NO 1274
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1274

aatcactccc ttttcaga 18

<210> SEQ ID NO 1275
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1275

taatcactcc cttttcag 18

<210> SEQ ID NO 1276
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1276

ataatcactc ccttttca 18

<210> SEQ ID NO 1277
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1277

aataatcact ccctttc 18

<210> SEQ ID NO 1278
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1278

caataatcac tcccttt 18

<210> SEQ ID NO 1279
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1279

acaataatca ctcccttt 18

<210> SEQ ID NO 1280
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1280

aacaataatc actccctt 18

<210> SEQ ID NO 1281
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1281

aaacaataat cactccct 18

<210> SEQ ID NO 1282
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1282

gaaacaataa tcactccc 18

<210> SEQ ID NO 1283
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1283

tgaaacaata atcactcc 18

<210> SEQ ID NO 1284
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1284

atgaaacaat aatcactc 18

<210> SEQ ID NO 1285
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1285

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aatgaaacaa taatcact 18

<210> SEQ ID NO 1286
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1286

taatgaaaca ataatcac 18

<210> SEQ ID NO 1287
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1287

ttaatgaaac aataatca 18

<210> SEQ ID NO 1288
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1288

attaatgaaa caataatc 18

<210> SEQ ID NO 1289
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1289

atcaaagatt aatgaaac 18

<210> SEQ ID NO 1290
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1290

catcaaagat taatgaaa 18

<210> SEQ ID NO 1291
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1291

ccatcaaaga ttaatgaa 18

<210> SEQ ID NO 1292

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1292

tccatcaaag attaatga 18

<210> SEQ ID NO 1293
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1293

ttccatcaaa gattaatg 18

<210> SEQ ID NO 1294
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1294

tttccatcaa agattaat 18

<210> SEQ ID NO 1295
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1295

gtttccatca aagattaa 18

<210> SEQ ID NO 1296
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1296

agtttccatc aaagatta 18

<210> SEQ ID NO 1297
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1297

cagtttccat caaagatt 18

<210> SEQ ID NO 1298
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1298

ccagtttcca tcaaagat 18

<210> SEQ ID NO 1299
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1299

tccagtttcc atcaaaga 18

<210> SEQ ID NO 1300
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1300

ttccagtttc catcaaag 18

<210> SEQ ID NO 1301
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1301

attccagttt ccatcaaa 18

<210> SEQ ID NO 1302
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1302

cattccagtt tccatcaa 18

<210> SEQ ID NO 1303
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1303

ccattccagt ttccatca 18

<210> SEQ ID NO 1304
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1304

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cccattccag tttccatc 18

<210> SEQ ID NO 1305
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1305

ccccattcca gtttccat 18

<210> SEQ ID NO 1306
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1306

tccccattcc agtttcca 18

<210> SEQ ID NO 1307
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1307

atccccattc cagtttcc 18

<210> SEQ ID NO 1308
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1308

gatccccatt ccagtttc 18

<210> SEQ ID NO 1309
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1309

cgatccccat tccagttt 18

<210> SEQ ID NO 1310
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1310

gcgatcccca ttccagtt 18

<210> SEQ ID NO 1311

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1311

tgcgatcccc attccagt 18

<210> SEQ ID NO 1312
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1312

ctgcgatccc cattccag 18

<210> SEQ ID NO 1313
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1313

gctgcgatcc ccattcca 18

<210> SEQ ID NO 1314
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1314

tgctgcgatc cccattcc 18

<210> SEQ ID NO 1315
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1315

gtgctgcgat ccccattc 18

<210> SEQ ID NO 1316
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1316

tgtgctgcga tccccatt 18

<210> SEQ ID NO 1317
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1317

atgtgctgcg atccccat 18

<210> SEQ ID NO 1318
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1318

tatgtgctgc gatcccca 18

<210> SEQ ID NO 1319
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1319

aagtataatt gatagtcc 18

<210> SEQ ID NO 1320
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1320

gaagtataat tgatagtc 18

<210> SEQ ID NO 1321
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1321

ggaagtataa ttgatagt 18

<210> SEQ ID NO 1322
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1322

tggaagtata attgatag 18

<210> SEQ ID NO 1323
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1323

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gtggaagtat aattgata 18

<210> SEQ ID NO 1324
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1324

tgtggaagta taattgat 18

<210> SEQ ID NO 1325
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1325

ctgtggaagt ataattga 18

<210> SEQ ID NO 1326
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1326

tctgtggaag tataattg 18

<210> SEQ ID NO 1327
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1327

gtctgtggaa gtataatt 18

<210> SEQ ID NO 1328
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1328

tgtctgtgga agtataat 18

<210> SEQ ID NO 1329
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1329

ctgtctgtgg aagtataa 18

<210> SEQ ID NO 1330

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1330

tctgtctgtg gaagtata 18

<210> SEQ ID NO 1331
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1331

ttctgtctgt ggaagtat 18

<210> SEQ ID NO 1332
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1332

gttctgtctg tggaagta 18

<210> SEQ ID NO 1333
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1333

agttctgtct gtggaagt 18

<210> SEQ ID NO 1334
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1334

aagttctgtc tgtggaag 18

<210> SEQ ID NO 1335
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1335

taagttctgt ctgtggaa 18

<210> SEQ ID NO 1336
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1336

ctaagttctg tctgtgga 18

<210> SEQ ID NO 1337
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1337

actaagttct gtctgtgg 18

<210> SEQ ID NO 1338
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1338

aactaagttc tgtctgtg 18

<210> SEQ ID NO 1339
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1339

aaactaagtt ctgtctgt 18

<210> SEQ ID NO 1340
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1340

gaaactaagt tctgtctg 18

<210> SEQ ID NO 1341
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1341

agaaactaag ttctgtct 18

<210> SEQ ID NO 1342
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1342

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tagaaactaa gttctgtc 18

<210> SEQ ID NO 1343
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1343

gtagaaacta agttctgt 18

<210> SEQ ID NO 1344
<211> LENGTH: 18
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<400> SEQUENCE: 1344

ggtagaaact aagttctg 18

<210> SEQ ID NO 1345
<211> LENGTH: 18
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1345

aggtagaaac taagttct 18

<210> SEQ ID NO 1346
<211> LENGTH: 18
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1346

gaggtagaaa ctaagttc 18

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1347

ggaggtagaa actaagtt 18

<210> SEQ ID NO 1348
<211> LENGTH: 18
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gggaggtaga aactaagt 18

<210> SEQ ID NO 1349

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<400> SEQUENCE: 1349

tgggaggtag aaactaag 18

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<211> LENGTH: 18
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1350

gtgggaggta gaaactaa 18

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<400> SEQUENCE: 1351

agtgggaggt agaaacta 18

<210> SEQ ID NO 1352
<211> LENGTH: 18
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<400> SEQUENCE: 1352

aagtgggagg tagaaact 18

<210> SEQ ID NO 1353
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<400> SEQUENCE: 1353

gaagtgggag gtagaaac 18

<210> SEQ ID NO 1354
<211> LENGTH: 18
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<400> SEQUENCE: 1354

tgaagtggga ggtagaaa 18

<210> SEQ ID NO 1355
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1355

atgaagtggg aggtagaa 18

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1356

tatgaagtgg gaggtaga 18

<210> SEQ ID NO 1357
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1357

ctatgaagtg ggaggtag 18

<210> SEQ ID NO 1358
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1358

tctatgaagt gggaggta 18

<210> SEQ ID NO 1359
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1359

ctctatgaag tgggaggt 18

<210> SEQ ID NO 1360
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1360

actctatgaa gtgggagg 18

<210> SEQ ID NO 1361
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1361

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cactctatga agtgggag 18

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1362

acactctatg aagtggga 18

<210> SEQ ID NO 1363
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1363

cacactctat gaagtggg 18

<210> SEQ ID NO 1364
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1364

acacactcta tgaagtgg 18

<210> SEQ ID NO 1365
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1365

cacacactct atgaagtg 18

<210> SEQ ID NO 1366
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1366

acacacactc tatgaagt 18

<210> SEQ ID NO 1367
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1367

ccctgatctt ccattctc 18

<210> SEQ ID NO 1368

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1368

accctgatct tccattct 18

<210> SEQ ID NO 1369
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1369

gaccctgatc ttccattc 18

<210> SEQ ID NO 1370
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1370

tgaccctgat cttccatt 18

<210> SEQ ID NO 1371
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1371

ctgaccctga tcttccat 18

<210> SEQ ID NO 1372
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1372

tctgaccctg atcttcca 18

<210> SEQ ID NO 1373
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1373

ctctgaccct gatcttcc 18

<210> SEQ ID NO 1374
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1374

actctgaccc tgatcttc 18

<210> SEQ ID NO 1375
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1375

tactctgacc ctgatctt 18

<210> SEQ ID NO 1376
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1376

atactctgac cctgatct 18

<210> SEQ ID NO 1377
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1377

aatactctga ccctgatc 18

<210> SEQ ID NO 1378
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1378

atgatttctt gtctggga 18

<210> SEQ ID NO 1379
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1379

catgatttct tgtctggg 18

<210> SEQ ID NO 1380
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1380

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ccatgatttc ttgtctgg 18

<210> SEQ ID NO 1381
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1381

gccatgattt cttgtctg 18

<210> SEQ ID NO 1382
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1382

ggccatgatt tcttgtct 18

<210> SEQ ID NO 1383
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1383

gggccatgat ttcttgtc 18

<210> SEQ ID NO 1384
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1384

aactaacatg taggcact 18

<210> SEQ ID NO 1385
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1385

gaactaacat gtaggcac 18

<210> SEQ ID NO 1386
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1386

ggaactaaca tgtaggca 18

<210> SEQ ID NO 1387

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1387

aggaactaac atgtaggc 18

<210> SEQ ID NO 1388
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1388

cttctgattc aagccatt 18

<210> SEQ ID NO 1389
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1389

gcttctgatt caagccat 18

<210> SEQ ID NO 1390
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1390

tgcttctgat tcaagcca 18

<210> SEQ ID NO 1391
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1391

gtgcttctga ttcaagcc 18

<210> SEQ ID NO 1392
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1392

agtgcttctg attcaagc 18

<210> SEQ ID NO 1393
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1393

aagtgcttct gattcaag 18

<210> SEQ ID NO 1394
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1394

aaagtgcttc tgattcaa 18

<210> SEQ ID NO 1395
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1395

taaagtgctt ctgattca 18

<210> SEQ ID NO 1396
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1396

ctaaagtgct tctgattc 18

<210> SEQ ID NO 1397
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1397

actaaagtgc ttctgatt 18

<210> SEQ ID NO 1398
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1398

gactaaagtg cttctgat 18

<210> SEQ ID NO 1399
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1399

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ggactaaagt gcttctga 18

<210> SEQ ID NO 1400
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1400

aggactaaag tgcttctg 18

<210> SEQ ID NO 1401
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1401

caggactaaa gtgcttct 18

<210> SEQ ID NO 1402
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1402

acaggactaa agtgcttc 18

<210> SEQ ID NO 1403
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1403

tacaggacta aagtgctt 18

<210> SEQ ID NO 1404
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1404

atacaggact aaagtgct 18

<210> SEQ ID NO 1405
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1405

gatacaggac taaagtgc 18

<210> SEQ ID NO 1406

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1406

agatacagga ctaaagtg 18

<210> SEQ ID NO 1407
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1407

cagatacagg actaaagt 18

<210> SEQ ID NO 1408
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1408

acagatacag gactaaag 18

<210> SEQ ID NO 1409
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1409

aacagataca ggactaaa 18

<210> SEQ ID NO 1410
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1410

gaacagatac aggactaa 18

<210> SEQ ID NO 1411
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1411

tgaacagata caggacta 18

<210> SEQ ID NO 1412
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1412

ctgaacagat acaggact 18

<210> SEQ ID NO 1413
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1413

actgaacaga tacaggac 18

<210> SEQ ID NO 1414
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1414

cactgaacag atacagga 18

<210> SEQ ID NO 1415
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1415

acactgaaca gatacagg 18

<210> SEQ ID NO 1416
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1416

gacactgaac agatacag 18

<210> SEQ ID NO 1417
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1417

tgacactgaa cagataca 18

<210> SEQ ID NO 1418
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1418

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ctgacactga acagatac 18

<210> SEQ ID NO 1419
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1419

gctgacactg aacagata 18

<210> SEQ ID NO 1420
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1420

ggctgacact gaacagat 18

<210> SEQ ID NO 1421
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1421

aggctgacac tgaacaga 18

<210> SEQ ID NO 1422
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1422

aaggctgaca ctgaacag 18

<210> SEQ ID NO 1423
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1423

aaaggctgac actgaaca 18

<210> SEQ ID NO 1424
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1424

gaaaggctga cactgaac 18

<210> SEQ ID NO 1425

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<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1425

tgaaaggctg acactgaa 18

<210> SEQ ID NO 1426
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1426

tgggatttaa aatgatgt 18

<210> SEQ ID NO 1427
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1427

atgggattta aaatgatg 18

<210> SEQ ID NO 1428
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1428

cttgagaaga aagccttc 18

<210> SEQ ID NO 1429
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1429

attaaggctc ttaggtta 18

<210> SEQ ID NO 1430
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1430

gtagacagtc tgttattt 18

<210> SEQ ID NO 1431
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1431

tgacatgtag agagatta 18

<210> SEQ ID NO 1432
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1432

tggtttaagg gcacaaac 18

<210> SEQ ID NO 1433
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1433

acttgagaag aaagcctt 18

<210> SEQ ID NO 1434
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1434

cctctgatac tccatcat 18

<210> SEQ ID NO 1435
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1435

aaatcttgtc ataggtga 18

<210> SEQ ID NO 1436
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1436

aattcttact tgagaaga 18

<210> SEQ ID NO 1437
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ggtgtataga gaattcag 18

<210> SEQ ID NO 1438
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1438

tactccatca tgagccta 18

<210> SEQ ID NO 1439
<211> LENGTH: 18
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1439

gctggatgga aaaagatc 18

<210> SEQ ID NO 1440
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1440

gtccctagaa caatctaa 18

<210> SEQ ID NO 1441
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1441

gaagaaattg acatgtag 18

<210> SEQ ID NO 1442
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<212> TYPE: DNA
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<400> SEQUENCE: 1442

catctacagt acaactta 18

<210> SEQ ID NO 1443
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<400> SEQUENCE: 1443

atctctgtct tggcaacagc 20

<210> SEQ ID NO 1444

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<211> LENGTH: 20
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aatctctgtc ttggcaacag 20

<210> SEQ ID NO 1445
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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caatctctgt cttggcaaca 20

<210> SEQ ID NO 1446
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1446

aagcaatctc tgtcttggca 20

<210> SEQ ID NO 1447
<211> LENGTH: 20
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1447

ttactctagg accaagaata 20

<210> SEQ ID NO 1448
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<220> FEATURE:
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<400> SEQUENCE: 1448

cttactctag gaccaagaat 20

<210> SEQ ID NO 1449
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1449

tgccttactc taggaccaag 20

<210> SEQ ID NO 1450
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgtgccttac tctaggacca 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1451

atgtgcctta ctctaggacc 20

<210> SEQ ID NO 1452
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cccaaatgtg ccttactcta 20

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gcccaaatgt gccttactct 20

<210> SEQ ID NO 1454
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<212> TYPE: DNA
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<400> SEQUENCE: 1454

gagcccaaat gtgccttact 20

<210> SEQ ID NO 1455
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1455

ggagcccaaa tgtgccttac 20

<210> SEQ ID NO 1456
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1456

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ttggagccca aatgtgcctt 20

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1457

tttggagccc aaatgtgcct 20

<210> SEQ ID NO 1458
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1458

ctgtctttgg agcccaaatg 20

<210> SEQ ID NO 1459
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1459

tgttctgtct ttggagccca 20

<210> SEQ ID NO 1460
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1460

cctgttctgt ctttggagcc 20

<210> SEQ ID NO 1461
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1461

acctgttctg tctttggagc 20

<210> SEQ ID NO 1462
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1462

agtacctgtt ctgtctttgg 20

<210> SEQ ID NO 1463

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1463

catcactgag aagtacctgt 20

<210> SEQ ID NO 1464
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1464

ctccatcact gagaagtacc 20

<210> SEQ ID NO 1465
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1465

ccactctctg catttcgaag 20

<210> SEQ ID NO 1466
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1466

accactctct gcatttcgaa 20

<210> SEQ ID NO 1467
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1467

gcaccactct ctgcatttcg 20

<210> SEQ ID NO 1468
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1468

atagcaccac tctctgcatt 20

<210> SEQ ID NO 1469
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1469

tatagcacca ctctctgcat 20

<210> SEQ ID NO 1470
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1470

acatctatag caccactctc 20

<210> SEQ ID NO 1471
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1471

tacatctata gcaccactct 20

<210> SEQ ID NO 1472
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1472

tttacatcta tagcaccact 20

<210> SEQ ID NO 1473
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1473

ctttacatct atagcaccac 20

<210> SEQ ID NO 1474
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1474

aactttacat ctatagcacc 20

<210> SEQ ID NO 1475
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1475

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aaactttaca tctatagcac 20

<210> SEQ ID NO 1476
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1476

ctccctttc agacaagaca 20

<210> SEQ ID NO 1477
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1477

actcccttt cagacaagac 20

<210> SEQ ID NO 1478
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1478

atcactccct tttcagacaa 20

<210> SEQ ID NO 1479
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1479

taatcactcc cttttcagac 20

<210> SEQ ID NO 1480
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1480

ataatcactc ccttttcaga 20

<210> SEQ ID NO 1481
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1481

atccccattc cagtttccat 20

<210> SEQ ID NO 1482

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1482

gatccccatt ccagtttcca 20

<210> SEQ ID NO 1483
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1483

gcgatcccca ttccagtttc 20

<210> SEQ ID NO 1484
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1484

tgcgatcccc attccagttt 20

<210> SEQ ID NO 1485
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1485

gctgcgatcc ccattccagt 20

<210> SEQ ID NO 1486
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1486

tgctgcgatc cccattccag 20

<210> SEQ ID NO 1487
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1487

tgtgctgcga tccccattcc 20

<210> SEQ ID NO 1488
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1488

atgtgctgcg atccccattc 20

<210> SEQ ID NO 1489
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1489

actaagttct gtctgtggaa 20

<210> SEQ ID NO 1490
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1490

gaaactaagt tctgtctgtg 20

<210> SEQ ID NO 1491
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1491

agaaactaag ttctgtctgt 20

<210> SEQ ID NO 1492
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1492

tgaccctgat cttccattct 20

<210> SEQ ID NO 1493
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1493

ctgaccctga tcttccattc 20

<210> SEQ ID NO 1494
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1494

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actctgaccc tgatcttcca 20

<210> SEQ ID NO 1495
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1495

atactctgac cctgatcttc 20

<210> SEQ ID NO 1496
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1496

aatactctga ccctgatctt 20

<210> SEQ ID NO 1497
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1497

ccatgatttc ttgtctggga 20

<210> SEQ ID NO 1498
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1498

ggccatgatt tcttgtctgg 20

<210> SEQ ID NO 1499
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1499

gggccatgat ttcttgtctg 20

<210> SEQ ID NO 1500
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1500

acttaagttc atctacagta 20

<210> SEQ ID NO 1501

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1501

tccactgctg gatggaaaaa 20

<210> SEQ ID NO 1502
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1502

atattattta tcttactcaa 20

<210> SEQ ID NO 1503
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1503

gttctaagtg ctttatatta 20

<210> SEQ ID NO 1504
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1504

aggaactaac atgtaggcac 20

<210> SEQ ID NO 1505
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1505

atataagata atacatgtaa 20

<210> SEQ ID NO 1506
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1506

catcatgagc ctaaaggaaa 20

<210> SEQ ID NO 1507
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1507

ccatcatgag cctaaaggaa 20

<210> SEQ ID NO 1508
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1508

cttctgattc aagccattaa 20

<210> SEQ ID NO 1509
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1509

tgcttctgat tcaagccatt 20

<210> SEQ ID NO 1510
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1510

gtgcttctga ttcaagccat 20

<210> SEQ ID NO 1511
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1511

agtgcttctg attcaagcca 20

<210> SEQ ID NO 1512
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1512

aagtgcttct gattcaagcc 20

<210> SEQ ID NO 1513
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1513

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aaagtgcttc tgattcaagc 20

<210> SEQ ID NO 1514
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1514

taaagtgctt ctgattcaag 20

<210> SEQ ID NO 1515
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1515

ctaaagtgct tctgattcaa 20

<210> SEQ ID NO 1516
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1516

actaaagtgc ttctgattca 20

<210> SEQ ID NO 1517
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1517

gactaaagtg cttctgattc 20

<210> SEQ ID NO 1518
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1518

ggactaaagt gcttctgatt 20

<210> SEQ ID NO 1519
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1519

aggactaaag tgcttctgat 20

<210> SEQ ID NO 1520

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<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1520

caggactaaa gtgcttctga 20

<210> SEQ ID NO 1521
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1521

acaggactaa agtgcttctg 20

<210> SEQ ID NO 1522
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1522

tacaggacta aagtgcttct 20

<210> SEQ ID NO 1523
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1523

gatacaggac taaagtgctt 20

<210> SEQ ID NO 1524
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1524

agatacagga ctaaagtgct 20

<210> SEQ ID NO 1525
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1525

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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acagatacag gactaaagtg 20

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aacagataca ggactaaagt 20

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ctgaacagat acaggactaa 20

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actgaacaga tacaggacta 20

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acactgaaca gatacaggac 20

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gacactgaac agatacagga 20

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tgacactgaa cagatacagg 20

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<212> TYPE: DNA
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ctgacactga acagatacag 20

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ggctgacact gaacagatac 20

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aggctgacac tgaacagata 20

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aaaggctgac actgaacaga 20

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tgaaaggctg acactgaaca 20

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<210> SEQ ID NO 1544
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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caaatgggat ttaaaatgat 20

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ttccaatgct tactggagaa gtga 24

<210> SEQ ID NO 1547
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ggaacactgt gtgatttcat agatga 26

<210> SEQ ID NO 1548
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tcctgtaatg gaactgc 17

What is claimed is:

1. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases of any of the nucleobase sequences of SEQ ID NOs: 20-401 and 441-1545.

2. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1107-1520 of SEQ ID NO: 2.

3. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1111-1200 of SEQ ID NO: 2.

4. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1211-1318 of SEQ ID NO: 2.

5. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1326-1540 of SEQ ID NO: 2.

6. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1331-1375 of SEQ ID NO: 2.

7. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1368-1391 of SEQ ID NO: 2.

8. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15,

at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1398-1424 of SEQ ID NO: 2.

9. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1411-1440 of SEQ ID NO: 2.

10. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1429-1481 of SEQ ID NO: 2.

11. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1502-1539 of SEQ ID NO: 2.

12. A compound, comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and comprising a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 consecutive nucleobases complementary to an equal length portion of nucleobases 1508-1539 of SEQ ID NO: 2.

13. The compound of claims **2-5**, wherein the nucleobase sequence of the modified oligonucleotide is at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% complementary to SEQ ID NO: 1.

14. The compound of any preceding claim, consisting of a single-stranded modified oligonucleotide.

15. The compound of any preceding claim, wherein at least one internucleoside linkage is a modified internucleoside linkage.

16. The compound of claim **8**, wherein at least one modified internucleoside linkage is a phosphorothioate internucleoside linkage.

17. The compound of claim **8**, wherein each modified internucleoside linkage is a phosphorothioate internucleoside linkage.

18. The compound of any preceding claim, wherein at least one internucleoside linkage is a phosphodiester internucleoside linkage.

19. The compound of any preceding claim, wherein at least one internucleoside linkage is a phosphorothioate linkage and at least one internucleoside linkage is a phosphodiester linkage.

20. The compound of any preceding claim, wherein at least one nucleoside comprises a modified nucleobase.

21. The compound of claim **13**, wherein the modified nucleobase is a 5-methylcytosine.

22. The compound of any preceding claim, wherein at least one nucleoside of the modified oligonucleotide comprises a modified sugar.

23. The compound of claim **15**, wherein the at least one modified sugar is a bicyclic sugar.

24. The compound of claim **16**, wherein the bicyclic sugar comprises a chemical link between the 2' and 4' position of the sugar 4'-CH2-N(R)—O-2' bridge wherein R is, independently, H, C1-C12 alkyl, or a protecting group.

25. The compound of claim **16**, wherein the bicyclic sugar comprises a 4'-CH2-N(R)—O-2' bridge wherein R is, independently, H, C1-C12 alkyl, or a protecting group.

26. The compound of claim **15**, wherein at least one modified sugar comprises a 2'-O-methoxyethyl group.

27. The compound of claim **15**, wherein the modified sugar comprises a 2'-$O(CH_2)_2$—OCH_3 group.

28. The compound of any preceding claim, wherein the modified oligonucleotide comprises:

a gap segment consisting of 10 linked deoxynucleosides;

a 5' wing segment consisting of 5 linked nucleosides; and

a 3' wing segment consisting of 5 linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

29. The compound of any preceding claim, wherein the modified oligonucleotide comprises:

a gap segment consisting of 8 linked deoxynucleosides;

a 5' wing segment consisting of 5 linked nucleosides; and

a 3' wing segment consisting of 5 linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

30. The compound of any preceeding claim, wherein the modified oligonucleotide comprises sugar modifications in any of the following patterns: eeekkddddddkkeee, eekkddddddddkkeee, ekddddddddekekeee, kekeddddddddekeke, and ekekddddddddkekee; wherein,

e=a 2'-O-methoxyethyl modified nucleoside

d=a 2'-deoxynucleoside, and

k=a cEt nucleoside.

31. The compound of any preceeding claim, wherein the modified oligonucleotide comprises internucleoside linkages in any of the following patterns: soooosssssssssssooss, sooossssssssooss, soossssssssooss, and sossssssssoooss; wherein,

s=a phosphorothioate linkage, and

o=a phosphodiester linkage.

32. The compound of any preceding claim, wherein the modified oligonucleotide consists of 20 linked nucleosides.

33. The compound of any preceding claim, wherein the modified oligonucleotide consists of 19 linked nucleosides.

34. The compound of any preceding claim, wherein the modified oligonucleotide consists of 18 linked nucleosides.

35. The compound of any preceding claim, wherein the modified oligonucleotide consists of 17 linked nucleosides.

36. A composition comprising the compound of any preceding claim or salt thereof and at least one of a pharmaceutically acceptable carrier or diluent.

37. A method comprising administering to an animal the compound or composition of any preceding claim.

38. The method of claim **37**, wherein the animal is a human.

39. The method of claim **38**, wherein administering the compound prevents, treats, ameliorates, or slows progression of a C9ORF72 associated disease, disorder or condition.

40. The method of claim **38**, wherein administering the compound prevents, treats, ameliorates, or slows progression of a C9ORF72 hexanucleotide repeat expansion associated disease, disorder or condition.

41. The method of claim **39** or **40**, wherein the disease, disorder or condition is amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), corticalbasal degeneration syndrome (CBD), atypical Parkinsonian syndrome, and olivopontocerellar degeneration (OPCD).

42. The method of claim **37**, wherein the administering reduces nuclear foci.

43. The method of claim **37**, wherein the administering reduces expression of C9ORF72 associated RAN translation products.

44. The method claim **43**, wherein the C9ORF72 associated RAN translation products are any of poly-(glycine-proline), poly-(glycine-alanine), and poly-(glycine-arginine).

45. Use of the compound or composition of any preceding claim for the manufacture of a medicament for treating a neurodegenerative disorder.

46. A method of selectively inhibiting a C9ORF72 pathogenic associated mRNA variant by administering an antisense compound targeting the region beginning at the start site of exon 1A to the start site of exon 1B of a C9ORF72 pre-mRNA.

* * * * *